Lecture Notes
STA2002 Probability and Statistics II
Mathematical Statistics/Statistical Inference

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*Correspondence email address: jiangsheng@cuhk.edu.cn . Sections marked by ☕ are not covered in the lecture. These are unpolished, incomplete course notes. Use with caution.

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1 Probability Theory Review

Reading(s) 1.1. Schervish and DeGroot (2014) Section 4.1 and 4.2 (expectation), 4.3 (variance), 4.4 (mgf), 6.2 (LLN), 6.3 (CLT); Wasserman (2004) Section 3.1, 3.2, 3.4, 5.2, 5.3.

The materials presented in this section are taken from Wasserman (2004). Most of these topics are covered in the pre-requisite probability theory class. If most materials are new to you, you probably are not ready to take this course.

1.1 Probability

Sample Spaces and Events

- Sample space Ω and some outcome $\omega \in \Omega$.
- Events: subsets of Ω
- Basic set operations: union $\cup$, intersection $\cap$, complement E^C
- Set relations: inclusion $\subset$, disjoint, mutually exclusive, ...
- $\sigma-$algebra, or $\sigma-$field
- Measurable space $(\Omega, \mathcal{F})$, probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Probability

- Probability (measure) $\mathbb{P}$ satisfies three axioms:
  - positivity
  - unity
  - countable additivity
- Useful properties
  - Union identity: for any two events A and B,
$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$
  - Continuity: for a sequence of events A_n such that $A_n \to A$, then, as $n \to \infty$,
$$\mathbb{P}(A_n) \to \mathbb{P}(A).$$
    Example: $A_n = [\frac{1}{n}, 1) \to (0, 1)$, $\mathbb{P}(\cdot)$ is the probability of uniform distribution on $[0, 1]$. $\mathbb{P}(A_n) = 1 - \frac{1}{n} \to 1$ which is the same as $\mathbb{P}((0, 1)) = 1$.

Counting methods

- Equally likely outcomes.
- Permutation and combination

Independence

- Two events A and B are independent, $A \perp B$, if and only if (*iff*) $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.
- Independence for a collection of events $\{A_i\}_{i \in I}$

Conditional Probability

- If $\mathbb{P}(B) > 0$, then the probability of A occurring given B occurs is

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

- Multiplicative rule:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A),$$

assuming both $\mathbb{P}(A)$ and $\mathbb{P}(B)$ are strictly positive.

Bayes' Theorem

- The Law of Total Probability: let $E_1, E_2, ..., E_k$ be a partition of Ω, then for any event B,

$$\mathbb{P}(B) = \sum_{i=1}^{k} \mathbb{P}(B|E_i)\mathbb{P}(E_i).$$

- Bayes' Theorem:

$$\mathbb{P}(E_i|B) = \frac{\mathbb{P}(B|E_i)\mathbb{P}(E_i)}{\sum_{i=1}^{k} \mathbb{P}(B|E_i)\mathbb{P}(E_i)}$$

1.2 The theory of random variables

Random Variables

- Definition: random variable X is a mapping

$$X : \Omega \to \mathbb{R}.$$

- For any (measurable) event A, $X(A) = \{X(\omega) : \omega \in A\} \subset \mathbb{R}$.
- For any subset B of $\mathbb{R}$, $X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}$ is an event.
- A random variable is discrete if it takes at most countably many values.
- A random variable is continuous if it admits a pdf (see definition below).

Distribution Functions and Probability Functions

- The cumulative distribution function, or CDF, of random variable X is defined as, for all $x \in \mathbb{R}$,

$$F_X(x) = \mathbb{P}(X \leq x).$$

A function $F : \mathbb{R} \to [0, 1]$ is a CDF of some random variable, *iff* F satisfies the following conditions

(i) non-decreasing

(ii) normalized

(iii) right-continuous.

- Suppose X is discrete, define the probability function or probability mass function (pmf) for X as follows,
$$f_X(x) = \mathbb{P}(X = x).$$

- A random variable X is continuous, if there exists a function $f_X(x)$ such that

  (i) $f_X(x) \geq 0$ for all $x \in \mathbb{R}$,

  (ii) $\int_{-\infty}^{\infty} f_X(x)dx = 1$

  (iii) for every $a \leq b$,
  $$\mathbb{P}(X \in (a, b)) = \int_a^b f_X(x)dx.$$

  The function $f_X(x)$ is the probability density function (pdf).

- Relation between pdf and CDF:
  - From pdf to CDF: $F_X(x) = \int_{-\infty}^{x} f_X(t)dt$.
  - From CDF to pdf: $f_X(x) = \frac{d}{dx}F_X(x)$ at all x where F_X is differentiable.

- Properties of CDF:
  - $\mathbb{P}(X = x) = F_X(x) - F_X(x^-)$ where $F_X(x^-) = \lim_{y \uparrow x} F_X(y)$.
  - $\mathbb{P}(X \in (x, y]) = F_X(y) - F_X(x)$
  - $\mathbb{P}(X > x) = 1 - F_X(x)$
  - If X is continuous, then
  $$F_X(b) - F_X(a) = \mathbb{P}(X \in (a, b)) = \mathbb{P}(X \in (a, b]) = \mathbb{P}(X \in [a, b)) = \mathbb{P}(X \in [a, b])$$

Important Discrete Random Variables

- Point mass at a point, Dirac δ function
- The discrete uniform distribution
- The Bernoulli distribution
- The Binomial distribution
- The Geometric distribution
- The Poisson distribution

Important Continuous Random Variables

- The Uniform distribution
- Normal (Gaussian) distribution
  - Standard Normal pdf and CDF are denoted as $\phi(\cdot)$ and $\Phi(\cdot)$ respectively.
- Exponential distribution
- Gamma distribution
- The Beta distribution
- The student$-t$ distribution
- Cauchy distribution
- The χ^2 distribution

Bivariate Distributions

- A pair of discrete random variables, joint mass function
- A pair of continuous random variables, joint pdf
- Joint CDF

Marginal Distributions

- Marginal mass functions
- Marginal density functions

Independent Random Variables

- Two random variables X and Y are independent, denoted as $X \perp Y$, if

  $$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$$

  holds for all A and B.
- If X and Y have joint pdf $f_{X,Y}(x, y)$. Then

  $$X \perp Y \Leftrightarrow f_{X,Y}(x, y) = f_X(x) f_Y(y), \forall x, y \in \mathbb{R}$$

- Factorization property: if the joint pdf of X and Y satisfies

  $$f_{X,Y}(x, y) = h(x)g(y)$$

  for all $x, y \in \mathbb{R}$, where $h(\cdot)$ and $g(\cdot)$ are two univariate functions, then $X \perp Y$.

Conditional Distributions

- The conditional pmf is

$$f_{X|Y}(x|y) = \mathbb{P}(X = x|Y = y) = \frac{f_{X,Y}(x,y)}{f_Y(y)},$$

  provided $f_Y(y) > 0$.

- The conditional pdf is

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)},$$

  provided $f_Y(y) > 0$.

Multivariate Distributions and iid Samples

- Multivariate distribution functions are defined in the same way as in the bivariate case.

- If $X_1, ..., X_n$ are independent and each has the same marginal distribution with CDF F, we say that $X_1, ..., X_n$ are iid (independent and identically distributed), and we write

$$X_1, ..., X_n \overset{iid}{\sim} F.$$

  If F has pdf f, it is equivalent to write $X_1, ..., X_n \overset{iid}{\sim} f$.

- Multivariate Normal

  - From iid standard Gaussian random variables to general multivariate Gaussian random vector: if $Z \sim N_d(0, I_d)$, and $X = \mu + \Sigma^{1/2}Z$, then $X \sim N_d(\mu, \Sigma)$.
    Here $\Sigma^{1/2}$ need not to be unique, but it is unique up to an orthonormal transformation.
  - Suppose $X \sim N_d(\mu, \Sigma)$ is partitioned to two blocks X_a and X_b:

$$X = \begin{bmatrix} X_a \\ X_b \end{bmatrix}, \mu = \begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix}, \Sigma = \begin{bmatrix} \Sigma_{aa}\Sigma_{ab} \\ \Sigma_{ba}\Sigma_{bb} \end{bmatrix}.$$

    Then,

    (1) The marginal distribution of X_a is also Gaussian, and $X_a \sim N_{d_a}(\mu_a, \Sigma_{aa})$.

    (2) The conditional distribution of X_b given $X_a = x_a$ is also Gaussian, and

$$X_b|X_a = x_a \sim N_{d_b}(\mu_b + \Sigma_{ba}\Sigma_{aa}^{-1}(x_a - \mu_a), \Sigma_{bb} - \Sigma_{ba}\Sigma_{aa}^{-1}\Sigma_{ab}).$$

    This expression can be understood from the perspective of regression.

    (3) Linear transformations: $c^T X$ is univariate Gaussian, and $c^T X \sim N(c^T\mu, c^T\Sigma c)$.

Transformations of Random Variables

- First principle method for transformations of a univariate random variable:

  (1) Suppose the transformed random variable $Y = r(X)$ and we know f_X. We find the CDF of Y by the definition of CDF:

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(r(X) \leq y) = \int_{\{x:r(x)\leq y\}} f_X(x)dx.$$

  This step holds for both discrete and continuous X's.

(2) If $F_Y(y)$ is differentiable, we obtain pdf $f_Y(y)$ by taking derivative: $f_Y(y) = F'_Y(y)$. If the transformation $r(\cdot)$ is strictly monotonic and has an inverse $s = r^{-1}$, then

$$f_Y(y) = f_X(s(y))|\frac{d}{dy}s(y)|.$$

If $F_Y(y)$ is not differentiable, we obtain pmf $f_Y(y)$ by calculating the jumps of $F_Y(y)$. Alternatively for discrete random variables, we could directly compute pmf $f_Y(y)$ by :

$$f_Y(y) = \mathbb{P}(Y = y) = \mathbb{P}(r(X) = y) = \mathbb{P}(X \in r^{-1}(y)).$$

- First principle method for transformations of n random variables:

(1) Suppose the transformed random variable $Y = r(X_1, ..., X_n)$, and we know the joint pdf $f_{X_1,...,X_n}$. Again, we could find the CDF of Y by

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(r(X_1, ..., X_n) \leq y) = \int_{\{\boldsymbol{x}: r(\boldsymbol{x}) \leq y\}} f_{X_1,...,X_n}(\boldsymbol{x}) dx_1 ... dx_n$$

where $\boldsymbol{x} \equiv (x_1, ..., x_n)$.

(2) If $F_Y(y)$ is differentiable, we obtain pdf $f_Y(y)$ by taking derivative: $f_Y(y) = F'_Y(y)$.

- Important cases: the sum/maximum/minimum of a set of random variables, the ratio/difference of two random variables, etc..

Example 1.1 (Normal)**.** Suppose $X \sim N(0, 1)$, find the distribution of $Y_1 = e^X$, and $Y_2 = X^2$. ♢

Example 1.2 (Poisson)**.** Suppose $X \sim Pois(\lambda)$, find the distribution of $Y_1 = X^2$. ♢

Example 1.3 (Uniform)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} U(0, \theta)$, find the distribution of $Y_1 = \max_{1 \leq i \leq n} \{X_i\}$, and $Y_2 = \min_{1 \leq i \leq n} \{X_i\}$. ♢

1.3 Moments

Definition 1.1 (Expectation)**.** For discrete random variable X,

$$\mathbb{E}[X] = \sum\nolimits_{x \in supp(X)} x f(x),$$

provided $\mathbb{E}[|X|] < \infty$.

For continuous random variable X,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) dx,$$

provided $\mathbb{E}[|X|] < \infty$.

Expectation of a Random Variable Expectation exists if $\mathbb{E}[|X|] < \infty$. The discrete version is essentially a weighted average of X, and the weights are the probabilities.

Theorem 1.1 (The Law of the Unconscious Statistician (LOTUS).)**.** *Suppose discrete $X \sim f$, function $h : \mathbb{R} \to \mathbb{R}$, then*

$$\mathbb{E}[h(X)] = \sum_{x \in supp(X)} h(x)f(x),$$

if the sum is finite.

Suppose continuous $X \sim f$, function $h : \mathbb{R} \to \mathbb{R}$, then

$$\mathbb{E}[h(X)] = \int_{-\infty}^{\infty} h(x)f(x)dx,$$

if the integral is finite.

Corollary 1.2. *The probability of an event E and expectation is related through*

$$\mathbb{P}(E) = \mathbb{E}[1_E],$$

where 1_E denotes the indicator function of E.

Theorem 1.3 (Linearity)**.** *Expectation of a linear function of X:*

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

Expectation of a linear function of $\mathbf{X}$*:*

$$\mathbb{E}[\sum_{i=1}^{n} a_i X_i + b] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i] + b$$

where $a_1, ..., a_n$ and b are constants.

Theorem 1.4. *Let $X_1, ...X_n$ be independent random variables. Then*

$$\mathbb{E}[\prod_{i=1}^{n} X_i] = \prod_{i=1}^{n} \mathbb{E}[X_i]$$

Definition 1.2 (Variance)**.** Variance is defined as

$$V(X) = \mathbb{E}[(X - \mathbb{E}(X))^2]$$

Variance For random variable X with mean μ, it often writes $\sigma_X^2 = \sigma^2 = V(X) = \mathbb{E}[(X-\mu)^2] = \int (x-\mu)^2 dF(x)$ provided that the expectations exist. The standard deviation is defined to be $sd(X) = \sqrt{V(X)}$ or simply σ and σ_X.

Lemma 1.5 (Alternative expression of variance)**.**

$$V(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Theorem 1.6. *Variance of a linear function of X:*

$$V[aX+b] = a^2 V[X]$$

Variance of a linear function of $\mathbf{X}$ *where* $\{X_i\}$ *are* ***independent****:*

$$V\left[\sum_{i=1}^n a_i X_i + b\right] = \sum_{i=1}^n a_i^2 V[X_i]$$

Corollary 1.7. *Important special cases of the above theorem under independence are as follows.*

- *The variance of the sum of independent random variables. For all $i = 1, ..., n$, $a_i = 1$ and $b = 0$,*

$$V\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n V[X_i].$$

- *The variance of the sample mean. For all $i = 1, ..., n$, $a_i = 1/n$ and $b = 0$,*

$$V\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \frac{1}{n^2}\sum_{i=1}^n V[X_i].$$

- *The variance of the sample mean under iid. For all $i = 1, ..., n$, $a_i = 1/n$ and $V(X_i) = \sigma^2$, and $b = 0$,*

$$V\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \frac{\sigma^2}{n}.$$

Covariance

- Let X and Y be two random variables with means μ_X and μ_Y, and standard deviations σ_X and σ_Y. The covariance between X and Y is defined to be

$$cov(X,Y) = \mathbb{E}[(X-\mu_X)(Y-\mu_Y)].$$

The correlation between X and Y is defined by

$$\rho = \rho_{X,Y} = \rho(X,Y) = \frac{cov(X,Y)}{\sigma_X \sigma_Y}.$$

- Properties:
  - $cov(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.
  - $\rho(X, Y) \in [-1, 1]$.
  - If $Y = aX + b$ for some constants a and b, then $\rho(X, Y) = sign(a)$.
  - If $X \perp Y$, then $cov(X, Y) = 0$.

Conditional Expectation

- Suppose the conditional pmf/pdf of X given $Y = y$ is given, the conditional expectation of X given $Y = y$ is

$$\mathbb{E}[X|Y = y] = \begin{cases} \sum_x x f_{X|Y}(x|y) & \text{discrete} \\ \int x f_{X|Y}(x|y)dx & \text{continuous} \end{cases}.$$

  $\mathbb{E}[X|Y]$ is a random variable and is a function of random variable Y.

- LOTUS for conditional expectation:

$$\mathbb{E}[r(X, Y)|Y = y] = \mathbb{E}[r(X, y)|Y = y] = \begin{cases} \sum_x r(x, y) f_{X|Y}(x|y) & \text{discrete} \\ \int r(x, y) f_{X|Y}(x|y)dx & \text{continuous} \end{cases}.$$

  Intuitively, once you condition on $Y = y$, the random variable Y becomes the constant y.

- The Law of Iterated Expectations: suppose all expectations exist,

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]].$$

  More generally,

$$\mathbb{E}[r(X, Y)] = \mathbb{E}[\mathbb{E}[r(X, Y)|X]] = \mathbb{E}[\mathbb{E}[r(X, Y)|Z]].$$

Conditional Variance

- Suppose the conditional pmf/pdf of X given $Y = y$ is given, the conditional variance of X given $Y = y$ is

$$V[X|Y = y] = \begin{cases} \sum_x (x - \mu(y))^2 f_{X|Y}(x|y) & \text{discrete} \\ \int (x - \mu(y))^2 f_{X|Y}(x|y)dx & \text{continuous} \end{cases},$$

  where $\mu(y) = \mathbb{E}[X|Y = y]$.

  The conditional variance is defined in the same spirit as the unconditional variance. Like conditional expectations, $V[X|Y]$ is a random variable and is a function of random variable Y.

- The Law of Total Variance: for two random variables X and Y,

$$V(Y) = \mathbb{E}[V(Y|X)] + V(\mathbb{E}[Y|X]).$$

  In some contexts, this law is called Eve's law. The alternative name makes it particularly easy to memorize the law.

Definition 1.3 (Moments)**.** $\mathbb{E}[X^k]$ for $k = 1, 2, ...$

Definition 1.4 (Central moments)**.** $\mathbb{E}[(X - \mathbb{E}(X))^k]$ for $k = 2, 3, ...$

Theorem 1.8. *If $\mathbb{E}[|X|^k] < \infty$ for some positive integer k, then*

$$\mathbb{E}[|X|^j] < \infty$$

for every positive integer $j \leq k$.

Definition 1.5. Moment generating function (mgf) of X: for real t,

$$\psi_X(t) = \mathbb{E}(e^{tX}) = \int e^{tX} dF(x),$$

provided that the integral is finite.

Moment Generating Functions The moment generating function (mgf) is also known as the Laplace transform. Immediately, $\psi_X(0) = 1$ for all random variable X.

Suppose the mgf is well-defined for all t in some open interval around $t = 0$, so we can interchange the operations of differentiation and expectation, that is

$$\psi'_X(t) = \frac{d}{dt}\psi_X(t) = \frac{d}{dt}\mathbb{E}[e^{tX}] = \mathbb{E}[\frac{d}{dt}e^{tX}] = \mathbb{E}[e^{tX}X].$$

Then $\psi'_X(0) = \mathbb{E}[X]$.

More generally, for any positive integer k, $\mathbb{E}[X^k] = \psi_X^{(k)}(0)$. Note the Taylor expansion of e^{tX} around $t = 0$ is

$$e^{tX} = 1 + Xt + \frac{1}{2}X^2t^2 + ... = \sum_{k=0}^{\infty} \frac{1}{k!}X^k t^k.$$

Then $\frac{d^k}{dt^k}e^{tX}|_{t=0} = X^k$. By taking expectation of both sides and interchanging the operations of differentiation and expectation, $\mathbb{E}[X^k] = \psi_X^{(k)}(0)$.

- Affine transformations: let $Y = aX + b$ for some constants a and b, then

$$\psi_Y(t) = \mathbb{E}[e^{tY}] = \mathbb{E}[e^{t(aX+b)}] = e^{bt}\psi_X(at).$$

- Independent sum: if $X_1, ..., X_n$ are independent, and $Y = \sum_{i=1}^n X_i$, then

$$\psi_Y(t) = \mathbb{E}[e^{tY}] = \mathbb{E}[e^{t(\sum_{i=1}^n X_i)}] = \prod_{i=1}^n \mathbb{E}[e^{tX_i}] = \prod_{i=1}^n \psi_{X_i}(t).$$

Theorem 1.9. *If $\psi_{X_1}(t) = \psi_{X_2}(t)$ for t in some neighborhood around $t = 0$, then*

$$X_1 \stackrel{d}{=} X_2.$$

That is, X_1 and X_2 have the same distribution (up to a null set).

Theorem 1.10. *If $\psi(t) < \infty$ for all values of t in some open interval around $t = 0$, then*

$$\mathbb{E}[X^k] < \infty,$$

and, for $k = 1, 2, ...,$

$$\mathbb{E}[X^k] = \psi^{(k)}(0).$$

1.4 Inequalities

1.4.1 Probability Inequalities

Theorem 1.11 (Markov)**.** *Let X be a non-negative random variable, and suppose $\mathbb{E}[X]$ exists. For any $t > 0$,*

$$\mathbb{P}(X > t) \leq \frac{\mathbb{E}[X]}{t}.$$

Without any doubt, Markov's inequality is *the* most fundamental probability inequality.

Theorem 1.12 (Chebyshev)**.** *For random variable X whose variance exists,*

$$\mathbb{P}(|X - \mathbb{E}[X]| > t) \leq V(X)/t^2.$$

Standardized X is defined to be $Z = \frac{X-\mu}{\sigma}$, and

$$\mathbb{P}(|Z| \geq t) \leq \frac{1}{t^2}.$$

- Hoeffding's inequality. Let $X_1, ..., X_n$ be independent random variables such that $\mathbb{E}[X_i] = 0$ and $X_i \in [a_i, b_i]$. For any $\varepsilon > 0$ and $t > 0$,

  $$\mathbb{P}(\sum_{i=1}^n X_i \geq \varepsilon) \leq e^{t^2\Delta_n - t\varepsilon}$$

  where $\Delta_n = \sum_{i=1}^n (b_i - a_i)^2/8$. Since the left-hand-side has no t, we can choose t on the right-hand-side to optimize the upper bound. The optimal choice of t is $t^* = \frac{\varepsilon}{2\Delta_n}$, and the optimal version is

  $$\mathbb{P}(\sum_{i=1}^n X_i \geq \varepsilon) \leq e^{-\frac{\varepsilon^2}{4\Delta_n}} = e^{-\frac{2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}}.$$

- Mill's ratio. Let $Z \sim N(0,1)$. Then

$$\mathbb{P}(|Z| > t) \leq \sqrt{\frac{2}{\pi}} \frac{e^{-\frac{1}{2}t^2}}{t}.$$

1.4.2 Inequalities for Expectations

- Cauchy-Schwartz inequality. For two random variables X and Y,

$$\mathbb{E}[|XY|] \leq \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}.$$

  If X or Y has infinite variance, then the above inequality holds trivially.

- Jensen's inequality. If $f(\cdot)$ is convex, then

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]).$$

  With Jensen's inequality, we see

  (1) $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$, where $f(x) = x^2$. Immediately, we see $V(X) \geq 0$.
  (2) if $X > 0$, $\mathbb{E}[1/X] \geq 1/\mathbb{E}[X]$, where $f(x) = 1/x$.
  (3) $\mathbb{E}[\log(X)] \leq \log\mathbb{E}[X]$, where $f(x) = -\log(x)$.

1.5 Convergence of Random Variables

Previously, the numer of random variables is fixed to some $n \in \mathbb{N}_+$. In many applications, the quantity of interest is a function of

$$\frac{1}{n}\sum\nolimits_{i=1}^{n} X_i,$$

a.k.a., sample mean, or sample average. Write $\bar{X}_n := \frac{1}{n}\sum_{i=1}^{n} X_i$.

What would happen as $n \to \infty$? In particular, is $\bar{X}_n$ convergent as $n \to \infty$? If so, in what sense and where does $\bar{X}_n$ converge to? Clearly, the question is too general. For our purpose, we would assume $X_i \overset{iid}{\sim} F$ such that its mean and variance exists.

Theorem 1.13 (Weak Law of Large Numbers)**.** *For iid sample $X_1, ..., X_n$ with mean μ and variance σ^2,*

$$\bar{X}_n \xrightarrow{p} \mu.$$

Intuitively, it says the sample mean is approaching the population mean "in probability". The qualifier "weak" is due to "$\xrightarrow{p}$", convergence in probability.

Definition 1.6 (Convergence in probability)**.** X_n converges to X in probability, written $X_n \xrightarrow{p} X$, if for every $\epsilon > 0$,

$$\mathbb{P}(|X_n - X| > \epsilon) \to 0,$$

as $n \to \infty$.

For $\epsilon_1 > \epsilon_2 > 0$, by the inclusion

$$\{|X_n - X| > \epsilon_1\} \subseteq \{|X_n - X| > \epsilon_2\},$$

it follows $\mathbb{P}(|X_n - X| > \epsilon_1) \leq \mathbb{P}(|X_n - X| > \epsilon_2)$. That is, we only need to check the case where ϵ is close to 0.

Example 1.4. Show the following by definition.

- $X_n \overset{ind}{\sim} Ber(\frac{1}{n})$, $X_n \xrightarrow{p} 0$.
- $X_n \overset{ind}{\sim} N(\mu, \sigma^2/n)$, $X_n \xrightarrow{p} \mu$.

♢

Theorem 1.14 (Strong Law of Large Numbers)**.** *If $X_i \overset{iid}{\sim} F$ with mean μ, then*

$$\bar{X}_n \xrightarrow{a.s.} \mu.$$

Definition 1.7 (Almost sure convergence)**.** X_n converges to X almost surely (a.s.), written $X_n \xrightarrow{a.s.} X$, if

$$\mathbb{P}(\lim_{n\to\infty} X_n = X) = 1.$$

The limiting quantity of LLN typically is a real number. Also, $n \to \infty$ never happens in reality. A more suitable of the limiting process is that n is finite but reasonably "large". Can we say anything distributional on the limit?

Theorem 1.15 (Lindeberg and Lévy)**.** *if $X_1, X_2, ..., X_n \overset{iid}{\sim} F$ with $\mathbb{E}[X_1] = \mu$ and $V(X_1) = \sigma^2$, then for each fixed number t,*

$$\lim_{n\to\infty} \mathbb{P}(\sqrt{n}(\bar{X}_n - \mu)/\sigma \leq t) = \Phi(t).$$

Write

$$\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0, 1).$$

Theorem 1.16. *Suppose r.v. $X_i \sim F_i$ with mgf ψ_i, and some r.v. X^* has mgf ψ^*. If $\psi_i(t) \to \psi^*(t)$ for all t in some interval around $t = 0$, then $X_i \xrightarrow{d} X^*$.*

Proof. The mgf of $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ with X_i's mgf ψ:

$$\psi_n(t) = \mathbb{E}[e^{t\sqrt{n}(\bar{X}_n - \mu)/\sigma}] = \mathbb{E}[e^{\frac{t}{\sqrt{n}} \sum_{i=1}^n (\frac{X_i - \mu}{\sigma})}].$$

By Theorem 1.16, the goal is to show $\lim_{n\to\infty} \psi_n(t) = e^{t^2/2}$ for t in some interval around $t = 0$.

Let $Y_i = \frac{X_i - \mu}{\sigma}$, so $Y_i \overset{iid}{\sim} F_Y$ with mean 0 and variance 1. With the Taylor expansion at 0:

$$\psi_Y(t) = \psi_Y(0) + \psi_Y^{(1)}(0)t + \frac{1}{2!}\psi_Y^{(2)}(0)t^2 + \frac{1}{3!}\psi_Y^{(3)}(0)t^3 + ...,$$

it follows

$$\psi_n(t) = \mathbb{E}[e^{\frac{t}{\sqrt{n}}\sum_{i=1}^n Y_i}] = [\psi_Y(t/\sqrt{n})]^n = \left(1 + \frac{1}{2}\frac{t^2}{n} + \frac{1}{3!}\psi_Y^{(3)}(0)\frac{t^3}{n^{3/2}} + ...\right)^n \to e^{t^2/2}.$$

□

Loosely speaking, the "standardized" random variable $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ with n held fixed is considered to "approximately" follow the standard Normal distribution. We postpone the proof of the CLT after we have studied moment generating functions.

Definition 1.8 (Convergence in distribution)**.** X_n converges to X in distribution, written $X_n \xrightarrow{d} X$, if

$$\lim_{n\to\infty} F_n(t) = F(t)$$

at all t for which F is continuous.

The Portmanteau theorem provides a few equivalent definitions of convergence in distribution.

Example 1.5. Show the following by definition. Pay attention to the points where F is discontinuous.

- $\mathbb{P}(X_n = 1/n) = 1$ and $\mathbb{P}(X = 0) = 1$.
- $F_n(x) = 1 - (1 - \frac{1}{n})^{nx}$ for $x > 0$.
- $X_n \overset{ind}{\sim} Bin(n, \lambda/n)$ for $\lambda > 0$.

♢

A direct implication: for n large enough,

$$\sqrt{n}(\bar{X}_n - \mu)/\sigma \approx N(0, 1).$$

As probability calculation for Normal random variables is well understood, the CLT provides a simple yet powerful way of approximating probabilities.

In many statistical applications, the law of large numbers and the central limit theorem cannot be applied directly, because the estimators are not sample averages. Instead, many estimators are continuous functions of sample averages.

The *continuous mapping theorem* (CMT) states that the convergence in probability and distribution is preserved under continuous transformations.

Theorem 1.17 (Continuous Mapping Theorem (CMT))**.** *Let $X_n, n = 1, ...,$ and X be random variables. Let $g(\cdot)$ be a continuous function.*

- *If $X_n \xrightarrow{p} X$, then $g(X_n) \xrightarrow{p} g(X)$.*
- *If $X_n \xrightarrow{d} X$, then $g(X_n) \xrightarrow{d} g(X)$.*

Proof. $\epsilon - \delta$ argument. □

Though Theorem 4.1 is stated for random variables, the same type of results also holds for random vectors and multivariate continuous functions. In the case of random vectors, $X_n \xrightarrow{p} X$ if their Euclidean distance $||X_n - X||$ (or in ℓ_2 norm) converges to 0 in probability.

Some convergence results are preserved under transformations. As arithmetic operations are continuous, the results are also known as continuous mapping theorem. The following can be implied by the random vector version of the CMT.

Theorem 1.18 (CMT under arithmetic operations)**.** *Let X_n, X, Y_n, Y be random variables, and c be some constant. Then*

- *If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then $X_n + Y_n \xrightarrow{p} X + Y$.*
- *If $X_n \xrightarrow{L_p} X$ and $Y_n \xrightarrow{L_p} Y$, then $X_n + Y_n \xrightarrow{L_p} X + Y$.*
- *If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then $X_n Y_n \xrightarrow{p} XY$.*
- *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$, then $X_n + Y_n \xrightarrow{d} X + c$. (Slutzky's theorem)*
- *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$, then $X_n Y_n \xrightarrow{d} cX$. (Slutzky's theorem)*

2 Descriptive Statistics

2.1 Descriptive Statistics

Definition 2.1. A statistic is a function of the data for some statistical purpose(s).

A descriptive statistic is used to summarize the sample data, e.g., the sample average. A test statistic is used in statistical hypothesis testing, e.g., reject the null hypothesis if the sample average, the test statistic, is greater than some threshold. In the examples, the sample average serves as both a descriptive statistic and a test statistic.

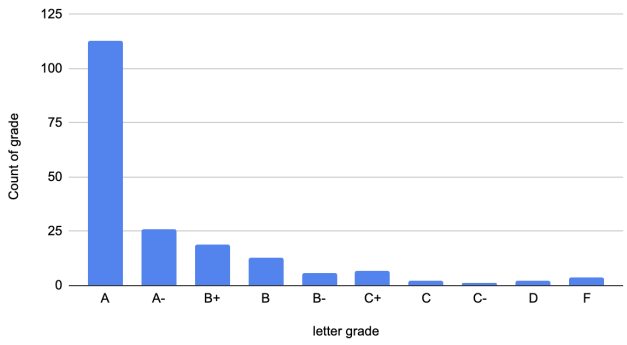
For univariate numeric data, e.g., stock prices, body weight, height, etc.,

- Maximum, minimum, range
- median, and other sample quantiles/percentiles
- sample average/mean, sample variance, sample standard deviation

For univariate discrete data, e.g., letter grades, penguin species, etc.. We can tabulate the `grades` data.

letter grade	A	A-	B	B-	B+	C	C-	C+	D	F
count	113	26	13	6	19	2	1	7	2	12

We can also plot their empirical frequency in a bar plot.




```r
library("ggplot2")
grades <- read.csv("stat131-data.csv")
head(grades)
ggplot(grades, aes(x=grade)) + geom_bar()
table(grades$grade )
```

We often prefer a nice visualization to a messy table of numbers.

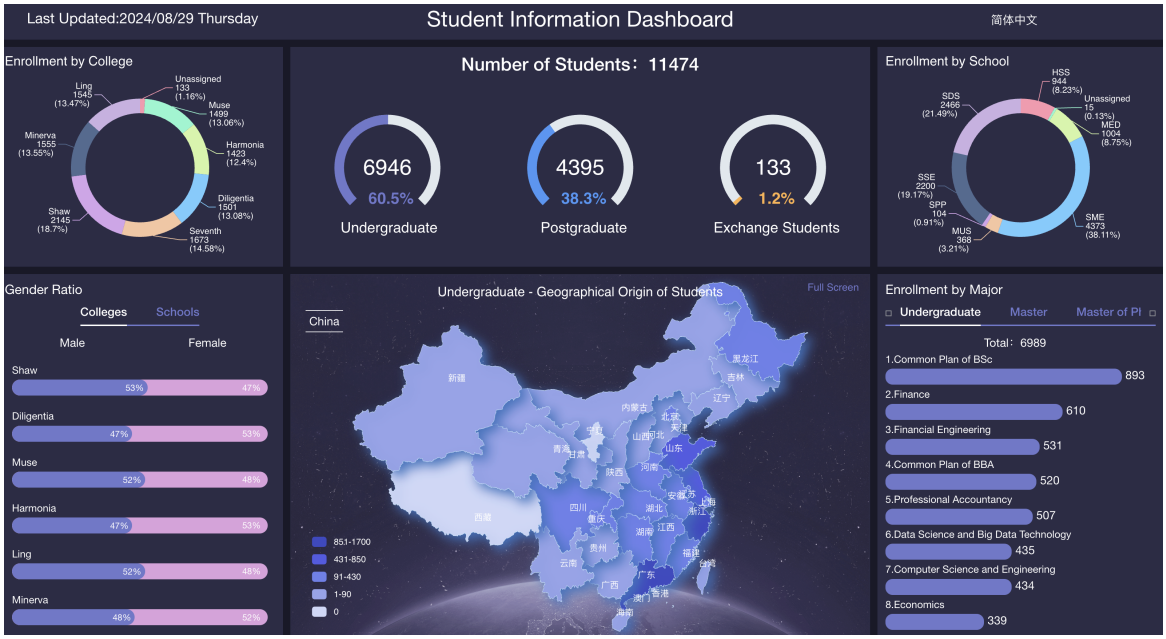


Figure 2.1: The student information dashboard of CUHK-SZ

Example 2.1 (Simpson's paradox)**.** Simpson's paradox refers to a phenomenon in which a trend appears in several groups of data but disappears or reverses when the groups are combined. Perhaps, the best-known example of Simpson's paradox is the study on the gender bias among graduate school admissions to University of California, Berkeley.

The following example of batting averages from the textbook **?**

	Player A			Player B		
season	AB	Hits	Average	AB	Hits	Average
1	500	126	0.252	300	75	0.250
2	300	90	0.300	500	145	0.290
totals	800	216	0.270	800	220	0.275

Ross (2007) demonstrates a similar phenomenon using the batting average of two baseball players, Derek Jeter and David Justice, during the years 1995 and 1996. In both 1995 and 1996, Justice had a higher batting average (in bold type) than Jeter did. However, when the two baseball seasons are combined, Jeter shows a higher batting average than Justice.

♢

2.2 Exploratory Data Analysis

Tukey (1977) advocated exploratory data analysis (EDA). EDA is an essential part of modern data science, as illustrated in the Box's loop.

Stem-and-leaf display Tukey (1977) introduces the stem-and-leaf display. In the first example (left of Figure 2.2), we simulate 10 standard Normal random numbers:

-0.626, 0.184, -0.836, 1.595, 0.330, -0.820, 0.487, 0.738, 0.576, -0.305.

After ordering and rounding, the data becomes

-0.836, -0.820, -0.626, -0.305, 0.184, 0.330, 0.487, 0.576, 0.738, 1.595.

```
The decimal point is at the |

-0 | 886
-0 | 3
 0 | 23
 0 | 567
 1 |
 1 | 6
```

10 standard Normal random numbers

```
The decimal point is 1 digit(s) to the right of the |

 0 | 2222223
 1 | 3
 2 |
 3 | 034
 4 | 6
 5 | 247
 6 | 1455668999
 7 | 0233456666677788899
 8 | 000001111222234444455555667777777788888999999999
 9 | 000000111111111222222333333444444555666666666777777777888
10 | 00000000111111122222333333444455666667888
11 | 55
```

numeric grades

Figure 2.2: Stem-and-leaf display

We can also "visualize" the numeric grades of the letter grades example with the stem-and-leaf display (right panel of Figure 2.2).

```
set.seed(1)
x <- rnorm(10)
print(x)
stem(x)
stem(grades$total)
```

The box plot A boxplot is a type of visual shorthand for measures of position (percentiles) that describe a distribution. It is also useful for identifying potential outliers.

- A box that indicates the range of the middle half of the data, a distance known as the *interquartile range* (IQR), the difference between the 25th percentile and the 75th percentile.
- Visual points that display observations that fall more than 1.5 times the IQR from either edge of the box. These are typically outliers.
- A line (or whisker) that extends from each end of the box and goes to the farthest nonoutlier point in the distribution.

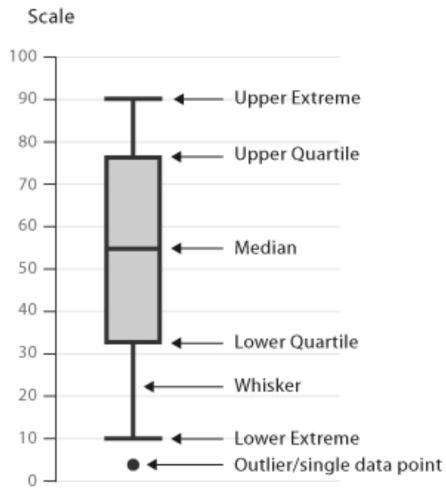


Figure 2.3: The composition of a box plot

Box plots are very effective in comparing the distribution of various groups. The R package `palmerpenguins` provides the great dataset `penguins` which has three species: Adelie, Chinstrap, and Gentoo.



Adelie Chinstrap Gentoo

Figure 2.4: The penguins

```
library(palmerpenguins)
library(tidyverse)
glimpse(penguins)
table(penguins$species)
```

Suppose we are interested in if the body masses of the species are different. We can visualize the body mass distributions by box plot.

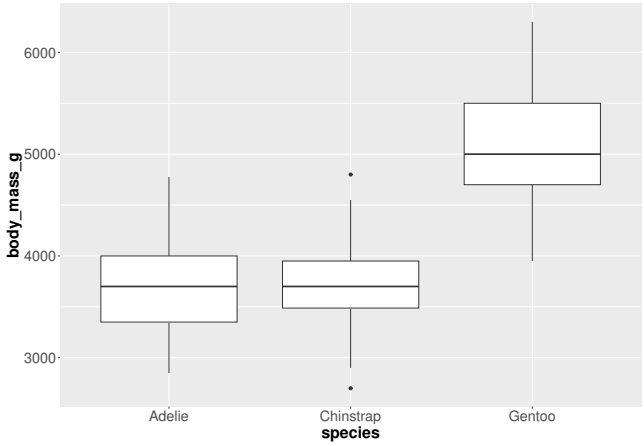


Figure 2.5: Compare the body mass of penguins

```
library(palmerpenguins)
library(tidyverse)
library(ggplot2)
ggplot(penguins, aes(x = species, y = body_mass_g)) + geom_boxplot()
```

3 Inferential Statistics

3.1 Statistical Inference

Reading(s) 3.1. Wasserman (2004) Section 6.1 and 6.2.

Statistical *inference*, informally speaking, is to make sense of data with certain *model*. This includes, but not limited to, the following aspects:

- identify *interpretable* patterns of the data
- help scientific endeavors by comparing scientific theory with empirical evidence, and planning experimental observations.

Statistical philosophy is that some *unknown* probability measures and/or processes generate the data, and statistical inference is to describe and understand the underlying data generating process. To this end, we define a statistical model as a set of distributions.

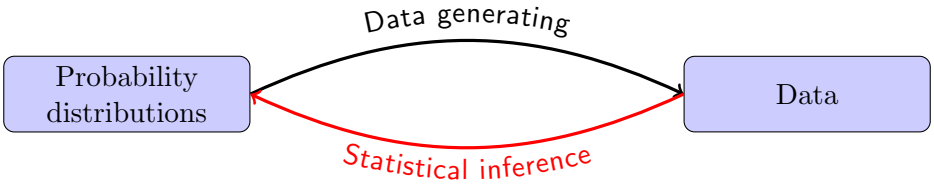


Figure 3.1: Statistical model and data

Parametric and nonparametric models Statistical models are generally classified into parametric models and non-parametric models. There are also semi-parametric models that have both parametric and non-parametric specification in the model.

A *parametric* model is a set of distributions that can be parameterized by a finite number of parameters. For instance, we could obtain a random sample from some univariate Gaussian distribution with unknown mean and variance. Here, the parametric model is univariate Gaussian, and the parameters are the mean and the variance. If you are *only* intersted in the mean, then the mean parameter is called *the parameter of interest* and the variance parameter is called a *nuisance* parameter.

A *non-parametric* model is a set of distributions that cannot be parameterized by a *fixed* number of parameters. For instance, in density estimation, the class of all pdf's cannot be parameterized by a fixed number of parameters, hence it is non-parametric. Similarly, in estimating CDF, the class of all CDF's is also non-parametric. In the regression setting, flexible regression functions are also considered as nonparametric, including splines, local polynomials, deep neural networks with ReLU activation function, and so on. Models that have a growing number of parameters as sample size grows are also called non-parametric, e.g., Bayesian nonparametric methods with Dirichlet process (mixture) priors.

3.2 The Box's Loop

Reading(s) 3.2. Blei (2014) and Box (1976).

Why *should* we study statistical inference? Statistical inference is the core and fundamental building block of modern data science.

Figure 3.2 presents a big picture of modern data analysis procedure/pipeline. The flow chart is essentially the Box's loop (Blei, 2014; Box, 1976). Box's loop illustrates the iterative nature of statistical analysis. Typically we cannot finish data analysis by fitting a single model. Instead, we strive to provide statistical analysis as accurate and convincing as possible by exploring various modeling options and alternatives.

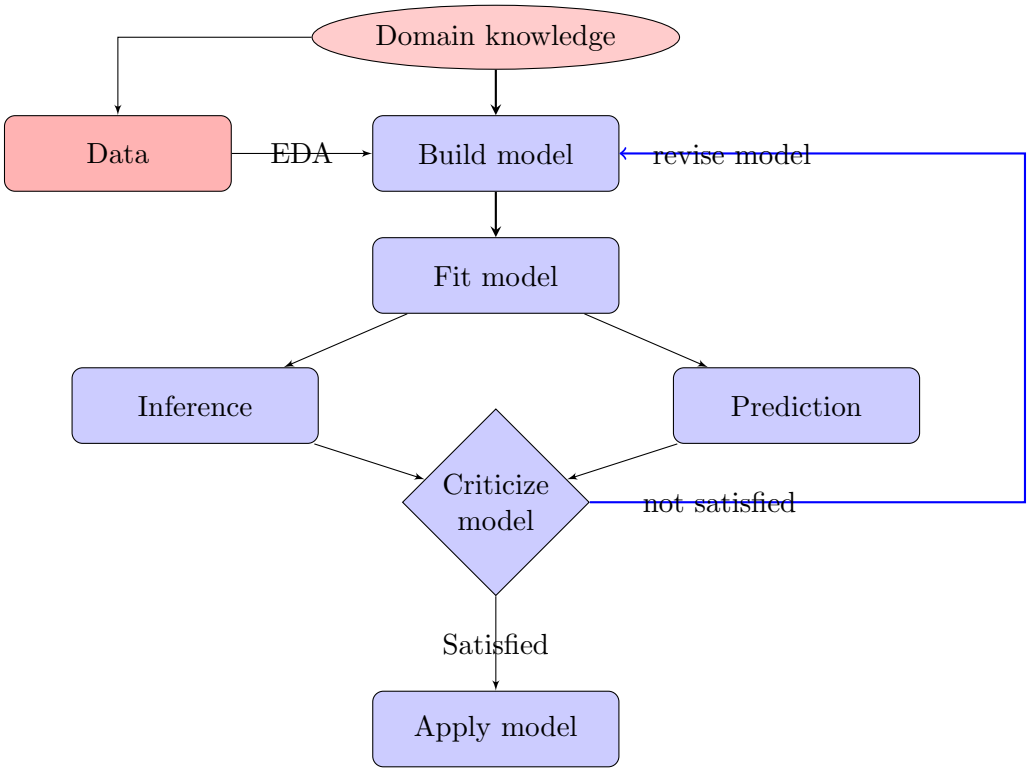


Figure 3.2: Box's Loop

Often times, statisticians/data scientists work with some domain experts. Data is often collected and cleaned under the guidance of certain domain knowledge.

We first summarize, visualize, and hypothesize about the data, using scatter plots, histograms, and other tools. At this stage, we also check the missingness, censorship and truncation of the data. This step is called exploratory data analysis (EDA).

After EDA, we build a probabilistic model with both domain knowledge and EDA results.

After model fitting, we make inference about the hidden quantities of the model and/or make predictions with the current model.

With the inference and/or prediction, we criticize the model by examining the model fit, checking the validity of the inference, and others. In assessing the model fit, we can ask the following questions,

- are we missing important features of the data?
- does our model work reasonably well on hold-out sample?
- is there a better way to model the data?

In checking the validity of the statistical analysis, we can ask the following questions:

- are the conclusions sensible to the domain experts?
- what are the important ways that the current model succeeds and fails?

If we are not satisfied with the current model, we revise it and repeat the steps of model fitting and criticism.

If we are satisfied with the current model, we can use this model to answer questions and make predictions.

Another important feature of modern data analysis is its collaborative nature. As more and more statistical tools are developed, domain experts may encounter data that are generated from a complicated process, but it might not be so efficient learning all the statistical tools themselves. A natural and more efficient alternative is that the domain experts come to statisticians with certain dataset and questions in mind. So the statisticians need to communicate with the domain experts on their goal(s) of inference and the data generating process.

3.3 Fundamental Tasks of Statistical Inference

Reading(s) 3.3. Wasserman (2004) Section 6.3.

We use parametric models to illustrate the fundamental concepts of statistical inference. Suppose the model is

$$\{\mathbb{P}_\theta : \theta \in \Theta\},$$

where Θ is the *parameter space*, and the observed data is independently generated from $\mathbb{P}_{\theta_0}$ where θ_0 is some "true" value. Our major interest is to learn θ_0.

3.3.1 Point Estimation

Point estimation of θ_0 is to provide an algorithm, say $t(\mathbf{X})$, such that the output is an estimate of θ_0. The output of the algorithm is data dependent, and the output is a number or vector that matches the dimension of θ_0. Since the output is essentially a point in Euclidean space, we call it point estimate. The conventional notation for point estimates is the *hat* notation: $\hat{\theta} = t(\mathbf{X})$.

Example 3.1 (Normal mean)**.** The model is $X_1, ..., X_n \overset{iid}{\sim} N(\theta, 1)$, and the goal is to provide a point estimator for the mean parameter θ.

Firstly, $\hat{\theta}$ could be $\frac{1}{n}\sum_{i=1}^n X_i$, the sample average. Secondly, $\hat{\theta}$ could be sample median. Thirdly, $\hat{\theta}$ could be X_1, the very first observation. There could be countless point estimators, some of which are more reasonable and "better" than others. ♢

Example 3.2 (Prediction)**.** Model-based prediction tries to say something beyond what we have observed with the statistical model. Continuing the Normal mean example, we can predict the next value to be observed with (some function of) the point estimate of θ. Its optimality requires decision-theoretic argument, which is out of the scope of this course. Essentially, we treat the next observation as an unknown parameter which can be a function of the model parameters. ♢

3.3.2 Confidence Sets

Point estimates are expected to be off the target in most cases, that is, the probability that the point estimate happens to be exactly the true value is 0:

$$\mathbb{P}_{\theta_0}(t(\mathbf{X}) = \theta_0) = 0.$$

A reasonable alternative is to replace the point estimate with a interval/set: $C(\mathbf{X})$. Like point estimation, interval/set estimation is to provide an algorithm, say $C(\mathbf{X})$, such that the output is a interval/set, with the hope that the interval/set contains the true value.

The inferential statement of the confidence interval/set is that the true value is in the interval/set $C(\mathbf{X})$ with at least $(1-\alpha)$ confidence:

$$\mathbb{P}_{\theta_0}(\theta_0 \in C(\mathbf{X})) \geq 1 - \alpha,$$

where we call $(1-\alpha)$ as the *coverage* of the confidence interval $C(\mathbf{X})$.

The interpretation of the confidence interval/set is a bit tricky. Firstly, despite the confidence interval/set is data dependent, for a particular dataset, the confidence interval/set is fixed. That is,

for a particular dataset, the confidence interval/set either contains the true value or does not contain it. Unfortunately, it is often impossible to know which scenario we are in.

Secondly, the probability $\mathbb{P}_{\theta_0}(\theta_0 \in C(\mathbf{X}))$ is approximately the relative frequency of the confidence interval containing the true value, if we *repeat* the data generating process/experiment sufficiently many times. That is, despite we cannot say much about the parameter θ given a particular dataset, the procedure of interval/set estimation is guaranteed to cover the true value with certain probability, even before any data is generated and observed. The "confidence" of the confidence interval/set is procedure-wise.

Example 3.3 (Bernoulli)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta)$. For certain values of θ_0, it is easy to show $\mathbb{P}_0(\hat{\theta} = \theta_0) = 0$. We can use the confidence interval $C(\boldsymbol{X}) = (\hat{\theta} - 2\sqrt{\frac{1}{n}\hat{\theta}(1-\hat{\theta})}, \hat{\theta} + 2\sqrt{\frac{1}{n}\hat{\theta}(1-\hat{\theta})})$, and it is easy to see that $\mathbb{P}(\theta_0 \in C(\boldsymbol{X})) > 0$. The exact coverage rate of $C(\boldsymbol{X})$ is often very challenging. $\diamondsuit$

3.3.3 Hypothesis Testing

Hypothesis testing addresses a slightly different problem than estimation. The focus is to evaluate the validity of certain statement which is called hypothesis. The hypothesis is typically formulated in terms of the parameter. The vanilla case of hypothesis testing has two hypotheses: the null hypothesis denoted as H_0, and the alternative hypothesis denoted as H_1.

For instance, we can have the following hypothesis testing problem:

$$H_0 : \theta_0 \geq 0, v.s.\ H_1 : \theta_0 < 0.$$

With certain reparameterization, such hypothesis testing problem is called A/B test. More generally, we can test the hypotheses

$$H_0 : \theta_0 \in \Theta_0, v.s.\ H_1 : \theta_0 \in \Theta_1.$$

In many genetics studies, we want to test a large collection of hypotheses all at once:

$$H_{0j} : \theta_{0j} \in \Theta_{0j}, v.s.\ H_{1j} : \theta_{0j} \in \Theta_{1j} \text{ for } j = 1, ..., N,$$

where j is the index of certain gene, and N can be tens of thousands.

3.4 The Paradigms of Statistical Inference

Reading(s) 3.4. Efron and Hastie (2021) 2.1, 3.1 and 3.4.

In statistical inference, there are two dominant paradigms: frequentist inference and Bayesian inference. Apart from the dominant paradigms, fiducial inference is also available in the literature. However, we will not discuss it in this course.

3.4.1 Frequentist Inference

Frequentist inference refers to inferential procedures whose accuracy is evaluated and guaranteed as the sample size grows to infinity or if the same data generating process is repeated infinitely many times. Here are a few examples:

Definition 3.1 (Bias)**.** The *bias* of the estimator $\hat{\theta}$ is defined as

$$Bias(\hat{\theta}) := \mathbb{E}_{\theta_0}[\hat{\theta}] - \theta_0.$$

As the bias also depends on data that are not observed, this is procedure-wise guarantee. In contrast, the actual error of estimation $\hat{\theta} - \theta_0$ is data dependent.

Example 3.4 (sample mean)**.** Suppose the data $X_1, ..., X_n \overset{iid}{\sim} F$ with mean θ_0, and the estimator for θ_0 is the sample mean:

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

By linearity of expectation and the iid assumption,

$$\mathbb{E}[\hat{\theta}] = \theta_0.$$

♢

Example 3.5 (sample variance)**.** Suppose the data $X_1, ..., X_n \overset{iid}{\sim} F$ with mean θ_0 and variance σ_0^2. The estimator for σ_0^2 is the sample variance:

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X}_n)^2,$$

where $\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$. It can be shown

$$\mathbb{E}[\hat{\sigma}^2] = \sigma_0^2.$$

Given $\hat{\sigma}^2$ is unbiased, the estimator $\frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X}_n)^2$ is biased.

♢

The *Mean Squared Error (MSE)* is another measure of accuracy.

Definition 3.2 (Mean Squared Error (MSE))**.** The Mean Squared Error (MSE) is defined as

$$MSE(\hat{\theta}) := \mathbb{E}_{\theta_0}[(\hat{\theta} - \theta_0)^2].$$

It can be shown that

$$MSE(\hat{\theta}) = bias(\hat{\theta})^2 + V_{\theta_0}(\hat{\theta}).$$

MSE depends on both the bias and the variability of the estimator. Often times, estimators are faced with the tradeoff between bias and variance.

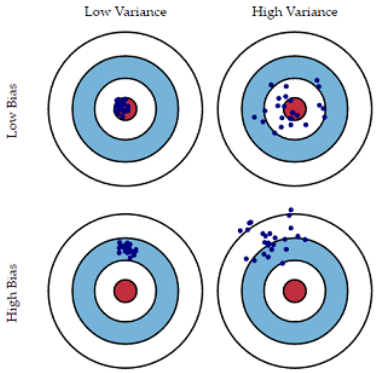


Figure 3.3: Bias and variance

Definition 3.3 (Consistency)**.** The (point) estimator $\hat{\theta}$ is *consistent* if

$$\hat{\theta} \to \theta_0 \text{ in probability-}\mathbb{P}_{\theta_0}.$$

The mode of convergence used above is convergence in probability. A sequence of random variables X_n converges to X in probability, written $X_n \xrightarrow{p} X$, if for every $\epsilon > 0$,

$$\mathbb{P}(|X_n - X| > \epsilon) \to 0,$$

as $n \to \infty$.

As sample size grows to infinity, the estimator gets arbitrarily close to the true value. It can be shown that $\hat{\theta}$ is consistent if its MSE goes to 0 as sample size grows to infinity.

Definition 3.4 (Rate of Convergence: Asymptotic Normality)**.** The (point) estimator $\hat{\theta}$ is *asymptotically Normal* if

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, V(\theta_0))$$

holds for some variance $V(\theta_0)$ which is potentially dependent of θ_0.

The mode of convergence used above is convergence in distribution. X_n converges to X in distribution, written $X_n \xrightarrow{d} X$, if

$$\lim_{n\to\infty} F_n(t) = F(t)$$

at all t for which F is continuous.

Asymptotic Normality is often used to construct asymptotic confidence intervals/sets. We will see more details later.

Example 3.6 (Frequentist properties of sample mean)**.** Suppose the data $X_1, ..., X_n \overset{iid}{\sim} F$ with mean θ_0 and variance σ_0^2. The estimator for θ_0 is the sample mean:

$$\hat{\theta} = \frac{1}{n}\sum\nolimits_{i=1}^{n} X_i.$$

Now we can examine the frequentist properties of the estimator:

- Bias
- MSE
- Consistency
- Asymptotic Normality

Remark: the above calculations only require the first two moments of X_i. In other words, the analysis is the same for the case $X_i \overset{iid}{\sim} Ber(\theta_0)$ and the case $X_i \overset{iid}{\sim} N(\theta_0, \sigma_0^2)$ if $\sigma_0^2 = \theta_0(1-\theta_0)$. $\diamondsuit$

It is worthwhile to emphasize that the aforementioned accuracy criteria also depend on the data that are not observed by the statistician. To be more specific, bias is the *averaged* error that not merely depends on the observed data, but also potentially observed data. This argument also applies to MSE which is the *averaged* squared error. Consistency depends on more hypothetical data than the observed data.

Limit theorems like LLN and CLT describe what would happen in *asymptopia*, and there seems a gap between the theoretical guarantee and the actual situation. In fact, asymptotic effects often kick in pretty fast, and moderate amount of observed data could provide reasonably accurate inference, making frequentist criteria sensible in practice.

3.4.2 Bayesian Inference

Bayesian inference refers to *probabilistic* inferential procedures based on Bayes theorem. In contrast to frequentist inference, Bayesian inference conditions on the observed data, hence, all inferential statements are more specific to the observed data.

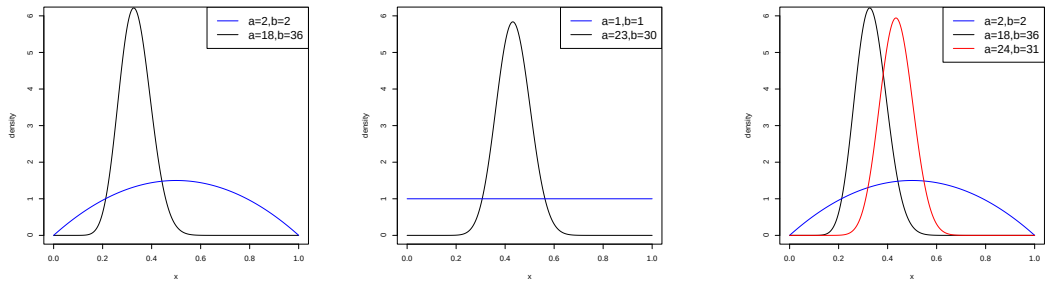
To be specific, with the probability model $\{\mathbb{P}_\theta : \theta \in \mathbb{R}\}$, we could specify a prior distribution on the parameter θ and the prior distribution admits a density function $\pi(\theta)$. With Bayes theorem, the posterior density of θ given data $\mathbf{X}$ is

$$\pi(\theta|\mathbf{X}) = \frac{\pi(\theta, \mathbf{X})}{\pi(\mathbf{X})} \propto p(\mathbf{X}|\theta)\pi(\theta)$$

where $p(\mathbf{X}|\theta)$ is the likelihood.

Example 3.7. Suppose $Y_1, ..., Y_n \overset{iid}{\sim} Ber(\theta)$ where $Y_i = 1$ denotes the success in i^{th} trial and n is the known number of trials. The model is equivalent to $X \sim Bin(n, \theta)$ where $X = \sum_{i=1}^{n} Y_i$ denotes the total number of successes. The unknown parameter is the probability of success θ.

Different prior distributions on θ and data produce different posterior distributions. Despite the impact of the prior specification, data often "speaks louder". As shown in Figure 3.4, the posterior inference is mainly driven by the data.



Prior:	$Beta(2,2)$	$Beta(1,1)$	$Beta(2,2)$
Data:	$(n, X) = (50, 16)$	$(n, X) = (51, 22)$	(50,16) and (51,22)
Posterior:	$Beta(18, 36)$	$Beta(23, 30)$	$Beta(18, 36)$ and $Beta(24, 31)$

Figure 3.4: From the prior to the posterior distribution

♢

Bayesian inference is made based on the posterior density $\pi(\theta|\mathbf{X})$. For instance, we can construct the point estimate using posterior mean $\mathbb{E}[\theta|\mathbf{X}] = \int \theta\pi(\theta|\mathbf{X})d\theta$, and we can construct interval/set estimate using the posterior quantiles

$$C_\alpha(\mathbf{X}) = [\pi^{-1}_{\theta|\mathbf{X}}(\alpha/2), \pi^{-1}_{\theta|\mathbf{X}}(1-\alpha/2)]$$

where $\pi^{-1}_{\theta|\mathbf{X}}(\alpha)$ denotes the αth quantile of $\pi(\theta|\mathbf{X})$. By construction, the posterior probability of $\{\theta \in C_\alpha(\mathbf{X})\}$ is $1-\alpha$. This is different from confidence intervals/sets, and it is called *credible* intervals/sets.

Some Bayesian estimators also have frequentist guarantees. For instance, Bayesian estimators can be consistent: $\mathbb{E}[\theta|\mathbf{X}] \to \theta_0$ in Probability-$\mathbb{P}_{\theta_0}$, and the credible interval can be *honest*:

$$\mathbb{P}_{\theta_0}(\theta_0 \in C_\alpha(\mathbf{X})) \geq 1-\alpha.$$

4 Point Estimation

4.1 The Method of Moments

Reading(s) 4.1. Wasserman (2004): 9.2. Hogg et al. (2018): 6.4.

4.1.1 The method

The basic idea of method of moments (MoM) is to estimate population moments by their sample analogues, resulting in moment conditions/equations with the parameters' estimates as the unknown quantities to be solved for. By solving the moment conditions/equations, we obtain the estimator of the method of moments.

Suppose our model is $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta)$ where $\theta \in \Theta \subset \mathbb{R}^p$. Further assume the existence of $p^{th}-$moment of X_1:

$$\begin{aligned} \mathbb{E}[X_1] &= m_1(\theta) \\ \mathbb{E}[X_1^2] &= m_2(\theta) \\ &... \\ \mathbb{E}[X_1^p] &= m_p(\theta), \end{aligned}$$

where the functions $m_j(\theta)$ for $j = 1, ..., p$ can be calculated in closed form with the model $X_1 \sim f(x;\theta)$. That is, the parameter θ determines the true values of the moments. Reversely, once the values of the moments are given, we can solve for θ as a function of the moments $\{m_j(\theta)\}$. As the system of equations have p equations and p unknowns, in principle there exists a solution.

Clearly, in practice we do not know the values of the moments. However, their sample analogues are readily available. Then, by replacing the population moments by their sample analogues, we obtain a system of p equations:

$$\begin{aligned} \frac{1}{n}\sum_{i=1}^n X_i &= m_1(\theta) \\ \frac{1}{n}\sum_{i=1}^n X_i^2 &= m_2(\theta) \\ &... \\ \frac{1}{n}\sum_{i=1}^n X_i^p &= m_p(\theta), \end{aligned}$$

which are used to solve for θ. These are called moment conditions. The method of moments estimates are obtained by solving the equations.

The intuition of the method of moments is that the sample analogues of population moments are reasonably close to the truth and the equation solving operation is continuous. Therefore, the solution to the sample version of the system of equations is close to the solution to the population version of the system of equations. More importantly, this implies that the method of moments implicitly requires a large sample size to provide reasonable estimates, otherwise the sample analogues could be quite different from their true values.

Example 4.1 (Right endpoint of the Uniform distribution)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} U(0,\theta)$, find the method of moments estimator for θ. This is done in three steps:

- Find the moment condition: $\mathbb{E}X_1 = \int_0^\theta x\frac{1}{\theta}dx = \frac{1}{2}\theta$.
- Plug in the sample analogue: $\frac{1}{n}\sum_{i=1}^n X_i = \frac{1}{2}\theta$.
- Solve the equation for MoM estimator: $\hat{\theta}_{MoM} = 2\frac{1}{n}\sum_{i=1}^n X_i$.

Suppose the sample is $\{1, 2, 9\}$ (three data points from discrete uniform on $\{1, 2, ..., 9\}$), then $\hat{\theta}_{MoM} = 2 \times 4 = 8$ which is less than 9. It should not surprise you seeing unreasonable MoM estimates in small samples. ♢

Example 4.2 (Univariate Normal mean)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\theta, 1)$, find the method of moments estimator for θ. This is done in three steps:

- Find the moment condition: $\mathbb{E}X_1 = \theta$.
- Plug in the sample analogue: $\frac{1}{n}\sum_{i=1}^n X_i = \theta$.
- Solve the equation for MoM estimator: $\hat{\theta}_{MoM} = \frac{1}{n}\sum_{i=1}^n X_i$.

♢

Example 4.3 (Univariate Normal mean and variance)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, find the method of moments estimator for (μ, σ^2). This is done in three steps:

- Find the moment conditions :
$$\begin{aligned} \mathbb{E}[X_1] &= \mu \\ \mathbb{E}[X_1^2] &= \sigma^2 + \mu^2. \end{aligned}$$
- Plug in the sample analogues:
$$\begin{aligned} \frac{1}{n}\sum_{i=1}^n X_i &= \mu \\ \frac{1}{n}\sum_{i=1}^n X_i^2 &= \sigma^2 + \mu^2. \end{aligned}$$
- Solve the equations for MoM estimator:
$$\begin{aligned} \hat{\mu}_{MoM} &= \frac{1}{n}\sum_{i=1}^n X_i, \\ \hat{\sigma}^2_{MoM} &= \frac{1}{n}\sum_{i=1}^n X_i^2 - \left(\frac{1}{n}\sum_{i=1}^n X_i\right)^2. \end{aligned}$$

♢

4.1.2 Implementation details

The existence of moment(s) The method of moments requires the existence of certain moment(s). If the desired moment does not exist, we can transform the data, e.g. logarithmic transformation, truncation, etc., such that the transformed data has desirable moments.

If we truncate the data to $[-M_1, M_2]$ for two positive numbers M_1 and M_2, then all moments of the truncated version exist. Consequently, specifying the truncation levels M_1 and M_2 becomes part of the analysis. It is worthwhile to mention that the truncation levels can grow with sample size.

Example 4.4 (Cauchy distribution)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Cauchy(\theta, 1)$, can we find the method of moments estimator for θ?

The pdf of $Cauchy(\theta, 1)$ is

$$f(x; \theta) = \frac{1}{\pi}\frac{1}{(x-\theta)^2 + 1},$$

and $\mathbb{E}[|X_1|] = \infty$. So the first moment cannot be used to construct an MoM estimator. Though fractional moments exist, they are not available in closed form for our location problem. (See, e.g., Rothenberg et al., 1964, and the follow-up work.))

On the other hand, for $\alpha \in (-1, 1)$ and $X_1 \sim Cauchy(0, \sigma)$, the following fractional moment exists $\mathbb{E}[|X_1|^\alpha] = \sigma^\alpha sec(\pi\alpha/2)$, and can be used to construct MoM estimators for the scale parameter. ♢

Algorithms to compute MoM estimates Solving a system of non-linear equations analytically is typically challenging and tedious. With the help of numerical solvers, there are basically two general strategies to compute MoM estimates.

The first strategy is to use the numerical solvers of nonlinear equations. In R, one such function is `nleqslv`. We showcase the use of the function in the following example.

Example 4.5 (Uniform)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Unif(\theta_1, \theta_2)$. We use the first two moments to construct the estimator:

$$\begin{aligned} \frac{1}{n}\sum_{i=1}^n X_i &= \frac{1}{2}(\hat{\theta}_1 + \hat{\theta}_2) \\ \frac{1}{n}\sum_{i=1}^n X_i^2 &= \frac{1}{3}(\hat{\theta}_1^2 + \hat{\theta}_1\hat{\theta}_2 + \hat{\theta}_2^2). \end{aligned}$$

The moment matching errors in presented in Figure 4.1:

$$\begin{aligned} \text{the first moment's matching error} &= \frac{1}{2}(\theta_1 + \theta_2) - \frac{1}{n}\sum_{i=1}^n X_i \\ \text{the second moment's matching error} &= \frac{1}{3}(\theta_1^2 + \theta_1\theta_2 + \theta_2^2) - \frac{1}{n}\sum_{i=1}^n X_i^2. \end{aligned}$$

The intersection of the two surfaces and the horizontal plane is the MoM estimate.

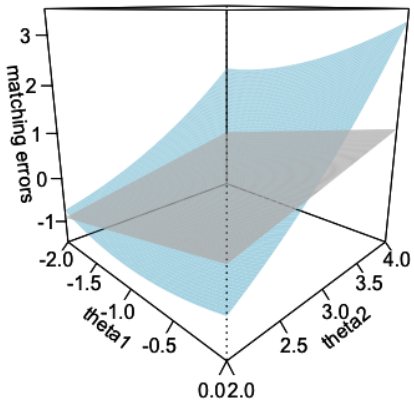


Figure 4.1: Surfaces of moment matching errors.

```
X <- runif(100, min = -1, max = 3)
angle = 45
persp(seq(-2,0,length.out = 100),
      seq(2,4, length.out = 100),
      outer(seq(-2,0,length.out = 100),
            seq(2,4, length.out = 100),
            function(x,y) {(x^2+y^2+x*y)/3 - mean(X^2)}),
      xlab = "theta1", ylab = "theta2", zlab = "matching errors",
      zlim = c(-1.5,3.5),
      col='lightblue',ticktype = "detailed",border = NA,theta = angle)
par(new=TRUE)
persp(seq(-2,0,length.out = 100),
      seq(2,4, length.out = 100),
      outer(seq(-2,0,length.out = 100),
            seq(2,4, length.out = 100),
            function(x,y) {(x+y)/2 - mean(X)}),
      xlab = "theta1", ylab = "theta2", zlab = "matching errors",
      zlim = c(-1.5,3.5),
      col='grey',ticktype = "detailed",border = NA,theta = angle)
```

We use `nleqslv` to solve the following equations:

$$\begin{aligned} \frac{1}{2}(\hat{\theta}_1 + \hat{\theta}_2) - \frac{1}{n}\sum_{i=1}^n X_i &= 0 \\ \frac{1}{3}(\hat{\theta}_1^2 + \hat{\theta}_1\hat{\theta}_2 + \hat{\theta}_2^2) - \frac{1}{n}\sum_{i=1}^n X_i^2 &= 0. \end{aligned}$$

```r
moments.eq <- function(theta){
    m1 <- (theta[1]+theta[2])/2
    m2 <- (theta[1]^2+theta[2]^2+theta[1]*theta[2])/3
    return(c(m1-mean(X),m2-mean(X^2)))
}
nleqslv(c(min(X)-10,max(X)+10), moments.eq)
```

♢

The second strategy is to turn the problem of equation solving to optimization. Instead of solving the moment equations directly, we choose the parameters to minimize the ℓ_2 loss of the moment matching errors.

Suppose we have p moment conditions, the optimization problem is to minimize

$$\sum_{k=1}^{p} \left(\frac{1}{n} \sum_{i=1}^{n} X_i^k - m_k(\theta) \right)^2$$

with respect to $\theta \in \Theta$. Since we have p parameters, the optimal value of the objective function should be 0.

In this case, we use optimization algorithms to solve for the MoM estimates. In R, the function `optim` is a general purpose optimization function.

Example 4.6 (Uniform, continued)**.** Under the same setup, the objective function to be minimized is

$$\left(\frac{1}{2}(\hat{\theta}_1 + \hat{\theta}_2) - \frac{1}{n} \sum_{i=1}^{n} X_i \right)^2 + \left(\frac{1}{3}(\hat{\theta}_1^2 + \hat{\theta}_1\hat{\theta}_2 + \hat{\theta}_2^2) - \frac{1}{n} \sum_{i=1}^{n} X_i^2 \right)^2.$$

Next we use `optim` to numerically solve for the MoM estimates.

```r
moments.opt <- function(theta, X){
    m1 <- (theta[1]+theta[2])/2
    m2 <- (theta[1]^2+theta[2]^2+theta[1]*theta[2])/3
    return((m1-mean(X))^2+(m2-mean(X^2))^2)
}
X <- runif(100, min = -3, max = 5)
optim(c(min(X)-1,max(X)+1), moments.opt,X=X)
```

♢

4.1.3 Asymptotic theory: Consistency

Section 4.1.2 describes a few algorithms to produce the point estimates of the method of moments. In the simulation examples, the point estimates are very close to the true values. One may expect that with more and more observations, the method of moments estimates will be more and more accurate. Indeed, the method of moments estimators are consistent if the solution of the moment conditions is continuous.

For many cases, consistency of the method of moments estimators is immediate by law of large numbers. For instance, the mean parameter of Bernoulli random variables, Gaussian random variables, and many others.

Example 4.7 (Bernoulli mean)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta_0)$. The MoM estimator for θ_0 is $\hat{\theta}_{M0M} := \frac{1}{n}\sum_{I=1}^{n} X_i$. By LLN, $\hat{\theta}_{MoM} \xrightarrow{p} \theta_0$. ♢

For more complicated problems, the method of moments estimators are not directly sample averages (e.g. the toy example of inferring the endpoints of a univariate uniform distribution), so LLN

cannot be applied directly to show consistency. In general, consistency of MoM estimators is established by both the law of large numbers and the *continuous mapping theorem* (CMT).

The continuous mapping theorem states that the convergence in probability and distribution is preserved under continuous transformations.

Theorem 4.1 (Continuous Mapping Theorem (CMT)). *Let $X_n, n = 1, ...,$ and X be random variables. Let $g(\cdot)$ be a continuous function. If $X_n \xrightarrow{p} X$, then $g(X_n) \xrightarrow{p} g(X)$.*

Though Theorem 4.1 is stated for random variables, the same type of results also hold for random vectors and multivariate continuous functions. In the case of random vectors, $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$ if their Euclidean distance $||\mathbf{X}_n - \mathbf{X}||$ (or in ℓ_2 norm) converges to 0 in probability.

Convergence results are preserved under arithmetic operations. As arithmetic operations are continuous, such results are also known as continuous mapping theorem. The following can be implied by the random vector version of the CMT.

Theorem 4.2 (CMT for arithmetic operations). *Let X_n, X, Y_n, Y be random variables, and c be some constant. Then*

- *If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then $X_n + Y_n \xrightarrow{p} X + Y$.*
- *If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then $X_n Y_n \xrightarrow{p} XY$.*
- *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$, then $X_n + Y_n \xrightarrow{d} X + c$. (Slutzky's theorem)*
- *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$, then $X_n Y_n \xrightarrow{d} cX$. (Slutzky's theorem)*

Now with the CMT, we can prove MoM estimators are consistent under mild conditions.

Theorem 4.3 (Consistency of MoM estimators). *Suppose $X_1, ..., X_n \overset{iid}{\sim} f(x; \theta_0)$, the $p-$dimensional parameter $\theta_0 \in \Theta \subseteq \mathbb{R}^p$, and the first p moments exist. That is, in the moment conditions*

$$\begin{aligned} \mathbb{E}_\theta[X_1] &= m_1(\theta) \\ \mathbb{E}_\theta[X_1^2] &= m_2(\theta) \\ &... \\ \mathbb{E}_\theta[X_1^p] &= m_p(\theta), \end{aligned}$$

the functions $\{m_j(\theta)\}_{j=1}^p$ are all well-defined. Assume there exists a unique solution to the system of moment equations:

$$\theta = g\left(\mathbb{E}_\theta[X_1], \mathbb{E}_\theta[X_1^2], ..., \mathbb{E}_\theta[X_1^p]\right),$$

and $g(\cdot)$ is continuous. Then, the method of moments estimator

$$\hat{\theta} = g\left(\frac{1}{n}\sum_{i=1}^n X_i, \frac{1}{n}\sum_{i=1}^n X_i^2, ..., \frac{1}{n}\sum_{i=1}^n X_i^p\right) \xrightarrow{p} \theta_0.$$

Proof. By LLN, $\frac{1}{n}\sum_{i=1}^n X_i^k \xrightarrow{p} \mathbb{E}_{\theta_0}[X_1^k]$ for all $k = 1, ..., p$. In fact, the convergence in probability for all p moments hold simultaneously. (A simple application of the union bound.)

By continuous mapping theorem and the assumption that $g(\cdot)$ is continuous, consistency of the method of moments estimator follows. □

The key ingredient is the assumption that $g(\cdot)$ is continuous. For many parametric models, the moments depend on the parameter values continuously, that is, the moments are continuous in the parameters. As the number of equations is the same as the number of unknowns/parameters, the inverse function $g(\cdot)$ can also be continuous under very mild conditions (Brouwer invariance of domain theorem). Indeed, the continuity assumption is a mild assumption. In real data applications, this assumption is verifiable.

4.1.4 The "plug-in" estimator

In general, practitioners are not always intersted in the model parameters. The parameter of interest, say η, is often a *continuous* function of the model parameters θ, say $\eta = g(\theta)$.

Example 4.8 (Effect size)**.** To evaluate the effect size of a vaccine, the treatment group is assigned the vaccine and the outcomes of the individuals of the treatment group are denoted as $\{X_1, ..., X_n\}$, and the control group is assigned the placebo and the outcomes are denoted as $\{Y_1, ..., Y_m\}$. Since the assignment of the treatment status and the control status is random, it is natural to assume $\{X_i\}$ and $\{Y_j\}$ are independent, and within each group, the outcomes are iid.

Suppose the data is the status of the infection with 1 denoting infected and 0 denoting not infected. (In other applications, the outcomes could be some other metrics, e.g., blood pressure.) Therefore, the data is modelled as two independent groups of Bernoulli random variables $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta_1)$, and $Y_1, ..., Y_m \overset{iid}{\sim} Ber(\theta_2)$ for some true parameters $(\theta_{0,1}, \theta_{0,2})$. The parameter of interest is the effect size:

$$\eta = g(\theta_1, \theta_2) = \theta_1 - \theta_2.$$

♢

Suppose we have a consistent estimator $\hat{\theta}$ for θ, we can *plug in* the estimator $\hat{\theta}$ to the continuous function $g(\cdot)$, and then the estimator for η,

$$\hat{\eta} := g(\hat{\theta}),$$

is a consistent estimator for η by continuous mapping theorem.

Example 4.9 (Effect size, continued)**.** The plug-in estimator for $\eta \equiv \theta_1 - \theta_2$ is

$$\hat{\eta} = g(\hat{\theta}_1, \hat{\theta}_2) = \hat{\theta}_1 - \hat{\theta}_2.$$

♢

As there is seldom lack of consistent estimators (the method of moments estimators are often consistent) and the parameter(s) of interest is often a continuous function of model parameters, the idea of "plug-in" is widely applicable.

Example 4.10 (Bernoulli variance)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta_0)$, and our main interest is in the variance of X_1:

$$V_{\theta_0}(X_1) \equiv \theta_0(1 - \theta_0).$$

If we write down the moment conditions for both the mean and the variance directly, then we would end up with two equations but only one unknown. In general, there is no solution to the system of equations with more unknowns than equations. (unless some equations are redundant.)

Since the variance of a Bernoulli random variable is a continuous function of the mean parameter, we can construct a consistent estimator for the variance using the method of moments estimator of the mean parameter. That is, a consistent estimator of the variance can be

$$\hat{V}_{plug-in} := \bar{X}_n(1 - \bar{X}_n)$$

where $\bar{X}_n = \frac{1}{n}\sum_{I=1}^{n} X_i$.

By LLN and continuous mapping theorem, $\hat{V}_{plug-in} \xrightarrow{p} V(\theta_0)$. $\diamondsuit$

4.2 Maximum Likelihood Estimation

Reading(s) 4.2. Wasserman (2004) 9.3, 9.6, 9.13.1 ; Hogg et al. (2018): 6.4.

4.2.1 Maximizing the Likelihood Function

The idea of maximum likelihood is pretty old. Stigler (2007) explains the history of maximum likelihood. The foundational theory of maximum likelihood estimation is mostly developed by R. A. Fisher, almost single-handedly.

Definition 4.1 (Likelihood function)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta)$ where $x \in \mathcal{X}$, $\theta \in \Theta$ is the parameter(s) of $f(x;\theta)$, and Θ is the parameter space. The *likelihood function* is defined by

$$\mathcal{L}_n(\theta) = \prod_{i=1}^n f(X_i;\theta).$$

The *log-likelihood function* is defined by $\ell_n(\theta) = \log\mathcal{L}_n(\theta)$.

While the likelihood function is the joint pdf/pmf of the whole dataset under the model $f(x;\theta)$, that is, $\mathcal{L}_n(\theta) : \mathcal{X}^n \times \Theta \to [0,\infty)$, it is treated as a function of θ:

$$\mathcal{L}_n(\theta) : \Theta \to [0,\infty),$$

because the data part is realized and fixed in practice.

Example 4.11. Suppose $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta)$. The likelihood function is

$$\mathcal{L}_n(\theta) = \prod_{i=1}^n \theta^{X_i}(1-\theta)^{1-X_i} = \theta^{\sum_{i=1}^n X_i}(1-\theta)^{n-\sum_{i=1}^n X_i}.$$

Suppose $X_1, ..., X_n \overset{iid}{\sim} Unif(0,\theta)$. The likelihood function is

$$\mathcal{L}_n(\theta) = \prod_{i=1}^n \left(\frac{1}{\theta}1\left(x_i \in (0,\theta)\right)\right) = \frac{1}{\theta^n}1\left(\max(\{x_i\}) < \theta\right)$$

where $\min(\{x_i\}) > 0$ holds with probability 1. ♢

The inferential goal is to infer θ_0 based on the data $\{X_i\}_{i=1}^n$. For the same data $\{X_i\}_{i=1}^n$, different parameters give different likelihoods. The likelihood function can be a measure of the "plausibility" of θ. To be more specific, let $\theta_1 \neq \theta_2 \in \Theta$ such that $\mathcal{L}_n(\theta) \neq 0$ for $\theta \in \{\theta_1, \theta_2\}$, the likelihood ratio

$$\frac{\mathcal{L}_n(\theta_1)}{\mathcal{L}_n(\theta_2)}$$

is a measure of the "plausibility" of θ_1 relative to θ_2. For instance, if the ratio is 2, then θ_1 is twice as "plausible" as θ_2.

Definition 4.2 (MLE)**.** The maximum likelihood estimator (MLE), denoted by $\hat{\theta}$, is the value of θ that maximizes $\mathcal{L}_n(\theta)$:

$$\hat{\theta}_{MLE} \in \arg\max_{\theta\in\Theta} \mathcal{L}_n(\theta) = \arg\max_{\theta\in\Theta} \ell_{\boldsymbol{x}}(\theta), \tag{1}$$

where we note logarithmic transformation is monotonic and does not change the solution to the optimization problem.

The solution to the optimization problem is not guaranteed to be unique, so we use the notation "$\in$" for generality. When the solution to the optimization problem is unique, we can replace "$\in$" with "$=$". Also, the solution can be non-existent. For the simplicity of presentation, we can safely assume the MLE is unique from now on. [reference for more careful presentation]

As we take a frequentist viewpoint, the likelihood, a function of the data, is intrinsically random: for different realizations of the dataset, their likelihoods can be different even given the same θ. Therefore, the maximizer of the likelihood is also random, which allows us to study its frequentist properties, e.g., bias, MSE, consistency, rate of convergence, etc.

For most parametric problems, we can safely assume $\ell_x(\theta)$ is differentiable in θ, then the *first order condition* (FOC) for the likelihood maximization problem is

$$\dot{\ell}_{\boldsymbol{x}}(\hat{\theta}) = 0, \tag{2}$$

where we follow the conventional notation for log-likelihood:

$$\begin{array}{rcl} \ell_{x_i}(\theta) & = & \log(f(x_i;\theta)), \text{ and} \\ \ell_{\boldsymbol{x}}(\theta) & = & \sum_{i=1}^n \ell_{x_i}(\theta) = \sum_{i=1}^n \log(f(x_i;\theta)), \end{array}$$

and the dot notation for self-differentiation:

$$\begin{array}{rcl} \dot{\ell}_x(\theta) & = & \frac{\partial}{\partial\theta}\ell_x(\theta), \text{ and} \\ \ddot{\ell}_x(\theta) & = & \frac{\partial^2}{\partial\theta^2}\ell_x(\theta) \text{ for twice differentiation.} \end{array}$$

More explicitly, MLE point estimate $\hat{\theta}$ is obtained by solving the equation $\sum_{i=1}^n \dot{\ell}_{x_i}(\theta) = 0$ for θ.

To make sure $\hat{\theta}$ indeed maximize the (log-)likelihood, we also need to verify the *second order condition* (SOC):

$$\ddot{\ell}_{\boldsymbol{x}}(\hat{\theta}) < 0,$$

provided that the second derivative exists.

Example 4.12 (Bernoulli)**.** $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta)$. The calculation is done in three steps.

1. The likelihood function

   $$\mathcal{L}_n(\theta) = \prod_{i=1}^n \theta^{X_i}(1-\theta)^{1-X_i} = \theta^{\sum_{i=1}^n X_i}(1-\theta)^{n-\sum_{i=1}^n X_i},$$

   and the log-likelihood function $\ell_{\boldsymbol{x}} = (\sum_{i=1}^n X_i)\log(\theta) + (n - \sum_{i=1}^n X_i)\log(1-\theta)$.

   [a plot of the Bernoulli likelihood function here]

2. The FOC is

   $$\dot{\ell}_{\boldsymbol{x}} = (\sum_{i=1}^n X_i)\frac{1}{\theta} - (n - \sum_{i=1}^n X_i)\frac{1}{1-\theta} = 0$$

   and $\hat{\theta} = \frac{1}{n}\sum_{i=1}^n X_i$.

3. Finally, we verify SOC $\ddot{\ell}_{\boldsymbol{x}}(\hat{\theta}) = -(\sum_{i=1}^n X_i)\frac{1}{\hat{\theta}^2} - (n - \sum_{i=1}^n X_i)\frac{1}{(1-\hat{\theta})^2} < 0$ is satisfied.

Now we can examine the frequentist properties of the estimator.

- Bias
- MSE
- Consistency

$\diamondsuit$

Example 4.13 (Gaussian)**.** $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$. The calculation is done in three steps.

1. Write down the likelihood

   [a plot of the Normal likelihood function here]
2. Write down the FOC and solve for the MLE.
3. Verify SOC given σ^2.

Now we can examine the frequentist properties of the estimator.

- Bias
- MSE
- Consistency

$\diamondsuit$

Example 4.14 (Uniform right endpoint)**.** $X_1, ..., X_n \overset{iid}{\sim} Unif(0, \theta)$. The calculation is done in three steps. Write down the likelihood:

$$\mathcal{L}_n(\theta) = \prod_{i=1}^n \left(\frac{1}{\theta} 1\left(x_i \in (0, \theta)\right) \right) = \frac{1}{\theta^n} 1\left(\max(\{x_i\}) < \theta\right)$$

where $\min(\{x_i\}) > 0$ holds with probability 1. Despite not differentiable, the likelihood function is still tractable.

[a plot of the Uniform likelihood function here]

To implement maximum likelihood estimation, two remedies are available. The first remedy is to modify the model: $X_1, ..., X_n \overset{iid}{\sim} Unif(0, \theta]$, so that the new likelihood function $\mathcal{L}'_n(\theta)$ becomes

$$\mathcal{L}'_n(\theta) = \frac{1}{\theta^n} 1\left(\max(\{x_i\}) \leq \theta\right)$$

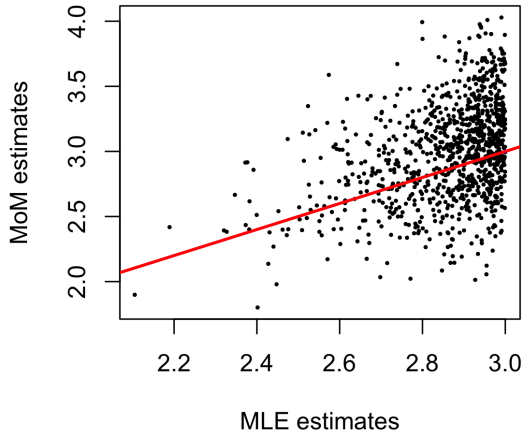
whose maximizer is well-defined.

Alternatively, we can solve for $\hat{\theta}$ such that the supremum of $\mathcal{L}_n(\theta)$ is achieved:

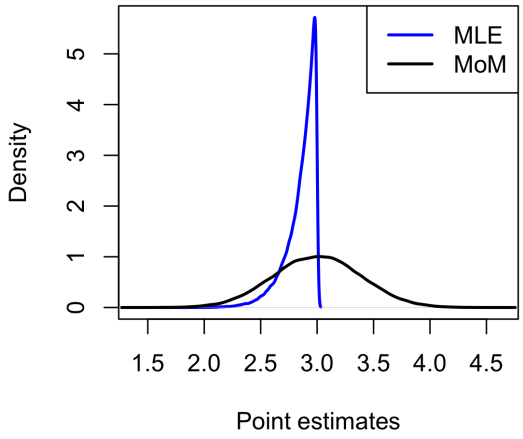
$$\hat{\theta} = \arg\sup_{\theta} \mathcal{L}_n(\theta)$$

instead of $\hat{\theta} = \arg\max_\theta \mathcal{L}_n(\theta)$.

Both remedies imply $\hat{\theta}_{MLE} = \max(\{X_i\})$. Recall $\hat{\theta}_{MoM} = 2\bar{X}_n$ for the same model. We can compare their (finite sample) sampling distributions by simulations. In the following simulation example, synthetic sample size is 20 and $\theta_0 = 3$.



The 45° line is in red. $N_{MC} = 10^3$



For a better density plot, $N_{MC} = 10^5$.

Figure 4.2: Compare the maximum likelihood estimator and the MoM estimator.

```r
Nmc <- 1e3 ## sampling distribution
n <- 20 ## synthetic sample size
theta <- 3 ## true value
hat.theta <- matrix(NA, nrow = Nmc, ncol = 2)
for(i in 1:Nmc){
    X <- runif(n,0,theta)
    hat.theta[i,1] <- max(X) ## MLE
    hat.theta[i,2] <- 2*mean(X) ## MoM
}

plot(hat.theta[,1], hat.theta[,2], pch=19, cex=0.25,
     xlab = "MLE estimates", ylab = "MoM estimates")
points(theta,theta, col = "blue", cex=0.25, pch=19)
abline(a=0,b=1,lwd=2,col = "red")

plot(density(hat.theta[,1]),
     xlim = c(min(hat.theta), max(hat.theta)),
     main = "", lwd = 2,col ="blue",
     xlab = "Point estimates")
lines(density(hat.theta[,2]),col = "black",lwd = 2)
legend("topright", c("MLE","MoM"),
       col = c("blue","black"), lty = 1,lwd = 2)
```

Now we can examine the frequentist properties of the estimator.

- Bias
- MSE
- Consistency

♢

4.2.2 Equivariance: the "plug-in" principle

In the above examples, we obtain maximum likelihood estimates for the parameters of the data generating processes. In many scientific and biostatistical applications, the main interest is some function of the parameters, e.g., half-life time, profit, etc.. How to obtain MLE for those functions?

By inspecting the optimization algorithm (1), it is immediate that MLE is equivariant. This is also called "plug-in".

Theorem 4.4 (Equivariance of MLE)**.** *Let $\tau = g(\theta)$ be a function of θ. Let $\hat{\theta}$ be the MLE of θ. Then, $\hat{\tau} = g(\hat{\theta})$ is the MLE of τ.*

Proof. We need to show $\hat{\tau} = g(\hat{\theta})$ is *a* maximizer of the likelihood function parameterized in τ. The reparameterization changes the model in the sense that

$$f'(x;\tau) = f(x; g^{-1}(\tau))$$

where $g^{-1}(\cdot)$ denotes the pre-image of $g(\cdot)$. Since $g(\cdot)$ needs not to be invertible, $g^{-1}(\tau)$ and $f'(x;\tau)$ can be two sets: $g^{-1}(\tau) \equiv \{\theta : g(\theta) = \tau\}$ and $f(x; g^{-1}(\tau)) \equiv \{f(x;\theta) : g(\theta) = \tau\}$. When $g(\cdot)$ is invertible, the two sets degenerate to singletons.

The likelihood maximization problem over Θ can be solved in two steps: first find the maximum on the slice $\{\theta : g(\theta) = \tau\}$ given some τ, then find the maximum in terms of τ over $g(\Theta)$:

$$f(\boldsymbol{X};\hat{\theta}) = \max_{\theta\in\Theta} f(\boldsymbol{X};\theta) = \max_{\tau\in g(\Theta)} \left\{ \max_{\{\theta: g(\theta)=\tau\}} f(\boldsymbol{X};\theta) \right\}.$$

Since $\hat{\theta} \in \{\theta : g(\theta) = g(\hat{\theta})\}$, $\hat{\tau} = g(\hat{\theta})$ also maximizes $f'(\boldsymbol{X};\tau)$:

$$f'(\boldsymbol{X};\hat{\tau}) = \max_{\tau\in g(\Theta)} \left\{ \max f'(\boldsymbol{X};\tau) \right\} = \max_{\tau\in g(\Theta)} \left\{ \max_{\{\theta: g(\theta)=\tau\}} f(\boldsymbol{X};\theta) \right\}.$$

Therefore, $\hat{\tau}$ is a maximum likelihood estimator of τ by definition.

□

Example 4.15 (Gamma distribution)**.** The Gamma distribution has two parameterizations: the shape-scale parameterization

$$f(x;k,\theta) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta},$$

and the shape-rate parameterization

$$f(x;\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}.$$

The two parameterizations differ in the scale/rate parameter and the scale parameter is reciprocal to the rate parameter.

Suppose $X_1, ..., X_n \overset{iid}{\sim} Ga(1,\theta)$ under the shape-scale parameterization, we can find the MLE for θ. On the other hand, suppose $X_1, ..., X_n \overset{iid}{\sim} Ga(1,\beta)$ under the shape-rate parameterization, we can find the MLE for β. Indeed, $\hat{\beta}\hat{\theta} = 1$. ♢

Example 4.16. Suppose the concentration of some chemical decays exponentially:

$$\mathbb{E}[Y(t)] = Y(0)e^{-\lambda t}$$

where t denotes time and λ is some rate parameter that controls the speed of the decay. The chemists are interested in the half life time which is the time point where the concentration is half of its initial value. By direct calculation, half time is $\frac{\log(2)}{\lambda}$. Then, its MLE is $\frac{\log(2)}{\hat{\lambda}}$ where $\hat{\lambda}$ is the MLE of λ.

♢

Example 4.17. Let $X_1, ..., X_n \overset{iid}{\sim} N(\theta, 1)$. The MLE for θ is $\hat{\theta} = \frac{1}{n}\sum_{i=1}^n X_i$.

- The MLE for $\tau = e^\theta$ is $\hat{\tau} = e^{\hat{\theta}} = e^{\frac{1}{n}\sum_{i=1}^n X_i}$.
- The MLE for $\tau = \theta^2$ is $\hat{\tau} = \hat{\theta}^2 = \left(\frac{1}{n}\sum_{i=1}^n X_i\right)^2$.
- The MLE for $\tau = 1/\theta$ is $\hat{\tau} = 1/\hat{\theta} = 1/\left(\frac{1}{n}\sum_{i=1}^n X_i\right)$.

♢

4.2.3 Consistency of MLE ☕

As discussed in the introduction, consistency is a basic requirement for all reasonable estimators. The first question to ask is if the MLE $\hat{\theta} \xrightarrow{p} \theta_0$ under certain conditions. The answer is affirmative and the *sufficient* conditions are expressed in Kullbakc-Leibler (KL) divergence.

KL divergence is a measure of "distance" between two probability distributions. For two probability distributions with pdf's being f and g supported on the same domain, the KL divergence between f and g is

$$KL(f||g) = \int f(x)\log\frac{f(x)}{g(x)}dx,$$

provided the ratio f/g and the integration are well-defined.

Clearly, if $g = f$, $KL(f||g) = KL(f||f) = 0$. It is also worthwhile to emphasize that $KL(f||g)$ can be different from $KL(g||f)$, so it is not a metric or distance in the sense of mathematical (real/functional) analysis.

For parametric models, it is convenient to write $KL(\theta||\eta) \equiv KL(f(x;\theta)||f(x;\eta))$. That is, we use the parameters to denote the corresponding pdf's.

Theorem 4.5 (Consistency of MLE)**.** *Suppose $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta_0)$ where $\theta_0 \in \Theta$, and X_i can be a vector. Let*

$$M_n(\theta) = \frac{1}{n}\sum_{i=1}^n \log\frac{f(x_i;\theta)}{f(x_i;\theta_0)}.$$

Suppose that

$$\sup_{\theta\in\Theta} |M_n(\theta) + KL(\theta_0||\theta)| \overset{\mathbb{P}_{\theta_0}}{\to} 0, \tag{3}$$

and that, for every $\epsilon > 0$,

$$\inf_{\theta:|\theta-\theta_0|\geq\epsilon} KL(\theta_0||\theta) > 0. \tag{4}$$

Then,

$$\hat{\theta} \xrightarrow{p} \theta_0 \text{ under } \mathbb{P}_{\theta_0}.$$

The assumption (3) asks for uniform convergence of $M_n(\theta)$ to $-KL(\theta_0||\theta)$. The assumption (4) requires that the optimum is uniquely identified. It is a mathematical exercise to show consistency.

Proof. For any positive constant ϵ, we hope to show that, as $n \to \infty$,

$$\mathbb{P}_{\theta_0}(|\hat{\theta} - \theta_0| \geq \epsilon) \to 0.$$

Pick any $\epsilon > 0$, and suppose $|\hat{\theta} - \theta_0| \geq \epsilon$. By assumption (4), there exists a constant $\delta > 0$ such that

$$KL(\theta_0||\hat{\theta}) > \delta.$$

Note $\hat{\theta}$ maximizes $M_n(\theta)$, $M_n(\theta_0) \leq M_n(\hat{\theta})$ by construction, then

$$\begin{aligned} KL(\theta_0||\hat{\theta}) &= M_n(\theta_0) + KL(\theta_0||\hat{\theta}) - M_n(\theta_0) - KL(\theta_0||\theta_0) \\ &\leq M_n(\hat{\theta}) + KL(\theta_0||\hat{\theta}) - M_n(\theta_0) - KL(\theta_0||\theta_0) \\ &\leq 2\sup_{\theta\in\Theta} |M_n(\theta) + KL(\theta_0||\theta)|. \end{aligned}$$

By assumption (3), $KL(\theta_0||\hat{\theta}) \overset{\mathbb{P}_{\theta_0}}{\to} 0$, and

$$\mathbb{P}_{\theta_0}(|\hat{\theta} - \theta_0| \geq \epsilon) \leq \mathbb{P}_{\theta_0}(KL(\theta_0||\hat{\theta}) > \delta) \to 0.$$

□

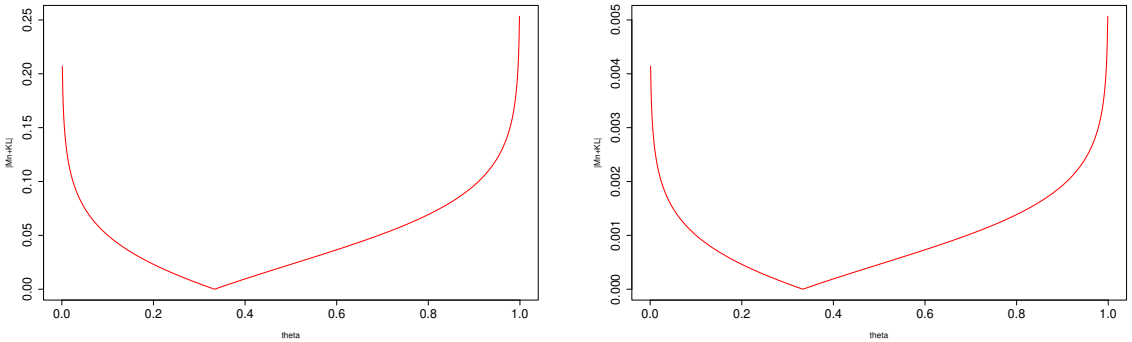
Example 4.18 (Bernoulli)**.** $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta)$. We know the MLE of θ is $\frac{1}{n}\sum_{i=1}^n X_i$. By LLN, we know MLE is consistent.

Now we calculate the quantities involved in the theorem:

$$\begin{aligned} M_n(\theta) = &\ \tfrac{1}{n}\textstyle\sum_{i=1}^n \log\frac{f(X_i;\theta)}{f(X_i;\theta_0)} = \left(\tfrac{1}{n}\textstyle\sum_{i=1}^n X_i\right)\log\left(\frac{\theta(1-\theta_0)}{\theta_0(1-\theta)}\right) + \log\frac{1-\theta}{1-\theta_0}, \\ -KL(\theta_0||\theta) = &\ \textstyle\sum_x f(x;\theta_0)\log\frac{f(x;\theta)}{f(x;\theta_0)} = \theta_0\log\left(\frac{\theta(1-\theta_0)}{\theta_0(1-\theta)}\right) + \log\frac{1-\theta}{1-\theta_0}. \end{aligned}$$

To check the assumption (3), we need to show the supremum (maximum) of the absolute difference $|M_n(\theta) + KL(\theta_0||\theta)| = |\left(\frac{1}{n}\sum_{i=1}^n X_i - \theta_0\right)\log\left(\frac{\theta(1-\theta_0)}{\theta_0(1-\theta)}\right)|$ over $\theta \in \Theta \equiv [0,1]$ converges to 0 in probability $\mathbb{P}_{\theta_0}$. Since $\log\left(\frac{\theta(1-\theta_0)}{\theta_0(1-\theta)}\right)$ is continuous and Θ is bounded, it is bounded on Θ. By continuous mapping theorem (or Slutsky theorem), the assumption (3) holds.

We can illustrate this by simulation. Let $\theta_0 = \frac{1}{3}$ and set the random seed as 1 in R. By increasing n, the maximum absolute difference $|M_n(\theta) + KL(\theta_0||\theta)|$ on $[0,1]$ decreases. (See Figure 4.3)



$n = 10$ $n = 1000$

Figure 4.3: $\sup_{\theta\in\Theta} |M_n(\theta) + KL(\theta_0||\theta)|$ decreases as n increases

To check the assumption (4), it is equivalent to solve the constrained optimization problem: for some $\epsilon > 0$,

$$\min_{\theta:|\theta-\theta_0|\geq\epsilon} KL(\theta_0||\theta),$$

and show that the optimum is strictly greater than 0. Since $KL(\theta_0||\theta) = \theta_0 \log\frac{\theta_0}{\theta} + (1-\theta_0)\log\frac{1-\theta_0}{1-\theta}$ is differentiable in θ and the derivative in the neighborhood of θ_0 is non-zero, it is follows that the minimum is strictly greater than zero.

$\diamondsuit$

Example 4.19 (Univariate Normal)**.** $X_1, ..., X_n \overset{iid}{\sim} N(\theta, 1)$. We know the MLE of θ is $\frac{1}{n}\sum_{i=1}^n X_i$. By LLN, we know MLE is consistent.

Now we calculate the quantities involved in the theorem:

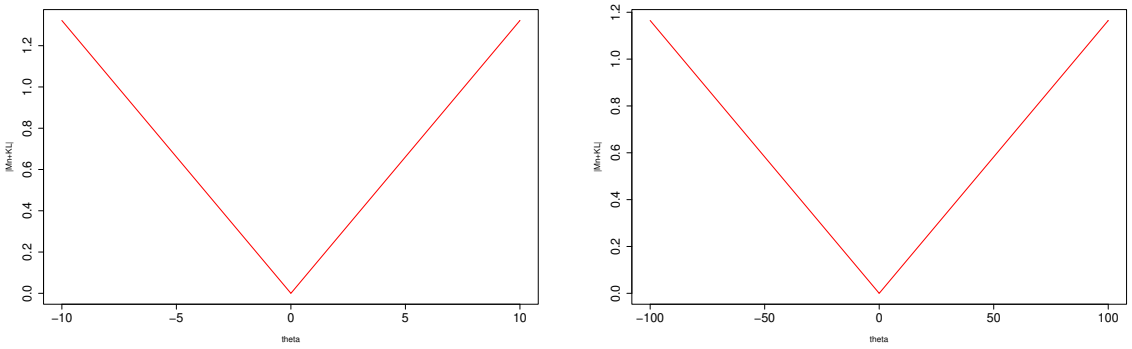
$$\begin{aligned} M_n(\theta) = &\ \tfrac{1}{n}\textstyle\sum_{i=1}^n \log\frac{f(X_i;\theta)}{f(X_i;\theta_0)} = \tfrac{1}{n}\sum_{i=1}^n \log\frac{e^{-\frac{1}{2}(X_i-\theta)^2}}{e^{-\frac{1}{2}(X_i-\theta_0)^2}} = -\tfrac{1}{2}\left(\theta^2 - 2\bar{X}_n(\theta-\theta_0) - \theta_0^2\right), \\ -KL(\theta_0||\theta) = &\ \textstyle\int f(x;\theta_0)\log\frac{f(x;\theta)}{f(x;\theta_0)}dx = \int f(x;\theta_0)\log\frac{e^{-\frac{1}{2}(x-\theta)^2}}{e^{-\frac{1}{2}(x-\theta_0)^2}}dx = -\tfrac{1}{2}(\theta-\theta_0)^2, \end{aligned}$$

where $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$.

To check the assumption (3), we need to show the supremum (maximum) of the absolute difference $|M_n(\theta) + KL(\theta_0||\theta)| = |(\bar{X}_n - \theta_0)(\theta - \theta_0)|$ over $\theta \in \Theta$ converges to 0 in probability $\mathbb{P}_{\theta_0}$. If we take $\Theta = \mathbb{R}$, then the supremum is ∞ with probability 1, and the assumption (3) does not hold. However, we know the MLE in this context is consistent.

For technical considerations, we would like to further assume Θ to be contained in $[-M, M]$ where $M > 0$ is some reasonably large constant. The assumption (3) in this case implicitly requires the parameter space to be bounded, which seems an artifact.

We can illustrate this by simulation. Let $\theta_0 = 0$ and set the random seed as 1 in R. By increasing n, the maximum absolute difference $|M_n(\theta) + KL(\theta_0||\theta)|$ on $[0, 1]$ decreases. (See Figure 4.4)



$n = 10, \Theta = [-10, 10]$ $\quad$ $n = 1000, \Theta = [-100, 100]$

Figure 4.4: $\sup_{\theta\in\Theta} |M_n(\theta) + KL(\theta_0||\theta)|$ depends on Θ

To check the assumption (4), the KL divergence takes a quadratic functional form in θ, so it is obvious that the assumption (4) holds.

$\diamondsuit$

4.2.4 Likelihood maximization is KL divergence minimization

To begin with, define the scaled log likelihood ratio as

$$M_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log\frac{f(x_i;\theta)}{f(x_i;\theta_0)}.$$

The trick here is to "normalize" the likelihood function by the likelihood evaluated at the true value which is constant with respect to θ (no need to worry about $f(x_i;\theta_0) = 0$). So $M_n(\theta)$ is a universal measure of the "plausibility" of θ.

Since we do not know θ_0 in practice, this metric is not feasible in real data analysis. Still, we can safely assume its existence in theoretical analysis.

With the scaling factor $\frac{1}{n}$, it follows by LLN,

$$M_n(\theta) \overset{\mathbb{P}_{\theta_0}}{\to} \mathbb{E}_{\theta_0}[\log\frac{f(X;\theta)}{f(X;\theta_0)}] = -\mathbb{E}_{\theta_0}[\log\frac{f(X;\theta_0)}{f(X;\theta)}] = -KL(\theta_0||\theta),$$

where $KL(\theta_0||\theta)$ denotes the Kullback-Leibler(KL) divergence between $\mathbb{P}_{\theta_0}$ and $\mathbb{P}_\theta$.

More generally, for two pdf's f and g supported on the same domain, the KL divergence between f and g is

$$KL(f||g) = \int f(x)\log\frac{f(x)}{g(x)}dx,$$

provided the ratio f/g and the integration are well-defined. KL divergence is roughly a distance metric for two distributions. It is worthwhile to emphasize that $KL(f||g)$ can be different from $KL(g||f)$, so it is not a metric or distance in (real/functional) analysis.

Loosely speaking, the convergence in probability means that when n is sufficiently large,

$$M_n(\theta) \approx -KL(\theta_0||\theta)$$

with sufficiently high probability. As $M_n(\theta)$ is $\ell_{\boldsymbol{x}}(\theta)$ up to an affine transformation, maximizing $\ell_{\boldsymbol{x}}(\theta)$ is the same as maximizing $M_n(\theta)$, hence, it is (approximately) the same as minimizing $KL(\theta_0||\theta)$ with respect to θ.

What is the solution of θ that minimizes $KL(\theta_0||\theta)$? To answer this question, note Jensen's inequality applied to $-\log(x)$,

$$KL(f||g) = \int f(x)(-\log\frac{g(x)}{f(x)})dx \geq -\log\left(\int f(x)\frac{g(x)}{f(x)}dx\right) = 0,$$

where "=" is obtained if $f(x) = g(x)$. In other words, if we fix $f(x)$ and minimize $KL(f||g)$ with respect to $g(x)$, the solution is to set $g(x) = f(x)$. Therefore, in our problem, the minimizer of $KL(\theta_0||\theta)$ is θ_0.

Intuitively speaking, by maximizing likelihood, we choose the distribution/parameter that is closest to the truth in the sense of KL divergence. Since the minimizer of the KL divergence is the truth itself, we *expect* the maximum likelihood estimates to be close to the true value.

Misspecification The perspective of KL divergence minimization helps us understand the properties of MLE for misspecified models. An inevitable reflection of MLE is what if our model specification is wrong? Indeed, in real data analysis, we postulate some models and do not think the models we fit are perfectly correct. When the models are misspecified, are the MLE results totally garbage? How vulnerable are the conclusions from MLE if assumptions are violated in real data?

Suppose the true data generating process is $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta_0)$, while our model is $X_1, ..., X_n \overset{iid}{\sim} g(x;\eta)$. Then, the MLE with respect to $g(x;\eta)$ is to choose the closest distribution to $f(x;\theta_0)$ from the model space $\{g(x;\eta) : \eta \in H\}$ in the sense of KL divergence. Therefore, if the model space is not too restrictive, we will have approximation to the true data generating process.

5 Confidence Intervals

5.1 Introduction

Reading(s) 5.1. Wasserman (2004): 6.3.2.

How accurate is our point estimator? We can evaluate the accuracy by calculating bias and MSE of the estimator, provided their existence. But these criteria are too concise and say little about the distribution of the (potential) estimation error.

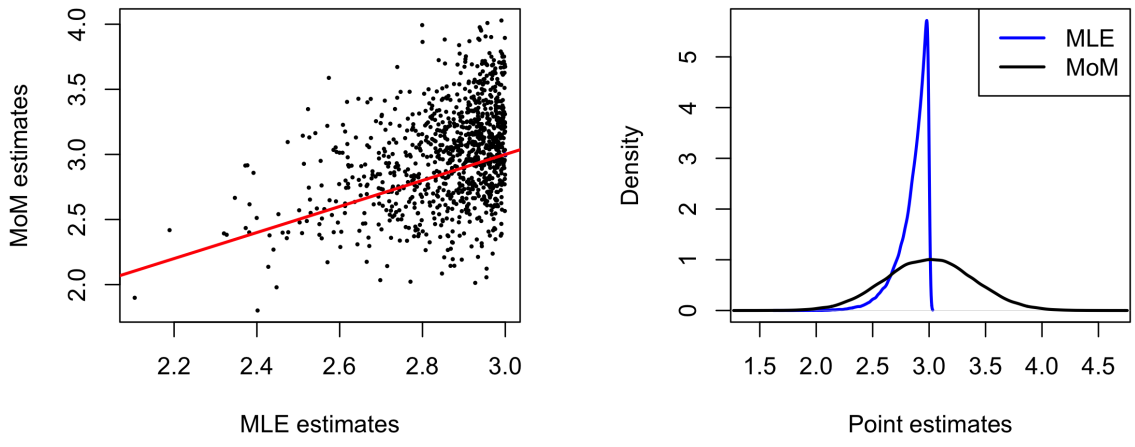
To go beyond the moment characteristics of the point estimator, classical statistics studies its sampling distribution. The definition of sampling distribution applies broadly to all statistics, including the point estimator. Recall the statistic is defined as a function of the data, containing no unknown quantities.

Definition 5.1. The sampling distribution of a random-sample-based *statistic* is its probability distribution.

Example 5.1 (sample mean)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ^2 known for simplicity. The estimator for μ, $\hat{\mu} \equiv \bar{X}_n$, is a statistic.

The sampling distribution of $\hat{\mu}$ is $N(\mu, \sigma^2/n)$, and the sampling distribution of estimation error, $\hat{\mu} - \mu$, follows $N(0, \sigma^2/n)$. Clearly, the sampling distribution of the estimation error provides much more information than the bias and the MSE of $\hat{\mu}$. ♢

Example 5.2 (Uniform right endpoint)**.** For $X_1, ..., X_{20} \overset{iid}{\sim} Unif(0, 3)$, the sampling distribution of $\hat{\theta}_{MoM} = 2\bar{X}_n$ and the sampling distribution of $\hat{\theta}_{MLE} = \max(\{X_i\}_i^n)$ are presented in the following figure.



The 45° line is in red. $N_{MC} = 10^3$

For a better density plot, $N_{MC} = 10^5$.

Figure 5.1: Sampling distributions of the maximum likelihood estimator and the MoM estimator.

♢

The sampling viewpoint is the core idea in confidence intervals/sets, as the goal is that the (data dependent) intervals/sets capture the true value with some pre-specified probability.

Definition 5.2 (Confidence set)**.** Suppose the random sample $X \sim \mathbb{P}_\theta$ and $\theta \in \Theta$. The set $C(X)$ is a γ confidence set for the parameter θ, if

$$\forall \theta \in \Theta, \mathbb{P}_\theta \left(\theta \in C(X)\right) \geq \gamma.$$

The parameter γ is called the confidence level or coefficient. In our definition, the covering probability $\mathbb{P}_\theta \left(\theta \in C(X)\right)$ is lower bounded by γ, "$\geq \gamma$", instead of strictly equal to γ, "$= \gamma$". As the covering probability needs to hold for all $\theta \in \Theta$, the "$\geq \gamma$" version is more convenient than the version with "$= \gamma$", allowing varying covering probabilities.

When $\Theta \subset \mathbb{R}$, that is, we work with a one parameter model, the confidence set is the confidence interval. For instance, $C(X)$ can be $[L(X), U(X)]$ a closed interval, $(L(X), U(X))$ an open interval, etc.. The Definition 5.2 is general, as it allows θ to be multi-dimensional, and even be a function.

Example 5.3 (CI for Gaussian mean)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ^2 known, and

$$C(X^n) \equiv (\bar{X}_n - 1.64\sigma/\sqrt{n}, \bar{X}_n + 1.64\sigma/\sqrt{n}).$$

It can be shown $\mathbb{P}_\mu(\mu \in C(X^n)) = 0.9$ for all $\mu \in \mathbb{R}$.

In the simulation example of Figure 5.2, each row represents one synthetic data: ten standard Normal random numbers are generated (the round dots); the triangular dot is the sample mean; the horizontal line segment is the confidence interval. If the confidence interval covers the true value 0, then its color is green; otherwise, its color is red.

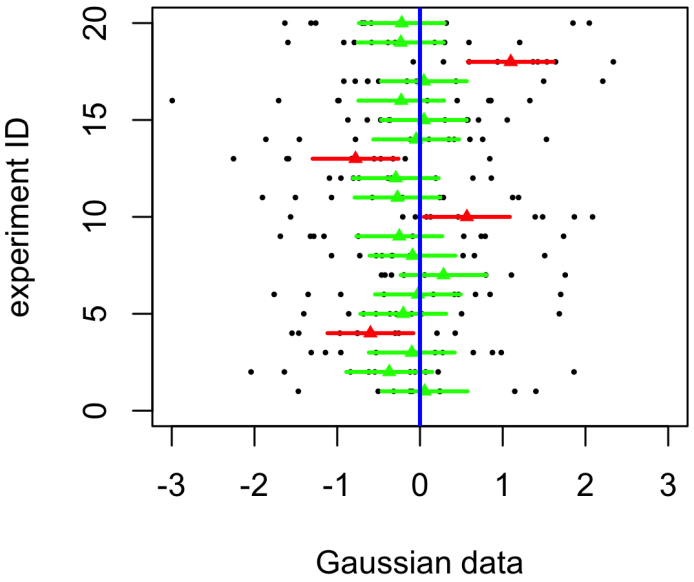


Figure 5.2: Coverage rate for 90% confidence interval

```r
mu <- 0
sigma <- 1
n <- 10
Nmc <- 20

y <- 1:Nmc
x <- matrix(rnorm(n*Nmc),nrow=Nmc)

rad <- 1.64
lower <- apply(x,1,function(xx) mean(xx) - rad/sqrt(n)*sigma)
upper <- apply(x,1,function(xx) mean(xx) + rad/sqrt(n)*sigma)
cover <- lower< mu & upper > mu
cols <- c("red","green")[as.numeric(cover)+1]

plot(x[1,], rep(y[1],n),pch=19,
     xlim = c(-1,1)*max(abs(x)),ylim = c(0,Nmc), cex=0.25,
     xlab = "Gaussian data",ylab = "experiment ID")
points(mean(x[1,]) , y[1], pch=17,cex=0.75, col=cols[1])
segments(lower[1],y[1], upper[1],y[1],lwd=2, col=cols[1])

for(ii in 2:Nmc){
    points(x[ii,],rep(y[ii],n),pch=19,cex=0.25)
    points(mean(x[ii,]) , y[ii], pch=17,cex=0.75, col=cols[ii])
    segments(lower[ii],y[ii], upper[ii],y[ii],lwd=2, col=cols[ii])
}

abline(v=mu, lwd=2,col = "blue")
```

♢

Like point estimators, the set $C(X)$ is often data dependent. As we treat the data X as random quantity in our modelling, it is meaningful to evaluate the covering probability under $\mathbb{P}_\theta$. In contrast, for a particular data x (a realization of X under $\mathbb{P}_\theta$), $C(x)$ is deterministic: it either contains θ (an unknown constant), or does not. In most applications, we do not know which case we are in, as we do not know the true θ.

Another drawback of confidence set is its inference is pre-experimental, and independent of the actual data at hand. The confidence level γ is not the assessment of the plausibility of the confidence interval at hand (given the observed data) containing the true parameter:

$$\mathbb{P}_\theta(\theta \in C(X)) \neq \mathbb{P}(\theta \in C(x)|X = x).$$

Example 5.4 (Berger and Wolpert (1988))**.** Let θ be a constant, and let ε_1 and ε_2 be two independent random variables such that $\mathbb{P}(\varepsilon_i = 1) = \mathbb{P}(\varepsilon_i = -1) = 1/2$ for $i = 1, 2$. Suppose we only observe $Y_i = \theta + \varepsilon_i$ for $i = 1, 2$.

Define the confidence interval for θ:

$$C(Y_1, Y_2) = \begin{cases} Y_1 - 1, & \text{if } Y_1 = Y_2 \\ (Y_1 + Y_2)/2, & \text{otherwise.} \end{cases}$$

It can be shown that $\mathbb{P}_\theta(\theta \in C(Y_1, Y_2)) = 0.75$ for all θ. But when we observe $Y_1 \neq Y_2$, we know $\theta = (Y_1 + Y_2)/2$ for sure. ♢

5.2 Exact confidence intervals with pivotal quantities

Confidence intervals with exact confidence level can be constructed based on pivotal quantities.

Definition 5.3 (Pivotal quantity)**.** A pivotal quantity is a function of the data *and* the parameters, such that its distribution is known.

Example 5.5 (Exponential model)**.** If $X \sim Exp(\lambda)$, then $\lambda X \sim Exp(1)$ is pivotal. This can be shown by deriving the distribution function of λX, e.g., pdf and CDF.

Alternatively, we can identify the distribution of λX by its moment generating function:

$$\varphi_{\lambda X}(t) = \mathbb{E}[e^{t\lambda X}] = \varphi_X(\lambda t) = \frac{\lambda}{\lambda - \lambda t} = \frac{1}{1-t} \text{ for } t < 1$$

is the moment generating function of $Exp(1)$. Therefore, $\lambda X \sim Exp(1)$. ♢

Remark. Recall the *statistic* is defined as a function of the data only and involves no parameters. With our definition, the pivotal quantity is *not* a statistic. Other authors, e.g. section 2.1 of Efron and Hastie (2021) , might take a slightly different viewpoint and define the pivotal *statistic* which is a function of data only and whose sampling distribution does not depend on the data generating process.

Suppose $T(\theta, X) \in \mathbb{R}$ is pivotal, then it has the same known CDF $F(t)$ for all $\theta \in \Theta$. For $\gamma \in (0,1)$, the probability

$$\mathbb{P}_\theta(L < T(\theta, X) \leq U) = \gamma$$

can be guaranteed for all $\theta \in \Theta$ by choosing L and U such that $F(U) - F(L) = \gamma$. The probability guarantee of the event $\{X : L < T(\theta, X) \leq U\}$ automatically produces a confidence interval/set $C(X)$ with desired confidence level by rewriting it in the form of $\{X : \theta \in C(X)\}$.

Example 5.6 (Exponential model, continued)**.** Suppose the statistician would like to obtain a 90%-confidence interval, s/he chooses $a = 0.05$ and $b = 0.95$, then

$$U = F^{-1}(0.95) = 2.996 \text{ and } L = F^{-1}(0.05) = 0.051,$$

where $F^{-1}(\cdot)$ is the quantile function of $Exp(1)$. As the CDF of $Exp(1)$ is $F(x) = 1 - e^{-x}$ for $x > 0$, its quantile function is $F^{-1}(t) = -\log(1-t)$ for $t \in (0,1)$.

Also, the quantiles are numerically available through R:

```
qexp(0.95,rate = 1) ## same as qgamma(0.05,1,rate = 1)
qexp(0.05,rate = 1) ## same as qgamma(0.95,1,rate = 1)
```

By pivotality, for all $\lambda > 0$,

$$\mathbb{P}_\lambda(0.051 < \lambda X \leq 2.996) = 0.9$$

which is equivalent to, for all $\lambda > 0$,

$$\mathbb{P}_\lambda\left(\frac{0.051}{X} < \lambda \leq \frac{2.996}{X}\right) = 0.9$$

where the event $\{X : \frac{0.051}{X} < \lambda \leq \frac{2.996}{X}\}$ is obtained by rewriting the event $\{X : 0.051 < \lambda X \leq 2.996\}$.

The confidence set/interval for λ, $C(X) \equiv \left(\frac{0.051}{X}, \frac{2.996}{X}\right]$, has the desired confidence level. We "check" if the theory holds true in the following simulation example.

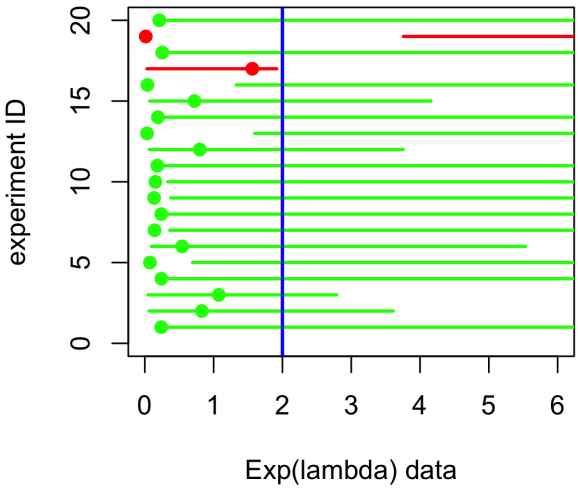


Figure 5.3: 90% confidence interval for λ where one data point is generated from $Exp(\lambda)$ for each synthetic data and $\lambda = 2$. Round dots are observed data points and line segments are confidence intervals.

```
lambda <- 2
Nmc <- 20
y <- 1:Nmc
x <- rgamma(Nmc, 1, rate = lambda)

lower <- qgamma(0.05,1,rate = 1)/x
upper <- qgamma(0.95,1,rate = 1)/x
cover <- lower< lambda & upper > lambda
cols <- c("red","green")[as.numeric(cover)+1]

plot(x[1],y[1], pch = 19,col=cols[1],
     xlim = c(0, lambda*3), ylim = c(0,Nmc),
     xlab = "Exp(lambda) data", ylab = "experiment ID")
segments(lower[1],y[1], upper[1],y[1],lwd=2, col=cols[1])
for(ii in 2:Nmc){
    points(x[ii] , y[ii], pch=19,cex=1, col=cols[ii])
    segments(lower[ii],y[ii], upper[ii],y[ii],lwd=2, col=cols[ii])
}

abline(v= lambda, lwd=2,col = "blue")
```

♢

As the bounds L and U are not uniquely identified, the rule of thumb is to choose (L, U) such that the length of $C(X)$ is reasonably short. Also, the convention is to pick symmetric quantiles, that is, $a = 1/2 - \gamma/2$ and $b = 1/2 + \gamma/2$, such that

$$P(T(\theta, X) > U) = P(T(\theta, X) \leq L) = (1 - \gamma)/2.$$

Theorem 5.1 presents a list of pivotal quantities related to Exponential distribution and Gamma distribution. We will study important pivotal quantities related to Gaussian distribution in the next section. However, pivotal quantities are not available in most statistical applications, especially for modern data analysis where complex models are built.

Theorem 5.1 (Pivotal quantities related to Gamma distribution)**.** *For Gamma distributions, we have the following pivotal quantities.*

- *Exponential distribution*
  - *If $X \sim Exp(\lambda)$, then $\lambda X \sim Exp(1)$ is pivotal.*
  - *If $X_1, ..., X_n \overset{iid}{\sim} Exp(\lambda)$, then $\lambda \bar{X}_n \sim Ga(n, n)$ and $2n\lambda \bar{X}_n \sim \chi^2_{2n}$ are pivotal.*
- *Gamma distribution*
  - *If $X \sim Ga(\alpha, \lambda)$ with α known , then $\lambda X \sim Ga(\alpha, 1)$ is pivotal.*
  - *If $X_1, ..., X_n \overset{iid}{\sim} Ga(\alpha, \lambda)$ with α known, then $\lambda \bar{X}_n \sim Ga(n\alpha, n)$ and $2n\lambda \bar{X}_n \sim \chi^2_{2n\alpha}$ are pivotal.*

The proof of the pivotality is based on the distribution theory of Gamma distribution. Under the shape-rate parameterization, the Gamma random variable $X \sim Ga(a, b)$ has the moment generating function $\varphi_X(t) = (1 - t/b)^{-a}$ for $t < b$.

Definition 5.4 (χ^2 distribution)**.** The random variable $X \sim \chi^2_n$, χ^2 distribution with n degree of freedom, if $X = \sum_{i=1}^n Z_i^2$ where $Z_1, ..., Z_n \overset{iid}{\sim} N(0, 1)$. The pdf of $X \sim \chi^2_k$ is

$$f(x; k) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}$$

where $\Gamma(\cdot)$ is the Gamma function.

By construction, the degree of freedom n should be a positive integer. In fact, χ^2 distribution is a special case of Gamma distribution whose shape parameter needs not to be some multiple of $1/2$: $\chi^2_k \overset{d}{=} Ga(\frac{k}{2}, \frac{1}{2})$ under the rate parameterization of Gamma distribution.

Lemma 5.2. *Suppose all Gamma distributions are under the shape-rate parameterization.*

- $\chi^2_1 \overset{d}{=} Ga(\frac{1}{2}, \frac{1}{2})$,
- $\chi^2_2 \overset{d}{=} Ga(1, \frac{1}{2}) \overset{d}{=} Exp(\frac{1}{2})$,
- $\chi^2_n \overset{d}{=} Ga(\frac{n}{2}, \frac{1}{2})$.
- $\chi^2_{2n} \overset{d}{=} Ga(n, \frac{1}{2})$.

Exercise 5.1. Derive $\varphi_{Z^2}(t)$, the mgf of Z^2 where $Z \sim N(0, 1)$. Show $\chi^2_n \overset{d}{=} Ga(\frac{n}{2}, \frac{1}{2})$ by the method of mgf. ☕

5.3 Confidence intervals for Gaussian mean μ and variance σ^2

Reading(s) 5.2. Wasserman (2004): 2.10 ; Hogg et al. (2018): 7.1, 7.2.

As the Gaussin model, where $X_1, .., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, plays a fundamental role in statistical theory. In this section, we study pivotal quantities related to Gaussian distribution, with which various confidence intervals are constructed.

Theorem 5.3 (Pivotal quantities related to Gaussian distribution)**.** *For Gaussian distributions, we have the following pivotal quantities.*

- *If $X \sim N(\mu, \sigma^2)$ with σ^2 known, then $X - \mu \sim N(0, \sigma^2)$ is pivotal.*
- *If $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ^2 known, then $\sqrt{n}(\bar{X}_n - \mu) \sim N(0, \sigma^2)$ is pivotal.*
- *If $X \sim N(\mu, \sigma^2)$ with μ known, then $(X - \mu)/\sigma \sim N(0, 1)$ is pivotal.*
- *If $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with μ known, then $\sum_{i=1}^n (X_i - \mu)^2/\sigma^2 \sim \chi^2_n$ is pivotal.*
- *If $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, then $\sum_{i=1}^n (X_i - \bar{X}_n)^2/\sigma^2 \sim \chi^2_{n-1}$ is pivotal.*
- *If $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, then $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sqrt{\frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2}} \sim t_{n-1}$ is pivotal.*

5.3.1 Multivariate Normal Distribution Theory

From a generative perspective, the multivariate Gaussian random vector can be defined as a linear transformation of iid standard Gaussian random variables.

Definition 5.5 (Multivariate Normal)**.** Suppose $Z_1, ..., Z_n \overset{iid}{\sim} N(0, 1)$, $Z \equiv (Z_1, Z_2, ..., Z_n)^T$ is a $n \times 1$ vector, and $X = \mu + \Sigma^{1/2} Z$, then $X \sim N_d(\mu, \Sigma)$.

Here $\Sigma^{1/2}$ needs not to be unique, but it is unique up to an orthonormal transformation such that $\Sigma \equiv \Sigma^{1/2}(\Sigma^{1/2})^T$ is well-defined. By the change of variable formula, the pdf of $X \sim N_d(\mu, \Sigma)$ is

$$f_X(x; \mu, \Sigma) = (2\pi)^{-d/2} |\Sigma|^{-1/2} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)}.$$

By completing the square (matrix version of the exercise $a^2 + 2xa + b^2 = (a + x)^2 + b^2 - x^2$), the mgf of $X \sim N_d(\mu, \Sigma)$ is, for $t \in \mathbb{R}^d$,

$$\varphi_X(t) = \mathbb{E}[e^{t^T X}] = e^{\mu^T t + \frac{1}{2} t^T \Sigma t}. \tag{5}$$

Theorem 5.4 (Linear transformation). *Suppose $X \sim N_d(\mu, \Sigma)$ is a multivariate Normal random vector, and $A \in \mathbb{R}^{p\times d}$ be a linear transformation, then AX is Gaussian, and*

$$AX \sim N_p(A\mu, A\Sigma A^T).$$

Proof. The mgf of AX is, for $t \in \mathbb{R}^p$,

$$\varphi_{AX}(t) = \mathbb{E}[e^{t^T(AX)}] = \mathbb{E}[e^{(A^Tt)^TX}] = \varphi_X(A^Tt) = e^{(A\mu)^Tt+\frac{1}{2}t^TA\Sigma A^Tt},$$

which implies the distribution of AX is another multivariate Normal, and we can read off the parameters of AX using (5). □

For a iid Normal sample, Theorem 5.4 implies the sample average follows the univariate Normal with mean μ and variance σ^2/n. To see it, let $A = (1/n, 1/n, ..., 1/n)$ which is a $1 \times n$ (row) matrix, and note the iid Normal sample implies its joint distribution $N_n(\mu\mathbf{1}_n, \sigma^2 I_n)$.

By letting $A = (0, .., 1, ...0)$ where 1 is in the ith entry and 0's appear elsewhere, the marginal distribution of X_i is also Normal with mean μ_i and variance Σ_{ii}.

Theorem 5.5 (conditional Normal). *Suppose $X \sim N_d(\mu, \Sigma)$ is a multivariate Normal random vector, and X is partitioned to two blocks X_a and X_b:*

$$X = \begin{bmatrix} X_a \\ X_b \end{bmatrix}, \mu = \begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix}, \Sigma = \begin{bmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{bmatrix}.$$

Then,

(1) The marginal distribution of X_a is also Gaussian, and

$$X_a \sim N_{d_a}(\mu_a, \Sigma_{aa}).$$

(2) The conditional distribution of X_b given $X_a = x_a$ is also Gaussian, and

$$X_b|X_a = x_a \sim N_{d_b}(\mu_b + \Sigma_{ba}\Sigma_{aa}^{-1}(x_a - \mu_a), \Sigma_{bb} - \Sigma_{ba}\Sigma_{aa}^{-1}\Sigma_{ab}).$$

This expression can be understood from the perspective of regression.

Proof. The first assertion follows by choosing an appropriate linear transformation A such that $X_a = AX$.

The second assertion is due to the factorization of the joint pdf:

$$f_{X_b|X_a}(x_b|x_a) \propto f_{X_a,X_b}(x_a, x_b) \propto e^{-\frac{1}{2}\left((x_b-\mu_b)^T\Omega_{bb}(x_b-\mu_b)+2(x_a-\mu_a)^T\Omega_{ab}x_b\right)}$$

where Ω_{ab} and Ω_{bb} are defined in the block matrix inverse identity

$$\begin{bmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{bmatrix}^{-1} = \begin{bmatrix} \Omega_{aa} & \Omega_{ab} \\ \Omega_{ba} & \Omega_{bb} \end{bmatrix},$$

and completing the square with respect to x_b with the following sub-matrices

$$\begin{aligned}\Omega_{aa} &= \Sigma_{aa}^{-1} + \Sigma_{aa}^{-1}\Sigma_{ab}\Omega_{bb}\Sigma_{ba}\Sigma_{aa}^{-1} \\ \Omega_{ab} &= -\Sigma_{aa}^{-1}\Sigma_{ab}\Omega_{bb} \\ \Omega_{ba} &= \Omega_{ab}^T = -\Omega_{bb}\Sigma_{ba}\Sigma_{aa}^{-1} \\ \Omega_{bb} &= \left(\Sigma_{bb} - \Sigma_{ba}\Sigma_{aa}^{-1}\Sigma_{ab}\right)^{-1}.\end{aligned}$$

□

Corollary 5.6. *Suppose $X \sim N_d(\mu, \Sigma)$ is a multivariate Normal random vector, and X is partitioned to two blocks X_a and X_b:*

$$X = \begin{bmatrix} X_a \\ X_b \end{bmatrix}, \mu = \begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix}, \Sigma = \begin{bmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{bmatrix}.$$

If $\Sigma_{ab} = \mathbf{0}$, then $X_a \perp X_b$

Proof. If $\Sigma_{ab} = \mathbf{0}$, the conditional distribution of $X_b|X_a$ specified by Theorem 5.5 does not depend on X_a. □

Remark. The conditional expectation $\mathbb{E}[X_b|X_a] = \mu_b + \Sigma_{ba}\Sigma_{aa}^{-1}(X_a - \mu_a)$ is linear in X_a, so it is also a multivariate Gaussian by Theorem 5.4.

Suppose $X_1, .., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, then the maximum likelihood estimator for (μ, σ^2) is

$$\hat{\mu}_{MLE} = \bar{X}_n, \hat{\sigma}^2_{MLE} = \frac{1}{n}\sum\nolimits_{i=1}^{n}(X_i - \bar{X}_n)^2.$$

The multivariate Normal distribution theory implies the joint sampling distribution of $(\hat{\mu}_{MLE}, \hat{\sigma}_{MLE})$.

Theorem 5.7 (Sample mean and variance)**.** *Suppose $X_1, .., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, then*

(1) the sample average $\bar{X}_n$ satisfies $\sqrt{n}(\bar{X}_n - \mu)/\sigma \sim N(0, 1)$,

(2) the scaled sample variance $\sum_{i=1}^{n}(X_i - \bar{X}_n)^2/\sigma^2 \sim \chi^2_{n-1}$,

(3) the sample mean, $\bar{X}_n$, and sample variance, $\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$, are independent.

Proof. The first assertion follows Theorem 5.4. Let $A = \frac{1}{\sigma\sqrt{n}}\mathbf{1}_n^T$, then

$$AX = \frac{1}{\sigma\sqrt{n}}\mathbf{1}_n^T X = \sqrt{n}\bar{X}_n/\sigma$$

where $X \equiv (X_1, ..., X_n)^T$. Theorem 5.4 implies $\sqrt{n}\bar{X}_n/\sigma \sim N(\sqrt{n}\mu/\sigma, 1)$, which is equivalent to $\sqrt{n}(\bar{X}_n - \mu)/\sigma \sim N(0, 1)$.

To prove the second assertion, note the following decomposition, in the same spirit of MSE,

$$\sum\nolimits_{i=1}^{n}(X_i - \mu)^2 = \sum\nolimits_{i=1}^{n}(X_i - \bar{X}_n)^2 + n(\bar{X}_n - \mu)^2. \tag{6}$$

This identity is essentially Gougu (Pythagorean) theorem. In matrix notation, the decomposition is

$$||X - \mu\mathbf{1}_n||_2^2 = ||X - \bar{X}_n\mathbf{1}_n||_2^2 + ||\mathbf{1}_n||_2^2 \left(\bar{X}_n - \mu\right)^2,$$

where the $||\cdot||_2$ norm for vectors is defined as $||a||_2^2 = a^T a$, the sample average $\bar{X}_n = \frac{1}{n}\mathbf{1}_n^T X$ is a scalar, and the cross product term is proportional to $(X - \bar{X}_n\mathbf{1}_n)^T\mathbf{1}_n = 0$.

By Definition 5.4, on the left hand side of (6), $\sum_{i=1}^n (X_i - \mu)^2/\sigma^2 \sim \chi_n^2$, and on the right hand side of (6) , the second term $n(\bar{X}_n - \mu)^2/\sigma^2 \sim \chi_1^2$. If the two terms on the right hand side of (6) are independent, $\sum_{i=1}^n (X_i - \bar{X}_n)^2 \perp n(\bar{X}_n - \mu)^2$ (proved next), then the relationship among their mgf's implies the second assertion. (Recall the additivity property of Gamma distribution.)

The third assertion implies $\sum_{i=1}^n (X_i - \bar{X}_n)^2 \perp n(\bar{X}_n - \mu)^2$, as scaling does not affect independence, and $n(\bar{X}_n - \mu)^2$ is a function of $\bar{X}_n$.

To see the third assertion, consider the linear transformation A such that

$$AX \equiv \begin{bmatrix} \bar{X}_n \\ X - \bar{X}_n\mathbf{1}_n \end{bmatrix}_{(n+1)\times 1} = \begin{bmatrix} \frac{1}{n}\mathbf{1}_n^T \\ I_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^T \end{bmatrix}_{(n+1)\times n} X,$$

where $X \sim N_n(\mu\mathbf{1}_n, \sigma^2 I_n)$. Direct calculation yields

$$AA^T = \begin{bmatrix} \frac{1}{n} & \mathbf{0}_n^T \\ \mathbf{0}_n & I_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^T \end{bmatrix}_{(n+1)\times(n+1)},$$

which implies, by Corollary 5.6, $\bar{X}_n \perp (X - \bar{X}_n\mathbf{1}_n)$. Then, the sample variance, a function of $(X - \bar{X}_n\mathbf{1}_n)$, is independent of the sample mean $\bar{X}_n$. □

5.3.2 Confidence Intervals for σ^2

Theorem 5.7 immediately implies a $(1-\alpha)$-confidence interval for σ^2 without the knowledge of μ:

$$\left(\sum_{i=1}^n (X_i - \bar{X}_n)^2/q_{n-1,1-\alpha/2}, \sum_{i=1}^n (X_i - \bar{X}_n)^2/q_{n-1,\alpha/2}\right) \tag{7}$$

where $q_{n-1,\alpha}$ denotes the αth quantile of χ_{n-1}^2. With the connection between χ^2 distribution and Gamma distribution, $q_{n-1,\alpha}$ is readily available with the R command `qgamma(`α`, (n-1)/2, 1/2)`.

When μ is known, the definition of χ^2 distribution implies the following $(1-\alpha)$-confidence set for σ^2:

$$\left(\sum_{i=1}^n (X_i - \mu)^2/q_{n,1-\alpha/2} \le \sigma^2 \le \sum_{i=1}^n (X_i - \mu)^2/q_{n,\alpha/2}\right), \tag{8}$$

where $q_{n,\alpha}$ denotes the αth quantile of χ_n^2.

Both intervals have the same confidence level. But the confidence interval (8) is generally (not always) shorter than the confidence interval (7). Intuitively, $\bar{X}_n$ is a noisy version of μ, and when you have more precise knowledge about the data, your inference and statements should be more precise.

Exercise 5.2. Verify the confidence level of the confidence intervals (7) and (8) is indeed $1-\alpha$. ☕

We "check" the theory in the following simulation study. Each synthetic data has ten observations from $N(\mu, \sigma^2)$ where $\mu = 100$ and $\sigma^2 = 4$. Then, we can compare the confidence interval (7) constructed without the knowledge of population mean μ with the confidence interval (8) with such knowledge.

Figure 5.4 provides a side-by-side comparison of the two intervals. Both empirically have desired coverage rate. The interval (8) is generally shorter than (7), but not always.

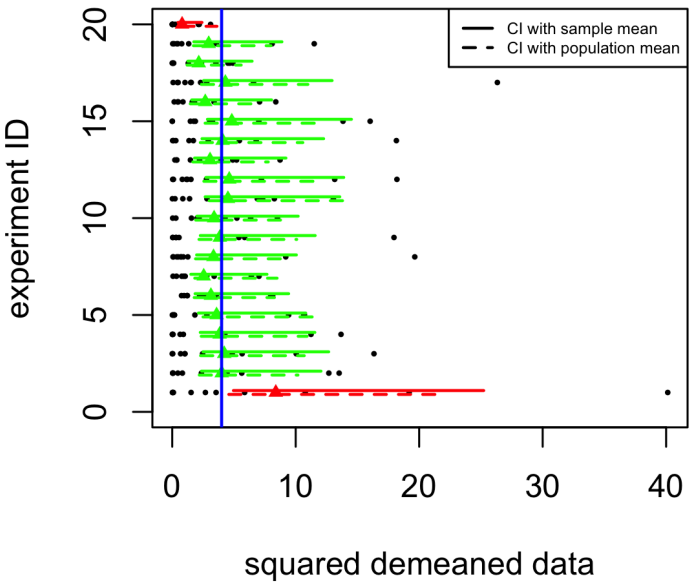


Figure 5.4: Confidence intervals for σ^2. Round dots are $(X_i - \bar{X}_n)^2$ and triangular dots are $\hat{\sigma}^2_{MLE}$.

```
mu <- 100
sigma <- 2
n <- 10
Nmc <- 20
y <- 1:Nmc
x <- matrix(rnorm(n*Nmc, mu,sigma),nrow=Nmc)

ss <- function(xx,mean = NULL){
    if(is.null(mean)) mean = mean(xx)
    sum((xx-mean)^2)
}
alpha <- 0.1
lower.samp <- apply(x,1,function(xx) ss(xx)/qgamma(1-alpha/2,(n-1)/2,1/2))
upper.samp <- apply(x,1,function(xx) ss(xx)/qgamma(alpha/2,(n-1)/2,1/2))
cover.samp <- lower.samp < sigma^2 & upper.samp > sigma^2
cols.samp <- c("red","green")[as.numeric(cover.samp)+1]
lower.pop <- apply(x,1,function(xx) ss(xx,mu)/qgamma(1-alpha/2,n/2,1/2))
upper.pop <- apply(x,1,function(xx) ss(xx,mu)/qgamma(alpha/2,n/2,1/2))
cover.pop <- lower.pop < sigma^2 & upper.pop > sigma^2
cols.pop <- c("red","green")[as.numeric(cover.pop)+1]

plot((x[1,]-mean(x[1,]))^2, rep(y[1],n),pch=19,
     xlim = c(0, max(apply(x,1,function(xx) (xx-mean(xx))^2))),
     ylim = c(0,Nmc), cex=0.25,
     xlab = "squared demeaned data",ylab = "experiment ID")
points(ss(x[1,])/n , y[1], pch=17,cex=0.75, col=cols.samp[1])
segments(lower.samp[1],y[1] + 0.1, upper.samp[1],y[1] + 0.1,
         lwd=1.5, lty = 1, col=cols.samp[1])
segments(lower.pop[1],y[1] - 0.1, upper.pop[1],y[1] - 0.1,
         lwd=1.5, lty = 2, col=cols.pop[1])
for(ii in 2:Nmc){
    points((x[ii,] - mean(x[ii,]))^2, rep(y[ii],n) , pch=19 , cex=0.25)
    points(ss(x[ii,])/n , y[ii], pch=17,cex=0.75, col=cols.samp[ii])
    segments(lower.samp[ii],y[ii] + 0.1, upper.samp[ii],y[ii] + 0.1,
             lwd=1.5, lty = 1, col=cols.samp[ii])
```

```
    segments(lower.pop[ii],y[ii] - 0.1, upper.pop[ii],y[ii] - 0.1,
             lwd=1.5, lty = 2, col=cols.pop[ii])
}
abline(v=sigma^2, lwd=1.5,col = "blue")
legend("topright", c("CI with sample mean", "CI with population mean"),
       lty = c(1,2), lwd=1.5, cex = 0.5)
```

5.3.3 Confidence Intervals for μ

When σ^2 is known, $\sqrt{n}(\bar{X}_n - \mu)$ is a pivotal quantity with sampling distribution $N(0, \sigma^2)$. Then, a $(1-\alpha)$-confidence interval for μ can be constructed as follows,

$$\bar{X}_n \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \tag{9}$$

where $\Phi(\cdot)$ denotes the CDF of $N(0,1)$ and $z_\alpha \equiv \Phi^{-1}(\alpha)$ is the αth quantile of $N(0,1)$.

Popular quantiles of standard Normal distribution are summarized in the following table. Suppose we are interested in a 95% confidence interval, then $\Phi^{-1}(1-0.05/2) = 1.96$ simplifies the interval as $\bar{X}_n \pm 1.96\frac{\sigma}{\sqrt{n}}$. For the 90% confidence level, the interval is $\bar{X}_n \pm 1.645\frac{\sigma}{\sqrt{n}}$.

π	0.9	0.95	0.975	0.99	0.995
z_α	1.282	1.645	1.960	2.326	2.576

Table 1: Some popular quantiles of $N(0,1)$

The length of the interval, $2z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}$, , is proportional to $1/\sqrt{n}$, which implies the more data we have, the more "accurate" is our estimation. Though the lengths of the confidence intervals (7) and (8) have similar relationship with n, their dependence on n is not so transparent.

When σ^2 is unknown, we have to estimate it from the data. The plug-in principle implies the plug-in confidence interval

$$\bar{X}_n \pm z_{\alpha/2}\frac{\hat{\sigma}}{\sqrt{n}} \tag{10}$$

where $\hat{\sigma}$ can be $\sqrt{\hat{\sigma}^2_{MLE}}$ or any other consistent estimators. However, confidence interval (10) covers the truth with a probability *lower* than the advertised level,

$$\mathbb{P}\left(\left|\frac{\bar{X}_n - \mu}{\hat{\sigma}/\sqrt{n}}\right| < z_{1-\alpha/2}\right) < \mathbb{P}\left(\left|\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}\right| < z_{1-\alpha/2}\right),$$

which seems a common feature of the plug-in principle when there is not enough data. Intuitively, directly plugging in the estimates as if the unknown parameters were known ignores the uncertainty in the unknowns, so the confidence intervals tend to under-cover. With enough data, the estimates are very close to the truth, and under-coverage is less problematic.

```
mu <- 0
sigma <- 2
n <- 10
Nmc <- 1e6
x <- matrix(rnorm(n*Nmc, mu,sigma),nrow=Nmc)
lower.pop <- apply(x,1,mean) - qnorm(0.95)*sigma/sqrt(n)
upper.pop <- apply(x,1,mean) + qnorm(0.95)*sigma/sqrt(n)
cover.pop <- lower.pop < mu & upper.pop > mu
```

```r
mean(cover.pop)  ## 0.9, as desired
sig.hat.MLE <- apply(x,1,function(xx) sqrt(mean((xx-mean(xx))^2)))
lower.plug <- apply(x,1,mean) - qnorm(0.95)*sig.hat.MLE/sqrt(n)
upper.plug <- apply(x,1,mean) + qnorm(0.95)*sig.hat.MLE/sqrt(n)
cover.plug <- lower.plug < mu & upper.plug > mu
mean(cover.plug) ## 0.846 < 0.9
```

To fix idea, let $\hat{\sigma}$ be $\sqrt{\hat{\sigma}^2_{MLE}}$. Theorem 5.7 says $\bar{X}_n \perp \hat{\sigma}^2_{MLE}$ so the coverage of the interval (10) relative to the interval (9) depends on the sampling distribution of $\hat{\sigma}$ relative to the truth. Theorem 5.7 also says $n\hat{\sigma}^2/\sigma^2 \sim \chi^2_{n-1}$ whose median is approximately $n-1$. Then, it is more likely to have the ratio $\hat{\sigma}^2/\sigma^2 < 1$ than the other way around.

We can examine the sampling distribution of $\hat{\sigma}/\sigma$ in more details by simulation. In the simulation study, $n = 10$ synthetic observations are generated from $N(\mu, \sigma^2)$ with $\mu = 0$ and $\sigma = 2$. As σ^2_{MLE} is downward biased, we may inflate $\hat{\sigma}$ a little bit to improve the coverage. The unbiased estimator of variance seems a good candidate. The under-coverage is less severe but still exists in the simulation.

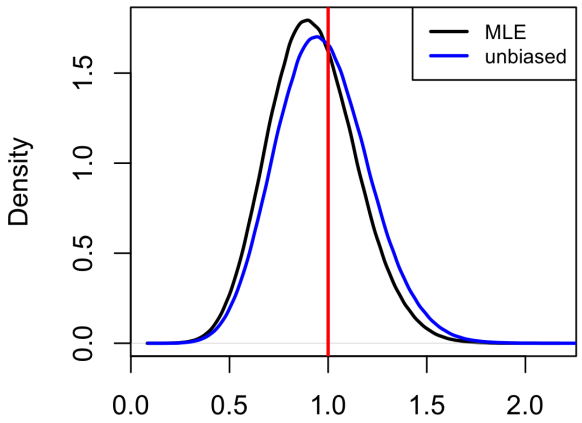


Figure 5.5: Sampling distribution of $\hat{\sigma}/\sigma$. "MLE" stands for $\hat{\sigma} = \sqrt{\hat{\sigma}^2_{MLE}}$, "unbiased" stands for $\hat{\sigma} = \sqrt{\frac{n}{n-1}\hat{\sigma}^2_{MLE}}$.

```r
mu <- 0
sigma <- 2
n <- 10
Nmc <- 1e6
x <- matrix(rnorm(n*Nmc, mu,sigma),nrow=Nmc)
sig.hat.MLE <- apply(x,1,function(xx) sqrt(mean((xx-mean(xx))^2)))
sig.hat.unb <- apply(x,1,function(xx) sqrt(sum((xx-mean(xx))^2)/(n-1)))
plot(density(sig.hat.MLE/sigma),lwd = 2, main="", xlab = "")
lines(density(sig.hat.unb/sigma),lwd = 2,col = "blue")
abline(v = 1, lwd = 2, col = "red")
legend("topright", c("MLE", "unbiased"),
       col = c("black","blue"), lwd=2, cex = 0.75)
```

As we have to estimate σ from the data, the sampling distribution of $\sqrt{n}(\bar{X}_n - \mu)/\hat{\sigma}$ needs more precise characterization. The sampling distribution turns out to be the Student's t distribution. The "Student" part is from William Sealy Gosset's 1908 paper in *Biometrika* under the pseudonym "Student". Gosset worked at the Guinness Brewery in Dublin, Ireland, and used the t test to determine the quality of raw material, e.g., barley, stout. It became well-known due to Ronald Fisher who called the distribution "Student's distribution" and represented the test value with the letter t.

Definition 5.6 (Student's t distribution)**.** The random variable $X \sim t_\nu$ if

$$X = \frac{Z}{\sqrt{S/\nu}}$$

where $Z \perp S$, $Z \sim N(0,1)$, and $S \sim \chi^2_\nu$. The parameter ν is called the degree of freedom.

We can derive the pdf of $X \sim t_\nu$ in first principles. As Z is symmetric about 0 and $Z \perp S$, X is also symmetric around 0. Then, for $t > 0$,

$$\mathbb{P}(X > t) = \frac{1}{2}\mathbb{P}\left(Z^2 > t^2 S/\nu\right) = \frac{1}{2}\int_0^\infty \int_{t^2 s/\nu}^\infty \frac{1}{\sqrt{2\pi}} q^{-1/2} e^{-q/2} dq \frac{2^{-\nu/2}}{\Gamma(\nu/2)} s^{\nu/2-1} e^{-s/2} ds$$

where $Z^2 \perp S$ and $Z^2 \sim \chi^2_1$ by Definition 5.4, noting $\Gamma(1/2) = \sqrt{\pi}$.

Taking derivative with respect to t,

$$f_X(t) = -\frac{d}{dt}\mathbb{P}(X > t) = \frac{1}{\sqrt{2\nu\pi}} \frac{2^{-\nu/2}}{\Gamma(\nu/2)} \int_0^\infty s^{(\nu+1)/2-1} e^{-s(1+t^2/\nu)/2} ds \propto \left(1 + t^2/\nu\right)^{-(\nu+1)/2}$$

where the integral is evaluated in close form $\frac{\Gamma((\nu+1)/2)}{((t^2/\nu+1)/2)^{(\nu+1)/2}}$, noting the Gamma pdf, and the normalizing constant is $\frac{\Gamma((\nu+1)/2)}{\Gamma(\nu/2)\sqrt{\nu\pi}}$ by direct calculation.

By construction, X is a stochastically scaled version of Z. So X inherits the bell-shape and the symmetry about 0 from Z but it is fatter in tail than Z. For large t, $f_X(t) \asymp |t|^{-(\nu+1)}$ which decays much slower than the standard Gaussian distribution $\phi(t) \asymp e^{-t^2/2}$.

Let $\nu = 1$, $f_X(t) \propto \frac{1}{1+t^2}$, that is, t_1 is also the standard Cauchy distribution. On the other extreme, let $\nu \to \infty$,

$$\lim_{\nu\to\infty} \left(1 + t^2/\nu\right)^{-(\nu+1)/2} = e^{-t^2/2},$$

that is, the limiting distribution of t_ν is standard Normal. Alternatively, write $S \equiv \sum_{i=1}^\nu Z_i^2$ where $Z_1, ..., Z_\nu \overset{iid}{\sim} N(0,1)$. By the law of large numbers, $S/\nu \xrightarrow{p} \mathbb{E}[Z_1^2] = 1$. Then by continuous mapping (or Slutsky) theorem, $X \xrightarrow{d} Z$ as $\nu \to \infty$.

Now we can calibrate the confidence interval (10) with the Student's t distribution. By Definition 5.6, Theorem 5.7 implies

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sqrt{\frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2}} = \frac{\sqrt{n}(\bar{X}_n - \mu)/\sigma}{\sqrt{\frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2/\sigma^2}} \sim t_{n-1}$$

is pivotal. Therefore, a $(1-\alpha)$-confidence interval for μ withtout knowing σ^2 is

$$\bar{X}_n \pm t_{n-1}(\alpha/2)\frac{\hat{\sigma}}{\sqrt{n}} \tag{11}$$

where $\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2$ (the unbiased estimator), and $t_{n-1}(\alpha)$ denotes the αth quantile of t_{n-1}. Compared to the plug-in confidence interval (10), the t distribution interval (11) uses $t_{n-1}(\alpha/2)$ as the cutoff quantile, instead of $z_{\alpha/2}$. For all $\alpha > 1/2$, $t_{n-1}(\alpha) > z_\alpha \equiv \Phi^{-1}(\alpha)$, and the difference decreases as the degree of freedom $n-1$ increases.

π	0.9	0.95	0.975	0.99	0.995
$t_1(\pi)$	3.078	6.314	12.706	31.821	63.657
$t_5(\pi)$	1.476	2.015	2.571	3.365	4.032
$t_{10}(\pi)$	1.372	1.812	2.228	2.764	3.169
$t_{30}(\pi)$	1.310	1.697	2.042	2.457	2.750
$t_{100}(\pi)$	1.290	1.660	1.984	2.364	2.626
z_π	1.282	1.645	1.960	2.326	2.576

Table 2: Compare the quantiles of $N(0,1)$ with those of t_ν

We check the coverage in the simulation. Both the confidence interval (9) knowing σ^2 and the confidence interval (11) not knowing σ^2 are honest.

```
hat.sig <- function(xx){
    sqrt(sum((xx-mean(xx))^2) / (length(xx)-1))
}
mu <- 0
sigma <- 2
n <- 10
Nmc <- 1e6
x <- matrix(rnorm(n*Nmc, mu,sigma),nrow=Nmc)
y <- 1:Nmc
alpha <- 0.1
rad <- qt(1-alpha/2, df = n-1) /sqrt(n)
lower.t <- apply(x,1,function(xx) mean(xx) - rad * hat.sig(xx))
upper.t <- apply(x,1,function(xx) mean(xx) + rad * hat.sig(xx))
cover.t <- lower.t < mu & upper.t > mu
mean(cover.t)
lower.pop <- apply(x,1,function(xx) mean(xx,mu) - qnorm(1-alpha/2)/sqrt(n)*sigma)
upper.pop <- apply(x,1,function(xx) mean(xx,mu) + qnorm(1-alpha/2)/sqrt(n)*sigma)
cover.pop <- lower.pop < mu & upper.pop > mu
mean(cover.pop)
```

The confidence interval (9) is on average shorter than the confidence interval (11). So when the information of σ^2 is available, the interval estimation of μ is more precise using the knowledge.

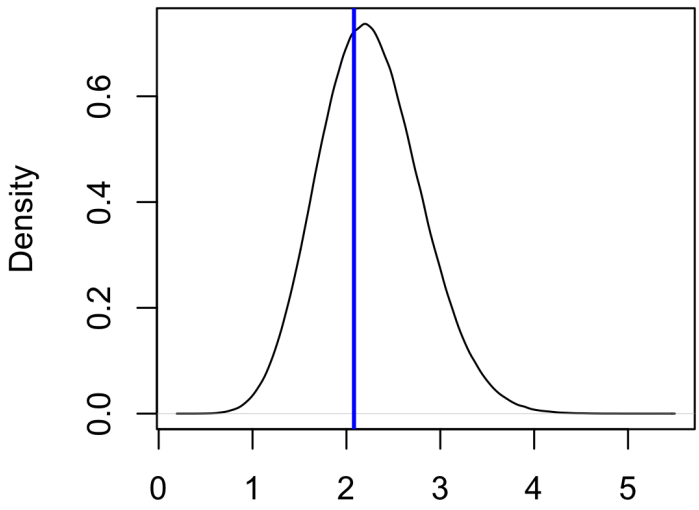


Figure 5.6: Sampling distribution of the length of (11) with the vertical line being the length of (9)

```r
mean((upper.t-lower.t) > 2*qnorm(1-alpha/2)*sigma/sqrt(n))
plot(density(upper.t-lower.t), main = "", xlab = "")
abline(v = 2*qnorm(1-alpha/2)*sigma/sqrt(n), lwd=2, col = "blue")
```

We compare the coverage of the two intervals in Figure 5.7.

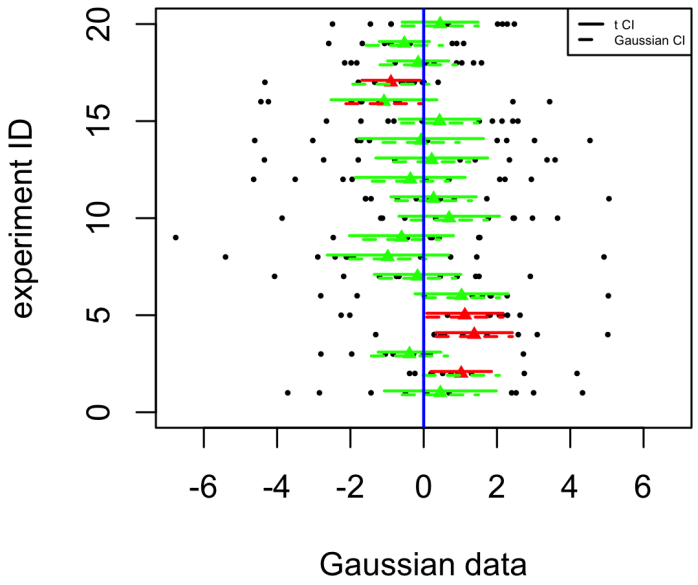


Figure 5.7: Confidence intervals for μ. The t CI is (11) and the Gaussian CI is (9).

```r
cols.t <- c("red","green")[as.numeric(cover.t)+1]
cols.pop <- c("red","green")[as.numeric(cover.pop)+1]
plot(x[1,], rep(y[1],n),pch=19,
     xlim = c(-1,1)*max(abs(x)),
     ylim = c(0,Nmc), cex=0.25,
     xlab = "Gaussian data",ylab = "experiment ID")
points(mean(x[1,]) , y[1], pch=17,cex=0.75, col=cols.t[1])
segments(lower.t[1],y[1] + 0.1, upper.t[1],y[1] + 0.1,
         lwd=1.5, lty = 1, col=cols.t[1])
segments(lower.pop[1],y[1] - 0.1, upper.pop[1],y[1] - 0.1,
         lwd=1.5, lty = 2, col=cols.pop[1])
for(ii in 2:Nmc){
    points(x[ii,], rep(y[ii],n) , pch=19 , cex=0.25)
    points(mean(x[ii,]) , y[ii], pch=17,cex=0.75, col=cols.t[ii])
    segments(lower.t[ii],y[ii] + 0.1, upper.t[ii],y[ii] + 0.1,
             lwd=1.5, lty = 1, col=cols.t[ii])
    segments(lower.pop[ii],y[ii] - 0.1, upper.pop[ii],y[ii] - 0.1,
             lwd=1.5, lty = 2, col=cols.pop[ii])
}
abline(v=mu, lwd=1.5,col = "blue")
legend("topright", c("t CI", "Gaussian CI"),
       lty = c(1,2), lwd=1.5, cex = 0.4)
```

5.3.4 Difference of two means

Comparing two samples is an important statistical task. For instance, in the new drug/COVID vaccine development, researchers need to compare the outcomes of the treatment group and the control group; Internet companies, like Meta Facebook, Google ads, Amazon, etc., have multiple versions of the same

variable/product. They can make the data-driven decision by running randomized experiments to see which version is better than others.



Figure 5.8: Example of two-sample comparison: Mobile A/B testing

Suppose the first sample is $X_1, ..., X_n \overset{iid}{\sim} N(\mu_1, \sigma_1^2)$, the second sample is $Y_1, ..., Y_m \overset{iid}{\sim} N(\mu_2, \sigma_2^2)$, and the two samples are independent. The parameter of interest is $\mu_1 - \mu_2$.

Since the two samples are independent, the joint distribution of $\{X_i\}$ and $\{Y_j\}$ is also multivariate Normal:

$$\begin{bmatrix} X \\ Y \end{bmatrix} \sim N_{n+m}\left(\begin{bmatrix} \mu_1 \mathbf{1}_n \\ \mu_2 \mathbf{1}_m \end{bmatrix}, \begin{bmatrix} \sigma_1^2 I_n & \mathbf{0}_{n\times m} \\ \mathbf{0}_{m\times n} & \sigma_2^2 I_m \end{bmatrix} \right),$$

where X denotes the column vector of $\{X_i\}$ and Y denotes the column vector of $\{Y_j\}$.

By Theorem 5.7, $\sqrt{n}(\bar{X}_n - \mu_1)/\sigma_1$ and $\sqrt{m}(\bar{Y}_m - \mu_2)/\sigma_2$ are independent and both follow the standard Normal distribution. For the parameter of interest,

$$\bar{X}_n - \bar{Y}_m - (\mu_1 - \mu_2) \sim N(0, \sigma_1^2/n + \sigma_2^2/m).$$

If we know σ_1^2 and σ_2^2, a $(1-\alpha)$-confidence interval is

$$\bar{X}_n - \bar{Y}_m \pm z_{\alpha/2}\sqrt{\sigma_1^2/n + \sigma_2^2/m},$$

where z_α denotes the αth quantile of $N(0,1)$.

If we do not know σ_1^2 or σ_2^2, they have to be estimated from the data. We first consider the simple case where $\sigma_1 = \sigma_2$, then consider the general case where $\sigma_1 \neq \sigma_2$.

Suppose $\sigma_1 = \sigma_2 = \sigma$, then $\bar{X}_n - \bar{Y}_m - (\mu_1 - \mu_2) \sim N\left(0, \frac{m+n}{mn}\sigma^2\right)$ by Theorem 5.4, and by Theorem 5.7 and the independence between the two samples,

$$\left(\sum_{i=1}^n (X_i - \bar{X}_n)^2 + \sum_{i=1}^m (Y_i - \bar{Y}_m)^2\right)/\sigma^2 \sim \chi^2_{m+n-2}.$$

By Definition 5.6,

$$\frac{\bar{X}_n - \bar{Y}_m - (\mu_1 - \mu_2)}{\sqrt{\frac{m+n}{mn}\frac{1}{m+n-2}\left(\sum_{i=1}^n (X_i - \bar{X}_n)^2 + \sum_{i=1}^m (Y_i - \bar{Y}_m)^2\right)}} \sim t_{m+n-2},$$

noting the independence between the sample means and sample variances. Then, a $(1-\alpha)$-confidence interval for $\mu_1 - \mu_2$ is

$$\bar{X}_n - \bar{Y}_m \pm t_{m+n-2}(\alpha/2)\sqrt{\frac{1}{m} + \frac{1}{n}}S_p,$$

where $S_p^2 = \frac{n-1}{m+n-2}S_X^2 + \frac{m-1}{m+n-2}S_Y^2$ is the pooled estimator for σ^2, $S_X^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2$ is the unbiased estimator for σ_1^2, and $S_Y^2 = \frac{1}{m-1}\sum_{i=1}^m (Y_i - \bar{Y}_m)^2$ is the unbiased estimator for σ_2^2. The factor $\sqrt{\frac{1}{m} + \frac{1}{n}}$ characterizes the shrinking speed of the interval as sample sizes m and n increase. In contrast, the corresponding factor of the one sample problem is $1/\sqrt{n}$.

When $\sigma_1^2 \neq \sigma_2^2$, exact confidence interval for $\mu_1 - \mu_2$ can be derived when $\sigma_1 = \ell\sigma_2$ for some known constant ℓ. Similar to the $\sigma_1 = \sigma_2$ case, $\bar{X}_n - \bar{Y}_m - (\mu_1 - \mu_2) \sim N\left(0, (\frac{\ell^2}{n} + \frac{1}{m})\sigma_2^2\right)$, and

$$\left(\ell^{-2}\sum_{i=1}^n (X_i - \bar{X}_n)^2 + \sum_{i=1}^m (Y_i - \bar{Y}_m)^2\right)/\sigma_2^2 \sim \chi^2_{m+n-2},$$

then a $(1-\alpha)$-confidence interval is

$$\bar{X}_n - \bar{Y}_m \pm t_{m+n-2}(\alpha/2)\sqrt{\frac{\ell^2}{n} + \frac{1}{m}}S_p(\ell),$$

where $S_p^2(\ell) = \frac{(n-1)\ell^{-2}}{m+n-2}S_X^2 + \frac{m-1}{m+n-2}S_Y^2$ is the pooled estimator for σ_2^2.

In the general case $\sigma_1^2 \neq \sigma_2^2$, exact confidence interval for $\mu_1 - \mu_2$ does not exist. The plug-in principle implies the following approximate confidence interval

$$\bar{X}_n - \bar{Y}_m \pm z_{\alpha/2}\sqrt{S_X^2/n + S_Y^2/m},$$

which, not surprisingly, suffers under-coverage when there is not enough data. Welch proposed, later modified by Aspin, the t distribution based confidence interval

$$\bar{X}_n - \bar{Y}_m \pm t_\nu(\alpha/2)\sqrt{S_X^2/n + S_Y^2/m}, \tag{12}$$

where the degree of freedom $\nu = \left(\frac{c^2}{n-1} + \frac{(1-c)^2}{m-1}\right)^{-1}$ and $c = \frac{S_X^2/n}{S_X^2/n + S_Y^2/m}$. That is, the sampling distribution of $\frac{\bar{X}_n - \bar{Y}_m}{\sqrt{S_X^2/n + S_Y^2/m}}$ is approximately t_ν. As t_ν's quantiles are larger than those of the standard Normal, Welch's interval has better coverage when m and n are moderate.

5.4 Exact confidence intervals by inverting the CDF

Theorem 5.8 suggests a general recipe to construct pivotal quantity for *univariate continuous* random variables by the probability integral transformation.

Theorem 5.8 (Probability integral transformation)**.** *Suppose X is a continuous random variable, then $F_X(X) \sim Unif[0,1]$.*

In one-parameter problems, exact confidence intervals are available if the CDF of the statistic is monotonic in the parameter. The procedure is illustrated in the following example.

Example 5.7 (Uniform right endpoint)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Unif(0, \theta)$. The sampling distribution of the maximum $X_{(n)} \equiv \max(\{X_i\})$ has CDF

$$F_\theta(t) = \mathbb{P}_\theta(X_{(n)} \leq t) = (t/\theta)^n 1(t > 0) \wedge 1.$$

For $0 < a \leq 1$, given the data $X_{(n)} = x$, let

$$\theta_a(x) = \arg\{\theta : F_\theta(x) = a\} = xa^{-1/n}.$$

For all $a \in (0, 1]$, $F_{\theta_a(x)}(x) = a$ by construction. In contrast, the quantile function, the inverse of the CDF, $F_\theta^{-1}(a) = \theta a^{1/n}$ is derived given θ.

Then, for $0 < b < a \leq 1$,

$$\mathbb{P}_\theta\left(\theta_a(X_{(n)}) \leq \theta \leq \theta_b(X_{(n)})\right) = \mathbb{P}_\theta(b \leq F_\theta(X_{(n)}) \leq a) = a - b$$

implies the $(a-b)$-confidence interval $[X_{(n)}a^{-1/n}, X_{(n)}b^{-1/n}]$. The first "=" of the proceeding display holds by plugging θ_a and θ_b into $F_\theta(X_{(n)})$ respectively, and $F_\theta(x)$ is monotonic in θ; the second "=" is due to Theorem 5.8.

As we wish the confidence interval also contains the maximum likelihood estimator, set $a = 1$ and the $(1-\alpha)$-confidence interval is

$$[X_{(n)}, X_{(n)}\alpha^{-1/n}].$$

```
n <- 5
theta <- 6
a <- 1
b <- 0.1
Nmc <- 1e6
X <- matrix(runif(Nmc*n, 0, theta), nrow = Nmc)
Xmax <- apply(X,1,max)
la <- Xmax
lb <- Xmax*b^(-1/n)
cover <- la<= theta & theta < lb
mean(cover)
```

♢

The covering probability guarantee in the Uniform right endpoint example is provided by Theorem 5.8. When the statistic is not a continuous random variable, which is often the case for count data and the like, Theorem 5.8 cannot be applied to provide guarantee for the exact coverage rates. Though the same idea applies, the construction is a bit involved and somewhat *ad hoc*, as the exact covering probability might not be available.

Example 5.8. Suppose $X_1, ..., X_n \overset{iid}{\sim} Pois(\lambda)$. Recall the mgf of $X \sim Pois(\lambda)$ is $\varphi_X(t) = e^{\lambda(e^t-1)}$. Then, $\sum_{i=1}^n X_i \sim Pois(n\lambda)$.

The CDF of $Pois(n\lambda)$ can be inferred from the relationship between Poisson process and Gamma distribution. Let $\{Y_t : t \geq 0\}$ denote the unit rate Poisson process, and let T_k denote the arrival time of the kth event of the stochastic process. By construction, $T_k \sim Ga(k, 1)$. For $k \in \mathbb{Z}_+$, $Y_t \geq k$ if and only if $T_k \leq t$: that at least k events have occurred by time t is equivalent to the arrival time of the kth event is not later than t. Then, the CDF of $Pois(n\lambda)$ is

$$F_{n\lambda}(k) = \mathbb{P}(Y_{n\lambda} \leq k) = 1 - \mathbb{P}(Y_{n\lambda} \geq k+1) = 1 - \mathbb{P}(T_{k+1} \leq n\lambda).$$

Clearly, the CDF $F_{n\lambda}(k)$ is monotonically decreasing in λ.

For $0 < a < 1$, let $\lambda_a(s) = \sup\{\lambda : F_{n\lambda}(s) \geq a\}$, and $F_{n\lambda_a(s)}(s) = a$ because the image of $F_{n\lambda}(s)$ as a function of λ (with s held fixed) is $(0, 1)$ for all $s \in \mathbb{Z}_+$. Then by monotonicity of $F_{n\lambda}(s)$ in λ,

$$\mathbb{P}_\lambda\left(\lambda \geq \lambda_a(S)\right) = \mathbb{P}_\lambda\left(F_{n\lambda}(S) \leq F_{n\lambda_a(S)}(S)\right) = \mathbb{P}_\lambda\left(F_{n\lambda}(S) \leq a\right),$$

where $S \equiv \sum_{i=1}^n X_i \sim Pois(n\lambda)$. For $0 < b < a < 1$, let $\lambda_b(s) = \inf\{\lambda : F_{n\lambda}(s) \leq b\}$, and $F_{n\lambda_b(s)}(s) = b$ for the same reason.

Therefore,

$$\mathbb{P}_\lambda\left(\lambda_a(S) \leq \lambda \leq \lambda_b(S)\right) = \mathbb{P}_\lambda\left(b \leq F_{n\lambda}(S) \leq a\right)$$

which can be smaller than $a - b$ due to the discreteness of S. One easy fix is to relax the left endpoint *or* right endpoint a bit:

$$\text{either } \lambda'_a(S) = \lambda_a(S-1) < \lambda_a(S), \text{or } \lambda'_b(S) = \lambda_b(S+1) > \lambda_b(S).$$

Indeed, relaxing both endpoints produces quite conservative coverage rates.

```
n <- 5
lambda <- 1
a <- 0.95
b <- 0.05

Smax <- qpois(1-1e-9, n*lambda)
probs <- dpois(0:Smax, n*lambda)
probs <- probs/sum(probs)

la <- qgamma(1-a, 0:Smax + 1, n)
lb <- qgamma(1-b, 0:Smax + 1, n)
cover <- la <= lambda & lambda <= lb
round(100 * sum(cover*probs), 2) ## 89.15

la <- qgamma(1-a, 0:Smax, n)
lb <- qgamma(1-b, 0:Smax + 1, n)
cover <- la <= lambda & lambda <= lb
round( 100 * sum(cover*probs), 2) ## 92.77

la <- qgamma(1-a, 0:Smax + 1, n)
lb <- qgamma(1-b, 0:Smax + 2, n)
cover <- la <= lambda & lambda <= lb
round( 100 * sum(cover*probs), 2)  ## 92.52

la <- qgamma(1-a, 0:Smax, n)
lb <- qgamma(1-b, 0:Smax + 2, n)
cover <- la <= lambda & lambda <= lb
round( 100 * sum(cover*probs), 2)  ## 96.14
```

♢

6 Asymptotic Confidence Intervals

6.1 MLE Confidence Intervals for One-Parameter Problems

Reading(s) 6.1. Wasserman (2004): 9.4 and 9.7; Hogg et al. (2018): 6.6; Efron and Hastie (2021): 4.1, 4.2, and 4.3; Brown et al. (2001).

The idea of maximum likelihood estimation provides a convenient and powerful class of point estimators, and point estimates are often computationally tractable. How to characterize the estimation accuracy of maximum likelihood estimators? In other words, how can we construct confidence sets/intervals with maximum likelihood estimators? To answer these questions, we examine basic properties of maximum likelihood estimators under certain regularity conditions. The success of maximum likelihood estimation (MLE) is largely due to its convenience in constructing confidence intervals that have asymptotic coverage guarantee.

Throughout the section, we assume $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta_0)$ where $\theta_0 \in \Theta \subset \mathbb{R}$, and X_i can be a vector. With the likelihood $\prod_{i=1}^n f(x_i;\theta)$, the maximum likelihood estimator for θ is

$$\hat{\theta} = \arg\max_{\theta\in\Theta} \prod_{i=1}^n f(x_i;\theta).$$

Under certain regularity conditions, $\hat{\theta}$ solves the *likelihood equation*

$$\dot{\ell}_{\boldsymbol{x}}(\theta) = 0$$

where $\ell_{\boldsymbol{x}}(\theta) \equiv \sum_{i=1}^n \log f(x_i;\theta)$ and $\dot{h}(t) \equiv \frac{d}{dt}h(t)$ denotes self-differentiation.

Consistency of the maximum likelihood estimator merely tells us that maximum likelihood estimates will be correct in the asymptopia, but does not tell us anything about the estimates in the process of $n \to \infty$.

To study the distribution of the maximum likelihood estimator in the process of $n \to \infty$, we expand $\dot{\ell}_{\boldsymbol{x}}(\hat{\theta})$ around θ_0 by mean-value theorem, further assuming $\ell_x(\theta)$ twice differentiable,

$$\dot{\ell}_{\boldsymbol{x}}(\hat{\theta}) = \dot{\ell}_{\boldsymbol{x}}(\theta_0) + \ddot{\ell}_{\boldsymbol{x}}(\tilde{\theta})(\hat{\theta} - \theta_0), \tag{13}$$

where $\tilde{\theta}$ is between $\hat{\theta}$ and θ_0. If $\ddot{\ell}_{\boldsymbol{x}}(\tilde{\theta}) \neq 0$, the likelihood equation $\dot{\ell}_{\boldsymbol{x}}(\hat{\theta}) = 0$ implies the estimation error of the maximum likelihood estimator is

$$\hat{\theta} - \theta_0 = -\frac{\dot{\ell}_{\boldsymbol{x}}(\theta_0)}{\ddot{\ell}_{\boldsymbol{x}}(\tilde{\theta})}. \tag{14}$$

As the log-likelihood function $\ell_{\boldsymbol{x}}(\theta)$ is the sum of the log pdf/pmf's of the iid observations, it is more insightful to write

$$\hat{\theta} - \theta_0 = -\frac{\frac{1}{n}\dot{\ell}_{\boldsymbol{x}}(\theta_0)}{\frac{1}{n}\ddot{\ell}_{\boldsymbol{x}}(\tilde{\theta})},$$

where both the numerator and the denominator are the averages of independent random variables.

For the numerator, firstly by LLN,

$$\frac{1}{n}\dot{\ell}_{\boldsymbol{X}}(\theta_0) = \frac{1}{n}\sum_{i=1}^n \dot{\ell}_{X_i}(\theta_0) \xrightarrow{p} \mathbb{E}_{\theta_0}[\dot{\ell}_{X_1}(\theta_0)].$$

We will see the limiting quantity $\mathbb{E}_{\theta_0}[\dot{\ell}_{X_1}(\theta_0)] = 0$ in Theorem 6.1. Secondly, by CLT,

$$\frac{1}{\sqrt{n}}\dot{\ell}_{\boldsymbol{X}}(\theta_0) = \sqrt{n}\left(\frac{1}{n}\sum\nolimits_{i=1}^{n}\dot{\ell}_{X_i}(\theta_0) - \mathbb{E}_{\theta_0}[\dot{\ell}_{X_1}(\theta_0)]\right) \xrightarrow{d} N(0, I(\theta_0))$$

where $I(\theta_0) \equiv V(\dot{\ell}_{X_1}(\theta))$ is the Fisher information defined in (15).

For the denominator, if the function $\ddot{\ell}_{\boldsymbol{x}}(\theta)$ is a deterministic continuous function of θ, then the consistency $\hat{\theta} \xrightarrow{p} \theta_0$ implies $\ddot{\ell}_{\boldsymbol{x}}(\tilde{\theta}) \xrightarrow{p} \ddot{\ell}_{\boldsymbol{x}}(\theta_0)$ under certain regularity conditions. In establishing asymptotic Normality, it is often convenient to assume that the estimator under consideration is consistent. As we almost pay exclusive attention to consistent estimators, consistency turns out to be a mild assumption.

However, $\ddot{\ell}_{\boldsymbol{X}}(\theta)$ is random for each θ, so the "limit" $\ddot{\ell}_{\boldsymbol{x}}(\theta_0)$ is not a appropriate limiting quantity to study. To get rid of the randomness in $\ddot{\ell}_{\boldsymbol{X}}(\theta_0)$ as $n \to \infty$, by LLN,

$$\frac{1}{n}\ddot{\ell}_{\boldsymbol{X}}(\theta_0) = \frac{1}{n}\sum\nolimits_{i=1}^{n}\ddot{\ell}_{X_i}(\theta_0) \xrightarrow{p} \mathbb{E}_{\theta_0}[\ddot{\ell}_{X_1}(\theta_0)].$$

We will see the limiting quantity $\mathbb{E}_{\theta_0}[\ddot{\ell}_{x_1}(\theta_0)] = -I(\theta_0)$ in Theorem 6.3.

In the end, under some regularity conditions, the denominator

$$\frac{1}{n}\ddot{\ell}_{\boldsymbol{X}}(\tilde{\theta}) \xrightarrow{p} -I(\theta_0).$$

So we expect that the estimation error $\hat{\theta} - \theta_0$ inflated by $\sqrt{n}$ converges to some Gaussian distribution. This is the asymptotic Normality we hope to establish.

Both the first and the second derivatives of the log likelihood function are involved in the expansion (13) and the estimation error (14). These two derivatives need further study and the following two important concepts are relevant: the score function and the Fisher information.

Definition 6.1 (Score function)**.** The score function is defined as

$$s(X;\theta) = \dot{\ell}_X(\theta) = \frac{\partial}{\partial\theta}\ell_X(\theta) = \frac{\partial}{\partial\theta}\log(f(X;\theta)).$$

The score function depends on both the observation X and the parameter θ. Under certain regularity conditions, the true value solves the population version of the likelihood equation.

Theorem 6.1. *Suppose the random variable $X \sim f(x;\theta_0)$, and there exists an $\epsilon > 0$ such that $\int \sup_{\theta\in[\theta_0-\epsilon,\theta_0+\epsilon]} |\frac{\partial}{\partial\theta} f(x;\theta)| dx < \infty$. Then, the expectation of the score function is 0:*

$$\mathbb{E}_{\theta_0}[s(X;\theta_0)] = 0.$$

Proof. First note $s(X;\theta) = \frac{1}{f(X;\theta)}\frac{\partial f(X;\theta)}{\partial\theta}$ by chain rule. Then, by LOTUS,

$$\mathbb{E}_{\theta_0}[s(X;\theta_0)] = \int \left.\frac{\partial\log(f(x;\theta))}{\partial\theta}\right|_{\theta=\theta_0} f(x;\theta_0)dx = \int \left.\frac{\partial f(x;\theta)}{\partial\theta}\right|_{\theta=\theta_0} dx,$$

where the differentiation is with respect to the parameter θ, and the integration is with respect to the random variable $X \sim f(x;\theta_0)$, the true data generating process.

Now swap the order of integration and differentiation of the last display, which yields

$$\mathbb{E}_{\theta_0}[s(X;\theta_0)] = \frac{\partial}{\partial\theta}\int f(x;\theta)dx\Big|_{\theta=\theta_0} = \frac{\partial}{\partial\theta}1 = 0.$$

□

The trick is to swap the order of integration and differentiation. The interchange of the two limit processes is essentially due to the dominated convergence theorem. Theorem 6.2 implies it suffices to have $|\frac{\partial}{\partial\theta}f(x;\theta)|$ uniformly dominated by a integrable function. In the assumptions of Theorem 6.1, the dominating function is chosen as $\sup_{\theta\in[\theta_0-\epsilon,\theta_0+\epsilon]}|\frac{\partial}{\partial\theta}f(x;\theta)|$ for some $\epsilon > 0$.

Theorem 6.2 (Theorem 2.27 of Folland (1999))**.** *Suppose that $f : \mathcal{X}\times[a,b] \to \mathbb{C}$ $(-\infty < a < b < \infty)$ and that $f(\cdot,t) : \mathcal{X} \to \mathbb{C}$ is integrable for each $t \in [a,b]$. Let $F(t) = \int_{\mathcal{X}} f(x,t)d\mu(x)$.*

a. Suppose that there exists $g \in L_1(\mu)$ such that $|f(x,t)| \le g(x)$ for all x,t. If $\lim_{t\to t_0} f(x,t) = f(x,t_0)$ for every x, then $\lim_{t\to t_0} F(t) = F(t_0)$; in particular, if $f(x,\cdot)$ is continuous for each x, then F is continuous.

b. Suppose that $\partial f/\partial t$ exists and there is a $g \in L_1(\mu)$ such that $|\frac{\partial}{\partial t}f(x,t)| \le g(x)$ for all x,t. Then, F is differentiable and

$$F'(t) \equiv \frac{\partial}{\partial t}\int_{\mathcal{X}} f(x,t)d\mu(x) = \int_{\mathcal{X}} \frac{\partial}{\partial t}f(x,t)d\mu(x).$$

Example 6.1 (Normal mean)**.** Suppose $X_1,...,X_n \overset{iid}{\sim} N(\mu,1)$. The score function for the dataset is

$$s(\{X_i\};\mu) = \sum\nolimits_{i=1}^{n}(X_i - \mu) = n(\bar{X}_n - \mu).$$

By direct calculation, $\mathbb{E}_\mu[s(\{X_i\};\mu)] = 0$.

Alternatively, direct calculation yields $\frac{\partial}{\partial\mu}f(x;\mu) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(x-\mu)^2}(x-\mu)$ and $\int\frac{\partial}{\partial\mu}f(x;\mu)dx = 0$ for all $\mu \in \mathbb{R}$. ♢

Example 6.2 (Uniform)**.** Suppose $X_1,...,X_n \overset{iid}{\sim} Unif[0,\theta]$. The score function does not exist due to the differentiability issue. The support of the observations depends on model parameters, which causes the problem. In the development of the general asymptotic theory, it is necessary to assume the support of the observations does not depend on model parameters. ♢

Remark. Theorem 6.1 also is a "moment" equation, and the method of moments estimator based on this moment condition is

$$\frac{1}{n}\sum\nolimits_{i=1}^{n} s(X_i;\hat{\theta}_{MM}) = 0,$$

which is the same as

$$\sum\nolimits_{i=1}^{n} s(X_i;\hat{\theta}_{MM}) = \dot{\ell}_n(\hat{\theta}_{MM}) = 0$$

whose solution is exactly the MLE, assuming its SOC also holds.

Definition 6.2 (The Fisher Information)**.** The Fisher information is defined as the variance of the score function.

$$I(\theta) = V_\theta[s(X;\theta)] = \mathbb{E}_\theta[s(X;\theta)^2] - (\mathbb{E}_\theta[s(X;\theta)])^2. \tag{15}$$

By Theorem 6.1, under the same regularity conditions, the Fisher information can be simplified as

$$I(\theta) = V_\theta[s(X;\theta)] = \mathbb{E}_\theta[s(X;\theta)^2].$$

That is, the Fisher information is the second moment of the score function.

Interestingly, the Fisher information admits an alternative expression.

Theorem 6.3. *Suppose the random variable $X \sim f(x;\theta)$, and there exists $\epsilon > 0$ such that $\int \sup_{\theta\in[\theta_0-\epsilon,\theta_0+\epsilon]} |\frac{\partial}{\partial\theta} f(x;\theta)|dx < \infty$ and $\int \sup_{\theta\in[\theta_0-\epsilon,\theta_0+\epsilon]} |\frac{\partial^2}{\partial\theta^2} f(x;\theta)|dx < \infty$. Then,*

$$I(\theta) = -\mathbb{E}_\theta[\ddot{\ell}_X(\theta)].$$

Proof. First note that, by chain rule,

$$\ddot{\ell}_X(\theta) = \frac{\partial^2}{\partial\theta^2}\log(f(X;\theta)) = \frac{\partial}{\partial\theta}\left(\frac{\frac{\partial}{\partial\theta}f(X;\theta)}{f(X;\theta)}\right) = \frac{\frac{\partial^2}{\partial\theta^2}f(X;\theta)}{f(X;\theta)} - (\dot{\ell}_X(\theta))^2.$$

By swapping the order of differentiation and integration, the first term on the right hand side satisfies

$$\mathbb{E}_\theta[\frac{\frac{\partial^2}{\partial\theta^2}f(X;\theta)}{f(X;\theta)}] = \int \frac{\partial^2}{\partial\theta^2} f(x;\theta)dx = \frac{\partial^2}{\partial\theta^2}\int f(x;\theta)dx = 0.$$

Then, taking expectation on both sides implies

$$\mathbb{E}_\theta[\ddot{\ell}_X(\theta)] = 0 - \mathbb{E}_\theta[(\dot{\ell}_X(\theta))^2] = -\mathbb{E}_\theta[s(X;\theta)^2] = -I(\theta).$$

□

With iid samples, the score function of the whole data is $\sum_{i=1}^n s(x_i;\theta)$, and the Fisher information for the whole dataset is

$$I_n(\theta) = V[\sum\nolimits_{i=1}^n s(x_i;\theta)] = \sum\nolimits_{i=1}^n V[s(x_i;\theta)] = nI(\theta).$$

The following asymptotic Normality theorem of MLE is fundamentally important to frequentist inference, because (1) it provides an automatic recipe to construct confidence intervals, and (2) the theorem applies to a wide range of parametric models. The conditions required for the asymptotic Normality of the MLE to hold are indeed mild.

Theorem 6.4 (Asymptotic Normality of the MLE, adapted from Theorem 9.14 of Keener (2010))**.** *Suppose the following regularity conditions hold.*

1. $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta_0)$ *where* $\theta_0 \in \Theta \subset \mathbb{R}$. *Let* $\ell_X(\theta) = \log f(X;\theta)$.
2. *The support* $\overline{\{x : f(x;\theta) > 0\}}$ *does not depend on* θ.
3. *For every* $x \in \mathcal{X}$, $\frac{\partial^2}{\partial\theta^2} f(x;\theta)$ *exists and is continuous in* θ.
4. *The Fisher information exists, is finite, and can be found by either* $I(\theta) = \mathbb{E}_\theta[\dot{\ell}_X(\theta)^2]$ *(Definition 6.2) or* $I(\theta) = -\mathbb{E}_\theta[\ddot{\ell}_X(\theta)]$ *(Theorem 6.3). Also,* $\mathbb{E}_\theta[\dot{\ell}_X(\theta)] = 0$.
5. *For every* θ' *in the interior of* Θ, *there exists* $\epsilon > 0$ *such that* $\mathbb{E}_\theta\left[\sup_{\theta\in[\theta'-\epsilon,\theta'+\epsilon]} |\ddot{\ell}_X(\theta)\right] < \infty$.
6. *The maximum likelihood estimator* $\hat{\theta}_n$ *is consistent.*

Then, the infeasible version of the theorem is

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1}) \text{ as } n \to \infty$$

under $\mathbb{P}_\theta$ *for every* θ *in the interior of* Θ.

Additionally, we have the feasible (plug-in) version,

$$\sqrt{nI(\hat{\theta}_n)}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0,1) \text{ as } n \to \infty$$

under $\mathbb{P}_\theta$ *for every* θ *in the interior of* Θ. *The key distinction from the infeasible version is the limiting distribution is standard Normal.*

Proof. The proof of Theorem 9.14 in Keener (2010) is for the infeasible version. The feasible (plug-in) version is due to the Slutsky theorem. □

In reality, we never have $n \to \infty$. With n being finite, we practically perform statistical inference using the approximate *sampling distribution*

$$\hat{\theta}_n \dot{\sim} N\left(\theta_0, \left(nI(\hat{\theta}_n)\right)^{-1}\right),$$

where $\dot{\sim}$ denotes "is approximately distributed as". That is, the true *sampling distribution* of the maximum likelihood estimator $\hat{\theta}$ is approximately distributed as the Normal distribution with mean θ_0 and variance $\left(nI(\hat{\theta}_n)\right)^{-1}$.

Though point estimators and point estimates are often readily available in close form and/or numerically, their exact sampling distributions are typically hard, if not impossible, to characterize. For instance, the sampling distribution of the difference of the two Gaussian mean with unequal variances has to be derived with some approximation. For another, the sampling distribution of the sample correlation in the bivariate Normal model is available in close form but it requires substantial mathematical skills to derive.

In contrast, Theorem 6.4 provides a universal method to approximate the sampling distributions for a wide range of problems. Simple yet powerful, Theorem 6.4 implies a general recipe to construct confidence intervals for θ_0 with (asymptotically) $(1-\alpha)$-confidence level:

$$\hat{\theta}_n \pm z_{\alpha/2} \frac{1}{\sqrt{n}} (I(\hat{\theta}_n))^{-1/2}.$$

Clearly, Theorem 6.5 is a consequence of Theorem 6.4. But due to its fundamental importance to statistical inference, it is named "theorem" rather than "corollary".

Theorem 6.5 (MLE CI)**.** *For $\alpha \in (0,1)$, let the confidence interval be*

$$C_\alpha = [\hat{\theta} - z_{\alpha/2}(nI(\hat{\theta}))^{-1/2}, \hat{\theta} + z_{\alpha/2}(nI(\hat{\theta}))^{-1/2}],$$

where z_α denotes the $\alpha-$upper quantile of the standard Normal distribution: $\mathbb{P}(Z \geq z_\alpha) = \alpha$ for $Z \sim N(0,1)$. Then, as $n \to \infty$,

$$\mathbb{P}_{\theta_0}(\theta_0 \in C_\alpha) \to 1 - \alpha.$$

Example 6.3 (Univariate Normal)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu_0, 1)$. The maximum likelihood estimator for μ is

$$\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^n X_i.$$

Under the Normal model, the sampling distribution of $\hat{\mu}_n$ is $N(\mu_0, \frac{1}{n})$ which is *exact*. Thus, exact confidence intervals exist.(see more details in Section 5.3.3)

To apply Theorem 6.4, we start with the score function

$$s(X, \mu) = \dot{\ell}_X(\mu) = X - \mu.$$

By Definition 6.2 and the linearity property of variance, the Fisher information is

$$I(\mu) = V_\mu(X - \mu) = V_\mu(X) = 1.$$

Then, Theorem 6.4 implies $\sqrt{n}(\hat{\mu} - \mu_0) \xrightarrow{d} N(0,1)$, and

$$\hat{\mu}_n \dot{\sim} N(\mu_0, 1/n).$$

The asymptotic Normal approximation due to Theorem 6.4 is exactly the same as the true sampling distribution. However, the approximation statement is weaker than its actual performance.

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Example 6.4 (Inferring the proportion)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta_0)$. The log-likelihood function of the dataset is

$$\ell_{\boldsymbol{X}}(\theta) = \sum_{i=1}^n X_i \log(\theta) + \sum_{i=1}^n (1 - X_i)\log(1 - \theta).$$

Then, by solving the FOC and verifying the SOC, the maximum likelihood estimator is

$$\hat{\theta}_n = \bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i.$$

The true sampling distribution of the maximum likelihood estimator $\hat{\theta}_n$ can be derived under the Bernoulli model:

$$\hat{\theta}_n \sim \frac{1}{n} \times Bin(n, \theta_0),$$

which is still a discrete random variable.

To apply the Normal approximation suggested in Theorem 6.4, we start with the score function

$$s(X, \theta) = \dot{\ell}_X(\theta) = \frac{X}{\theta} - \frac{1 - X}{1 - \theta} = \frac{X}{\theta(1 - \theta)} - \frac{1}{1 - \theta}.$$

Then, there are two ways to compute the Fisher information: either by Definition 6.2, or by Theorem 6.3. By Definition 6.2 and the linearity property of variance,

$$I(\theta) = V_\theta(s(X,\theta)) = \frac{V_\theta(X)}{\theta^2(1-\theta)^2} = \frac{1}{\theta(1-\theta)}.$$

Alternatively, by Theorem 6.3,

$$I(\theta) = -\mathbb{E}_\theta[\ddot{\ell}_X(\theta)] = \mathbb{E}_\theta\left[\frac{X}{\theta^2} + \frac{1-X}{(1-\theta)^2}\right] = \frac{1}{\theta(1-\theta)}.$$

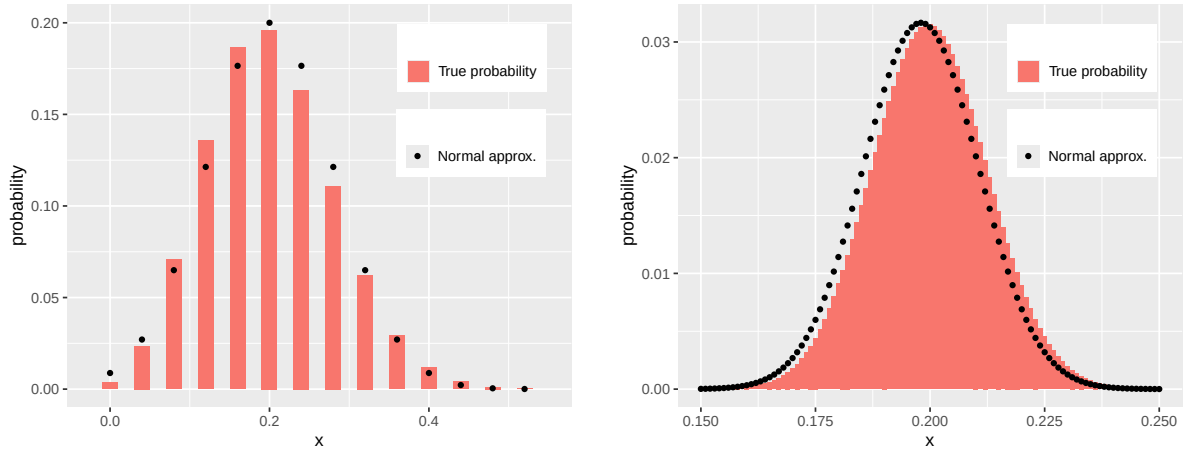
Clearly, the two methods arrive at the same answer.

Hence, Theorem 6.4 implies

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \theta_0(1-\theta_0)).$$

Note the maximum likelihood estimator $\hat{\theta}_n$ is the average of the iid sample points, CLT and the Slutsky theorem also imply the same result.

We examine the approximation in the following simulation study where $\theta_0 = 0.2$. We first set $n = 25$ and then set $n = 1000$.



$n = 25$ $n = 1000$

Figure 6.1: True sampling distribution v.s. Normal approximation

```r
n <- 1000 ## sample size
theta0 <- 0.2 ## true value
pp <- dbinom(0:n,n,theta0) ## true sampling distribution
set.seed(1)
X <- rbinom(n,1,theta0) ## one sample
hat.theta <- mean(X)  ## MLE point estimate
qq <- dnorm(0:n/n, hat.theta, sqrt(hat.theta*(1-hat.theta)/n))
qq <- qq/sum(qq)
df <- tibble("x" = 0:n/n, "pp" = pp, "qq"= qq)

ggplot(data = df, aes(x=x)) +
    geom_bar(stat="identity", aes(y = pp, fill = "True probability")) +
    geom_point(aes(y = qq, shape = "Normal approx.")) +
    labs(y = "probability", fill = "",shape = "") + xlim(0.15, 0.25)+
    theme(legend.key.size = unit(0.5, 'cm'),
```

```
          legend.position = c(.95, .95),
          legend.justification = c("right", "top"),
          legend.box.just = "right",
          legend.margin = margin(6, 6, 6, 6))
```

Further by Theorem 6.5, the approximate $(1-\alpha)$-confidence interval for θ_0 is

$$\hat{\theta}_n \pm z_{\alpha/2}\frac{1}{\sqrt{n}}\sqrt{\hat{\theta}_n(1-\hat{\theta}_n)},$$

where z_α is the $\alpha-$upper quantile of the standard Normal distribution. This interval is also called the "Wald confidence interval" in the literature.

We examine the coverage of the above CI numerically. Again, we set $n = 25$ and $\theta_0 = 0.2$. The actual coverage rate in the simulation study is roughly 88.4%, which is lower than the 95% nominal rate.

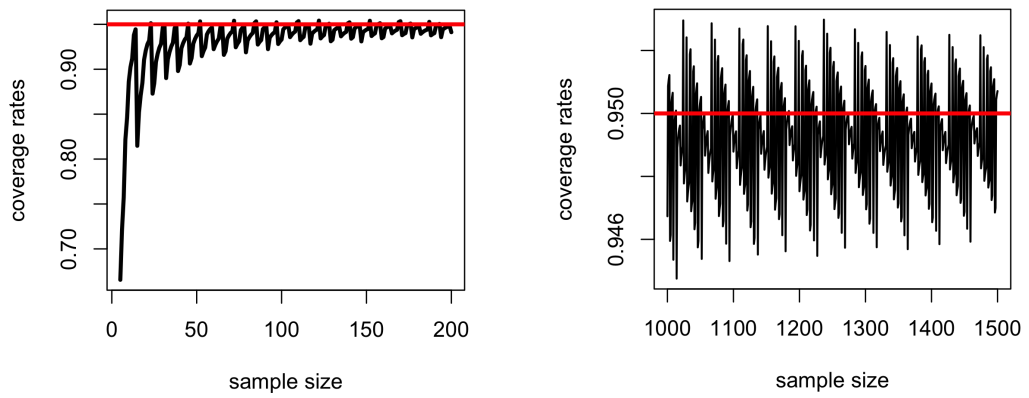
```
n <- 25 ## sample size
theta0 <- 0.2 ## true value
CI.cover.rate <- function(truth,n){
    pp <- dbinom(0:n, n, truth)
    xx <- 0:n/n
    rad <- qnorm(0.975)*sqrt(xx*(1-xx)/n)
    return(sum((xx + rad >= truth & xx - rad <= truth) * pp))
}
CI.cover.rate(theta0, n) ## 0.8844
```

The covering of the Wald confidence interval is chaotic in two ways. (See more details in Brown et al. (2001) and the references therein.)

(1) for fixed θ_0, the coverage rates of the same nominal rate CI oscillate as n increases;

(2) for fixed n, the coverage rates of the same nominal rate CI also oscillate as θ_0 changes, and the coverage rates when θ_0 being close to 0 and 1 drop dramatically.

We first look at the chaotic behaviors of the first type.

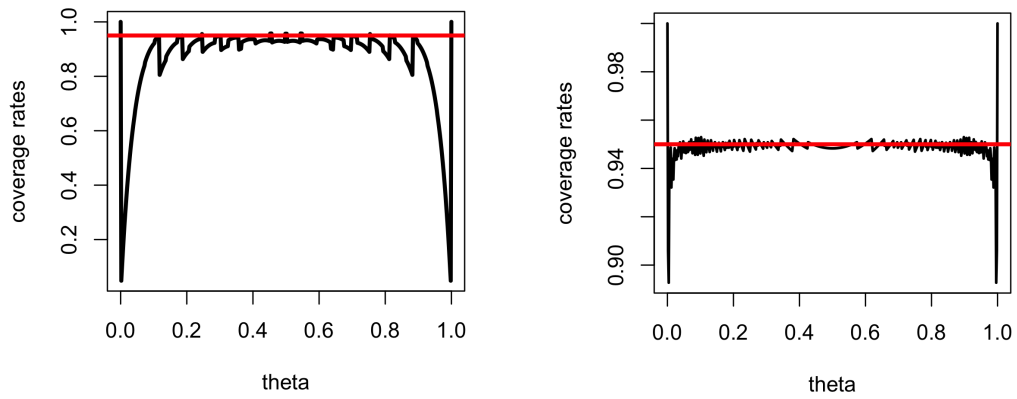


$n \in \{10, 11, ..., 200\}$ $n \in \{1000, 1001, ..., 1500\}$

Figure 6.2: Chaotic behaviors of the Wald CI with $\theta_0 = 0.2$

```
n.seq <- 1000:1500
cover.rates <- sapply(n.seq, function(nn){CI.cover.rate(theta0,nn)})
plot(n.seq, cover.rates, type = "l", lwd=1.5,
     xlab="sample size", ylab = "coverage rates")
abline(h=0.95,col="red",lwd=3)
```

Then, we first look at the chaotic behaviors of the second type.



$n = 25$ $n = 2000$

Figure 6.3: Chaotic behaviors of the Wald CI with $\theta_0 \in [0, 1]$

```
n <- 2000
theta.seq <- seq(from = 0, to = 1, length.out = 501)
cover.rates <- sapply(theta.seq, function(theta){CI.cover.rate(theta, n)})
plot(theta.seq, cover.rates, type = "l",lwd=2, xlab="theta", ylab = "coverage rates")
abline(h=0.95,col="red",lwd=3)
```

Brown et al. (2001) suggest the following alternatives to the Wald confidence interval.

(1) For small n ($n \leq 40$), Brown et al. (2001) recommend the Wilson interval

$$CI_{Wilson} = \tilde{\theta} \pm \frac{z_{\alpha/2}}{\sqrt{n} + z_{\alpha/2}^2/\sqrt{n}} \sqrt{\hat{\theta}(1-\hat{\theta}) + z_{\alpha/2}^2/(4n)},$$

where $\tilde{\theta} = \frac{\sum_{i=1}^n X_i + z_{\alpha/2}^2/2}{n + z_{\alpha/2}^2}$, or the equal-tailed Jeffreys' prior interval

$$\left[q_{B(a_n,b_n)}(\alpha/2), q_{B(a_n,b_n)}(1-\alpha/2)\right]$$

where $a_n = \sum_{i=1}^n X_i + 1/2$, $b_n = n - \sum_{i=1}^n X_i + 1/2$, and $q_{B(a,b)}(\alpha)$ denotes the $\alpha-$quantile of the distribution $Beta(a, b)$.

(2) For larger n, Brown et al. (2001) recommend the Agresti–Coull interval:

$$CI_{AC} = \tilde{\theta} \pm \frac{z_{\alpha/2}}{\sqrt{n + z_{\alpha/2}^2}} \sqrt{\tilde{\theta}(1-\tilde{\theta})},$$

where $\tilde{\theta} = \frac{\sum_{i=1}^n X_i + z_{\alpha/2}^2/2}{n + z_{\alpha/2}^2}$. If $\alpha = 0.05$, $z_{\alpha/2} \approx 1.96$, and the Agresti–Coull interval roughly "adds two successes and two failures" to the original dataset.

Hogg et al. (2018) call $\varepsilon \equiv z_{\alpha/2} \frac{1}{\sqrt{n}} \sqrt{\hat{\theta}_n(1-\hat{\theta}_n)}$ the *maximum error* of the point estimate $\hat{\theta}_n$ with $(1-\alpha)$-confidence. Once the maximum error is specified, the required sample size for the estimation error being less than the maximum error is

$$n = \frac{z_{\alpha/2}^2 \hat{\theta}_n(1-\hat{\theta}_n)}{\varepsilon^2}$$

Note $ab \leq (\frac{a+b}{2})^2$, the sample size that works for all $\hat{\theta}_n$ is $\frac{z^2_{\alpha/2}}{4\varepsilon^2}$ and it suffices to pick $n = \left\lceil \frac{z^2_{\alpha/2}}{4\varepsilon^2} \right\rceil$.

♢

Definition 6.3 (The Observed Fisher Information)**.** The observed Fisher information is defined as

$$I_{obs} = -\ddot{\ell}_{\boldsymbol{x}}(\theta)|_{\theta=\hat{\theta}}.$$

It turns out the plug-in Fisher information estimator is sub-optimal in small samples. Fisher suggested that the following gives a better idea of $\hat{\theta}$'s accuracy

$$\hat{\theta}_n \dot{\sim} N\left(\theta_0, I_{obs}^{-1}\right),$$

and the (asymptotic) $(1-\alpha)-$confidence interval based on the *observed Fisher information* is

$$\hat{\theta}_n \pm z_{\alpha/2} I_{obs}^{-1/2}.$$

(See Efron and Hinkley, 1978, for more details.) When sample size grows to infinity, both Fisher information estimators converge to the same quantity. Asymptotically, they are equivalent. In finite sample, they are identical for many parametric families including the exponential family.

Example 6.5. For models in the exponential family, $f(x;\theta) = h(x)e^{\theta T(x) - A(\theta)}$, using the observed Fisher information is exactly the same as the plug-in Fisher information:

$$I_{obs} = nI(\hat{\theta}_n).$$

The log-likelihood function is $\ell_X(\theta) = \log(h(X)) + \theta T(X) - A(\theta)$. By definition, $I_{obs} = -\ddot{\ell}_{\boldsymbol{x}}(\hat{\theta}) = n\ddot{A}(\hat{\theta}_n)$, and $I(\theta) = -\mathbb{E}_\theta[\ddot{\ell}_X(\theta)] = \ddot{A}(\theta)$. ♢

Another (perhaps stronger) reason to use the observed Fisher information is that it is readily available through optimization routines in R/Python. If the numerical algorithm minimizes the negative log-likelihood function, then the Hessian (matrix), which is part of the solver output, is the observed Fisher information (matrix). That is, there is no need to derive the Fisher information and then apply the plug-in.

Example 6.6 (Cauchy location)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta) = \frac{1}{\pi(1+(x-\theta)^2)}$ where $\theta \in \mathbb{R}$. The log-likelihood function $\ell_{\boldsymbol{X}}(\theta) = -\sum_{i=1}^n \log(1+(X_i-\theta)^2) - n\log(\pi)$. The maximum likelihood estimator $\hat{\theta}_n$ solves

$$\sum_{i=1}^n \frac{2(X_i-\theta)}{1+(X_i-\theta)^2} = 0.$$

The solution is only available numerically.

```
theta <- 3
n <- 100
X <- theta + rt(n,1)
nloglik <- function(theta,dat){
    pdf <- sapply(dat, function(xx) log(dt(xx-theta,1)))
    return(-sum(pdf))
}
fit <- nlm(nloglik, p = median(X), dat=X, hessian = TRUE)
fit
```

The score function $s(X;\theta) = \dot{\ell}_X(\theta) = \frac{2(X-\theta)}{1+(X-\theta)^2}$. Indeed, $\mathbb{E}_\theta[s(X;\theta)] = 0$. Further, by definition,

$$I(\theta) = \mathbb{E}_\theta[s(X;\theta)^2] = \int_{-\infty}^{\infty} \frac{4(x-\theta)^2}{\pi(1+(x-\theta)^2)^3} dx = \int_{-\infty}^{\infty} \frac{4x^2}{\pi(1+x^2)^3} dx = 1/2.$$

Theorem 6.4 implies $\hat{\theta}_n \dot{\sim} N(\theta_0, 2/n)$ whose variance is independent of the parameter θ.

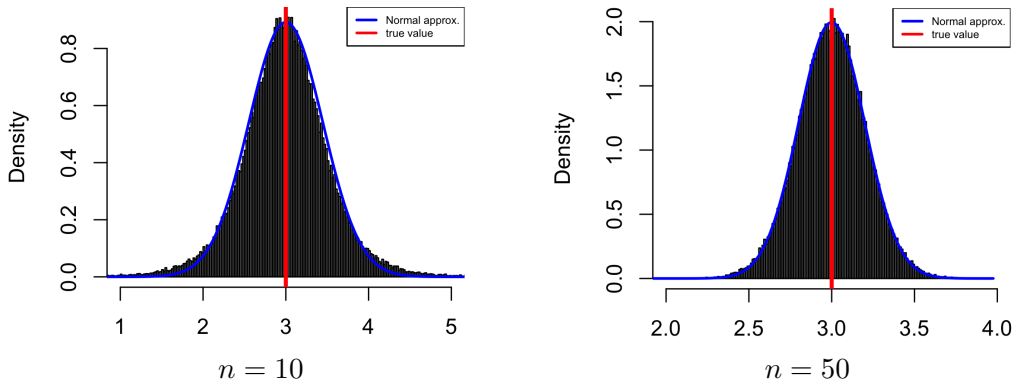


Figure 6.4: True sampling distribution of $\hat{\theta}_n$ overlaid by $N(\theta_0, 2/n)$

```
n <- 10
Nmc <- 1e5
theta <- 3
X <- theta + rt(n*Nmc,1) %>% matrix(., nrow = Nmc)
fit.list <- apply(X,1, function(xx)
    nlm(nloglik, p = median(xx), dat = xx, hessian = TRUE))
hat.theta.mat <- lapply(fit.list, function(yy) yy$estimate) %>% unlist()

h1 <- hist(hat.theta.mat,breaks = "FD", probability = TRUE,
           main = "",xlab = "", xlim = c(1,5))
qq <- dnorm(h1$mids, theta, sqrt(2/n))
qq <- qq/sum(qq) * sum(h1$density)
lines(h1$mids, qq, lwd=2,col="blue")
abline(v=theta, col = "red", lwd=3)
legend("topright", c("Normal approx.", "true value"),
       col = c("blue", "red"),lty = c(1,1),lwd=2,cex=0.5)
```

On the other hand, the observed Fisher information is

$$I_{obs} = -\ddot{\ell}_{\boldsymbol{X}}(\hat{\theta}_n) = \sum_{i=1}^{n} \frac{2(1-(X_i-\hat{\theta}_n)^2)}{(1+(X_i-\hat{\theta}_n)^2)^2},$$

which is already computed in the process of numerical optimization and can be accessed through the Hessian (matrix) in the optimzation routine output. The alternative approximation to the sampling distribution is $\hat{\theta}_n \dot{\sim} N(\theta_0, I_{obs}^{-1})$ whose variance is data dependent and depends on the accuracy of $\hat{\theta}_n$.

In the simulation, we compare the coverage rates of the confidence intervals implied by the two versions of Fisher information. The confidence interval using the observed Fisher information offers better coverage *and* is shorter on average.

- When $n = 10$, the coverage is 91.6% while the coverage of the confidence interval using the Fisher information is 91.2%. The average standard error using the observed Fisher information is 0.422, while $\sqrt{2/n} \approx 0.447$.

- When $n = 50$, the coverage is 94.5% while the coverage of the confidence interval using the Fisher information is 94.2%. The average standard error using the observed Fisher information is 0.198, while $\sqrt{2/n} = 0.2$.

```
sapply(hat.theta.mat, function(xx) xx-1.96*sqrt(2/n) <=theta &
                                   xx + 1.96*sqrt(2/n) >= theta) %>% mean()
hess.mat <- lapply(fit.list, function(yy) yy$hessian) %>% unlist()
mapply(function(xx, yy) xx-1.96/sqrt(yy) <=theta &
                        xx + 1.96/sqrt(yy) >= theta,
       xx = hat.theta.mat, yy = hess.mat) %>% mean()

1/sqrt(hess.mat)%>% mean()
1/sqrt(n/2)
```

♢

Estimation Optimality: Cramér-Rao lower bound Optimality is the mark of the maturity of statistical methods. In terms of variance, maximum likelihood estimators have attained the minimum.

Theorem 6.6 (Cramér-Rao)**.** *Suppose* $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta)$, $\theta \in \Theta \subset \mathbb{R}$, *and* $\mathbb{E}_{\theta_0}[s(X_1;\theta_0)] = 0$. *For any unbiased estimator* $\tilde{\theta}(\boldsymbol{X})$ *such that* $\mathbb{E}_{\theta_0}[\tilde{\theta}(\boldsymbol{X})^2] < \infty$,

$$V_\theta(\tilde{\theta}) \geq I_n(\theta)^{-1}.$$

Proof. Note the Cauchy-Schwartz inequality implies the covariance inequality: for any two random variables X and Y,

$$\text{cov}(X, Y)^2 \leq V(X)V(Y)$$

where the "=" is attained when X is linear in Y (and vice versa). Then, the covariance between the score function of the whole dataset and the unbiased estimator satisfies

$$\text{cov}_\theta(s(\boldsymbol{X};\theta), \tilde{\theta})^2 \leq V_\theta(s(\boldsymbol{X};\theta))V_\theta(\tilde{\theta}),$$

where the covariance on the left hand side is

$$\text{cov}_\theta(s(\boldsymbol{X};\theta), \tilde{\theta}) = \mathbb{E}_\theta[s(\boldsymbol{X};\theta)\tilde{\theta}] = \int_{\mathcal{X}^n} \frac{\partial}{\partial\theta} f(\boldsymbol{x};\theta)\tilde{\theta}(\boldsymbol{x})d\boldsymbol{x} = \frac{\partial}{\partial\theta}\int_{\mathcal{X}^n} f(\boldsymbol{x};\theta)\tilde{\theta}(\boldsymbol{x})d\boldsymbol{x} = \frac{\partial}{\partial\theta}\mathbb{E}_\theta[\tilde{\theta}(\boldsymbol{x})] = 1,$$

and on the right hand side, $V_\theta(s(\boldsymbol{X};\theta)) = I_n(\theta)$ by definition of Fisher information. Rearranging the terms, the result follows.

In deriving the covariance, we use the assumption $\mathbb{E}_{\theta_0}[s(X_1;\theta_0)] = 0$ to simplify the expression of the covariance; also, the technical condition(s) required for swapping the order of integration and differentiation is(are) implied by $\mathbb{E}_{\theta_0}[s(X_1;\theta_0)] = 0$, because $\tilde{\theta}(\boldsymbol{x})$ contains no θ. Lastly, the unbiasedness of $\tilde{\theta}(\boldsymbol{X})$ means $\mathbb{E}_\theta[\tilde{\theta}(\boldsymbol{x})] = \theta$. □

The Cramér-Rao Theorem provides a lower bound for the variance of all *unbiased* estimators. As the bound is inverse to the Fisher information, it naturally compares the variance of all unbiased estimators with the MLE which attains the lower bound. That is, if MLE is unbiased, there is no strictly better unbiased estimators that can beat MLE in variance.

Unfortunately, MLE's are often biased. For instance, the MLE's of uniform distribution's endpoints are biased. Nonetheless, the bias of the MLE's are often in the order of $\frac{1}{n}$, hence, are asymptotically

unbiased. Therefore, the Cramér-Rao Theorem is still meaningful in MLE's asymptotically superior MSE relative to other estimators.

In the same argument as in Theorem 6.6, we have the following corollary for general estimators. When $\tilde{\theta}$ is unbiased, $h(\theta) = \theta$, reproducing the result of Theorem 6.6.

Corollary 6.7. *Suppose* $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta)$, $\theta \in \Theta \subset \mathbb{R}$, *and* $\mathbb{E}_{\theta_0}[s(X_1;\theta_0)] = 0$. $\tilde{\theta}$ *is an estimator such that* $\mathbb{E}_\theta[\tilde{\theta}] = h(\theta)$ *and* $h(\cdot)$ *is differentiable, and* $\mathbb{E}_{\theta_0}[\tilde{\theta}(\boldsymbol{X})^2] < \infty$, *then*

$$V_\theta(\tilde{\theta}) \geq \frac{(\dot{h}(\theta))^2}{I_n(\theta)}.$$

6.2 MLE Confidence Sets for Multiparameter Problems

Reading(s) 6.2. Wasserman (2004): 9.10

Similar to the score function in the one-parameter case, define the score function for the multi-parameter case.

Definition 6.4 (The Score function)**.** Suppose $X \sim f(x;\theta)$ where $\theta \in \Theta$, and θ is $p-$dimensional. The score function is defined as

$$s(X,\theta) = \nabla \ell_X(\theta) = \begin{bmatrix} \frac{\partial}{\partial \theta_1}\ell_X(\theta) \\ \vdots \\ \frac{\partial}{\partial \theta_p}\ell_X(\theta) \end{bmatrix}.$$

Lemma 6.8. *Suppose $X \sim f(x;\theta)$ and $\theta \in \Theta$ is $p-$dimensional. Assume $\int |\frac{\partial}{\partial \theta_j} f(x;\theta)|dx < \infty$ for all $j = 1, ..., p$ and all $\theta \in \Theta$. Then*

$$\mathbb{E}_\theta[\nabla \ell_X(\theta)] = \mathbf{0},$$

where $\mathbf{0}$ is the $p \times 1$ vector of zeros.

Proof. Swap the order of differentiation and integration, and note the unity property of pdf/pmf $f(x;\theta)$. □

Definition 6.5 (The Fisher Information Matrix)**.** The Fisher information matrix is defined as

$$I(\theta) = \text{cov}_\theta[s(X,\theta)] = \mathbb{E}_\theta[\nabla \ell_X(\theta) \nabla \ell_X(\theta)^T].$$

In the univariate case, the Fisher information is the variance of the score function. The multivariate case is similar. Since the score function is a vector, the Fisher information $I(\theta)$ becomes a matrix and the $(i,j)^{th}$ entry of $I(\theta)$ is

$$\mathbb{E}_\theta[\frac{\partial}{\partial \theta_i}\ell_X(\theta)\frac{\partial}{\partial \theta_j}\ell_X(\theta)].$$

Similar to the univariate case, the Fisher information matrix also admits an alternative expression. In multivariate calculus, the (square) matrix of the second order partial derivatives of a scalar-valued function is called the Hessian matrix, or Hessian. It describes the local curvature of a function of many variables.

Lemma 6.9 (The Fisher Information Matrix)**.** *The Fisher Information Matrix is the same as*

$$I(\theta) = -\mathbb{E}_\theta[H]$$

where H is the Hessian matrix of $\ell_X(\theta)$ whose $(i,j)^{th}$ entry is

$$H_{ij} = \frac{\partial^2}{\partial\theta_i \partial\theta_j}\ell_X(\theta).$$

Example 6.7 (Univariate Normal mean and variance: the Fisher information matrix)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$. The data log-likelihood is

$$\ell_n(\mu, \sigma^2) = const - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^n (X_i - \mu)^2.$$

The log-likelihood of the data point X:

$$\ell_X(\mu, \sigma^2) = const - \frac{1}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}(X - \mu)^2.$$

The score function is

$$s(X; \mu, \sigma^2) = \begin{bmatrix} \frac{\partial}{\partial\mu}\ell_X(\mu, \sigma^2) \\ \frac{\partial}{\partial\sigma^2}\ell_X(\mu, \sigma^2) \end{bmatrix} = \begin{bmatrix} \frac{1}{\sigma^2}(X - \mu) \\ -\frac{1}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}(X - \mu)^2 \end{bmatrix}.$$

The Hessian matrix with respect to the log likelihood function $\ell_X(\mu, \sigma^2)$ is

$$H_X(\mu, \sigma^2) = \begin{bmatrix} \frac{\partial^2}{\partial\mu^2}\ell_X(\mu, \sigma^2) & \frac{\partial^2}{\partial\mu\partial\sigma^2}\ell_X(\mu, \sigma^2) \\ \frac{\partial^2}{\partial\mu\partial\sigma^2}\ell_X(\mu, \sigma^2) & \frac{\partial^2}{(\partial\sigma^2)^2}\ell_X(\mu, \sigma^2) \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sigma^2} & -\frac{1}{(\sigma^2)^2}(X - \mu) \\ -\frac{1}{(\sigma^2)^2}(X - \mu) & \frac{1}{2(\sigma^2)^2} - \frac{1}{(\sigma^2)^3}(X - \mu)^2 \end{bmatrix}.$$

The Fisher information Matrix is

$$I(\mu, \sigma^2) = -\mathbb{E}_{\mu,\sigma^2}[H_X(\mu, \sigma^2)] = \begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2(\sigma^2)^2} \end{bmatrix}.$$

♢

Theorem 6.10 (Asymptotic Normality of MLE)**.** *Suppose $X_1, ..., X_n \overset{iid}{\sim} f(x; \theta)$ for some true value θ_0. The parameter $\theta \in \Theta \subset \mathbb{R}^p$ is p-dimensional. $\hat{\theta}$ is the MLE. Under some regularity conditions, ... Then, as $n \to \infty$,*

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N_p(\mathbf{0}, I(\theta_0)^{-1}),$$

where $I(\theta)$ is the Fisher information matrix.

Due to the Slutzky's theorem, Theorem 6.10 implies

$$\hat{\theta} - \theta_0 \overset{d}{\approx} N_p(\mathbf{0}, I_n(\hat{\theta})^{-1}),$$

where $I_n(\hat{\theta}) = nI(\hat{\theta})$ is the *plug-in* Fisher information matrix.

Further, due to the Slutzky's theorem, Theorem 6.10 also implies the following feasible version:

$$\sqrt{n}I^{1/2}(\hat{\theta})(\hat{\theta} - \theta_0) \xrightarrow{d} N_p(\mathbf{0}, I_p)$$

where $I^{1/2}(\theta)$ satisfies $I^{1/2}(\theta)I(\theta)^{-1}(I^{1/2}(\theta))^T = I_p$ and the limiting distribution is the standard *multivariate Normal distribution*. The "rescaling"/"standardization" matrix $I^{1/2}(\theta)$ can be identified by the spectral/eigen decomposition of $I(\theta)$ if $I(\theta)$ is a positive (semi-)definite square matrix:

$$I^{1/2}(\theta) = V\sqrt{\Lambda}$$

where $I(\theta) = V\Lambda V^T$ and V is an orthonormal matrix.

Example 6.8 in the following works out the two feasible versions, and indeed, the version with the plug-in Fisher information is more concise.

Example 6.8 (Univariate Normal mean and variance: the plug-in Fisher information matrix)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$. The MLE for (μ, σ^2) is obtained by solving FOC and verifying SOC:

$$\begin{bmatrix} \hat{\mu} \\ \widehat{\sigma}^2 \end{bmatrix} = \begin{bmatrix} \bar{X}_n \\ \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X}_n)^2 \end{bmatrix}$$

With the Fisher information matrix derived in Example 6.7, its asymptotic sampling distribution by Theorem 6.10 is

$$\begin{bmatrix} \hat{\mu} \\ \widehat{\sigma}^2 \end{bmatrix} - \begin{bmatrix} \mu_0 \\ \sigma_0^2 \end{bmatrix} \overset{d}{\approx} N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{\widehat{\sigma}^2}{n} & 0 \\ 0 & \frac{2(\widehat{\sigma}^2)^2}{n} \end{bmatrix} \right).$$

In this example, $I^{1/2}(\theta)$ is readily available as $I(\theta)$ is diagonal:

$$\sqrt{n} \begin{bmatrix} \frac{1}{\sqrt{\widehat{\sigma}^2}} & 0 \\ 0 & \frac{1}{\sqrt{2\widehat{\sigma}^2}} \end{bmatrix} \left(\begin{bmatrix} \hat{\mu} \\ \widehat{\sigma}^2 \end{bmatrix} - \begin{bmatrix} \mu_0 \\ \sigma_0^2 \end{bmatrix} \right) \overset{d}{\approx} N\left(\mathbf{0}, I_2\right),$$

where $\mathbf{0}$ is 2×1 vector of 0's, and I_2 is the identity matrix of rank 2. ♢

Definition 6.6 (The observed Fisher information matrix)**.** The observed Fisher information in the multiparameter caseis defined as

$$I_{obs} = -\ddot{\ell}_{\boldsymbol{x}}(\theta)|_{\theta=\hat{\theta}} = -\left[\frac{\partial^2}{\partial\theta_i \partial\theta_j}\ell_{\boldsymbol{x}}(\theta)\right]\Big|_{\theta=\hat{\theta}}.$$

Similarly, by continuous mapping theorem, the following asymptotic Normality also holds

$$\hat{\theta} - \theta_0 \overset{d}{\approx} N_p(\mathbf{0}, I_{obs}^{-1}),$$

where $I_{obs} = -H_n(\boldsymbol{X}; \theta)|_{\theta=\hat{\theta}}$ is the *observed* Fisher information matrix.

Example 6.9 (Univariate Normal mean and variance: the observed Fisher information matrix)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$. The MLE for (μ, σ^2) is obtained by solving FOC and verifying SOC:

$$\begin{bmatrix} \hat{\mu} \\ \widehat{\sigma}^2 \end{bmatrix} = \begin{bmatrix} \bar{X}_n \\ \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X}_n)^2 \end{bmatrix}.$$

By linearity of differentiation and the additivity of the log-likelihood function,

$$H_n(\mu, \sigma^2) = \sum\nolimits_{i=1}^{n} H_{X_i}(\mu, \sigma^2).$$

Then the Hessian with respect to $\ell_{\boldsymbol{x}}(\mu, \sigma^2)$ is

$$\begin{aligned} H_n(\mu, \sigma^2) &= \begin{bmatrix} \frac{\partial^2}{\partial \mu^2}\ell_n(\mu, \sigma^2) & \frac{\partial^2}{\partial \mu \partial \sigma^2}\ell_n(\mu, \sigma^2) \\ \frac{\partial^2}{\partial \mu \partial \sigma^2}\ell_n(\mu, \sigma^2) & \frac{\partial^2}{(\partial \sigma^2)^2}\ell_n(\mu, \sigma^2) \end{bmatrix} \\ &= \begin{bmatrix} -\frac{n}{\sigma^2} & \frac{-1}{(\sigma^2)^2}\sum_{i=1}^{n}(X_i - \mu) \\ \frac{-1}{(\sigma^2)^2}\sum_{i=1}^{n}(X_i - \mu) & \frac{1}{2(\sigma^2)^2} - \frac{1}{(\sigma^2)^3}\sum_{i=1}^{n}(X_i - \mu)^2 \end{bmatrix}. \end{aligned}$$

With $(\hat{\mu}, \hat{\sigma}^2)$, the observed Fisher information is

$$I_{obs} = -H_n(\hat{\mu}, \hat{\sigma}^2) = I_n(\hat{\mu}, \hat{\sigma}^2).$$

In this example, the plug-in Fisher information matrix is exactly the same as the observed Fisher information matrix. ♢

Note Theorem 6.10 requires $X_1, ..., X_n \overset{iid}{\sim} f(x; \theta)$ which might fail to hold in practice. Suppose the log-likelihood function for the whole dataset is $\ell_n(\theta)$, and some integrability conditions hold for its partial derivatives, then the following asymptotic Normality still holds:

$$\hat{\theta} - \theta_0 \overset{d}{\approx} N_p(\mathbf{0}, I_n(\hat{\theta})^{-1}),$$

where $I_n(\hat{\theta}) = -\mathbb{E}_\theta[H_n(\theta)]|_{\theta=\hat{\theta}}$ is the *plug-in* Fisher information, and H_n is the Hessian matrix of $\ell_n(\theta)$. The asymptotic Normality does not require the entire data to be iid.

Similar to the univariate case, when the expected Fisher information matrix is not available or it is tedious to derive, we can replace the plug-in Fisher information matrix with the observed Fisher information matrix, and the asymptotic Normality also holds:

$$\hat{\theta} - \theta_0 \overset{d}{\approx} N_p(\mathbf{0}, I_{obs}^{-1}),$$

where $I_{obs} = -H_n(X^n; \theta)|_{\theta=\hat{\theta}}$ is the *observed* Fisher information.

Example 6.10 (Groups of Bernoulli's)**.** Two independent groups of Bernoulli random variables $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta_1)$, and $Y_1, ..., Y_m \overset{iid}{\sim} Ber(\theta_2)$. The Bernoulli model is equivalent to observe two independent datapoints such that $X \sim Bin(n, \theta_1)$, and $Y \sim Bin(m, \theta_2)$.

The log-likelihood for the data is

$$\ell_{n,m}(\theta_1, \theta_2) = \sum\nolimits_{i=1}^{n} X_i \log(\theta_1) + \sum\nolimits_{i=1}^{n}(1 - X_i)\log(1 - \theta_1) + \sum\nolimits_{j=1}^{m} Y_j \log(\theta_2) + \sum\nolimits_{j=1}^{m}(1 - Y_j)\log(1 - \theta_2).$$

By solving the FOC and verifying SOC, the MLE for (θ_1, θ_2) is

$$\begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix} = \begin{bmatrix} \bar{X}_n \\ \bar{Y}_m \end{bmatrix}.$$

To find the asymptotic sampling distribution for $\hat{\theta} = (\hat{\theta}_1, \hat{\theta}_2)$, we derive the Hessian

$$H_{n,m} = \begin{bmatrix} -\frac{\sum_{i=1}^{n} X_i}{\theta_1^2} - \frac{\sum_{i=1}^{n}(1 - X_i)}{(1-\theta_1)^2} & 0 \\ 0 & -\frac{\sum_{j=1}^{m} Y_j}{\theta_2^2} - \frac{\sum_{j=1}^{m}(1 - Y_i)}{(1-\theta_2)^2} \end{bmatrix},$$

the Fisher information matrix

$$I_{n,m}(\theta_1, \theta_2) = -\mathbb{E}_{\theta_1,\theta_2}[H_{n,m}] = \begin{bmatrix} \frac{n}{\theta_1(1-\theta_1)} & 0 \\ 0 & \frac{m}{\theta_2(1-\theta_2)} \end{bmatrix},$$

and both the plug-in Fisher information matrix and the observed Fisher information matrix are

$$I_{obs} = I_{n,m}(\hat{\theta}_1, \hat{\theta}_2) = \begin{bmatrix} \frac{n}{\hat{\theta}_1(1-\hat{\theta}_1)} & 0 \\ 0 & \frac{m}{\hat{\theta}_2(1-\hat{\theta}_2)} \end{bmatrix}.$$

In the end,

$$\begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix} - \begin{bmatrix} \theta_{0,1} \\ \theta_{0,2} \end{bmatrix} \overset{d}{\approx} N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{n}\hat{\theta}_1(1-\hat{\theta}_1) & 0 \\ 0 & \frac{1}{m}\hat{\theta}_2(1-\hat{\theta}_2) \end{bmatrix} \right).$$

See Example 6.12 for more follow-up analysis. ♢

6.3 The Delta Method

Reading(s) 6.3. Wasserman (2004): 9.9 and 9.10.

6.3.1 The Univariate Delta Method

In many problems, the parameter of interest is a function of the model parameter(s). How can we construct confidence intervals for transformations of maximum likelihood estimators?

Theorem 6.11. *Suppose* $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta)$ *for some true value* θ_0. *Let* $\hat{\theta}$ *be the MLE for* θ. *Suppose it is asymptotically Normal*

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1}),$$

where $I(\theta)$ *is the Fisher information. The parameter of interest is*

$$\eta = g(\theta)$$

and $g(\cdot)$ *is differentiable with* $g'(\theta_0) \neq 0$.

Then as $n \to \infty$,

$$\frac{\sqrt{nI(\hat{\theta})}(\hat{\eta} - \eta_0)}{|g'(\hat{\theta})|} \xrightarrow{d} N(0,1).$$

where $\hat{\eta} = g(\hat{\theta})$, *and* $\eta_0 = g(\theta_0)$.

Remark. The above idea can be used to derive the univariate Delta method for the method of moments estimators.

Proof. First order Taylor expansion of $g(\hat{\theta})$ around $g(\theta_0)$:

$$g(\hat{\theta}) - g(\theta_0) = g'(\theta_0)(\hat{\theta} - \theta_0) + O((\hat{\theta} - \theta_0)^2).$$

Then the result follows by the Slutzky's theorem. □

By equivariance, $\hat{\eta} = g(\hat{\theta})$ is MLE for η. By continuous mapping theorem, $\hat{\eta}$ is a consistent estimator for η. With the Delta method, we could obtain asymptotic Normal approximation to the sampling distribution of $\hat{\eta}$.

Example 6.11 (Log Odds)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta)$ with the true value θ_0. The parameter of interest is the log odds

$$\eta = g(\theta) = \log \frac{\theta}{1-\theta}$$

which is also known as the *logit* of θ. Direct calculation yields

$$g'(\theta) = \frac{1}{\theta(1-\theta)}.$$

By Theorem 6.4,

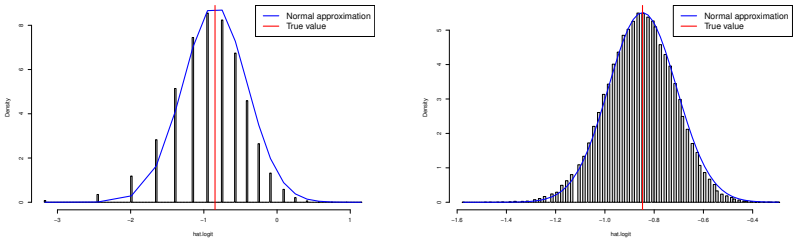
$$\sqrt{n}(\hat{\theta} - \theta_0) \overset{d}{\approx} N(0, \hat{\theta}(1-\hat{\theta}))$$

where $\hat{\theta} = \bar{X}_n$. (See Example 6.4 for more details.)

Then by Theorem 6.11,

$$\sqrt{n}(\hat{\eta} - \eta_0) \overset{d}{\approx} N\left(0, \frac{1}{\hat{\theta}(1-\hat{\theta})}\right).$$

The following simulation studies illustrate the approximation quality. The true value $\theta_0 = 0.3$. Two different sample sizes are compared.



$n = 25$ $n = 250$

Figure 6.5: True sampling distribution v.s. asymptotic approximation

```r
n <- 25 ## n <- 250
theta0 <- 0.3
hat.theta <- sapply(1:1e5, function(x) mean(rbinom(n,1,theta0)))
hat.logit <- log(hat.theta/(1-hat.theta))

h2 <- hist(hat.logit,breaks = "FD", probability = TRUE, main = "")
mids <- h2$mids[h2$counts >0]
dd <- dnorm(mids,log(theta0/(1-theta0)),sqrt(1/(theta0*(1-theta0))/n))
lines(mids,dd*sum(h2$density)/sum(dd),lwd=3,col="blue")
abline(v=log(theta0/(1-theta0)), col="red",lwd=3)
legend("topright",c("Normal approximation","True value"),col=c("blue","red"),
       lty = c(1,1),lwd=3,cex=1.5)
```

♢

Exercise 6.1 (Log arrival rate)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Pois(\lambda)$. The parameter of interest is the log rate parameter

$$\eta = g(\lambda) = \log(\lambda).$$

Direct calculation yields $g'(\lambda) = \frac{1}{\lambda}$. Find the asymptotic Normal theory for η.

☕

Exercise 6.2 (Standard deviation)**.** $X_1, ..., X_n \overset{iid}{\sim} N(0, \sigma^2)$, the parameter of interest is

$$g(\sigma^2) = \sigma = \sqrt{\sigma^2}.$$

Find the asymptotic Normal theory for σ.

☕

6.3.2 The multi-parameter Delta method

For general parametric models with multiple parameters, we have the following Delta method. which is a multi-parameter extension of the univariate version Delta method.

Theorem 6.12. *Suppose $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta)$ for some true value θ_0. θ is $p-$dimensional. Let $\hat{\theta}$ be the MLE for θ. Suppose it is asymptotically Normal*

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N_p(\mathbf{0}, I(\theta_0)^{-1}),$$

where $I(\theta)$ is the Fisher information matrix. The parameter of interest is

$$\eta = g(\theta),$$

and $g(\cdot)$ is differentiable with its gradient $\nabla_\theta g(\theta_0)$ such that $\nabla_\theta g(\theta_0)^T I(\theta_0)^{-1} \nabla_\theta g(\theta_0) \neq 0$.

Then as $n \to \infty$,

$$\sqrt{n}(\hat{\eta} - \eta_0) \xrightarrow{d} N(0, \nabla_\theta g(\theta_0)^T I(\theta_0)^{-1} \nabla_\theta g(\theta_0)).$$

Feasible version:

$$\frac{\sqrt{n}(\hat{\eta} - \eta_0)}{\widehat{\sigma}_\eta} \xrightarrow{d} N(0,1)$$

where $\widehat{\sigma}_\eta = \sqrt{\nabla_\theta g(\hat{\theta})^T I(\hat{\theta})^{-1} \nabla_\theta g(\hat{\theta})}$.

Proof. First order Taylor expansion of $g(\hat{\theta})$ around θ_0, and then apply the Slutzky's theorem. □

The approximate variance $\nabla_\theta g(\hat{\theta})^T I(\hat{\theta})^{-1} \nabla_\theta g(\hat{\theta})$ is also called the "sandwich" form. Theorem 6.12 also implies

$$\hat{\eta} - \eta_0 \overset{d}{\approx} N(0, \frac{1}{n} \nabla_\theta g(\hat{\theta})^T I(\hat{\theta})^{-1} \nabla_\theta g(\hat{\theta})).$$

Theorem 6.13 (CI of η)**.** *The asymptotic $(1-\alpha)-$confidence interval for $\eta_0 = g(\theta_0)$:*

$$C_\alpha = \hat{\eta} \pm z_{\alpha/2} \widehat{se}$$

where $\hat{\eta} = g(\hat{\theta})$, z_α denotes the $\alpha-$upper quantile of the standard Normal distribution, and $\widehat{se} = \frac{1}{\sqrt{n}} \sqrt{\nabla_\theta g(\hat{\theta})^T I(\hat{\theta})^{-1} \nabla_\theta g(\hat{\theta})}$.

Then, as $n \to \infty$,

$$\mathbb{P}_{\theta_0}(\eta_0 \in C_\alpha) \to 1 - \alpha.$$

Proof. TBD □

Example 6.12 (Effect size)**.** To evaluate the effect size of a vaccine, the treatment group is assigned the vaccine and the outcomes of the individuals of the treatment group are denoted as $\{X_1, ..., X_n\}$, and the control group is assigned the placebo and the outcomes are denoted as $\{Y_1, ..., Y_m\}$. Since the assignment of the treatment status and the control status is random, it is natural to assume $\{X_i\}$ and $\{Y_j\}$ are independent, and within each group, the outcomes are iid.

Suppose the data is the status of the infection with 1 denoting infected and 0 denoting not infected. (In other applications, the outcomes could be some other metrics, e.g., blood pressure.) Therefore, the data is modelled as two independent groups of Bernoulli random variables $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta_1)$, and $Y_1, ..., Y_m \overset{iid}{\sim} Ber(\theta_2)$ for some true parameters $(\theta_{0,1}, \theta_{0,2})$. The parameter of interest is the effect size:

$$\eta = g(\theta_1, \theta_2) = \theta_1 - \theta_2.$$

With the asymptotic Normality result derived in Example 6.10 and the gradient of $g(\cdot, \cdot)$ being

$$\nabla g(\theta_1, \theta_2) = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

Theorem 6.12 implies

$$\hat{\eta} - \eta_0 \overset{d}{\approx} N\left(0, \frac{1}{n}\hat{\theta}_1(1-\hat{\theta}_1) + \frac{1}{m}\hat{\theta}_2(1-\hat{\theta}_2)\right),$$

where $\hat{\eta} = g(\hat{\theta}_1, \hat{\theta}_2) = \hat{\theta}_1 - \hat{\theta}_2$ and $\eta_0 = g(\theta_{0,1}, \theta_{0,2}) = \theta_{0,1} - \theta_{0,2}$.

We examine the quality of the asymptotic approximation in the following simulation study. $n = 30$ and $m = 25$ are set to be moderate. The true $\theta_0 = (0.4, 0.2)$ so that the true effect size is 0.2 which is also moderate. With the moderate sample size, the approximation is overall accurate up to some continuity correction.

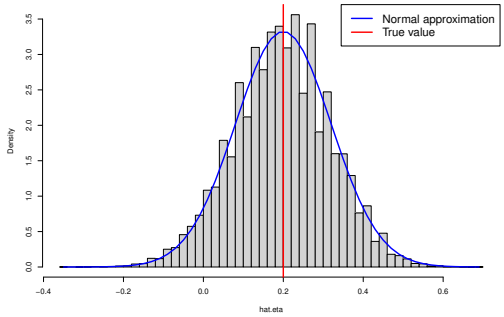


Figure 6.6: True sampling distribution v.s. asymptotic approximation

```r
n <- 30; m <- 25
theta0 <- c(0.4,0.2)
eta0 <- theta0[1] - theta0[2]

hat.eta <- sapply(1:1e5,function(x)
    mean(rbinom(n,1,theta0[1]))-mean(rbinom(m,1,theta0[2])))

h1 <- hist(hat.eta,breaks = 60, probability = TRUE, main="")
mids <- h1$mids[h1$counts >0]
v1 <- theta0[1]*(1-theta0[1])/n
v2 <- theta0[2]*(1-theta0[2])/m
dd <- dnorm(mids,eta0,sqrt(v1+v2))
lines(mids,dd*sum(h1$density)/sum(dd),lwd=3,col="blue")
abline(v=eta0, col="red",lwd=3)
legend("topright",c("Normal approximation","True value"),col=c("blue","red"),
       lty = c(1,1),lwd=3,cex=1.5)
```

The $(1-\alpha)$ asymptotic CI for the effect size η_0 is

$$\hat{\eta} \pm z_{\alpha/2}\widehat{se},$$

where $\widehat{se} = \sqrt{\frac{1}{n}\hat{\theta}_1(1-\hat{\theta}_1) + \frac{1}{m}\hat{\theta}_2(1-\hat{\theta}_2)}$, and z_α denotes the $\alpha-$upper quantile of the standard Normal distribution.

We also examine the actual coverage rate of the asymptotically 95% CI. With $n = 30$ and $m = 25$, and the true $\theta_0 = (0.4, 0.2)$, the coverage rate is 93.6% which is smaller than the 95% nominal rate.

```
CI.covers <- function(est,truth,n){
    v1 <- est[1]*(1-est[1])/n[1]
    v2 <- est[2]*(1-est[2])/n[2]
    se <- sqrt(v1+v2)
    CI.left <- est[1]-est[2] - 1.96*se
    CI.right <- est[1]-est[2] + 1.96*se
    eta0 <- truth[1] -truth[2]
    return((CI.right-eta0)*(CI.left-eta0)<=0)
}
n <- 30; m <- 25
theta0 <- c(0.4,0.2)
cover <- sapply(1:1e5, function(x)
    CI.covers(c(mean(rbinom(n,1,theta0[1])),mean(rbinom(m,1,theta0[2]))),
              theta0, c(n,m)))
mean(cover)
```

It is well known that the Wald CI for the proportion is erratic and chaotic. We could also assess it using simulation. In the following simulation study, $\theta_{0,2} = 0.2$ and $\theta_{0,1}$ varies from 0 to 1. To see the finite sample performance, we set $n = 30$ and $m = 25$ as before. Figure 6.7 shows that (1) the CI implied by the Delta method consistently under covers the truth than the nominal rate; (2) the coverage rates also oscillate as $\theta_{0,1}$ changes. One can imagine the oscillation patterns would be more complicated when both $\theta_{0,1}$ and $\theta_{0,2}$ are allowed to change.

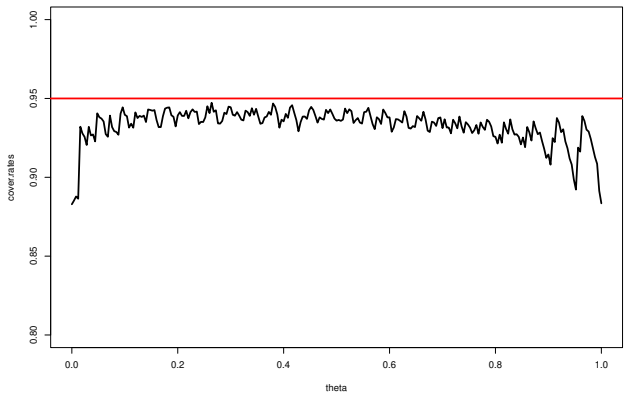


Figure 6.7: Coverage rates v.s. true proportion

```
theta.seq <- seq(from = 0, to = 1, length.out = 251)
theta <- theta0
cover.rates <- NULL
for(i in 1:length(theta.seq)){
    theta[1] <- theta.seq[i]
    cover <- sapply(1:1e5, function(x)
        CI.covers(c(mean(rbinom(n,1,theta[1])),mean(rbinom(m,1,theta[2]))),
                  theta, c(n,m)))
    cover.rates[i] <- mean(cover)
}
plot(theta.seq, cover.rates, type = "l",lwd=3,xlab="theta",ylim = c(0.8,1))
```

```
abline(h=0.95,col="red",lwd=3)
```

♢

Example 6.13 (Log Odds Ratio)**.** Suppose the probability of the outcome is p, then the odds of the outcome is defined as

$$Odds = \frac{p}{1-p},$$

which is the ratio of the probability that the outcome occurs and the probability that the outcome does not occur. Odds also provides a metric of how likely is the outcome.

For Bernoulli random variable $X \sim Ber(\theta_1)$, define its odds as $\theta_1/(1-\theta_1)$. For two Bernoulli random variables $X \sim Ber(\theta_1)$ and $Y \sim Ber(\theta_2)$, define the odds ratio as

$$OR = \frac{\theta_1/(1-\theta_1)}{\theta_2/(1-\theta_2)},$$

which is simply the ratio of two odds. By definition, the odds ratio captures the relative likelihood of the outcome occurs.

Suppose in the same setup of Example 6.12, the parameter of interest is the log odds ratio:

$$\eta = g(\theta_1, \theta_2) = \log\left(\frac{\theta_1/(1-\theta_1)}{\theta_2/(1-\theta_2)}\right).$$

Immediately, we know the gradient of the transformation $g(\cdot,\cdot)$ is

$$\nabla g(\theta_1, \theta_2) = \begin{bmatrix} \frac{1}{\theta_1(1-\theta_1)} \\ \frac{1}{\theta_2(1-\theta_2)} \end{bmatrix}$$

By the Delta method, we could derive its asymptotic sampling distribution.

$$\hat{\eta} - \eta_0 \overset{d}{\approx} N\left(0, \frac{1}{n\hat{\theta}_1(1-\hat{\theta}_1)} + \frac{1}{m\hat{\theta}_2(1-\hat{\theta}_2)}\right),$$

where $\hat{\eta} = g(\hat{\theta}_1, \hat{\theta}_2)$ and $\eta_0 = g(\theta_{0,1}, \theta_{0,2})$,

♢

Exercise 6.3 (Rescaled mean)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, Find a 95% CI for $g(\mu, \sigma) = \mu/\sigma$. ☕

Exercise 6.4 (Rate difference)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Exp(\eta_1)$ and $Y_1, ..., Y_m \overset{iid}{\sim} Exp(\eta_2)$ are two independent samples. The quantity of interest is $g(\eta_1, \eta_2) = \eta_1 - \eta_2$.

Now the quantity of interest is $g(\eta_1, \eta_2) = |\eta_1 - \eta_2|$. ☕

Exercise 6.5 (the length of the support)**.** $X_1, ..., X_n \overset{iid}{\sim} U(\theta_1, \theta_2)$, $g(\theta_1, \theta_2) = \theta_2 - \theta_1$. ☕

7 Hypothesis Testing

7.1 Testing Statistical Hypotheses

Reading(s) 7.1. Wasserman (2004) 10.0; Hogg et al. (2018) 8.1, 8.2.

Previously, we have seen estimation methods, e.g., method of moments, maximum likelihood estimation, and confidence intervals. The problem of parameter estimation directly provides description of the parameters. However, in many scientific applications, we are interested in more qualitative aspects of the parameters. For instance, scientists are interested in if two factors/genes are independent; biostatisticians hope to assess if the new drug is effective and/or has any side effects; economists and political scientists hope to evaluate if the new policy alleviates poverty of some regions.

Example 7.1 (The lady tasting tea)**.** Cambridge, England, 1920. In a gathering, Lady Ottoline claimed that she could tell the difference between milk tea with first milk then tea, and the milk tea with first tea then milk. As the crowd did not believe it. They quickly arranged an experiment of eight cups of milk tea, with four milk first and four tea first, to test the claim. Lady Ottoline correctly identified the 8 cups out of 8 cups. Is this sheer guess work?

Despite being a little bit detour for our purpose, RA Fisher had the following food for thought:

- In what order should the cups be presented?
- How many cups must be correctly identified to conclude that the lady can truly tell the difference?

♢

Example 7.2 (A/B testing)**.** A/B testing is a recent buzz word for randomized controlled trials (RCT). It compares the average outcomes of two proposals. For instance, the size of the "Add to Cart" button can be either small or large. By intuition or instinct, it is hard to tell which one is better in terms of the profit and the sales. The online companies now can randomly assign the two options to random consumers and empirically see which one is better. ♢

The statistical inference assessing the validity of those statements is hypothesis testing. In statistics, the statements first need to be "translated" into certain probability processes. That is, we specify a model that could possibly generate the data we observe. Under suitable parameterizations, the qualitative statements are expressed in terms of the parameters of the model.

Definition 7.1 (Hypothesis)**.** A hypothesis is a statement about a population parameter.

Example 7.3. In the case of evaluating effect size, suppose the outcome follows the Normal distribution $N(\theta, 1)$, and we could parameterize the mean effect size as θ. Then, the hypothesis that the new drug has no effects becomes $\theta = 0$. Similarly, the hypothesis that the new drug has no adverse effects becomes $\theta \geq 0$. ♢

By the functional forms of the hypotheses, we classify the hypotheses into different categories. The first classification is *simple* hypothesis v.s. *composite* hypothesis.

- Simple hypotheses refer to the hypotheses that specify the parameter values completely. Such hypotheses are often defined by $\theta = \theta_0$ for some constant θ_0. For instance, $\theta = 0$.
- Composite hypotheses refer to the hypotheses that allow the parameter(s) to take at least one value. For instance, $\theta \in [0, 1]$.

The second classification is *two-sided* test v.s. *one-sided* test. Testing the simple hypothesis $\theta = \theta_0$ is also called the two sided test, as its alternative is in the form $\theta \neq \theta_0$. Testing the composite hypothesis of the form $\theta \geq \theta_0$ or $\theta \leq \theta_0$ is called one-sided test, as its complement is in the form $\theta > \theta_0$ or $\theta < \theta_0$.

To set up the hypothesis testing problem more generally, suppose the data generating process (model) is parameterized by θ. For some sets Θ_0 and Θ_1,

$$\text{the null hypothesis is } \theta \in \Theta_0, \text{ v.s., the alternative hypothesis is } \theta \in \Theta_1.$$

The convention is to specify $\Theta_0 \cap \Theta_1 = \emptyset$ and $\Theta_0 \cup \Theta_1 = \Theta$. In some cases, we may have $H_0 : \theta = \theta_0$ and $H_1 : \theta = \theta_1$ for two different values θ_0 and θ_1. We can think of this case as $\Theta = \{\theta_0, \theta_1\}$.

Example 7.4 (Hypotheses to be tested in the lady tasting tea)**.** We can write down the hypotheses for the examples above. In the example of Lady tasting tea,

$$H_0\text{: the lady } \textit{cannot} \text{ tell the difference. v.s. } H_1\text{: the lady } \textit{can} \text{ tell some difference.}$$

♢

Example 7.5 (Hypotheses to be tested in the A/B testing)**.** In the example of A/B testing, when the question asked is if the option A is significantly *better than* the option B,

$$H_0\text{: } \mu_A \leq \mu_B, \text{ v.s. } H_1\text{: } \mu_A > \mu_B.$$

When the question asked is if the option A is *different* from the option B,

$$H_0\text{: } \mu_A = \mu_B, \text{ v.s. } H_1\text{: } \mu_A \neq \mu_B.$$

Different questions lead to different null hypotheses. ♢

Example 7.6 (Keith Harward)**.** A true lifestory of Keith Harward was reported on the website wtkr.com:

> The legal process to free a wrongfully convicted former sailor in a 1982 rape and murder in Newport News has begun. Keith Allen Harward was convicted in the September 19, 1982 murder of Jesse Perron and rape of Perron's wife.
>
> According to the Innocence Project, Perron was beaten to death with a crowbar and his wife was repeatedly raped and bitten in their Newport News home. Forensic odontologists said that bite mark evidence proved that Harward, then a 26-year-old sailor, was the perpetrator.
>
> Harward has been in custody since 1983 and was sentenced to life in prison in 1986. He has maintained his innocence ever since. He was convicted of capital murder, rape, forcible sodomy and robbery. He was initially sentenced to 65 years. It was then changed to first degree murder follow a successful appeal but he was still sentenced to life.
>
> New DNA testing has proven that Harward was not the perpetrator.

Though the error rate could be small, the adverse impact on the innocent people can be enormous. We should be very careful in rejecting the null hypothesis, especially for those life-changing decisions. ♢

The general principle of specifying the couple of hypotheses is to put discoveries in the alternative and no discoveries in the null hypotheses, so that we reject the null only when we have enough evidence against it.

$$H_0\text{: no discovery, v.s. } H_1\text{: some discovery.}$$

Hypothesis testing is essentially to make a decision. We decide to reject or retain the null hypothesis based on the data (evidence):

$$\delta : \boldsymbol{X} \to \{0, 1\}$$

where $\boldsymbol{X} \equiv \{X_i\}_{i=1}^n$ denotes the data at hand, $\delta(\cdot)$ denotes the decision function such that $\delta = 1$ means rejecting H_0, and that $\delta = 0$ means retaining H_0. For instance, the decision rule is to reject the null hypothesis if

$$T(\boldsymbol{X}) > c$$

where $T(\boldsymbol{X})$ is the *test statistic* containing no unknown parameters, and c is the *critical value*, and to retain the null hypothesis if otherwise.

With the decision rule, the set

$$\mathcal{R} \equiv \{\boldsymbol{X} : \delta(\boldsymbol{X}) = 1\}$$

is called *rejection region*. That is, if the data $\boldsymbol{X}$ falls in this region, the action is to reject the null hypothesis. Some authors also call the rejection region as *critical region*.

As the decision making is data dependent, making errors is typically inevitable. In the convention of statistics, the error of rejecting the null while the null is true is called the *Type I error*; the error of retaining the null while the alternative is true is called the *Type II error*. All possible outcomes are summarized in Table 3.

		Decision on H_0	
		reject	retain
truth:	H_0	Type I error	✓
	H_1	✓	Type II error

Table 3: Two types of errors

Example 7.7 (Gaussian mean)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\theta, 1)$. The hypothesis testing problem is

$$H_0 : \theta = 0 \text{ v.s. } H_1 : \theta \neq 0.$$

Suppose the test statistic is $|\bar{X}_n|$ with $\bar{X}_n \equiv \frac{1}{n}\sum_{i=1}^n X_i$, and the decision rule is to reject the null if

$$\sqrt{n}|\bar{X}_n| > 1,$$

where 1 is the critical value to fix idea. The rejection region of the test is $\{\boldsymbol{x} : \sqrt{n}|\bar{x}_n| > 1\}$.

The Type I error rate is

$$\mathbb{P}_0(\sqrt{n}|\bar{X}_n| > 1) = \mathbb{P}_0(\sqrt{n}\bar{X}_n > 1) + \mathbb{P}_0(\sqrt{n}\bar{X}_n < -1) = 1 - \Phi(1) + \Phi(-1) = 2\Phi(-1),$$

where the sampling distribution $\sqrt{n}\bar{X}_n \sim N(0, 1)$ under the null hypothesis.

♢

For *simple* null hypotheses, Type I error probabilities can be computed or approximated analytically. To extend the same idea to *composite* null hypotheses, the concept of size and/or level are used as a single number summary of the Type I error rates.

Definition 7.2 (size and level)**.** Suppose the hypotheses are $H_0 : \theta \in \Theta_0$ v.s. $H_1 : \theta \in \Theta_1$, and the rejection region is $\mathcal{R}$. The test is of size α if

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(\boldsymbol{X} \in \mathcal{R}) = \alpha.$$

The test is of level α if

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(\boldsymbol{X} \in \mathcal{R}) \leq \alpha.$$

The parameter space under the null Θ_0 is allowed to contain only a point, and then $\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(\cdot) = \mathbb{P}_{\theta_0}(\cdot)$ trivially. The sup operator reflects the largest error rate under the null, which is the worst case under the null. Designing statistical tests of certain size/level automatically "guards" against the worst scenarios under the null hypotheses.

Example 7.8 (Example 7.7 continued)**.** As the null hypothesis is simple, both the size and the level of the test are $2\Phi(-1)$. ♢

The action of retaining the null could also be interpreted as failing to reject. There is a subtle difference between retaining the null and accepting the null, as failing to reject could also due to the test itself is not powerful. For a trivial illustration, consider the test that never rejects, then its Type I error would be 0 by definition.

Definition 7.3 (power function)**.** Suppose the rejection region is $\mathcal{R}$, the data generating process $\boldsymbol{X} \sim \mathbb{P}_{n,\theta}$, and the hypotheses are parameterized by $\theta \in \Theta$. The power function is defined as

$$\beta(\theta) = \mathbb{P}_{n,\theta}(\boldsymbol{X} \in \mathcal{R}).$$

Remark. The power function is the Type I error rate when $\theta \in \Theta_0$. The Type II error is $1 - \beta(\theta)$ for $\theta \in \Theta_1$.

Example 7.9 (Example 7.7 continued)**.** The power function $\beta(\theta)$ is

$$\begin{aligned} \beta(\theta) &= \mathbb{P}_\theta(\sqrt{n}|\bar{X}_n| > 1) \\ &= \mathbb{P}_\theta(\sqrt{n}(\bar{X}_n - \theta) > 1 - \sqrt{n}\theta) + \mathbb{P}_\theta(\sqrt{n}(\bar{X}_n - \theta) < -1 - \sqrt{n}\theta) \\ &= 1 - \Phi(1 - \sqrt{n}\theta) + \Phi(-1 - \sqrt{n}\theta) \end{aligned}$$

where $\sqrt{n}(\bar{X}_n - \theta) \sim N(0,1)$ under the model $X_1, ..., X_n \overset{iid}{\sim} N(\theta, 1)$.

For general critical value c, the power function is

$$\beta(\theta) = 1 - \Phi(c - \sqrt{n}\theta) + \Phi(-c - \sqrt{n}\theta).$$

Figure 7.1 overlays three different power functions for critical values 1,2, and 3.

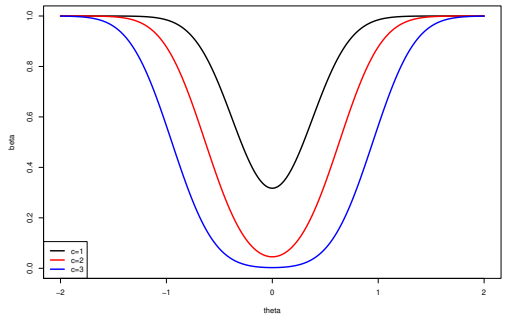


Figure 7.1: The power functions under different critical values

Under the alternative, the associated Type II error rates are $1 - \beta(\theta)$ for $\theta \neq 0$. It is also easy to see that $\lim_{n\to\infty} \beta(\theta) = 1$ for all $\theta \neq 0$, which implies a zero asymptotic Type II error rate.

As the error rates depend on the critical value, we can change the critical value to achieve desirable error rates. However, there is a tradeoff between the Type I error rate and the Type II error rate. To be conservative, people usually set the critical value such the Type I error rate is under 5%. That is, the level of the test is 5%.

♢

7.2 Neyman-Pearson Hypothesis Testing

Reading(s) 7.2. Wasserman (2004) 10.0; Hogg et al. (2018) 8.1, 8.2.

The Neyman-Pearson's approach to hypothesis testing is to determine a procedure whose error probabilities are verified before examining the data. Once the size/level α is specified and the data is observed, Neyman-Pearson will report either one of the following,

- reject H_0 with size/level α,
- retain H_0 with size/level α.

Since the type I error and type II error probabilities can be computed theoretically without looking at the data, the testing procedure is fully specified by the error probabilities. For instance, the size α can be used to pin down the critical value. The justification for this approach is the frequentist principle: the empirical error rates of the procedure will be close to the error probabilities and, hence, are controlled.

The criticism also stems from the frequentist principle. Instead of making a decision for the data at hand, Neyman-Pearson seeks for a pre-experimental decision-making procedure. As the Neyman-Pearson approach reports a decision (reject or retain H_0) and the pre-specified error probabilities (irrelevant to the data at hand), the data with a z-score 2.01 receives exactly the same conclusion and decision as the data with a z-score of 100: rejecting the null hypothesis with size α.

It is tempting to choose the size with the data at hand, either to reject the null hypothesis with a larger size, or to "improve" the "implausibility" of the null hypothesis with a smaller size. However, the *post hoc* procedure violates the frequentist principle and the error probability guarantees no longer hold true. We will return to a similar idea in Section 7.4 Fisher's significance testing.

Example 7.10. Suppose the data $X_1, ..., X_n \overset{iid}{\sim} N(\theta, 4)$, and the hypotheses are

$$H_0\text{: } \theta = \theta_0 = 0, \text{ v.s., } H_1\text{: } \theta = \theta_1 = 2.5.$$

To illustrate the idea of Type I error, Type II error and the rejection region, consider the test statistic $T(\boldsymbol{X}) = \bar{X}_n$ and the decision rule that rejects the null hypothesis if

$$T(\boldsymbol{X}) = \bar{X}_n > 2.$$

By construction, the rejection region is $\mathcal{R} = \{\boldsymbol{X} : \bar{X}_n > 2\}$. Figure 7.2 plots the sampling distributions of the test statistic $T(\boldsymbol{X}) = \bar{X}_n$ under the null and the alternative hypotheses respectively, and the rejection region is the region to the right of the vertical bar $x = 2$.

Under H_0, the sampling distribution of $T(\boldsymbol{X})$ is $N(0, \frac{4}{n})$. The Type I error is the grey area of Figure 7.2. Under H_1, the sampling distribution of $T(\boldsymbol{X})$ is $N(\theta_1, \frac{4}{n})$. The Type II error is the blue area of Figure 7.2. With $n = 5$, the Type II error is sizable.

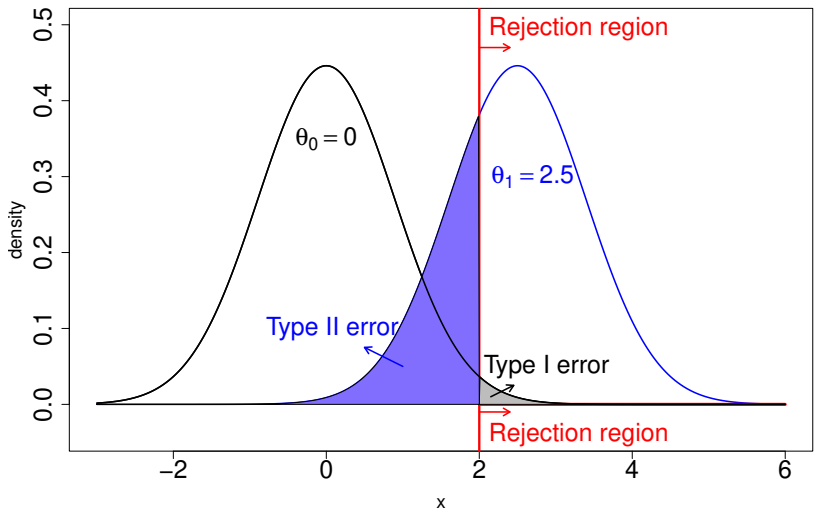


Figure 7.2: Sampling distributions of $T(\boldsymbol{X})$ and the error rates, sample size $n = 5$

Once the critical value is changed from 2 to other values, the errors will change accordingly. Interestingly, there is a tradeoff between the Type I error and the Type II error. If the critical value increases, then the Type I error would decrease, but the Type II error would increase; if the critical value decreases, then the Type II error would decrease, but the Type I error would increase.

How to reduce the error rates simultaneously? An effective method, yet potentially costly, is to increase the sample size. Figure 7.3 repeats Figure 7.2 but the sample size is 20 times larger. When $n = 100$, the sampling distributions under the null and the alternative hypotheses are both much skinnier than those under $n = 5$. Consequently, the Type I error becomes negligible and the Type II error is much smaller.

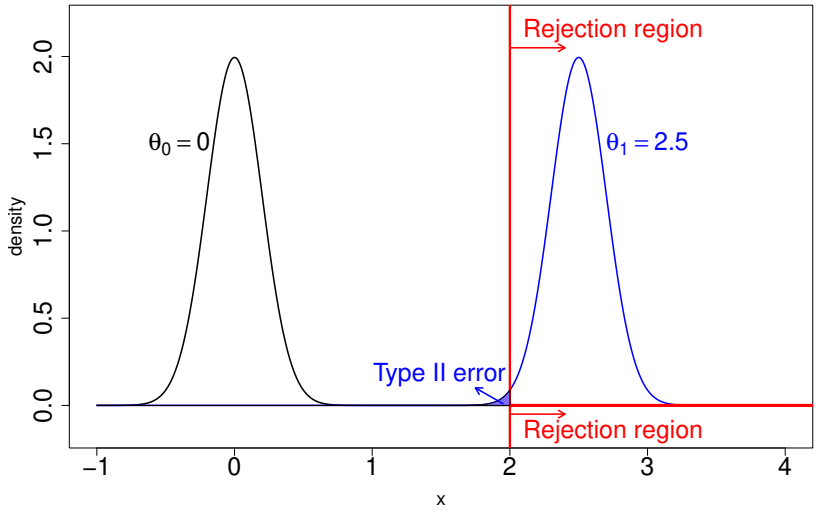


Figure 7.3: Sampling distributions of $T(\boldsymbol{X})$ and the error rates, sample size $n = 100$

♢

Example 7.11 (One sample t-test)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ where both μ and σ^2 are unknown. The hypothesis testing problem is to test

$$H_0\text{: } \mu = \mu_0, \text{ v.s., } H_1\text{: } \mu \neq \mu_0.$$

Student discovered that

$$T(\boldsymbol{X}) \equiv \frac{\bar{X}_n - \mu_0}{\widehat{se}} \sim t_{n-1}$$

where $\widehat{se} = \sqrt{S^2/n}$ and $S^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2$, which is the standard student t distribution with the degree of freedom being $n-1$.

Then the decision rule is to reject the null if

$$|T(\boldsymbol{X})| > t_{n-1}(1 - \alpha/2)$$

for some α, e.g., $\alpha = 5\%$. The size of the test is α. $\diamondsuit$

7.3 The Wald Test

Reading(s) 7.3. Wasserman (2004): 10.1; Hogg et al. (2018): 8.4

Following the Neyman-Pearson school, to characterize the error rates of the test requires the sampling distributions of the test statistic. Exact sampling distributions are often technically challenging to derive. The next best option for exact sampling distributions is the approximate sampling distribution. For instance, the approximate t-distribution that we have seen in the two Gaussian means problem.

Example 7.12 (Two sample t-test)**.** Suppose the first sample is $X_1, ..., X_n \overset{iid}{\sim} N(\mu_1, \sigma_1^2)$, the second sample is $Y_1, ..., Y_m \overset{iid}{\sim} N(\mu_2, \sigma_2^2)$, and the two samples are independent. The parameters (μ_i, σ_i^2) for $i = 1, 2$ are all unknown.

The hypothesis testing problem is to test

$$H_0\text{: } \mu_1 = \mu_2, \text{ v.s., } H_1\text{: } \mu_1 \neq \mu_2.$$

Using the same sampling distribution for the Welch confidence interval, the test statistic is

$$T(\boldsymbol{X}, \boldsymbol{Y}) \equiv \frac{\bar{X}_n - \bar{Y}_m}{\sqrt{S_X^2/n + S_Y^2/m}}$$

where $S_X^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2$ is the unbiased estimator for σ_1^2 and $S_Y^2 = \frac{1}{m-1}\sum_{i=1}^m (Y_i - \bar{Y}_m)^2$ is the unbiased estimator for σ_2^2. The decision rule is to reject H_0 if

$$|T(\boldsymbol{X}, \boldsymbol{Y})| > t_\nu(1 - \alpha/2)$$

where the degree of freedom $\nu = \left(\frac{c^2}{n-1} + \frac{(1-c)^2}{m-1}\right)^{-1}$ and $c = \frac{S_X^2/n}{S_X^2/n + S_Y^2/m}$, so the size of the test is approximately α. ♢

Towards a general class of tests to test *simple* null hypotheses, the Wald test (Wald, 1943) is constructed based on asymptotic Normal theory, hence has asymptotic guarantee on the error rates. The generality of the Wald test is due to the asymptotic Normality of test statistics which holds for a variety of estimators, e.g., maximum likelihood estimators and method of moments estimators. It is often effortless to write down the test statistic, and derive its asymptotic sampling distribution.

Suppose $X_1, ..., X_n \overset{iid}{\sim} f(x; \theta)$, and the null hypothesis is a simple hypothesis

$$H_0 : \theta = \theta_0 \text{ v.s. } H_1 : \theta \neq \theta_0,$$

where θ is a *scalar* . Assume the estimator $\hat{\theta}$ is asymptotically Normal *under the null hypothesis*:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \sigma^2),$$

where σ^2 can depend on θ_0. By Slutsky theorem,

$$\frac{\sqrt{n}(\hat{\theta} - \theta_0)}{\hat{\sigma}} \xrightarrow{d} N(0, 1),$$

where the $\hat{\sigma}$ term is a consistent estimator of σ.

Asymptotically Normal estimators are not difficult to construct. For instance, both the MLE and the MoM estimators are asymptotically Normal under mild conditions. Also, if σ^2 depends on θ_0 smoothly, then by continuous mapping theorem, the plug-in estimator $\hat{\sigma} = \sqrt{\sigma^2(\hat{\theta})}$ is consistent.

The test statistic of the Wald test is

$$W(\boldsymbol{X}) = \frac{\sqrt{n}(\hat{\theta} - \theta_0)}{\hat{\sigma}}.$$

Alternatively, the Wald statistic could also be $W(\boldsymbol{X}) = \frac{\sqrt{n}(\hat{\theta}-\theta_0)}{\sigma(\theta_0)}$, as it also asymptotically follows the standard Normal distribution *under the null hypothesis*.

The critical value is chosen to be $z_{\alpha/2}$ which is the $\alpha/2$ upper quantile of the standard Normal distribution. In `R`, the α upper quantile is obtained by `qnorm(1-`α`)`.

In summary, the decision rule of the Wald test is to reject the null hypothesis H_0 if

$$|W| > z_{\alpha/2}.$$

Theorem 7.1 (Type I error)**.** *The Wald test has asymptotic size α, that is,*

$$\mathbb{P}_{\theta_0}(|W| > z_{\alpha/2}) \to \alpha.$$

Proof. By the asymptotic Normality assumption of $\hat{\theta}$ or W under the null hypothesis,

$$\mathbb{P}_{\theta_0}(|W| > z_{\alpha/2}) \to \mathbb{P}(|Z| > z_{\alpha/2}) = \alpha,$$

where Z follows the standard Normal distribution. □

Example 7.13 (Treatment effect)**.** The LA Times news reported the following.

> The study, led by Dr. Supachai Rerks-Ngarm of the Thai Ministry of Public Health's Department of Disease Control, involved more than 16,000 volunteers in Thailand, all from the general population rather than from a pool of high-risk homosexuals and intravenous drug users used in past studies. Half received four priming doses of Alvac and two boost doses of Aidsvax over a six-month period; the other half received placebo shots.
>
> After three years of follow-up, new HIV infections were observed in 74 of the 8,198 people who received the placebo, but in only 51 of the 8,197 given the vaccine, a statistically significant 31% reduction.

The news describes a new vaccine called Alvac and Aidsvax. The scientific question is if Alvac as a primer and Aidsvax as a booster would be effective in reducing HIV infection. A study was conducted in Thailand with 16,395 individuals from the general (not high-risk) population:

- 51 HIV cases reported in the 8197 individuals receiving the treatment
- 74 HIV cases reported in the 8198 individuals receiving placebos

Let p_1 and p_2 denote the probability of HIV infection in the placebo and treatment populations respectively. Let $X \sim Bin(n, p_1)$ be the number of infected cases in the treatment group with $n = 8197$, and let $Y \sim Bin(m, p_2)$ be the number of infected cases in the control group with $m = 8198$. As the assignment of the treatment and the control group status is random, X and Y are independent.

Suppose the hypothesis testing is to test

$$H_0 : p_1 = p_2 \text{ v.s. } H_1 : p_1 \neq p_2.$$

To construct the Wald test statistic, we need univariate estimators that are asymptotically Normal under the null hypothesis. To this end, the theory of MLE is very useful by the fundamental theorem of MLE.

The MLE of the difference $\delta \equiv p_1 - p_2$ is

$$\hat{\delta} \equiv \hat{p}_1 - \hat{p}_2 = \frac{X}{n} - \frac{Y}{m},$$

where $\hat{p}_j$ is the MLE of p_j for $j = 1, 2$.

By the fundamental theorem of MLE (or multivariate CLT when $m = n$),

$$\begin{bmatrix} \hat{p}_1 \\ \hat{p}_2 \end{bmatrix} - \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} \dot{\sim} N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{n}\hat{p}_1(1-\hat{p}_1) & 0 \\ 0 & \frac{1}{m}\hat{p}_2(1-\hat{p}_2) \end{bmatrix} \right)$$

where the variance-covariance is the observed Fisher information matrix.

By the delta method, the asymptotic distribution of $\hat{\delta}$ under the null distribution is

$$\hat{\delta} - \delta \dot{\sim} N\left(0, \frac{1}{n}\hat{p}_1(1-\hat{p}_1) + \frac{1}{m}\hat{p}_2(1-\hat{p}_2)\right).$$

The Wald test statistic is

$$W = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{1}{n}\hat{p}_1(1-\hat{p}_1) + \frac{1}{m}\hat{p}_2(1-\hat{p}_2)}},$$

and under the null hypothesis, as both $n \to \infty$ and $m \to \infty$

$$W \to N(0, 1).$$

Then the Wald test of size α is to reject the null hypothesis if

$$|W| \geq z_{\alpha/2}$$

where z_α denotes the upper α quantile of the standard Normal distribution.

In the HIV vaccine example, $\hat{p}_1 = 51/8197 \approx 0.006$, $\hat{p}_2 = 74/8198 \approx 0.009$, and $W \approx -2.065$. For $z_{2.5\%} \approx 1.96$, the decision is to reject the null hypothesis. In contrast, for $z_{0.5\%} \approx 2.58$, the decision is to retain the null hypothesis.

```
X <- 51 ; n <- 8197 ## treatment group
Y <- 74; m <- 8198 ## control group
p1  <- X/n
p2 <- Y/m
W <- (p1-p2)/sqrt(p1*(1-p1)/n+p2*(1-p2)/m)
```

The Wald statistic W is based on the delta method. An alternative to this is the parametric bootstrap. That is, the standard error can be obtained by the parametric bootstrap. A sample code is as follows. The Wald statistic based on the bootstrap is very close to the Wald statistic based on the delta method.

```
### parametric bootstrap se
B <- 1e6
hat.eta <- sapply(1:B, function(x) rbinom(1,n,p1)/n-rbinom(1,m,p2)/m)
W <- (p1-p2)/sd(hat.eta); W
```

Suppose the hypothesis testing is to test

$$H_0\colon\ p_1 \geq p_2,\ \text{v.s.}\ H_1\colon\ p_1 < p_2.$$

For the one-sided testing problem, the Wald test of size α is to reject the null hypothesis if

$$W < -z_\alpha$$

where z_α denotes the upper α quantile of the standard Normal distribution.

For $z_{5\%} \approx 1.64$, the decision is to reject the null hypothesis. For $z_{1\%} \approx 2.33$, the decision is to retain the null hypothesis.

♢

Example 7.14 (COVID-19 vaccine)**.** Similar to the above HIV vaccine example, the Science article reported the following.

> Continuing the spate of stunning news about COVID-19 vaccines, the biotech company Moderna announced the final results of the 30,000-person efficacy trial for its candidate in a press release today: Only 11 people who received two doses of the vaccine developed COVID-19 symptoms after being infected with the pandemic coronavirus, versus 185 symptomatic cases in a placebo group. That is an efficacy of 94.1%, the company says, far above what many vaccine scientists were expecting just a few weeks ago.

The outcomes of the experiment are summarized as follows.

- Treatment group: 11 individuals are symptomatic, out of 15,000 individuals.
- Control group: 185 individuals are symptomatic, out of 15,000 individuals.

With the same Wald test, the Wald statistic is $W = -12.5$. At both 1% and 5% significance level, the decision is to reject the null.

```
X <- 11 ; n <- 15000 ## treatment group
Y <- 185; m  <- 15000 ## control group
p1  <- X/n
p2 <- Y/m
W <- (p1-p2)/sqrt(p1*(1-p1)/n+p2*(1-p2)/m)
### parametric bootstrap se
B <- 1e6
hat.eta <- sapply(1:B, function(x) rbinom(1,n,p1)/n-rbinom(1,m,p2)/m)
W <- (p1-p2)/sd(hat.eta); W
```

♢

Example 7.15 (paired comparison)**.** A special case of evaluating the treatment effect with two groups is paired comparison. The basis of causal inference. ♢

Theorem 7.2 (Power function)**.** *For $\theta \neq \theta_0$, the power function of the Wald test is*

$$\beta_W(\theta) =$$

Proof. direct calculation. □

[plot of the power function]

Theorem 7.3 (Duality between confidence interval and hypothesis testing)**.** *Suppose the null hypothesis is $\theta = \theta_0$, and the alternative hypothesis is $\theta \neq \theta_0$, where $\theta \in \Theta$ can be a vector. Let $\eta = g(\theta)$ be univariate.*

Let the interval $C_\alpha = [a(\boldsymbol{X}), b(\boldsymbol{X})]$ be a $(1-\alpha)-$confidence interval of η under the null hypothesis, that is,

$$\mathbb{P}_0(\eta \in C_\alpha) = 1 - \alpha,$$

where $\mathbb{P}_0$ denotes the probability (measure) under the null hypothesis.

The test based on the confidence interval is to reject the null hypothesis if $\eta_0 \notin C_\alpha$, that is, the rejection region is

$$\mathcal{R}_\alpha \equiv \{\boldsymbol{x} : \eta_0 \leq a(\boldsymbol{x})\} \cup \{\boldsymbol{x} : \eta_0 \geq b(\boldsymbol{x})\}.$$

Then the size of the test is α.

On the other hand, suppose the size-α test has the rejection region $\mathcal{R}_\alpha \equiv \{\boldsymbol{x} : T(\boldsymbol{x}; \eta_0) \geq c\}$, where $T(\boldsymbol{x}; \eta_0)$ is the test statistic and potentially depends on η_0. That is,

$$\mathbb{P}_0(\boldsymbol{X} \in \mathcal{R}_\alpha) = \alpha,$$

where $\mathbb{P}_0$ denotes the probability (measure) under the null hypothesis.

Then, a $(1-\alpha)-$confidence interval of η_0 can be constructed as follows

$$\{\eta : T(\boldsymbol{X}; \eta) \leq c\}.$$

Proof. From CI to HT:

From HT to CI:

□

What if the parameter is not scalar? The best arena of the Wald test is when the parameter in the hypotheses is a scalar. When the parameter is a vector, asymptotically Normal estimators are also readily available by MLE and/or MoM, but the limiting distribution is a multivariate Normal.

In general, specifying the critical value vector for the vector of estimators is quite challenging. Instead, it is possible to construct a test statistic as a function of the estimators. By the multivariate delta method, the limiting distribution of the test statistic is univariate Normal. Then the z score critical values are useful again. However, choosing the function can be tricky, as the theory does not imply the optimal choice.

A little bit more about Abraham Wald Perhaps Wald is better known for his contribution to aircraft survivability. During World War II, Wald was a member of the Statistical Research Group (SRG) at Columbia University. One of the wartime problems that Wald worked on was to examine the distribution of damage to aircraft returning after flying missions, and to provide advice on how to minimize bomber losses to enemy fire. Instead of recommending adding armor to areas with the most bullet holes, he recommended adding armor to the areas that suffered the least damage. The bullet holes in the returning aircraft represented areas where a bomber could take damage and still fly well enough to return safely to base. This observation is later coined as the *survivorship bias* which has broader applications in finance, business and many others.

7.4 The $p-$values: Significance Testing

Reading(s) 7.4. Wasserman (2004): 10.2; Wasserstein and Lazar (2016)

7.4.1 Introduction

Hypothesis testing with the error rates controlled belongs to Neyman(-Pearson) School. The approach provides error rates guarantees for the decision making procedure: for repeated experiments, the number of the two types of errors made is controlled. However, for the observed dataset, we do not know if we make a correct decision on H_0.

Fisher thinks differently. His idea is to come up with a *data dependent*, single number summary of the "plausibility" of H_0. The metric used for this purpose is p value and hypothesis testing using p values is called *significance* testing. *Significance* testing using *p-values* is by far the dominant method of testing in science. The proper use of p-values remains challenging in scientific communities; it is viewed by many as being the major source of the problems of reproducibility of science. (Nuzzo, 2014; Wasserstein and Lazar, 2016)

Definition 7.4 (p value)**.** Suppose a size α test is associated with the rejection region $\mathcal{R}_\alpha$ for every $\alpha \in (0,1)$. The p value is the smallest level at which the test rejects the null hypothesis H_0:

$$p - value(\boldsymbol{x}) = \inf\{\alpha : \boldsymbol{x} \in \mathcal{R}_\alpha\},$$

where $\boldsymbol{x} \equiv \{x_i\}_{i=1}^n$ denotes the observed data.

Intuitively speaking, p value is a measure of the compatibility of the data and H_0. However, despite being some probability, the p value is not the probability that the null hypothesis is true.

Example 7.16 (Lady tasting tea)**.** The experiment is to present eight cups in total, with four milk added to tea and four tea added to milk. The lady was aware of the experiment design so her answer would take this into account.

The hypotheses are

$$H_0\text{: the lady cannot tell the difference; v.s. } H_1\text{: the lady can tell the difference.}$$

Heuristically, the more cups correctly identified, the less convincing is H_0. So the test is to reject the null hypothesis if

$$X \geq c$$

where X denotes the number of cups correctly identified, and c is the critical value.

Given the experiment design, under the null hypothesis, the pmf of X is tabulated as follows.

X	0	1	2	3	4	5	6	7	8
probability	$\frac{1}{\binom{8}{4}}$	0	$\frac{\binom{4}{1}\binom{4}{1}}{\binom{8}{4}}$	0	$\frac{\binom{4}{2}\binom{4}{2}}{\binom{8}{4}}$	0	$\frac{\binom{4}{3}\binom{4}{3}}{\binom{8}{4}}$	0	$\frac{1}{\binom{8}{4}}$
$\approx$	0.014	0	0.229	0	0.514	0	0.229	0	0.014

Then we tabulate the associated sizes for different critical values.

$c=$	0	1	2	3	4	5	6	7	8
size $\mathbb{P}_0(X \geq c) \approx$	1	0.986	0.986	0.757	0.757	0.243	0.243	0.014	0.014

Suppose the observed outcome is x. To reject the null hypothesis, the critical value c must satisfy

$$x \geq c \geq 0.$$

Since the size/level is decreasing in c, the smallest level is achieved at the largest possible critical value x, and the smallest level is $\mathbb{P}_0(X \geq x)$. That is,

$$p \text{ value} = \mathbb{P}_0(X \geq x)$$

which is data dependent. Intuitively, it is the probability of what is observed and what is more extreme than the observed.

It turned out the lady correctly identified eight cups. The p value of rejecting H_0 that the outcome was sheer guesswork is 0.014.

Questions: what is the p value if the Lady does not know the experiment design? what is the p value if the experiment design is randomly serving milk added to tea cups and tea added to milk cups? ♢

The p value is widely used by scientific communities. Typically, researchers in various fields use the following evidence scale.

$p-$value	significance
< 0.01	very strong evidence against H_0
$(0.01, 0.05)$	strong evidence against H_0
$(0.05, 0.10)$	weak evidence against H_0
> 0.1	little or no evidence against H_0

Table 4: Statistical significance levels

In computing the p value, the Definition 7.4 is not very operational. The following theorem simplifies the p value calculation for a very special yet common class of tests.

Theorem 7.4 (p value)**.** *Suppose the size α test is to reject H_0 if and only if $T(\boldsymbol{X}) \geq c_\alpha$. Then, the p value given the data $\boldsymbol{x}$ is*

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(T(\boldsymbol{X}) \geq T(\boldsymbol{x})).$$

In particular, if $\Theta_0 = \{\theta_0\}$, then

$$p - value = \mathbb{P}_{\theta_0}(T(\boldsymbol{X}) \geq T(\boldsymbol{x})).$$

Proof. By assumption, the rejection region of the size α test is $R_\alpha = \{\boldsymbol{X} : T(\boldsymbol{X}) \geq c_\alpha\}$, and by the definition of size,

$$\alpha = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta(T(\boldsymbol{X}) \geq c_\alpha).$$

Observe the rejection region is monotonic in critical value c_α: for two sizes α and α', if $c_\alpha \geq c_{\alpha'}$, $T(\boldsymbol{X}) \geq c_\alpha$ implies $T(\boldsymbol{X}) \geq c_{\alpha'}$, that is,

$$\mathbb{P}_\theta(T(\boldsymbol{X}) \geq c_\alpha) \leq \mathbb{P}_\theta(T(\boldsymbol{X}) \geq c_{\alpha'})$$

where the "$\leq$" still holds by taking supremum over Θ_0 with respect to θ. Therefore, the size α is also monotonic in the critical value c_α: as critical value decreases, the size increases; and vice versa.

Recall the definition of the p value is

$$p - value = \inf\{\alpha : T(\boldsymbol{X}) \geq c_\alpha\}.$$

By the monotonic relationship between the critical value and the size, the infimum critical value given data $\boldsymbol{x}$ is attained at $T(\boldsymbol{x})$, and the associated size, by the definition of size, is

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(T(\boldsymbol{X}) \geq T(\boldsymbol{x})).$$

□

We only use the information of H_0.

The p value is the probability of observing a value of the test statistic the same as or more extreme than what was actually observed.

Example 7.17 (The Higgs boson particle)**.** A major goal of the Large Hadron Collider (LHC) at the European Organization for Nuclear Research (CERN) is to determine if the Higgs boson particle actually exists.

Physicists record the number of the events that are characteristic of Higgs boson production in LHC particle collisions within some time window T. Denote the number as X. The statistical model is

$$X \sim Pois(\theta + b)$$

where θ is the mean rate of production of Higgs events in time T, and b is the (known) mean rate of production of events with the same characteristics from background sources in time T.

The hypothesis testing problem is

$$H_0 : \theta = 0 \text{ v.s. } H_1 : \theta > 0,$$

where H_0 corresponds to no Higgs. The rejection region is of the form

$$\{X : X \geq c\}$$

for some critical value c.

The $p-$value is

$$\mathbb{P}(X \geq x | \theta = 0, b) = \sum\nolimits_{k=x}^{\infty} \frac{b^k}{k!} e^{-b} = 1 - F_0(x-1),$$

where x denotes the actual observation, and $F_0(\cdot)$ denotes the CDF under the null hypothesis. The p value is the probability of the observed value x *and* the values that are more extreme than x.

We can compute the p values based on the following experiment data:

(1) With $x = 7$ and $b = 1.2$, the `R` function `1- ppois(7-1,1.2)` gives 0.0003 as the p value.

(2) With $x = 6$ and $b = 2.2$, the `R` function `1- ppois(6-1,2.2)` gives 0.025 as the p value.

♢

Theorem 7.5 (p value of the Wald test)**.** *Suppose the Wald test has the test statistic $W(\boldsymbol{X})$. The p value, given data $\boldsymbol{x}$, converges to $2\Phi(-W(\boldsymbol{x}))$.*

Proof. direct calculation □

Example 7.18 (HIV vaccine)**.** In the example of LA Times news,

- 51 HIV cases reported in the 8197 individuals receiving the treatment
- 74 HIV cases reported in the 8198 individuals receiving placebos.

The Wald test statistic is -2.065. With Theorem 7.5, the p value of the (two-sided) Wald test is about 0.039. In `R`, the function `2*pnorm(-2.065)` gives the answer. ♢

Theorem 7.6 (Monte Carlo based p value)**.** *Suppose the test is to reject the null hypothesis if the test statistic*

$$T(\boldsymbol{X}) \geq c$$

for some critical value c.

If H_0 fully specifies the data generating process, for $j = 1, ..., N_{mc}$, independently sample

$$\boldsymbol{X}_{(j)} \sim f(\boldsymbol{x}; \theta_0).$$

If H_0 does not fully specify the data generating process, then for $j = 1, ..., N_{mc}$, independently sample

$$\boldsymbol{X}_{(j)} \sim f(\boldsymbol{x}; \hat{\theta}_0)$$

where $\hat{\theta}_0$ is a consistent estimator constrained to Θ_0.

With the simulated synthetic data, obtain the Monte Carlo replications of the test statistic

$$\left\{T\left(X^n_{(1)}\right), ..., T\left(X^n_{(N_{mc})}\right)\right\}.$$

Then, the Monte Carlo based p-value is

$$\frac{1}{N_{mc}} \sum_{j=1}^{N_{mc}} \mathbb{1}\left(T(X^n_{(j)}) \geq T(\boldsymbol{x})\right)$$

where $\boldsymbol{x}$ is the observed data.

$\hat{\theta}_0$ can be the MLE constrained to Θ_0, as done in the (generalized) likelihood ratio test. Non parametric bootstrap estimates would be misleading if the data is from the alternative hypotheis. In contrast, parametric bootstrap estimates are less biased.

Example 7.19. Suppose the data $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, and the hypotheses are

$$H_0\text{: } \mu = 0, \text{ v.s. } H_1\text{: } \mu \neq 0.$$

The test is to reject the null hypothesis H_0 if

$$|\bar{X}_n| \geq c$$

for some critical value c.

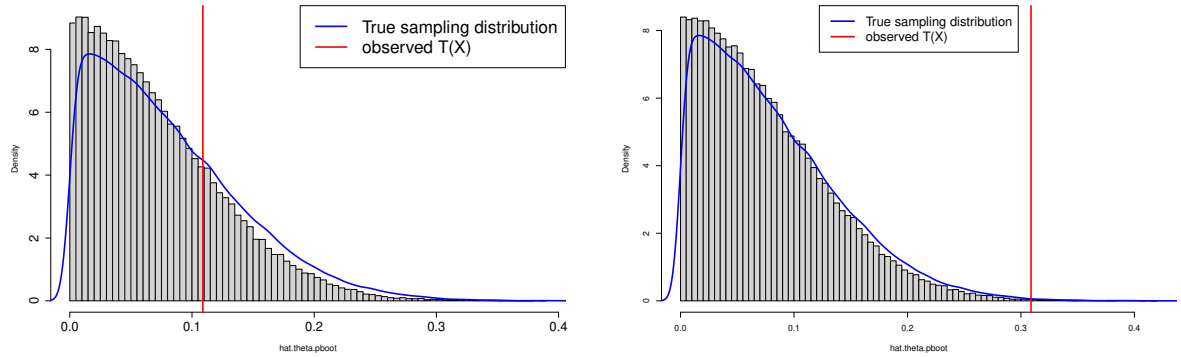
Suppose $\sigma^2 = 1$ is known, the null hypothesis, together with the model assumption, fully specifies the data generating process. Then we can directly simulate the sampling distribution of the test statistic $|\bar{X}_n|$ under the null hypothesis. The true p value can be well approximated by the Monte Carlo simulated p value.

Suppose σ^2 is unknown, the null hypothesis only specifies the mean with the variance of the data generating process unspecified. We have to estimate it from the data, and simulate the sampling distribution of the test statistic $|\bar{X}_n|$ from the estimated model under the null hypothesis. Under the null hypothesis H_0: $\mu = 0$, a consistent estimator of the variance can be

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2.$$

Figure 7.4 presents two cases. On the left panel, the data is 100 random observations from $N(0,1)$ and the observed test statistic is 0.109. To simulate the sampling distribution of the test statistic under the null hypothesis, the test statistics are obtained from random samples of $N(0, \hat{\sigma}^2)$ where $\hat{\sigma}^2$ is defined above. The blue curve is the estimated density of the sampling distribution under the null hypothesis, the histogram and the blue curve are very close, which implies that the simulated sampling distribution is similar to the true sampling distribution under the null hypothesis. The p value based on the true data generating process is 0.278, and the Monte Carlo simulated/bootstrapped p value is 0.224.

On the right panel, the data is 100 random observations from $N(0.2,1)$ which is a model from the alternative hypothesis, and the observed test statistic is 0.309. As the synthetic data is drawn from the alternative hypothesis, we expect the p value to be small as the evidence against the null hypothesis. The p value based on the data generating process $N(0,1)$ is approximately 0.002, while the Monte Carlo simulated/bootstrapped p value is 0.001.



DGP is $N(0,1)$ DGP is $N(0.2,1)$

Figure 7.4: Bootstrap/Monte Carlo simulation based p value

```r
set.seed(1)
n <- 100
X <- rnorm(n,0,1)
### sample from H0
```

```r
hat.theta <- sapply(1:1e5, function(x) abs(mean(rnorm(n,0,1))))
### sample from estimated H0
sd <- sqrt(mean((X-0)^2)) ## sd est under H0
hat.theta.pboot <- sapply(1:1e5, function(x) abs(mean(rnorm(n,0,sd))))
hist(hat.theta.pboot,breaks = "FD",main = "", probability = TRUE, cex.axis = 1.5)
lines(density(hat.theta),lwd = 3,col="blue")
abline(v= abs(mean(X)),lwd=3,col="red")
legend('topright',c("True sampling distribution","observed T(X)"), lty = 1,
       col = c("blue","red"), lwd = 3,cex = 2)

mean(hat.theta> abs(mean(X))) ## true p val
mean(hat.theta.pboot> abs(mean(X))) ## estimated p val

set.seed(1)
Y <- rnorm(n,0.2,1)
sd <- sqrt(mean(Y^2))
hat.theta.pboot <- sapply(1:1e5, function(x) abs(mean(rnorm(n,0,sd))))
hist(hat.theta.pboot,breaks = "FD",main = "", probability = TRUE)
lines(density(hat.theta),lwd = 3,col="blue")
abline(v= abs(mean(Y)),lwd=3,col="red")
legend('top',c("True sampling distribution","observed T(X)"), lty = 1,
       col = c("blue","red"), lwd = 3,cex = 1.5)

mean(hat.theta> abs(mean(X))) ## true p val
mean(hat.theta.pboot> abs(mean(X))) ## estimated p val
```

♢

7.4.2 Misuse of the p values

By frequentist principle, choose α first and then perform hypothesis testing. As α is independent of the data, this procedure has the frequentist guarantee for the Type I error control.

On the other hand, we can compute the p values. The p values provide more information than rejecting or retaining at the pre-specified significance level. Therefore, given the α and the observed test statistic, the p value*should not* affect our decision on rejecting or retaining the null hypothesis H_0.

In practice, this order could be (incorrectly) reversed. Practitioners often first compute the p value and then choose a favorable α to make their results statistically significant. For instance, if the p value is 0.03, then the α is chosen to be 0.05; if the p value is 0.07, then the α is chosen to be 0.10; if the p value is 0.005, then the α is chosen to be 0.01. Choosing α after the analysis violates the frequentist principle, and hence the frequentist guarantee for Type I error rate control no longer holds.

ASA Statement on Statistical Significance and P-Values The p-value is often misused and misinterpreted. The American Statistical Association (ASA) published an official statement on p-values in 2016. (Wasserstein and Lazar, 2016)

The principles addressed in Wasserstein and Lazar (2016) are as follows.

1. P-values indicate how incompatible the data are with a specified statistical model.
2. P-values do not measure the probability that the studied hypothesis is true, or the probability that the data were produced by random chance alone.
3. Scientific conclusions and business or policy decisions should not be based only on whether a p-value passes a specific threshold.

4. Proper inference requires full reporting and transparency.

5. A p-value, or statistical significance, does not measure the size of an effect or the importance of a result.

6. By itself, a p-value does not provide a good measure of evidence regarding a model or hypothesis.

7.4.3 Hypothesis Testing with p values

Testing with p values is legit as long as the size α is pre-specified. The test based on the p value is to reject the null hypothesis H_0 if

$$p < \alpha.$$

In this case, the test statistic is the p value and the critical value is the pre-specified size. This test combines the idea of Fisher and Neyman, but probably would be disliked by both Fisher and Neyman.

Theorem 7.7 (Uniform distribution under the null)**.** *Suppose the test statistic has a continuous distribution, then under the null hypothesis,*

$$p - value \sim Unif[0, 1].$$

Proof. Denote the test statistic as $T(X)$ and the observation is x. By continuity, the CDF of the test statistic is strictly increasing and its inverse exists. Denote the CDF of $T(X)$ under the null hypothesis as $F_T(t)$. Then, the p-value is

$$p(x) = \mathbb{P}_0(T(X) \geq T(x)) = 1 - F_T(T(x)).$$

Clearly, $p(x) \in [0, 1]$. For random observation X, its p-value $p(X)$ is also random.

Denote the CDF of $p(X)$ under the null hypothesis as $F_0(p)$. Pick $s \in [0, 1]$,

$$F_0(s) = \mathbb{P}_0(p(X) \leq s) = \mathbb{P}_0(1 - F_T(T(X)) \leq s) = \mathbb{P}_0(F_T(T(X)) \geq 1 - s) = s.$$

It is easy to check the case where $s \notin [0, 1]$. Therefore, $F_0(p)$ is the CDF of $Unif[0, 1]$ and $p(X)$ follows the Uniform distribution under the null hypothesis. □

Intuitively, Theorem 7.7 says that if the null hypothesis is true, the p value obtained from the data is like a random draw from $Unif[0, 1]$. Therefore, if we reject H_0 as long as the p value is less than α, the probability of type I error is α. The popular choice of the threshold is 0.05.

Example 7.20 (The sampling distribution of the p value under H_0)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, 1)$, and the hypotheses are

$$H_0\text{: } \mu \leq 0, \text{ v.s., } H_1\text{: } \mu > 0.$$

The test is to reject the null hypothesis H_0, if

$$median(\boldsymbol{X}) \geq c$$

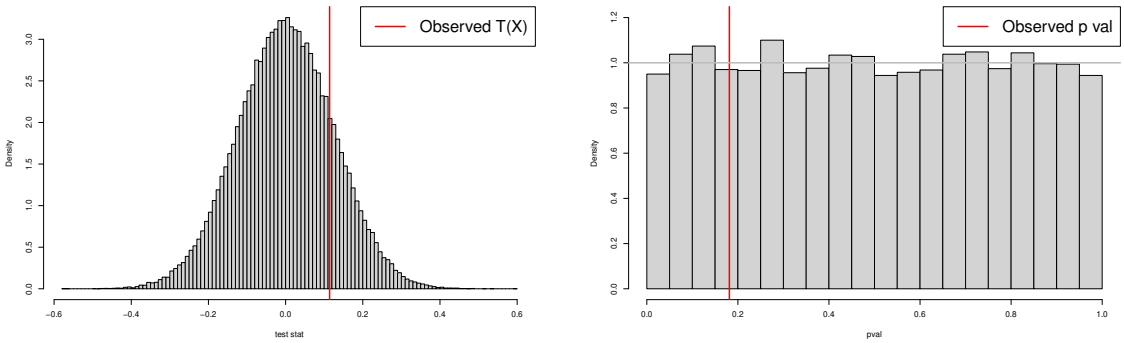
for some critical value c. Then, the p value is

$$\sup_{\mu \leq 0} \mathbb{P}_\mu \left(median(\boldsymbol{X}) \geq median(\boldsymbol{x}) \right) = \mathbb{P}_0 \left(median(\boldsymbol{X}) \geq median(\boldsymbol{x}) \right)$$

where $\boldsymbol{x}$ denotes the observed data.

To compute the p value numerically, we first simulate the sampling distribution of the test statistic $median(\boldsymbol{X})$ where $\boldsymbol{X}$ is drawn from $N(0,1)$. The sampling distribution is presented in the left panel of Figure 7.5. Then, the p value is 0.181. In `R`, the function is `mean(TS>=median(X))` where `X` denotes the synthetic data, and `TS` denotes the Monte Carlo samples of the test statistic.

To examine the sampling distribution of the p value, we need to numerically compute the p values for random synthetic samples. The sampling distribution of the p value is presented in the right panel of Figure 7.5. The histogram implies the sampling distribution is close to the uniform distribution on $[0,1]$.



Sampling distribution of $median(\boldsymbol{X})$ Sampling distribution of the p value

Figure 7.5: Sampling distributions under H_0

```
set.seed(1)
n <- 100
X <- rnorm(n,0,1)
#### sampling distribution of the test statistic
TS <- sapply(1:1e5,function(x) median(rnorm(n,0,1)))
hist(TS,breaks = "FD",probability = TRUE,main="",xlab = "test stat")
abline(v=median(X),lwd=3,col="red")
legend("topright","Observed T(X)",lty=1,lwd=3,col="red",cex=2)
#### p value of the data
mean(TS>=median(X))
#### sampling distribution of the p value
pval <- sapply(1:1e4, function(x)
    mean(abs(TS) >= abs(median(rnorm(n,0,1)))))
hist(pval,breaks = "FD",probability = TRUE,main="",ylim = c(0,1.2))
abline(v=mean(TS >=median(X)),lwd=3,col="red")
abline(h=1,lwd=3,col="grey")
legend("topright","Observed p val",lty=1,lwd=3,col="red",cex=2)
```

♢

Example 7.21 (The test statistic is not continuous)**.** Recall Theorem 7.7 requires that the test statistic is a *continuous* random variable under the null hypothesis. This assumption is crucial to the conclusion that the p value follows a uniform distribution.

Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, 1)$, and the hypotheses are

$$H_0\text{: } \mu \leq 0, \text{ v.s., } H_1\text{: } \mu > 0.$$

The test is to reject the null hypothesis H_0, if

$$T(\boldsymbol{X}) \geq c$$

for some critical value c, and the test statistic is defined as

$$T(\boldsymbol{X}) = \bar{X}_n \mathbb{1}(\bar{X}_n \leq h) + 2h\mathbb{1}(\bar{X}_n > h)$$

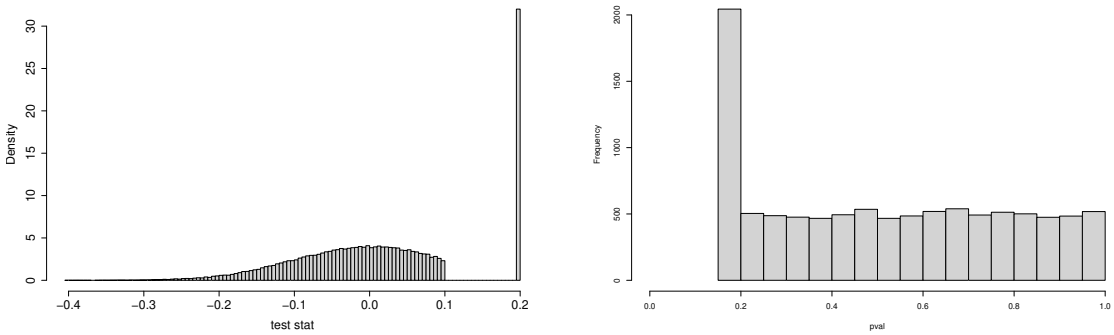
where $h = 0.1$ in our simulation example. The indicator functions in $T(\boldsymbol{X})$ create a jump of $T(\boldsymbol{X})$ at $\bar{X}_n = h$, so the test statistic itself is not a continuous function of $\bar{X}_n$.

The test statistic is not a continuous random variable, as it has a point mass at $2h$:

$$\mathbb{P}_0\left(T(\boldsymbol{X}) = 2h\right) = \mathbb{P}_0(\bar{X}_n > h) = 1 - \Phi(\sqrt{n}h)$$

which is 0.159 in our simulation example. Indeed, the sampling distribution of $T(\boldsymbol{X})$ in the simulation example is presented in the left panel of Figure 7.6, and it has a point mass at 0.2.

Then, we can examine the sampling distribution of $T(\boldsymbol{X})$ under the null hypothesis. The sampling distribution of the p value is presented in the right panel of Figure 7.5. The histogram implies the sampling distribution is clearly not a uniform distribution on $[0, 1]$.



Sampling distribution of $T(\boldsymbol{X})$ Sampling distribution of the p value

Figure 7.6: Sampling distributions under H_0

```r
set.seed(1)
n <- 100
X <- rnorm(n,0,1)
test.stat <- function(x){
    m <- mean(x)
    c <- 0.1
    return(m*(m <= c)+ 2*c*(m > c))
}
test.stat(X)
### sampling distribution of T(X)
TS <- sapply(1:1e5,function(x) test.stat(rnorm(n,0,1)))
hist(TS,breaks = "FD",probability = TRUE,main="",xlab="test stat",
     cex.lab=1.5,cex.axis=1.5)
### p value
mean(TS >= test.stat(X))
### sampling distribution of the p value
pval <- sapply(1:1e4, function(x) mean(TS >= test.stat(rnorm(n,0,1))))
hist(pval,breaks = "FD",main="",xlim = c(0,1))
```

♢

Towards the Conditional Inference A key drawback of the hypothesis testing idea of the Neyman school is that the error rates are irrelevant to the data observed. Despite knowing the genuine size/level, we still have no idea about if the decision on H_0 is correct or not for the data at hand. (The same type of problem with confidence intervals.) While p value is data dependent, it is only suggestive of the validity of H_0 without taking the alternative H_1 into account.

In fact, we need a theory of hypothesis testing whose conclusions are more relevant to the data at hand. To this end, we will study Bayesian statistics, to see if the Bayesian method can resolve the issues.

7.5 The Neyman-Pearson lemma

Reading(s) 7.5. Hogg et al. (2018): 8.6, 8.7

The Neyman-Pearson Lemma implies that for simple v.s. simple hypothesis testing problems, the likelihood ratio test is the most powerful (MP) test. That is, there exists no test that has smaller size than the likelihood ratio test and that is more powerful than the likelihood ratio test.

Lemma 7.8 (Neyman-Pearson)**.** *Suppose* $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta)$*, and the hypotheses are*

$$H_0\text{: } \theta = \theta_0, \text{ v.s., } H_1\text{: } \theta = \theta_1.$$

Let

$$T(X^n) = \frac{\mathcal{L}_n(\theta_1)}{\mathcal{L}_n(\theta_0)}.$$

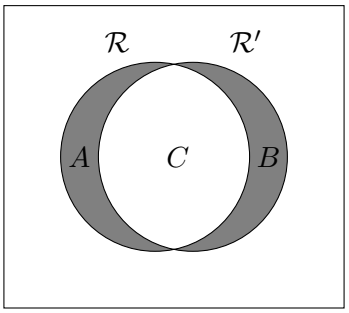
The likelihood ratio test is to reject the null hypothesis if

$$T(X^n) \geq c$$

for some critical value c*. Let* $\alpha = \mathbb{P}_{\theta_0}(T(X^n) \geq c)$ *be the size of the likelihood ratio test.*

Then, the likelihood ratio test maximizes the power $\beta(\theta_1)$ *among all tests of size* α*.*

Proof. Define the rejection region of the competing test as $\mathcal{R}'$ and the rejection region of the Likelihood ratio test as $\mathcal{R}$. Let $A = \mathcal{R}\backslash\mathcal{R}'$, $B = \mathcal{R}'\backslash\mathcal{R}$, and $C = \mathcal{R} \cap \mathcal{R}'$. The sets A and B are illustrated in the Venn diagram below.



By the assumption that the competing test has the same size,

$$\mathbb{P}_{\theta_0}(\mathcal{R}) = \mathbb{P}_{\theta_0}(\mathcal{R}') = \alpha.$$

Note the decomposition identities $\mathbb{P}_{\theta_0}(\mathcal{R}) = \mathbb{P}_{\theta_0}(A) + \mathbb{P}_{\theta_0}(C)$ and $\mathbb{P}_{\theta_0}(\mathcal{R}') = \mathbb{P}_{\theta_0}(B) + \mathbb{P}_{\theta_0}(C)$, it follows

$$\mathbb{P}_{\theta_0}(A) = \mathbb{P}_{\theta_0}(B).$$

The goal is to show the likelihood ratio test is the most powerful, that is, $\mathbb{P}_{\theta_1}(\mathcal{R}) \geq \mathbb{P}_{\theta_1}(\mathcal{R}')$, it is equivalent to show

$$\mathbb{P}_{\theta_1}(A) + \mathbb{P}_{\theta_1}(C) \geq \mathbb{P}_{\theta_1}(B) + \mathbb{P}_{\theta_1}(C).$$

On $A \subseteq \mathcal{R}$, $T(X^n) \geq c$ by definition. Then we can integrate both sides on A with respect to $\mathbb{P}_{\theta_0}$ while maintaining the inequality:

$$\int_A T(x^n) f(x^n; \theta_0) dx^n \geq c \int_A f(x^n; \theta_0) dx^n = c\mathbb{P}_{\theta_0}(A).$$

Note $T(x^n)$ is the likelihood ratio, $\int_A T(x^n) f(x^n; \theta_0) dx^n = \int_A f(x^n; \theta_1) dx^n = \mathbb{P}_{\theta_1}(A)$. That is,

$$\mathbb{P}_{\theta_1}(A) \geq c\mathbb{P}_{\theta_0}(A).$$

On the other hand, on $B \subset \mathcal{R}^c$, $T(X^n) < c$ by definition. Then we can also integrate both sides on B with respect to $\mathbb{P}_{\theta_0}$ while maintaining the inequality:

$$\int_B T(x^n) f(x^n; \theta_0) dx^n \leq c \int_B f(x^n; \theta_0) dx^n = c\mathbb{P}_{\theta_0}(B),$$

where "=" holds if B is a null set with respect to $\mathbb{P}_{\theta_0}$ (as it forces the non-negative LHS to be 0 as well). Again note $T(x^n)$ is the likelihood ratio, $\int_B T(x^n) f(x^n; \theta_0) dx^n = \int_B f(x^n; \theta_1) dx^n = \mathbb{P}_{\theta_1}(B)$. That is,

$$\mathbb{P}_{\theta_1}(B) \leq c\mathbb{P}_{\theta_0}(B).$$

Now we can combine the two inequalities $\mathbb{P}_{\theta_1}(A) \geq c\mathbb{P}_{\theta_0}(A)$ and $\mathbb{P}_{\theta_1}(B) \leq c\mathbb{P}_{\theta_0}(B)$, and the equation $\mathbb{P}_{\theta_0}(A) = \mathbb{P}_{\theta_0}(B)$ to conclude

$$\mathbb{P}_{\theta_1}(A) \geq \mathbb{P}_{\theta_1}(B)$$

where "=" holds when both A and B are null set with respect to $\mathbb{P}_{\theta_0}$. □

By inspecting the proof, the two inequalities $\mathbb{P}_{\theta_1}(A) \geq c\mathbb{P}_{\theta_0}(A)$ and $\mathbb{P}_{\theta_1}(B) \leq c\mathbb{P}_{\theta_0}(B)$ imply that

$$\mathbb{P}_{\theta_1}(A) - \mathbb{P}_{\theta_1}(B) \geq c\left(\mathbb{P}_{\theta_0}(A) - \mathbb{P}_{\theta_0}(B)\right).$$

If the competing test has smaller size than the likelihood ratio test, then $\mathbb{P}_{\theta_0}(A) - \mathbb{P}_{\theta_0}(B)$ is strictly positive. By the above inequality, $\mathbb{P}_{\theta_1}(A) > \mathbb{P}_{\theta_1}(B)$ and the likelihood ratio test is strictly more powerful.

In contrast, the Wald test could be sub-optimal to the likelihood ratio test in terms of power. In some cases, the test statistic of the Wald test happens to be some monotone transformation of the likelihood ratio statistic. In such cases, the Wald test is essentially a likelihood ratio test and hence most powerful at the given size α.

The likelihood ratio statistic $T(X^n)$ can often be simplified by various monotonic transformations. For instance, the logarithmic function preserves monotonicity, the rejection region $\{X^n : T(X^n) > c\}$ is equivalent to the rejection region

$$\{X^n : \log T(X^n) > \log(c)\} = \{X^n : \log T(X^n) > c'\}$$

with the new test statistic being the log likelihood ratio and the new critical value c' being another critical value to be determined later. In fact, we only need to care about the data dependent part of the likelihood ratio statistic $T(X^n)$.

Example 7.22 (Gaussian mean)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, 1)$, and the hypotheses are

$$H_0\text{: } \mu = \mu_0, \text{ v.s., } H_1\text{: } \mu = \mu_1.$$

Derive the likelihood ratio test statistic. ♢

Example 7.23 (Binomial mean)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta)$, and the hypotheses are

$$H_0\text{: } \theta = \theta_0, \text{ v.s., } H_1\text{: } \theta = \theta_1.$$

Derive the likelihood ratio test statistic. ♢

Example 7.24 (Poisson LRT: simple v.s. simple)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Pois(\lambda)$, and the hypotheses are

$$H_0\text{: } \lambda = \lambda_0, \text{ v.s., } H_1\text{: } \lambda = \lambda_1.$$

The likelihood ratio statistic in logarithm is

$$\log\Lambda_n(X^n) = \log\frac{\mathcal{L}_n(\lambda_1)}{\mathcal{L}_n(\lambda_0)} = \left(\sum\nolimits_{i=1}^n X_i\right)\log(\lambda_1/\lambda_0) - n(\lambda_1 - \lambda_0),$$

which is a monotonic function of $S_n = \sum_{i=1}^n X_i$.

If $\theta_1 > \theta_0$, $\log\Lambda_n(X^n)$ is monotonically increasing in S_n, and the rejection region is

$$S_n \geq c_+$$

with c_+ to be determined and dependent on the size α. By additivity of independent Poisson random variables, under the null hypothesis,

$$S_n \sim Pois(n\lambda_0)$$

and the size-critical value relationship is

$$\alpha = \mathbb{P}_0(S_n \geq c_+) = 1 - F_0(c_+ - 1)$$

where $F_0(\cdot)$ denotes the CDF of $Pois(n\lambda_0)$. With a pre-specified α, we can determine the critical value c_+. Due to the discrete nature of Poisson, the terminology level is more appropriate than size.

Similarly, under the alternative hypothesis,

$$S_n \sim Pois(n\lambda_1).$$

With c_+ chosen, the power is

$$\mathbb{P}_1(S_n \geq c_+) = 1 - F_1(c_+ - 1)$$

where $F_1(\cdot)$ denotes the CDF of $Pois(n\lambda_1)$.

If $\theta_1 < \theta_0$, $\log\Lambda_n(X^n)$ is monotonically decreasing in S_n, and the rejection region is

$$0 \leq S_n \leq c_-$$

with c_- to be determined and dependent on the size α. The size-critical value relationship is

$$\alpha = \mathbb{P}_0(0 \leq S_n \leq c_-) = F_0(c_-)$$

where $F_0(\cdot)$ denotes the CDF of $Pois(n\lambda_0)$. With c_- chosen, the power is

$$\mathbb{P}_1(0 \leq S_n \leq c_-) = F_1(c_-)$$

where $F_1(\cdot)$ denotes the CDF of $Pois(n\lambda_1)$.

♢

The Neyman-Pearson lemma applies to the simple v.s. simple hypothesis testing problems. The Karlin-Rubin theorem is an extension of the Neyman-Pearson lemma for composite hypotheses.

Definition 7.5 (MLR)**.** A family of density functions $\{f_\theta(x) : \theta \in \Theta\}$ indexed by a parameter θ taking values in an ordered set Θ has a monotone likelihood ratio (MLR) in the statistic $T(X^n)$ if for any $\theta_1 < \theta_2$, the likelihood ratio $\frac{f_{\theta_1}}{f_{\theta_2}}(x^n)$ is non-decreasing function of $T(x^n)$.

Remark. MLR is a property for families of distributions. A special case is the one observation case $T(X^n) = X$.

Example 7.25. The following distributions have MLR in $\sum_{i=1}^n X_i$.

- Exponential
- Poisson
- Gaussian mean $N(\mu, 1)$.

♢

Theorem 7.9 (Karlin-Rubin)**.** *Suppose $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta)$ where $\theta \in \Theta \subset \mathbb{R}$, and the likelihood ratio $\lambda(X^n) \equiv \frac{\mathcal{L}_n(\theta_1)}{\mathcal{L}_n(\theta_0)}$ is non-decreasing in $T(X^n)$ for any $\theta_1 \geq \theta_0$, then the likelihood ratio test is the uniformly most powerful (UMP) test for testing*

$$H_0 : \theta \leq \theta_0 \text{ v.s. } H_1 : \theta > \theta_0.$$

Remark. UMP tests do not exist for vector parameters or for two-sided tests. The above likelihood ratio test is exactly the same test as testing

$$H_0 : \theta = \theta_0 \text{ v.s. } H_1 : \theta > \theta_0.$$

Example 7.26 (Poisson LRT: one-sided)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Pois(\lambda)$, and the hypotheses are

$$H_0\text{: } \lambda \leq \lambda_0, \text{ v.s., } H_1\text{: } \lambda > \lambda_0.$$

By the Karlin-Rubin theorem, the UMP test is to reject the null if

$$S_n \geq c_\alpha$$

for some critical value c_α. ♢

7.6 The (Generalized) Likelihood Ratio test

Reading(s) 7.6. Wasserman (2004): 10.6
YouTube video: How Mendel's pea plants helped us understand genetics

Suppose the hypothesis testing problem is to test

$$H_0 : \theta \in \Theta_0 \text{ v.s. } H_1 : \theta \in \Theta_1,$$

where θ is potentially a vector and H_0 can be composite. For multi-dimensional parameters, various Wald test statistics can be constructed by the multiparameter delta method. However, in general, the Wald test is less powerful than the likelihood ratio test.

The likelihood ratio test statistic is based on the log likelihood ratio

$$\lambda_n = 2\log\left(\frac{\sup_{\theta\in\Theta}\mathcal{L}_n(\theta)}{\sup_{\theta\in\Theta_0}\mathcal{L}_n(\theta)}\right) = 2\log\left(\frac{\mathcal{L}_n(\hat{\theta})}{\mathcal{L}_n(\hat{\theta}_0)}\right),$$

where $\hat{\theta}$ is the MLE and $\hat{\theta}_0$ is the MLE when the parameter space is constrained to Θ_0. Since $\Theta_0 \subset \Theta$, $\mathcal{L}_n(\hat{\theta}) \geq \mathcal{L}_n(\hat{\theta}_0)$ and $\lambda_n \geq 0$ by construction.

By consistency of MLE, we expect $\hat{\theta}$ to be approximately the truth. If $\hat{\theta}_0$ deviates too much from $\hat{\theta}$ in terms of likelihood, H_0 is not compatible with the observed data. Therefore, the likelihood ratio test is to reject the null hypothesis if

$$\lambda_n \geq c_\alpha$$

where c_α is some critical value such that the size is α.

Towards a general method of choosing the critical values, the Wilks' theorem (Wilks, 1938) pins down the critical value c_α such that the *asymptotic* size of the likelihood ratio test is α.

Theorem 7.10 (Wilks)**.** *Suppose the dimension of Θ_0 is q and the dimension of Θ is r, then under the null hypothesis,*

$$\lambda_n \xrightarrow{d} \chi^2_{r-q},$$

where χ^2_{r-q} denotes the χ^2 distribution with the degree of freedom being $r-q$.

Moreover, the p-value for the test is $\mathbb{P}(\chi^2_{r-q} \geq \lambda_n)$.

Proof. See Section 17.2 of Keener (2010) for an accessible proof of the Wilks' theorem. □

Example 7.27 (Normal)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$. The hypotheses are

$$H_0 : \mu = \mu_0, \text{ v.s., } H_1 : \mu \neq \mu_0.$$

Then the parameter spaces are $\Theta = \{(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 \in \mathbb{R}_+\}$, and $\Theta_0 = \{(\mu_0, \sigma^2) : \sigma^2 \in \mathbb{R}_+\}$. Clearly, Θ contains two free parameters and $dim(\Theta) = 2$; Θ_0 contains only one free parameter and $dim(\Theta_0) = 1$. Therefore, by Wilks' theorem, under the null hypothesis,

$$\lambda_n \xrightarrow{d} \chi^2_1.$$

Based on the asymptotic distribution theory, we can perform the asymptotically size α likelihood ratio test:

reject the null H_0 if $\lambda_n \geq \chi^2_{1,\alpha}$

where $\chi^2_{1,\alpha}$ denotes the upper α quantile of χ^2_1. Additionally, the p value based on the likelihood ratio test statistic is approximately $\mathbb{P}(\chi^2_1 \geq \lambda_n)$.

To compute the test statistic λ_n, we first find the MLEs. The MLEs are straightforward to compute: $\hat{\mu} = \frac{1}{n}\sum_{i=1}^n X_i$, $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \hat{\mu})^2$, and $\hat{\sigma}^2_0 = \frac{1}{n}\sum_{i=1}^n (X_i - \mu_0)^2$. Then by direct calculation, the test statistic is simply

$$\lambda_n = 2\left(\ell_n(\hat{\mu}, \hat{\sigma}^2) - \ell_n(\mu_0, \hat{\sigma}^2_0)\right) = n\left(\log(\hat{\sigma}^2_0) - \log(\hat{\sigma}^2)\right)$$

where $\ell_n(\cdot)$ denotes the log likelihood function.

We can examine the finite sample performance in the simulation where $n = 30$ and $X_1, .., X_n \overset{iid}{\sim} N(0, 4)$. Figure 7.7 implies that the asymptotic distribution well matches the finite sample sampling distribution.

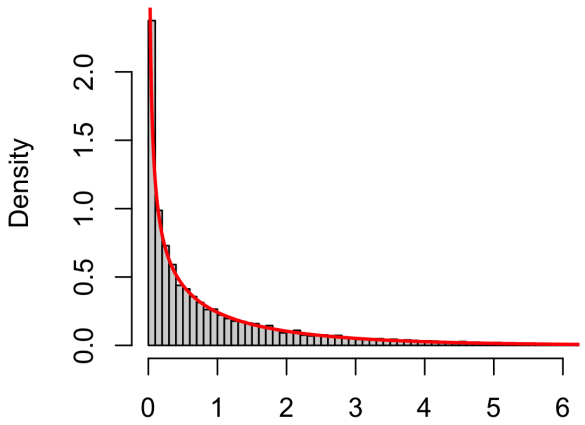


Figure 7.7: Sampling distribution of λ_n under H_0 overlaid by the pdf of χ^2_1 (in red)

```r
mu <- 0
sig <- 2
mu0 <- 0
LR.gaussian <- function(X,mu0){
    n <- length(X)
    sig2.hat <- mean((X-mean(X))^2)
    sig2.hat.0 <- mean((X-mu0)^2)
    return((sig2.hat/sig2.hat.0)^(-n/2))
}
Nmc <- 1e4
n <- 30
x <- matrix(rnorm(Nmc*n, mu,sig), nrow = Nmc)
LRs <- apply(x, 1, function(xx) LR.gaussian(xx,0))
lambda <- 2*log(LRs)
hist(lambda,breaks = "FD", probability = TRUE, main = "", xlim = c(0,6))
lines(sort(lambda), dgamma(sort(lambda), 1/2,rate = 1/2), col= "red", lwd=2)
```

♢

Example 7.28 (Mendel's peas)**.** Gregor Mendel bred peas of four types of progeny: $\{green, yellow\} \times \{round, wrinkled\}$. The number of peas in each type follows a multinomial distribution with the probability vector $p \equiv (p_1, p_2, p_3, p_4)$ denoting the probabilities for round yellow, wrinkled yellow,

round green, and wrinkled green, respectively. Denote the numbers of the peas in each progeny as $X \equiv (X_1, X_2, X_3, X_4)$, and

		color	
		yellow	green
shape	round	315	108
	wrinkled	101	32

Table 5: The number of peas in each type

The numbers of peas in each type naturally follow a multinomial distribution $X \sim MN(n, p)$ where n is the total number of peas and p is the probability vector. His theory of inheritance predicts that the probability vector is $p_0 \equiv (\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16})$. The same information is presented in the table below.

		color	
		yellow	green
shape	round	9/16	3/16
	wrinkled	3/16	1/16

Table 6: H_0 for the four types of progeny

The hypothesis testing problem is formulated to test

$$H_0 : p = p_0 \text{ v.s. } H_1 : p \neq p_0.$$

To perform the likelihood ratio test, we need to pin down the dimension difference between the unconstrained parameter space and the null hypothesis parameter space, and compute the likelihood ratio test statistic λ_n with MLEs.

The unconstrained parameter space is $\Theta = \{p : \sum_{i=1}^4 p_i = 1, p_j \in [0,1] \text{ for } j = 1, ..., 4\}$ and $dim(\Theta) = 3$ due to the equality constraint; $\Theta_0 = \{p = p_0\}$ and $dim(\Theta_0) = 0$ as Θ_0 only contains one point. Therefore, by Wilks' theorem, under the null hypothesis,

$$\lambda_n \xrightarrow{d} \chi^2_3.$$

Therefore, we can perform the asymptotically size α test:

$$\text{reject } H_0 \text{ if } \lambda_n \geq \chi^2_{3,\alpha}$$

where $\chi^2_{3,\alpha}$ denotes the upper α quantile of χ^2_3. Additionally, the p value of the likelihood ratio test is approximately $\mathbb{P}(\chi^2_3 \geq \lambda_n)$.

To compute the test statistic, we need to find the MLE of p. The MLE of p is the empirical proportion:

$$\hat{p} = X/n.$$

With data $X = (315, 101, 108, 32)$ and $n = 556$, we can compute the test statistic:

$$\lambda_n = 2\log\left(\frac{\prod_{i=1}^4 \hat{p}_i^{X_i}}{\prod_{i=1}^4 p_{0,i}^{X_i}}\right) = 2\sum_{i=1}^4 X_i \log\left(\frac{X_i/n}{p_{0,i}}\right) \approx 0.48.$$

With the $\alpha = 5\%$ chosen before the experiment, we retain the null hypothesis. Additionally, the p value is approximately 0.92.

We can examine the approximation accuracy of the asymptotic sampling distribution. Figure 7.8 implies that χ_3 well matches the finite sample sampling distribution. The support of $MN(n, p_0)$ contains $\binom{n+3}{3} \approx 28.96 \times 10^6$ outcomes, so it is computationally more efficient to sample from the null distribution than brute-force calculation.

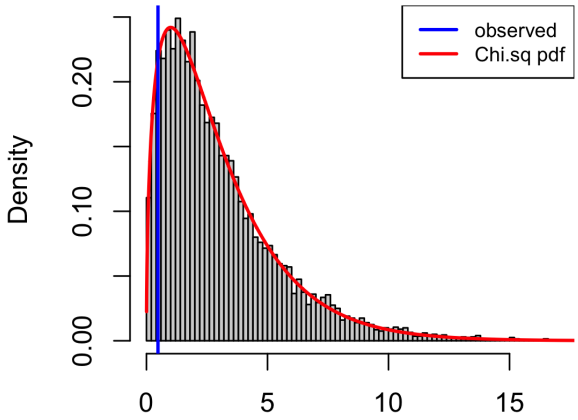


Figure 7.8: Sampling distribution of λ_n under H_0 overlaid by the pdf of χ^2_3 (in red). The observed λ_n is in blue.

```
LR.mn <- function(X, pp){
    n <- sum(X)
    pp.hat <- X/n
    return( prod((pp.hat/pp)^X))
}
X <- c(315,101,108,32) ## data
p0 <- c(9/16, 3/16,3/16,1/16) ## H_0
x <- rmultinom(Nmc, sum(X), prob = p0) %>%t()
LRs <- apply(x,1,function(xx) LR.mn(xx, p0))
lambda <- 2*log(LRs)

hist(lambda,breaks = "FD", probability = TRUE, main = "", xlim = c(0,17))
lines(sort(lambda), dgamma(sort(lambda), 3/2,rate = 1/2), col= "red", lwd=2)
abline(v = 2*log(LR.mn(X,p0)),col="blue",lwd=2)
legend("topright", c("observed", "Chi.sq pdf"), col = c("blue","red"), lwd = 2)
```

♢

What if there is no dimension difference? The Wilks' theorem is most useful when the null hypothesis imposes some equality constraint(s). In many hypothesis testing problems, the dimension of the parameter space under H_0 is the same as the dimension of the unconstrained parameter space, e.g., one sided tests. In such cases, the Wilks theorem is not applicable. Unfortunately, there is no nice general solution to it.

We can start with the (generalized) likelihood ratio test statistic, and see if the sampling distribution under H_0 is available. For one-parameter one-sided problems, we would like to explore the possibility of MLR property to apply the UMP theory.

Example 7.29 (One-sided Poisson)**.** Suppose $X_1, ..., X_n \overset{iid}{\sim} Pois(\lambda)$, and the hypotheses are

$$H_0\text{: } \lambda \leq \lambda_0, \text{ v.s., } H_1\text{: } \lambda > \lambda_0.$$

The generalized likelihood ratio test statistic is

$$\lambda_n(\boldsymbol{X}) = \log\frac{\sup_{\lambda>0}\mathcal{L}_n(\lambda)}{\sup_{\lambda\leq\lambda_0}\mathcal{L}_n(\lambda)} = \left(\sum_{i=1}^n X_i\right)\log(\hat{\lambda}/\hat{\lambda}_0) - n(\hat{\lambda}-\hat{\lambda}_0),$$

where $\hat{\lambda} = \bar{X}_n$ is the unconstrained MLE, and $\hat{\lambda}_0 = \lambda_0 \wedge \bar{X}_n$ is the constrained MLE. Note the statistic drops the 2 factor.

The (generalized) likelihood ratio test rejects the null hypothesis if

$$\lambda_n(\boldsymbol{X}) \geq c$$

for some positive critical value c.

If $\bar{X}_n \leq \lambda_0$, then $\hat{\lambda} = \hat{\lambda}_0$ and $\log\Lambda_n(\boldsymbol{X}) = 0$. If $\bar{X}_n > \lambda_0$, the (generalized) log likelihood ratio λ_n is not trivial and

$$\lambda_n(\boldsymbol{X}) = \log\frac{\mathcal{L}_n(\bar{X}_n)}{\mathcal{L}_n(\lambda_0)} = n\bar{X}_n\log(\bar{X}_n/\lambda_0) - n(\bar{X}_n-\lambda_0) = n\bar{X}_n\left(\log(\bar{X}_n) - \log(\lambda_0) - 1\right) + n\lambda_0$$

which is a monotonic function of $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$. To see why, check the sign of the first derivative of $\lambda_n(\boldsymbol{X})$ with respect to $\bar{X}_n$: $\frac{\partial}{\partial \bar{X}_n}\lambda_n = n\left(\log(\bar{X}_n) - \log(\lambda_0)\right) > 0$ by the assumption that $\bar{X}_n > \lambda_0$.

Therefore, the (generalized) likelihood ratio test rejects the null hypothesis if

$$\bar{X}_n \geq c$$

for some positive critical value c, which is equivalent to reject the null hypothesis if

$$S_n \geq nc$$

for some positive critical value c.

Note $\{F_\lambda(t) : \lambda \leq \lambda_0\}$, the CDF's of the Poisson distributions $\{Pois(\lambda) : \lambda \leq \lambda_0\}$, are also monotonic in λ for fixed $t > 0$, the size of the test is

$$\sup_{\lambda\leq\lambda_0}\mathbb{P}_\lambda(\bar{X}_n \geq c) = \mathbb{P}_{\lambda_0}(\bar{X}_n \geq c)$$

where $S_n|\lambda \sim Pois(n\lambda)$ and $\bar{X}_n|\lambda \sim \frac{1}{n}Pois(n\lambda)$ is a scaled Poisson random variable. The size and critical value relationship is

$$\alpha = \mathbb{P}_{\lambda_0}(S_n \geq nc) = 1 - F_0(nc-1)$$

where $F_0(\cdot)$ denotes the CDF of $Pois(n\lambda_0)$.

Alternatively, by CLT, the test statistic is asymptotically Normal under λ_0:

$$\bar{X}_n \dot{\sim} N(\lambda_0, \lambda_0/n),$$

and the critical value for size α test is

$$\lambda_0 + z_\alpha\sqrt{\lambda_0/n}$$

where z_α denotes the $\alpha-$upper quantile of the standard Normal distribution.

$\diamondsuit$

7.7 The score test and the triad

The (generalized) likelihood ratio test is a powerful testing procedure, but evaluating the maximum likelihoods can be computationally challenging and/or expensive in some applications. In contrast, evaluating the gradient of the likelihood function is often more feasible. C.R. Rao (Rao, 1948) examines the gradient of the log-likelihood function evaluated at the null hypothesis. Recall the expectation of the score function under the true data generating process is 0. The realized score function under H_0 should be close to 0 if the null hypothesis H_0 is true.

Suppose $X_1, ..., X_n \overset{iid}{\sim} f(x;\theta)$, and the null hypothesis is simple, $H_0 : \theta = \theta_0$, where θ is a scalar. Then by CLT, under H_0,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} s(X_i;\theta_0) \xrightarrow{d} N(0, I(\theta_0)), \text{ as } n \to \infty,$$

where $I(\theta_0) \equiv V(s(X_1;\theta_0))$ by definition of Fisher information. By Slutsky theorem, it also holds

$$\frac{1}{\sqrt{nI(\theta_0)}} \sum_{i=1}^{n} s(X_i;\theta_0) \xrightarrow{d} N(0,1), \text{ as } n \to \infty.$$

The test statistic of the score test is $T_s(\boldsymbol{X}) \equiv \frac{1}{\sqrt{nI(\theta_0)}} \sum_{i=1}^{n} s(X_i;\theta_0) = \frac{\dot{\ell}_n(\theta_0)}{\sqrt{I_n(\theta_0)}}$, and its approximate sampling distribution is standard Normal. For an approximately level/size α test, the critical value $c_\alpha = z_{\alpha/2}$ for two-sided alternatives and $c_\alpha = z_\alpha$ for one-sided alternatives.

Example 7.30. Suppose $X_1, ..., X_n \overset{iid}{\sim} Ber(\theta)$, the null hypothesis is simple, $H_0 : \theta = \theta_0 = \frac{1}{2}$, and the alternative hypothesis is two-sided, $H_1 : \theta \neq \theta_0$.

The likelihood function is $\mathcal{L}_n(\theta) = \theta^{\sum_{i=1}^n X_i}(1-\theta)^{n-\sum_{i=1}^n X_i}$, the log-likelihood function is

$$\ell_n(\theta) = (\sum_{i=1}^{n} X_i)\log(\theta) + (n - \sum_{i=1}^{n} X_i)\log(1-\theta),$$

the score function (for the whole data) evaluated at θ_0 is

$$\dot{\ell}_n(\theta_0) = \frac{1}{\theta_0}\sum_{i=1}^{n} X_i - \frac{1}{1-\theta_0}(n - \sum_{i=1}^{n} X_i) = \frac{n}{\theta_0(1-\theta_0)}(\bar{X}_n - \theta_0),$$

and the Fisher information (for a single data point) is $I(\theta_0) = -\mathbb{E}_{\theta_0}[\ddot{\ell}_X(\theta_0)] = \frac{1}{\theta_0(1-\theta_0)}$.

Therefore, the test statistic of the score test is

$$T_s(\boldsymbol{X}) = \frac{\dot{\ell}_n(\theta_0)}{\sqrt{I_n(\theta_0)}} = \sqrt{\frac{n}{\theta_0(1-\theta_0)}}\left(\bar{X}_n - \theta_0\right) = \frac{2}{\sqrt{n}}\left(\bar{X}_n - \frac{1}{2}\right).$$

Note $\bar{X}_n$ is exactly the maximum likelihood estimator for θ in the last display, the fundamental theorem of maximum likelihood estimation also guarantees the asymptotic distribution of $\sqrt{\frac{n}{\theta_0(1-\theta_0)}}\left(\bar{X}_n - \theta_0\right)$ is the standard Normal.

♢

The Wald test, the likelihood ratio test, and the Lagrange multiplier (score) test are asymptotically equivalent, so they are called the triad of hypothesis testing in the literature. (Buse, 1982; Engle, 1984) If the log-likelihood is quadratic in the parameter(s), the three tests' test statistics are identical. We illustrate their finite-sample differences and connections in the following univariate problem where θ is the parameter of interest.

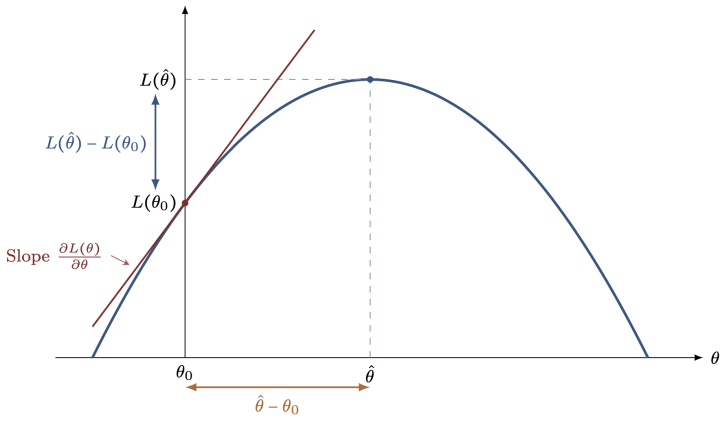


Figure 7.9: The triad of hypothesis testing

Theorem 7.11. *Suppose the d-dimensional random vector $X \in \mathbb{R}^d$ follows the multivariate Normal distribution $N_d(\mathbf{0}, \Sigma)$ where Σ is a positive definite covariance matrix, then*

$$X^T \Sigma^{-1} X \sim \chi^2_d.$$

Proof. Since Σ is a positive definite covariance matrix, it has the eigen-decomposition

$$\Sigma = V \Lambda V^T$$

where V is an orthonormal matrix such that $V^T V = I_d$, and $\Lambda = \text{diag}(\lambda_1, \lambda_2, ..., \lambda_d)$ is a diagonal matrix with $\lambda_i > 0$ for all $i = 1, ..., d$. Under the eigen-decomposition, $\Sigma^{-1} = V \Lambda^{-1} V^T$. Define

$$\Sigma^{-1/2} \equiv V \Lambda^{-1/2} V^T$$

where $\Lambda^{-1/2} \equiv \text{diag}(\lambda_1^{-1/2}, \lambda_2^{-1/2}, ..., \lambda_d^{-1/2})$. Due to orthonormality of V, $\Sigma^{-1/2}\Sigma^{-1/2} = \Sigma^{-1}$ and $\Sigma^{-1/2}\Sigma\Sigma^{-1/2} = I_d$. By linearity of multivariate Normal distribution,

$$\Sigma^{-1/2} X \sim N_d(\mathbf{0}, I_d).$$

Recall the defintion of χ^2_d distribution is $\sum_{i=1}^d Z_i^2$ where $Z_1, ..., Z_d \overset{iid}{\sim} N(0, 1)$. That is, $\boldsymbol{Z} \sim N(0, I_d)$, then $\boldsymbol{Z}^T \boldsymbol{Z} \sim \chi^2_d$. Therefore,

$$X^T \Sigma^{-1} X = (\Sigma^{-1/2} X)^T (\Sigma^{-1/2} X) \sim \chi^2_d.$$

□

Section 6.3 of Strang (2022) provides an excellent introduction to *the* fundamental property of symmetric square matrices. For our purpose, it is more convenient to work with orthonormal matrices than orthogonal matrices, so that we obtain the unique "sandwich" factorization.

For multiparameter problems where $H_0 : \theta = \theta_0$ and $\theta \in \Theta \subset \mathbb{R}^p$, the limiting distribution implied by (multivariate) CLT

$$\frac{1}{\sqrt{n}} \nabla \ell_n(\theta_0) \xrightarrow{d} N_p(\mathbf{0}, I(\theta_0))$$

implies the approximate sampling distribution

$$\nabla \ell_n(\theta_0)^T I_n(\theta_0)^{-1} \nabla \ell_n(\theta_0) \overset{d}{\approx} \chi^2_p.$$

Therefore, the test statistic of the score test is $T_s(\boldsymbol{X}) = \nabla \ell_n(\theta_0)^T I_n(\theta_0)^{-1} \nabla \ell_n(\theta_0)$, and the approximate α level/size test is to reject the null hypothesis if $T_s(\boldsymbol{X}) > \chi^2_{p,\alpha}$.

The Wald test introduced in Section 7.3 works with scalar parameters, both test statistics and critical values being scalars. The same idea can be extended to multivariate cases. Suppose under H_0,

$$\sqrt{n}(g(\hat{\theta}_n) - g(\theta_0)) \xrightarrow{d} N_p(0, K(\theta_0))$$

where $g(\cdot)$ is some transformation, and $K(\theta_0)$ is a $p \times p$ positive definite square matrix which potentially depends on θ_0. Then, by Slutsky theorem, the "sandwich" form

$$n(g(\hat{\theta}_n) - g(\theta_0))^T K(\theta_0)^{-1} (g(\hat{\theta}_n) - g(\theta_0)) \xrightarrow{d} \chi^2_p.$$

Similar testing procedure to the score test can be constructed.

7.8 The χ^2 tests

The χ^2 tests refer to tests whose limiting distributions of the test statistics are χ^2.

7.8.1 Pearson's χ^2 test

Pearson's χ^2 test is used for multinomial data. Suppose we observe $X = (X_1, ..., X_k)$ where each X_i denotes the number of observations from the category i. Naturally, the model is $X \sim MN(\pi)$ where $\pi = (\pi_1, ..., \pi_k)$ is the parameter vector.

The hypothesis testing problem is to test

$$H_0 : \pi = \pi_0 \text{ v.s. } H_1 : \pi \neq \pi_0$$

where $\pi_0 = (\pi_{0,i}, ..., \pi_{0,k})$ is some fixed vector.

The test statistic of Pearson's χ^2 test is

$$T = \sum_{j=1}^{k} \frac{(X_j - E_j)^2}{E_j}$$

where $E_j = n\pi_{0,j}$ is the expected value of X_j under H_0.

Theorem 7.12 (Pearson's χ^2 test)**.** *Under the null H_0, as $n = \sum_{i=1}^{k} X_i \to \infty$,*

$$T \xrightarrow{d} \chi^2_{k-1}.$$

Proof. TBD □

Remark. In the asymptotic analysis, k, the number of categories, is fixed, while $n \to \infty$.

Therefore, an (asymptotic) size-α Pearson's χ^2 test is to reject the null H_0 if $T \geq \chi^2_{k-1,\alpha}$. Also, the p-value of the test is approximately $\mathbb{P}(\chi^2_{k-1} \geq t_{obs})$ where t_{obs} denotes the observed test statistic.

Example 7.31 (Mendel's peas)**.** Recall in Mendel's experiments, Mendel bred four different types of peas. The number of peas in each type naturally follows a multinomial distribution. So apart from the likelihood ratio test, we could also perform Pearson's χ^2 test.

Mendel's hypotheses are

$$H_0 : p = p_0 \text{ v.s. } H_1 : p \neq p_0$$

where $p_0 \equiv (\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16})$.

With sample size $n = 556$ and p_0, the expected numbers are

$$E = n \times p_0 = 556 \times (\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16}) = (312.75, 104.25, 104.25, 34.75).$$

With data $X = (315, 101, 108, 32)$, we can compute the test statistic

$$T = \frac{(315 - 312.75)^2}{312.75} + \frac{(101 - 104.25)^2}{104.25} + \frac{(108 - 104.25)^2}{104.25} + \frac{(32 - 34.75)^2}{34.75} = 0.47.$$

If we choose $\alpha = 0.05$, the upper quantile $\chi^2_{3,\alpha} = 7.815$ which is greater than the observed test statistic. So we retain the null hypothesis.

Recall the likelihood ratio (LR) test uses the same limiting distribution—χ^2_3. Also, the test statistic of the LR test is 0.48, which is close to Pearson's χ^2 test statistic.

♢

7.8.2 The Goodness-of-fit test

In the previous testing problems, the null hypothesis is a subset of the parameter space. In some cases, we are more intersted in the distributional aspect of the data than the parameters. The hypothesis testing problem seeks to test

$$H_0\text{: } X_i \overset{iid}{\sim} f(x;\theta), \text{ v.s. } H_1\text{: } X_i \overset{iid}{\sim} g(x;\eta) \text{ for some } g \neq f,$$

where f and g are two pdf/pmf's. For instance, H_0: $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$ and we are more interested in if the data is Gaussian than knowing (μ, σ^2).

Testing such null hypotheses is to test the goodness-of-fit of the distributions of the null hypotheses to the data. This can be done in the spirit of Pearson's χ^2 test.

The first step is to discretize the distribution of the null hypothesis and to bin the data. In this step, we cast the original problem to a problem that can be solved by Pearson's χ^2 test.

- Choose a partition of the support:

  $$\text{supp}(X) = \cup_{i=1}^k I_i.$$

  For instance, if $\text{supp}(X) = \mathbb{R}$, then we can partition $\mathbb{R}$ with $\cup_{i=1}^3 I_i$ where $I_1 = (-\infty, 0]$, $I_2 = (0, 1]$, and $I_3 = (1, \infty)$. Of course, we can choose a finer partition $\cup_{i=1}^k I_i$ with a large k.

- Define

  $$p_i(\theta) = \mathbb{P}_{0,\theta}(X_1 \in I_i) = \int_{I_i} f(x;\theta)dx$$

  where θ is allowed to be unspecified by H_0.

- Define Y_i for $i = 1, ..., k$ as follows,

  $$Y_i = \sum\nolimits_{j=1}^n \mathbb{1}(X_j \in I_i)$$

  where the original data $\{X_i\}_{i=1}^n$ can be continuous.

After the discretization, $Y \sim MN(\pi)$ for some π, making Pearson's χ^2 test applicable.

Precise null hypotheses Suppose the null hypothesis H_0 also specifies a true value $\theta_0 \in \Theta_0$ that generates $\{X_i\}_{i=1}^n$ and by construction,

$$Y = (Y_1, ..., Y_k) \sim MN(p(\theta_0))$$

where $p(\theta_0) = (p_{\mathbb{1}}(\theta_0), ..., p_k(\theta_0))$. Pearson's χ^2 test statistic is immediately

$$T(p(\theta_0)) = \sum\nolimits_{i=1}^k \frac{(Y_i - np_i(\theta_0))^2}{np_i(\theta_0)},$$

and we reject the null hypothesis H_0 if $T(p(\theta_0)) \geq \chi^2_{k-1,\alpha}$ for some size α.

Composite null hypotheses In many cases, θ_0 is left unspecified in H_0. Therefore, $T(p(\theta_0))$ contains unknown parameter and is unfeasible. Following the plug-in principle, we would like to replace the unknown truth $p(\theta_0)$ with its MLE $\hat{p} \equiv (p_{\mathbb{1}}(\hat{\theta}_D), ..., p_k(\hat{\theta}_D))$ maximizing the discretized likelihood:

$$\hat{\theta}_D = \arg\max_{\theta} \prod_{i=1}^{k} p_i(\theta)^{Y_i}$$

where the subscript "D" denotes the MLE estimates associated with the discretized likelihood.

With $\hat{\theta}_D$, the test statistic that contains only data is

$$T(\hat{p}) = \sum_{i=1}^{k} \frac{(Y_i - n\hat{p}_i)^2}{n\hat{p}_i},$$

where $\hat{p}_i = p_i(\hat{\theta}_D)$. Under the null hypothesis H_0, as $n \to \infty$,

$$T(\hat{p}) \xrightarrow{d} \chi^2_{k-1-s}$$

where s is the dimension of θ.

Based on the asymptotic theory, we reject the null hypothesis H_0 if

$$T(\hat{p}) \geq \chi^2_{k-1-s,\alpha}$$

for some size α.

It is tempting to plug in the MLE of θ under H_0:

$$T(p(\hat{\theta})) = \sum_{i=1}^{k} \frac{(Y_i - np_i(\hat{\theta}))^2}{np_i(\hat{\theta})},$$

where $\hat{\theta} = \arg\max_\theta \prod_{i=1}^{n} f(X_i; \theta)$ is the MLE estimates associated with the original likelihood. However, the limiting distribution of $T(p(\hat{\theta}))$ is not in the χ^2 family. (Chernoff and Lehmann, 1954)

Example 7.32 (Uniform distribution with *known* support)**.** Suppose the hypotheses are

$$H_0\text{: } X_i \overset{iid}{\sim} Unif[0,1], \text{ v.s. } H_1\text{: } X_i \overset{iid}{\sim} f(x;\eta) \text{ for some } f \neq Unif[0,1].$$

We first discretize the H_0 distribution and the data with the partition $\cup_{i=1}^{k} I_i$ where $I_i = (\frac{i-1}{k}, \frac{i}{k}]$. Under the null hypothesis and the partition,

$$Y \sim MN(p_0)$$

where $p_0 = (1/k, ..., 1/k)$, which involves no unknown parameters. Therefore, we obtain the test statistic by Pearson's χ^2 test directly:

$$T = \sum_{i=1}^{k} \frac{(Y_i - n/k)^2}{n/k},$$

and we reject the null hypothesis if $T \geq \chi^2_{k-1,\alpha}$ for some size α.

♢

Example 7.33 (Uniform distribution with *unknown* support)**.** Suppose the hypotheses are

$$H_0\text{: } X_i \overset{iid}{\sim} Unif[\theta_1, \theta_2], \text{ v.s. } H_1\text{: } X_i \overset{iid}{\sim} f(x;\eta) \text{ for some } f \neq Unif[\theta_1, \theta_2].$$

We first discretize the support of the H_0 distribution with the partition $\cup_{i=a}^{b} I_i$ where $I_i = ((i-1)\ell, i\ell]$. The lengths of the intervals are chosen to be ℓ uniformly for simplicity. The hyper-parameters a and b are two integers chosen such that all data points are contained in $[(a-1)\ell, b\ell]$.

Under the null hypothesis and the partition,

$$Y \sim MN(p_0(\theta))$$

where $\theta = (\theta_1, \theta_2)$, $p_0(\theta) = (p_{0,a}(\theta), ..., p_{0,b}(\theta))$ and

$$p_{0,j}(\theta) = \mathbb{P}(X_1 \in I_j) = \int_{(j-1)\ell}^{j\ell} \frac{\mathbb{1}(x \in [\theta_1, \theta_2])}{\theta_2 - \theta_1} dx = \frac{0 \vee (\theta_2 \wedge j\ell - \theta_1 \vee (j-1)\ell)}{\theta_2 - \theta_1}.$$

Then the discretized likelihood function is the following multinomial pmf

$$\prod_{j=a}^{b} p_{0,j}(\theta)^{Y_j}.$$

```
dlik <- function(par,Y,loc){### discretized likelihood in log
    thetal=par[1]
    thetar=par[2]
    K <- length(loc)
    pj <- pmax(pmin(loc[-1],thetar)-pmax(loc[-K],thetal),0)/(thetar-thetal)
    return(sum(Y[pj>0]*log(pj[pj>0])))
}
```

Next, we solve for the MLE of the discretized likelihood

$$\hat{\theta}_D = \arg\max_{\theta} \prod_{i=a}^{b} p_{0,i}(\theta)^{Y_i}.$$

The test statistic is

$$T = \sum_{i=1}^{k} \frac{(Y_i - n\hat{p}_i)^2}{n\hat{p}_i}$$

where $\hat{p}_i = p_{0,i}(\hat{\theta}_D)$, and we reject the null hypothesis if $T \geq \chi^2_{k-2,\alpha}$ for some size α.

To illustrate, we work out the χ^2 uniformity test with the synthetic data generated from some Normal distribution.

```
X <- rnorm(1e3,0,5)
endpts <- seq(from = floor(min(X)), to= ceiling(max(X)),by=0.5)
Y <- table(cut(X,endpts))
res.opt <-optim(c(min(X)-0.1,max(X)+0.1), dlik, method = "L-BFGS-B", Y=Y, loc=endpts,
                lower=c(-Inf, max(X)-1/1e7), upper=c(min(X)+1/1e7, Inf),
                control=list(fnscale=-1))
opt.par <- res.opt$par
K <- length(endpts)
thetal <- opt.par[1]
thetar <- opt.par[2]
hat.p <- pmax(pmin(endpts[-1],thetar)-pmax(endpts[-K],thetal),0)/(thetar-thetal)
EY <- length(X)*hat.p ### expected number of observations
test.stat <- sum((Y[EY>0]-EY[EY>0])^2/EY[EY>0]) ### test statistic
1-pchisq(test.stat,df=K-2-length(opt.par)) ## p value
```

♢

Example 7.34 (Testing Normality)**.** We have been assuming the data is from some Normal distribution without justifying the assumption. Now we can test this assumption statistically via the χ^2 test.

Now the hypotheses are

$$H_0\text{: } X_i \overset{iid}{\sim} N(\mu,\sigma^2), \text{ v.s. } H_1\text{: } X_i \overset{iid}{\sim} f(x;\eta) \text{ for some } f(\cdot;\eta) \neq N(\mu,\sigma^2).$$

We first discretize the support of the H_0 distribution by the partition $\cup_{i=a}^{b} I_i$ where $I_i = (i-1, i] \times \ell$ and ℓ is a constant. The lengths of the intervals are chosen to be ℓ uniformly for simplicity. The hyper-parameters a and b are chosen such that all observations are contained in $[(a-1)\ell, b\ell]$, and the hyper-parameter ℓ is chosen such that the number of intervals is neither too big nor too small.

Then, we discretize the data $\{X_i\}$ to $\{Y_j\}_{j=a}^{b}$:

$$Y_j = \sum_{i=1}^{n} \mathbb{1}(X_i \in I_j).$$

Under the null and the partition,

$$\boldsymbol{Y} \sim MN(\boldsymbol{p}_0)$$

where $p_{0,j}(\mu,\sigma^2) = \mathbb{P}(X_1 \in I_j) = \Phi\left(\frac{\ell j-\mu}{\sigma}\right) - \Phi\left(\frac{\ell(j-1)-\mu}{\sigma}\right)$. Then the discretized likelihood function is the following multinomial pmf

$$\prod_{j=a}^{b} p_{0,j}(\mu,\sigma^2)^{Y_j}.$$

```
dlik <- function(par,Y,loc){ ## discretized likelihood in log
    mu=par[1]
    sig2=par[2]
    cdfs <- pnorm(loc,mu,sqrt(sig2))
    K <- length(cdfs)
    pj <- cdfs[-1]-cdfs[-K]
    return(sum(Y*log(pj)))
}
```

So we can solve for the MLE of the discretized likelihood

$$(\hat{\mu},\hat{\sigma}^2) = \arg\max \prod_{j=a}^{b} p_{0,j}(\mu,\sigma^2)^{Y_j}.$$

The test statistic is

$$T = \sum_{j=a}^{b} \frac{(Y_j - n\hat{p}_j)^2}{n\hat{p}_j}$$

where $\hat{p}_j = p_{0,j}(\hat{\mu},\hat{\sigma}^2)$, and we reject the null hypothesis if $T \geq \chi^2_{b-a-2,\alpha}$ for some size α.

We first simulate 1000 synthetic data points from $N(0,1)$ with random seed being 11. The range of the data is about $[-3.3, 3.7]$. So we set (a,b,ℓ) to produce the partition $(-4.0,-3.5]\cup(-3.5,-3.0]\cup \ldots \cup (3.5,4.0]$. The discretized MLE estimates are $(\hat{\mu},\hat{\sigma}^2) \approx (0.01, 0.99)$. The p value of the χ^2 test is about 0.723. So we do not reject the null hypothesis that the data is from some normal distribution at 5% level.

```
set.seed(11)
X <- rnorm(1e3,0,1)
range(X)
endpts <- seq(from = floor(min(X)), to= ceiling(max(X)),by=0.5)
Y <- table(cut(X,endpts))
res.opt <- optim(c(mean(X),var(X)), dlik, method = "L-BFGS-B", Y=Y, loc=endpts,
    control=list(fnscale=-1))
opt.par <- res.opt$par
cdfs <- pnorm(endpts,opt.par[1],sqrt(opt.par[2]))
K <- length(cdfs)
hat.p <- cdfs[-1]-cdfs[-K]
EY <- length(X)*hat.p ### expected number of observations
test.stat <- sum((Y-EY)^2/EY) ### test statistic
1-pchisq(test.stat,df=K-2-length(opt.par)) ## p value
```

We use the synthetic data of Example for illustration. A histogram of the synthetic data is presented in Figure 7.10.

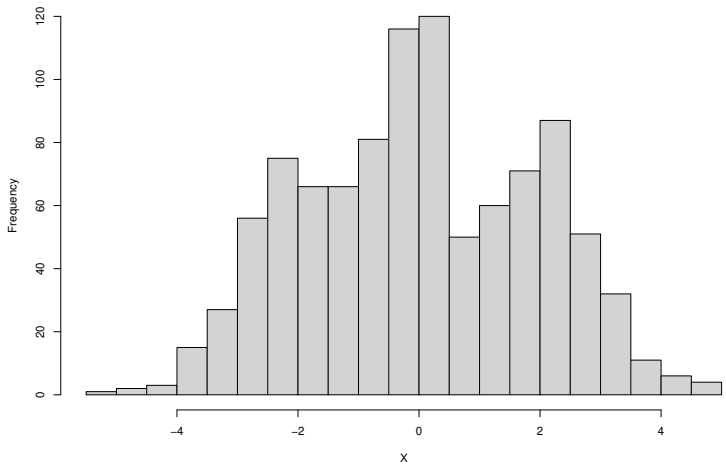


Figure 7.10: The synthetic data of Example

The support of the data is about $[-5.27, 4.93]$, so we set $a = -10$ $b = 10$, and $\ell = 0.5$, ending up with 21 intervals. Following the same procedure, the resulting p-value is about 4.7×10^{-13}. So we reject the null hypothesis at 5% level. ♢

Optimal choice of the length ℓ?

Clearly, extension of the idea to high-dimensional data would be difficult, as data points are scattered and sparsely distributed.

7.8.3 Testing independence

8 Linear Regression

8.1 Regression

Reading(s) 8.1. Wasserman (2004) 13.1, 13.2 Hogg et al. (2018)

The idea of regression dates back to the method of least squares which Legendre and Gauss both applied to determine the orbits of bodies about the Sun. The term regression was first introduced in Galton (1886) to describe the phenomenon that the heights of descendants of tall ancestors tend to be relatively short and vice versa. The descendants' heights would *regress* towards the average.

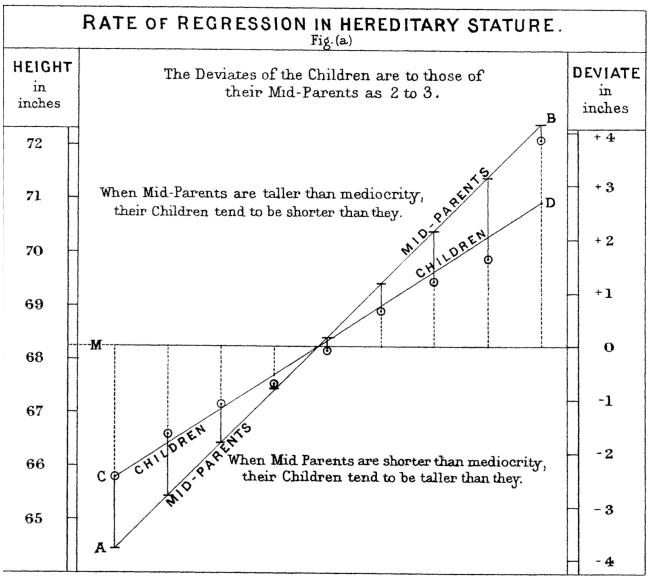


Figure 8.1: Plate IX of Galton (1886)

Regression analysis studies the relationship between a response variable Y and a covariate X. More specifically, the focus is on how Y changes in response to the change in X (with all the others unchanged). In different contexts, the terminology for "response" and "covariate" pair might be different. Table 7 collects a few versions.

covariate X	response Y
independent	dependent
input	output
regressor	regressand
predictor	predicted
explanatory	explained
exogenous	endogenous
manipulated	measured
exposure	outcome
feature	label or target

Table 7: Names used in regression analysis

Our statistical model for the dependence is

$$\mathbb{E}[Y|X = x] = m(x)$$

where $m(\cdot)$ denotes the *mean* response function to be learned/estimated from data. We do not strive for a model to explain every individual observation. Instead, we focus on the mean response $m(x)$, which is more reasonable because X is only a partial list of variables that can explain the variation in Y.

With data $\{(Y_i, X_i)\}_{i=1}^n$, the equivalent data-generating model for observation i is

$$Y_i = m(X_i) + \varepsilon_i,$$

where $\mathbb{E}[\varepsilon_i|X_i] = 0$, and ε_i is the observational error that cannot be "explained" by $m(X_i)$. One convenient interpretation of ε_i is measurement error plus unobserved disturbances that are "uncorrelated" with X_i.

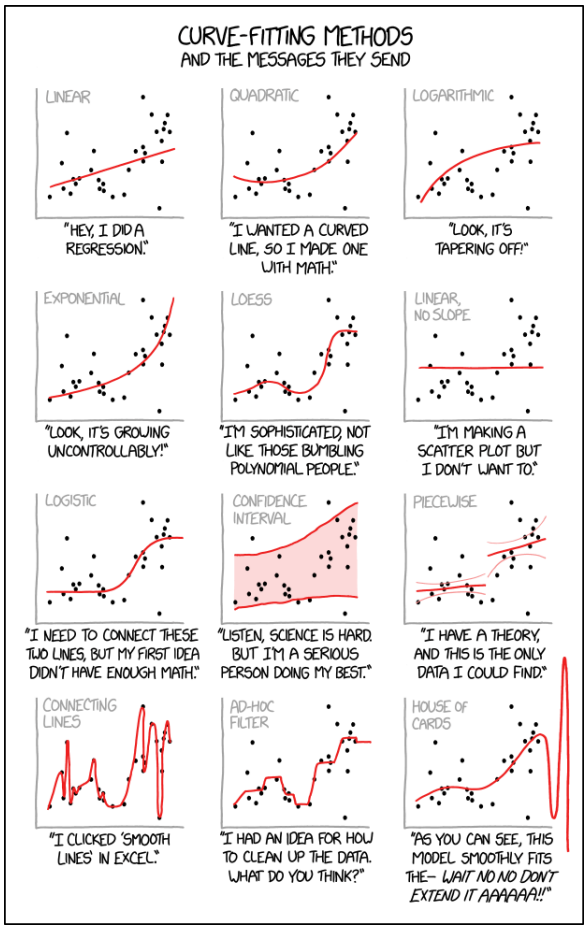


Figure 8.2: Curve-fitting methods

8.2 Linear regression

To begin the study of regression, we focus on the most fundamental regression model

$$m(x) = x^T\beta \text{ for } x \in \mathbb{R}^p$$

where $x = (x_1, x_2, ..., x_p)$ contains p features/covariates, and the error distribution satisfies

$$\varepsilon_i | X_i \overset{iid}{\sim} N(0, \sigma^2).$$

We will return to the error distribution assumption later.

With data $\{(Y_i, X_i)\}_{i=1}^n$, linear regression in matrix form is

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}_{n\times 1} = \begin{bmatrix} X_{1,1} & X_{1,2} & ... & X_{1,p} \\ X_{2,1} & X_{2,2} & ... & X_{2,p} \\ & & \vdots & \\ X_{n,1} & X_{1,2} & ... & X_{n,p} \end{bmatrix}_{n\times p} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}_{p\times 1} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}_{n\times 1},$$

which can be compactly written as

$$Y = X\beta + \varepsilon,$$

where β is called regression coefficients, and the covariates are collected in X which is called the *design* matrix.

The "linear" of linear regression refers to the mean response function $m(x)$ is linear in β. Linear models are expressive because they could also capture non-linear dependence of Y on X.

Example 8.1 (Polynomial regression)**.** In polynomial regression, regressors are polynomials of some covariates. For instance, quadratic regression takes the form

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \varepsilon_i,$$

and the design matrix is

$$\begin{bmatrix} 1 & X_1 & X_1^2 \\ 1 & X_2 & X_2^2 \\ \vdots & \vdots & \vdots \\ 1 & X_n & X_n^2 \end{bmatrix}_{n\times 3}.$$

More generally, polynomial regression can have higher order terms

$$Y_i = \sum\nolimits_{i=0}^{K} \beta_1 X_i^k + \varepsilon_i.$$

As $m(X_i)$ is linear in β, the quadratic regression still belongs to linear models. ♢

In general, linear regression

$$Y_i = \sum\nolimits_{j=0}^{p} \beta_j f_j(X_i) + \varepsilon_i, \text{ for } i = 1, ..., n,$$

allows versatile non-linear functions $f_j(\cdot)$ to flexibly model the mean response function.

Example 8.2 (one-way ANOVA)**.** In one-way analysis of variance (ANOVA), multiple samples are compared to see if there is any difference in the mean. For instance,

- in the first sample, $Y_i = \beta_1 + \varepsilon_i$ for $i = 1, ..., n$;

- in the second sample, $Y_i = \beta_2 + \varepsilon_i$ for $i = n+1, ..., n+m$;
- and in the third sample, $Y_i = \beta_3 + \varepsilon_i$ for $i = n+m+1, ..., n+m+k$.

Further assume $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$ for $i = 1, ..., n+m+k$. Then, the one-way ANOVA is essentially a linear regression

$$Y = X\beta + \varepsilon$$

with design matrix being

$$\begin{bmatrix} 1 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 1 \end{bmatrix}_{(n+m+k)\times 3}.$$

♢

There are basically two types of designs: random design and fixed design. In the fixed design, X is deterministic and treated as fixed. In the random design, X is generated from a base distribution $\mathbb{P}_X$. As the focus is on the regression coefficients, likelihood based inference ignores $\mathbb{P}_X$ because of the factorization:

$$\mathcal{L}_n(\theta) = \prod_{i=1}^n f_{Y|X}(y_i|x_i;\theta) f_X(x_i) \propto \prod_{i=1}^n f_{Y|X}(y_i|x_i;\theta)$$

where the model parameter(s) θ is (are) unrelated to $\prod_{i=1}^n f_X(x_i)$.

8.3 Least squares

Reading(s) 8.2. Wasserman (2004) 13.3; Hogg et al. (2018)

8.3.1 Ordinary least squares

Adrien-Marie Legendre (Legendre, 1805) and Carl Friedrich Gauss (Gauss, 1809) respectively applied the method of least squares to determine the orbits of bodies about the Sun. Despite the dispute on its first inventor (Stigler, 1981), the importance of least squares is of no doubt. As interestingly put in the beginning sentence of Stigler (1981),

> The method of least squares is the automobile of modern statistical analysis: despite its limitations, occasional accidents, and incidental pollution, it and its numerous variations, extensions, and related conveyances carry the bulk of statistical analyses, and are known and valued by nearly all.

The method of least squares seeks for the best β to minimize the ℓ_2-loss between the observation Y and the *fitted/predicted* value $X\beta$:

$$\min_{\beta\in\mathbb{R}^p} \sum_{i=1}^n (Y_i - X_i^T\beta)^2. \tag{16}$$

The "least" refers to the minimization. The "squares" refers to the sums of the squared residuals (to be formally defined later) whose matrix form is $(Y-X\beta)^T(Y-X\beta) = ||Y-X\beta||_2^2$.

As the objective function is quadratic, it suffices to solve for β with the FOC:

$$\frac{\partial}{\partial\beta}\left[(Y-X\beta)^T(Y-X\beta)\right] = 0 \implies (X^TX)\beta = X^TY.$$

If X^TX is invertible, then

$$\hat{\beta}_{OLS} = (X^TX)^{-1}X^TY. \tag{17}$$

When the covariates are linearly dependent, that is, at least one of the covariates can be written as a linear combination of other covariates, X^TX is not invertible, and the least squares solution is not well-defined. In fact, X^TX is invertible *if and only if* the columns of X are linearly independent. (See Section 4.2 of Strang (2022))

Example 8.3 (Simple linear regression)**.** In the simple linear regression,

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \text{ for } i = 1, 2, ..., n.$$

The design matrix is

$$\begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix}_{n\times 2}.$$

Then, X^TY is

$$X^TY = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ X_1 & X_2 & \cdots & X_n \end{bmatrix}_{2\times n} \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}_{n\times 1} = \begin{bmatrix} \sum_{i=1}^n Y_i \\ \sum_{i=1}^n X_iY_i \end{bmatrix}_{2\times 1} = n\begin{bmatrix} \overline{Y}_n \\ \overline{XY}_n \end{bmatrix}_{2\times 1},$$

X^TX is

$$X^TX = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ X_1 & X_2 & \cdots & X_n \end{bmatrix}_{2\times n} \begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix}_{n\times 2} = \begin{bmatrix} n & \sum_{i=1}^n X_i \\ \sum_{i=1}^n X_i & \sum_{i=1}^n X_i^2 \end{bmatrix}_{2\times 2} = n \begin{bmatrix} 1 & \overline{X}_n \\ \overline{X}_n & \overline{X^2}_n \end{bmatrix}_{2\times 2}$$

and $(X^TX)^{-1}$ can be calculated by the formula

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

By direct calculation,

$$\hat{\beta}_{OLS} = \frac{1}{\overline{X^2}_n - \left(\overline{X}_n\right)^2} \begin{bmatrix} \overline{X^2}_n & -\overline{X}_n \\ -\overline{X}_n & 1 \end{bmatrix}_{2\times 2} \begin{bmatrix} \overline{Y}_n \\ \overline{XY}_n \end{bmatrix}_{2\times 1}.$$

If the sample mean $\overline{X}_n = 0$, the expression for the solution becomes quite simple

$$\hat{\beta}_{OLS} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix}_{OLS} = \begin{bmatrix} \overline{Y}_n \\ \left(\overline{X^2}_n\right)^{-1} \overline{XY}_n \end{bmatrix}_{2\times 1}.$$

This motivates the modeller to demean the covariate X_i to $X_i - \overline{X}_n$ such that the new covariate has sample mean 0, and consequently the simple linear regression becomes

$$Y_i = \tilde{\beta}_0 + \beta_1(X_i - \overline{X}_n) + \varepsilon_i, \text{ for } i = 1, 2, ..., n,$$

where $\tilde{\beta}_0 = \beta_0 + \beta_1\overline{X}_n$. As the intercept is often a nuisance parameter, demeaning is often done in data pre-processing.

♢

Example 8.4 (one-way ANOVA)**.** Recall the design matrix in the one-way ANOVA example. Then, X^TY is

$$X^TY = \begin{bmatrix} 1 & \cdots & 1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 1 & \cdots & 1 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 0 & 1 & \cdots & 1 \end{bmatrix}_{3\times N} \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}_{N\times 1} = \begin{bmatrix} \sum_{i=1}^n Y_i \\ \sum_{i=n+1}^{n+m} Y_i \\ \sum_{i=n+m+1}^{n+m+k} Y_i \end{bmatrix}_{3\times 1} = \begin{bmatrix} n\overline{Y}_{(n)} \\ m\overline{Y}_{(m)} \\ k\overline{Y}_{(k)} \end{bmatrix}_{3\times 1},$$

where $N = m + n + k$, X^TX is

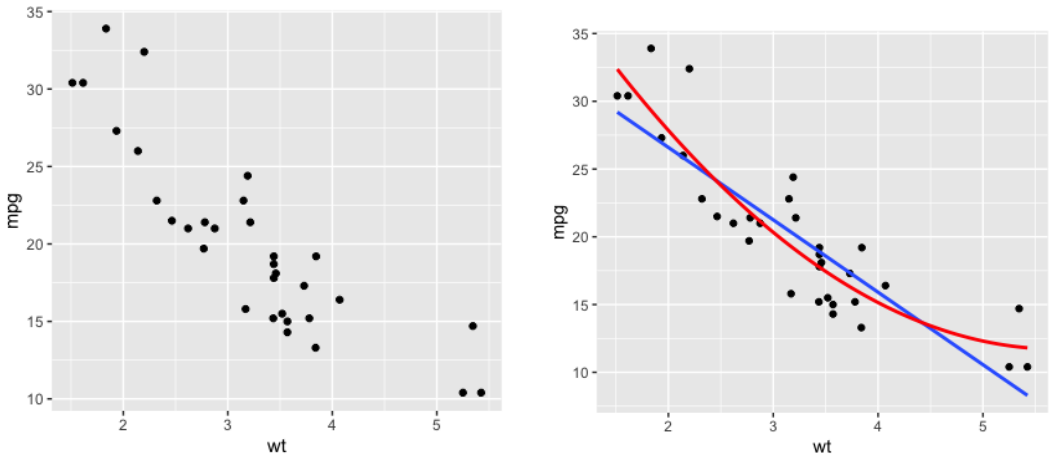
$$X^TX = \begin{bmatrix} 1 & \cdots & 1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 1 & \cdots & 1 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 0 & 1 & \cdots & 1 \end{bmatrix}_{3\times N} \begin{bmatrix} 1 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 1 \end{bmatrix}_{N\times 3} = \begin{bmatrix} n & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & k \end{bmatrix}_{3\times 3},$$

and by direct calculation,

$$\hat{\beta}_{OLS} = \begin{bmatrix} \overline{Y}_{(n)} \\ \overline{Y}_{(m)} \\ \overline{Y}_{(k)} \end{bmatrix}_{3\times 1}.$$

♢

Example 8.5. With the help of computer, we can obtain OLS fit for quadratic regression and other fitted curves easily.



scatter plot scatter plot with fitted line/curve

Figure 8.3: Linear models of `wt` and `mpg`

```
mtcars
ggplot(mtcars,aes(x = wt, y = mpg)) + geom_point()

ggplot(mtcars,aes(x = wt, y = mpg)) + geom_point()+
  geom_smooth(method = "lm", formula = y~x, se = FALSE) +
  geom_smooth(method = "lm", formula = y~ poly(x,2,raw = TRUE),
              se = FALSE, level=0.99, color = "red")

design = cbind(intp = 1, x = mtcars$wt)
response = mtcars$mpg
beta = solve(crossprod(design)) %*% t(design) %*% response
print(beta)

design = cbind(intp = 1, x = mtcars$wt, x2 = mtcars$wt^2)
beta = solve(crossprod(design)) %*% t(design) %*% response
print(beta)
```

♢

8.3.2 Understanding least squares

The point estimate for $\mathbb{E}[Y_i|X_i] = X_i^T\beta$ is $X_i^T\hat{\beta}_{OLS}$, following the plug-in principle. The *fitted/predicted* value for Y_i is $X_i^T\hat{\beta}_{OLS}$ for $i = 1, 2, ..., n$. In matrix notation, the fitted values of Y are essentially the part of Y *projected* onto $\mathcal{C}(X)$, the *column space spanned* by X:

$$\hat{Y} \equiv X\hat{\beta}_{OLS} = X(X^TX)^{-1}X^TY = P_XY,$$

where $P_X \equiv X(X^TX)^{-1}X^T$ denotes the *projection* matrix onto $\mathcal{C}(X)$.

Definition 8.1 (Column space)**.** Suppose X is a $n \times p$ matrix, the *column space spanned* by X is defined as

$$\mathcal{C}(X) = \{Xa : a \in \mathbb{R}^p\}.$$

The column space $\mathcal{C}(X) \subset \mathbb{R}^n$ contains all combinations of the columns of X. For our purpose, it is reasonable to assume the columns of X are linearly independent, and the column space $\mathcal{C}(X)$ is precisely the amount of variations that can be "explained" by X.

In our least squares application, the response vector

$$Y = P_X Y + (I - P_X)Y,$$

where $P_X Y$, the fitted/predicted part of Y, belongs to $\mathcal{C}(X)$. The remainder variations in Y, $(I-P_X)Y$, shall belong to the complement subspace of $\mathcal{C}(X)$.

The *null space* $\mathcal{N}(X^T)$ is the complement of $\mathcal{C}(X)$, which is one of the fundamental results of linear algebra:

$$\mathbb{R}^n = \mathcal{C}(X) \oplus \mathcal{N}(X^T)$$

where $\mathcal{C}(X)$ and $\mathcal{N}(X^T)$ are perpendicular.

Definition 8.2 (Null space)**.** Suppose X is a $n \times p$ matrix, the *null space* of X is defined as

$$\mathcal{N}(X) = \{a \in \mathbb{R}^p : Xa = 0_{n\times 1}\}.$$

The null space $\mathcal{N}(X)$ is also called the *kernel* of X.

All elements in $\mathbb{R}^n$ can be written as a sum of one element in $\mathcal{C}(X)$ and one element in $\mathcal{N}(X^T)$. In our least squares application, the response vector

$$Y = P_X Y + (I - P_X)Y,$$

where $P_X Y \in \mathcal{C}(X)$ and $(I - P_X)Y \in \mathcal{N}(X^T)$.

The *residuals* are defined as the difference between the observed value of Y_i and its fitted/predicted value

$$\hat{\varepsilon}_i = Y_i - X_i^T \hat{\beta}_{OLS}.$$

By the construction of least squares (16), the *residual sums of squares* (RSS) at $\hat{\beta}_{OLS}$,

$$RSS(\hat{\beta}_{OLS}) = \sum\nolimits_{i=1}^{n} \hat{\varepsilon}_i^2 = \hat{\varepsilon}^T \hat{\varepsilon},$$

achieves the minimum. The residuals in matrix form are

$$\hat{\varepsilon} = Y - X\hat{\beta}_{OLS} = Y - X(X^T X)^{-1} X^T Y = (I_n - P_X)Y$$

where $I_n - P_X \equiv I_n - X(X^T X)^{-1} X^T$ is the *projection* matrix onto $\mathcal{N}(X^T) = \mathcal{C}(X)^{\perp}$, the complement subspace to the *column space* spanned by X.

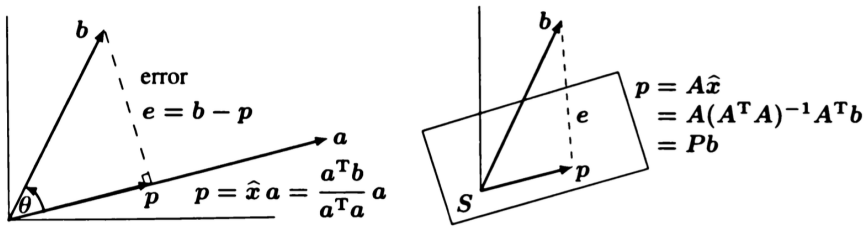


Figure 8.4: Projections in low dimensions, Figure 4.3 of Strang (2022)

Theorem 8.1 (projection matrix)**.** *Suppose X^TX is invertible, then the projection matrix $P_X \equiv X(X^TX)^{-1}X^T$ satisfies the following properties.*

- *P_X is symmetric: $P_X = P_X^T$.*
- *P_X is idempotent: $P_X^2 = P_XP_X = P_X$.*
- *For all $z \in \mathcal{C}(X)$, $P_Xz = z$.*
- *For all $z \in \mathcal{N}(X^T)$, $P_Xz = 0$.*
- *$I - P_X$ is idempotent: $(I - P_X)^2 = (I - P_X)(I - P_X) = I - P_X$.*

Proof. Direct calculations by definition. □

Theorem 8.2. *The residual matrix belongs to the null space of X^T:*

$$\hat{\varepsilon} \in \mathcal{N}(X^T).$$

Proof. By direct calculation,

$$X^T\hat{\varepsilon} = X^T(I - P_X)Y = ((I - P_X)X)^TY = \mathbf{0}_{p\times 1}.$$

□

8.3.3 Properties of least squares

Theorem 8.3. *In the linear model $Y_i = X_i^T\beta + \varepsilon_i$, suppose $\mathbb{E}[\varepsilon_i|X] = 0$ for $i = 1, 2, ..., n$. Then $\hat{\beta}_{OLS}$ is an unbiased estimator of β:*

$$\mathbb{E}[\hat{\beta}_{OLS}|X] = \beta.$$

Further assume $\mathbb{E}[\varepsilon\varepsilon^T|X] = \sigma^2I_n$, then

$$V(\hat{\beta}_{OLS}|X) = \sigma^2(X^TX)^{-1}.$$

Proof. Recall $Y = X\beta + \varepsilon$, by definition,

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T (X\beta + \varepsilon) = \beta + (X^T X)^{-1} X^T \varepsilon. \tag{18}$$

Then, by linearity of (conditional) expectation,

$$\mathbb{E}[\hat{\beta}_{OLS}|X] = \mathbb{E}[\beta + (X^T X)^{-1} X^T \varepsilon|X] = \beta + (X^T X)^{-1} X^T \mathbb{E}[\varepsilon|X] = \beta.$$

Let $A_X = (X^T X)^{-1} X^T$, then

$$V[\hat{\beta}_{OLS}|X] = V[\beta + A_X \varepsilon|X] = A_X V[\varepsilon|X] A_X^T = \sigma^2 A_X A_X^T = \sigma^2 (X^T X)^{-1}.$$

□

Definition 8.3. The OLS estimator of σ^2 is

$$s^2 \equiv \frac{\hat{\varepsilon}^T \hat{\varepsilon}}{n-p}.$$

Recall $\hat{\varepsilon} \equiv Y - \hat{Y} = (I_n - P_X)Y = (I_n - P_X)\varepsilon$, $\text{rank}(\hat{\varepsilon}\hat{\varepsilon}^T) = \text{rank}(I_n - P_X) = n - p$. The denominator of s^2 reflects the "effective" average of the squared residuals. As p is the number of regression coefficients, we can intuitively think p observations are used to estimate regression coefficients, with the remaining $n - p$ observations used to estimate observation error's variance.

Theorem 8.4. *In the linear model $Y_i = X_i^T \beta + \varepsilon_i$, assume $\mathbb{E}(\varepsilon^T \varepsilon|X) = \sigma^2 I_n$, and $\mathbb{E}(\varepsilon|X) = \mathbf{0}_{n\times 1}$. Then s^2 is an unbiased estimator for σ^2:*

$$\mathbb{E}[s^2|X] = \sigma^2.$$

Proof. Note $\hat{\varepsilon} = (I_n - P_X)Y = (I_n - P_X)(X\beta + \varepsilon) = (I_n - P_X)\varepsilon$ and $I_n - P_X$ is idempotent,

$$\hat{\varepsilon}^T \hat{\varepsilon} = \varepsilon^T (I_n - P_X)\varepsilon.$$

Recall the basic property of trace $\text{tr}(AB) = \text{tr}(BA)$,

$$\varepsilon^T (I_n - P_X)\varepsilon = \text{tr}\left(\varepsilon^T (I_n - P_X)\varepsilon\right) = \text{tr}\left((I_n - P_X)\varepsilon\varepsilon^T\right).$$

Then, by linearity of (conditional) expectation and trace,

$$\mathbb{E}[\hat{\varepsilon}^T \hat{\varepsilon}|X] = \text{tr}\left((I_n - P_X)\mathbb{E}[\varepsilon\varepsilon^T|X]\right) = \sigma^2 \text{tr}(I_n - P_X) = (n-p)\sigma^2,$$

where $\text{tr}(I_n) = n$, $\text{tr}(P_X) = \text{tr}(X(X^T X)^{-1} X^T) = \text{tr}((X^T X)^{-1} X^T X) = \text{tr}(I_p) = p$.

□

8.3.4 Asymptotics and inference of OLS

The mean and variance of $\hat{\beta}_{OLS}$ does not provide enough ingredients for statistical inference. The following asymptotic theorem, together with plug-in estimates of σ^2 and $\left(\mathbb{E}\left[X_1 X_1^T\right]\right)^{-1}$, allows us to construct confidence intervals that have asymptotic coverage guarantee.

Theorem 8.5 (Consistency of OLS)**.** *In the linear model $Y_i = X_i^T\beta + \varepsilon_i$, $i = 1, 2, ..., n$, the covariate and the error term $\{(X_i, \varepsilon_i)\}$ are iid with the second moment $\mathbb{E}[X_1X_1^T]$ invertible and $\mathbb{E}[\varepsilon_1|X_1] = 0$. Then $\hat{\beta}_{OLS}$ is a consistent estimator of β.*

Proof. Equation (18) can be further written as

$$\hat{\beta}_{OLS} = \beta + \left(\frac{1}{n}\sum\nolimits_{i=1}^{n} X_iX_i^T\right)^{-1}\left(\frac{1}{n}\sum\nolimits_{i=1}^{n} X_i\varepsilon_i\right).$$

Since $\{(X_i, \varepsilon_i)\}$ are iid, LLN implies $\frac{1}{n}\sum_{i=1}^{n} X_iX_i^T \xrightarrow{p} \mathbb{E}[X_iX_i^T]$, which is finite by assumption, and

$$\frac{1}{n}\sum\nolimits_{i=1}^{n} X_i\varepsilon_i \xrightarrow{p} \mathbb{E}[X_i\varepsilon_i] = 0$$

where $\mathbb{E}[X_i\varepsilon_i] = \mathbb{E}[X_i\mathbb{E}[\varepsilon_i|X_i]] = 0$ by law of iterated expectation. By continuous mapping theorem,

$$\hat{\beta}_{OLS} \xrightarrow{p} \beta + \left(\mathbb{E}[X_iX_i^T]\right)^{-1}\mathbb{E}[X_i\varepsilon_i] = \beta.$$

□

Theorem 8.6 (Asymptotic Normality of OLS)**.** *In the linear model $Y_i = X_i^T\beta + \varepsilon_i$, $i = 1, 2, ..., n$, the covariate and the error term $\{(X_i, \varepsilon_i)\}$ are iid with the second moment $\mathbb{E}[X_1X_1^T]$ invertible, $\mathbb{E}[\varepsilon_1|X_1] = 0$, and $\mathbb{E}[\varepsilon_i^2|X_i] = \sigma^2$, then $\hat{\beta}_{OLS}$ is asymptotically Normal:*

$$\sqrt{n}(\hat{\beta}_{OLS} - \beta) \xrightarrow{d} N\left(0, \left(\mathbb{E}\left[X_1X_1^T\right]\right)^{-1}\sigma^2\right).$$

Proof. Since $\{(X_i, \varepsilon_i)\}$ are iid and $\mathbb{E}[X_i\varepsilon_i] = 0$, CLT implies

$$\sqrt{n}\left(\frac{1}{n}\sum\nolimits_{i=1}^{n} X_i\varepsilon_i\right) \xrightarrow{d} N\left(0, V\left(X_1\varepsilon_1\right)\right)$$

where $V\left(X_1\varepsilon_1\right) = \mathbb{E}\left[X_1X_1^T\varepsilon_1^2\right] - \left(\mathbb{E}[X_1\varepsilon_1]\right)\left(\mathbb{E}[X_1\varepsilon_1]\right)^T = \mathbb{E}\left[X_1X_1^T\mathbb{E}[\varepsilon_1^2|X_1]\right] = \mathbb{E}\left[X_1X_1^T\right]\sigma^2$ by law of iterated expectation and the homoscedasticity assumption.

Lastly, by Slutsky theorem and linear transformation property of multivariate Normal,

$$\sqrt{n}(\hat{\beta}_{OLS} - \beta) \xrightarrow{d} N\left(0, \left(\mathbb{E}\left[X_1X_1^T\right]\right)^{-1}\sigma^2\right).$$

□

Theorem 8.7. *In the linear model $Y_i = X_i^T\beta + \varepsilon_i$, the covariate and the error term $\{(X_i, \varepsilon_i)\}$ are iid with the second moment $\mathbb{E}[X_1X_1^T]$ invertible, $\mathbb{E}[\varepsilon_1|X_1] = 0$, and $\mathbb{E}[\varepsilon_1^2] = \sigma^2$. Then s^2 is a consistent estimator for σ^2:*

$$s^2 \xrightarrow{p} \sigma^2.$$

Proof. The estimator is proportional to the average of squared errors

$$s^2 = \frac{n}{n-p}\frac{1}{n}\sum_{i=1}^{n}\hat{\varepsilon}_i^2.$$

Note $\hat{\varepsilon}_i = Y_i - \hat{Y}_i = X_i^T\beta + \varepsilon_i - X_i^T\hat{\beta}_{OLS} = X_i^T(\beta - \hat{\beta}_{OLS}) + \varepsilon_i$,

$$\hat{\varepsilon}_i^2 = \varepsilon_i^2 + (\beta - \hat{\beta}_{OLS})^T\left(X_iX_i^T\right)(\beta - \hat{\beta}_{OLS}) + 2\left(\varepsilon_iX_i^T\right)(\beta - \hat{\beta}_{OLS}),$$

and their average $\frac{1}{n}\sum_{i=1}^{n}\hat{\varepsilon}_i^2$ is

$$\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i^2 + (\beta - \hat{\beta}_{OLS})^T\left(\frac{1}{n}\sum_{i=1}^{n}X_iX_i^T\right)(\beta - \hat{\beta}_{OLS}) + 2\left(\frac{1}{n}\sum_{i=1}^{n}\varepsilon_iX_i^T\right)(\beta - \hat{\beta}_{OLS}).$$

Since the ε_i's are iid, by LLN,

$$\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i^2 \xrightarrow{p}\sigma^2.$$

Since the X_i's are iid, by LLN,

$$\frac{1}{n}\sum_{i=1}^{n}X_iX_i^T \xrightarrow{p}\mathbb{E}[X_1X_1^T].$$

Since (X_i, ε_i)'s are iid, by LLN,

$$\frac{1}{n}\sum_{i=1}^{n}\varepsilon_iX_i^T \xrightarrow{p}\mathbb{E}[\varepsilon_1X_1^T] = \mathbb{E}[\mathbb{E}[[\varepsilon_1|X_1]X_1^T] = 0.$$

Note Theorem 8.5 says $\beta - \hat{\beta}_{OLS}\xrightarrow{p}0$. Slutsky theorem implies

$$s^2\xrightarrow{p}\sigma^2.$$

□

Remark. By inspecting the proof, $\hat{\beta}_{OLS}$ can be replaced with any consistent estimator of β, and s^2 is still a consistent estimator of σ^2.

Combining Theorem 8.6 and Theorem 8.7, Slutsky theorem implies

$$\frac{1}{\sqrt{s^2/n}}S_{XX}^{1/2}(\hat{\beta}_{OLS} - \beta)\xrightarrow{d}N\left(\mathbf{0}_{p\times 1}, I_p\right),$$

where $S_{XX} = \frac{1}{n}\sum_{i=1}^{n}X_kX_k^T = \frac{1}{n}X^TX$ is a consistent estimator for $\mathbb{E}[X_1X_1^T]$ by LLN, holds under the same conditions for Theorem 8.6 and Theorem 8.7. The approximate sampling distribution of $\hat{\beta}_{OLS}$ is

$$\hat{\beta}_{OLS}\overset{d}{\approx}N\left(\beta, s^2(X^TX)^{-1}\right).$$

Since the marginal distribution of a multivariate Normal is also Gaussian, the approximate sampling distribution of $\hat{\beta}_{OLS,k}$ is, for $k = 1, 2, ..., p$,

$$\hat{\beta}_{OLS,k}\overset{d}{\approx}N\left(\beta_k, s^2[(X^TX)^{-1}]_{k,k}\right)$$

where $[M]_{k,k}$ denotes the $(k,k)^{th}$ entry of matrix M. With the approximate sampling distribution, the approximate $(1-\alpha)$-confidence *interval* of β_k for $k = 1, 2, ..., p$, is

$$\hat{\beta}_{OLS,k} \pm \sqrt{s^2[(X^TX)^{-1}]_{k,k}} \times z_{\alpha/2},$$

and the approximate $(1-\alpha)$-confidence *interval* of the fitted value $X_i^T\beta$ is

$$X_i^T\hat{\beta}_{OLS} \pm \sqrt{s^2[P_X]_{i,i}} \times z_{\alpha/2},$$

where $P_X \equiv X(X^TX)^{-1}X^T$ is the projection matrix onto $\mathcal{C}(X)$.

Similar to the fitted value, the point prediction (a point estimator) for a new observation Y_* with the new covariate X_* is $X_*^T\hat{\beta}_{OLS}$, and the *asymptotic* $(1-\alpha)$-confidence interval for the mean response $X_*^T\beta$ is

$$X_*^T\hat{\beta}_{OLS} \pm \sqrt{s^2X_*^T(X^TX)^{-1}X_*} \times z_{\alpha/2}.$$

However, the *asymptotic* $(1-\alpha)$-confidence (prediction) interval for Y_* is

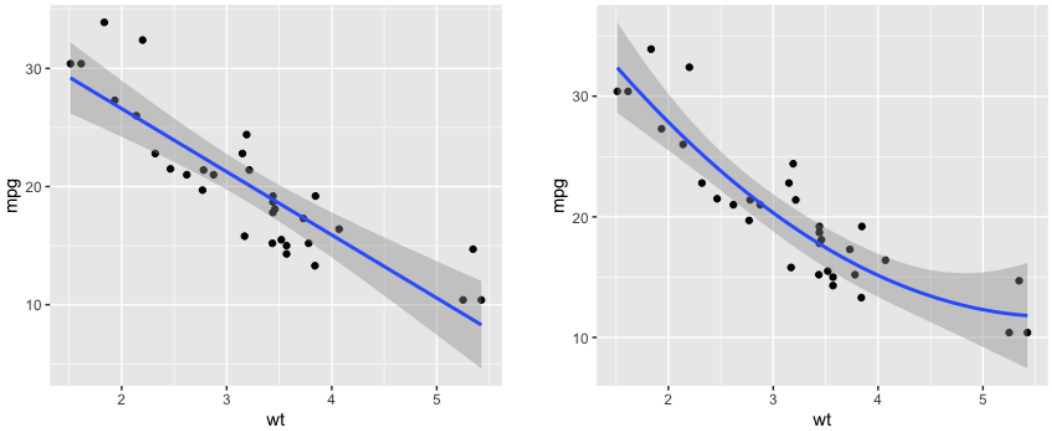
$$X_*^T\hat{\beta}_{OLS} \pm \sqrt{s^2\left(1 + X_*^T(X^TX)^{-1}X_*\right)} \times z_{\alpha/2}$$

which is a wider interval than the prediction interval of the mean response $X_*^T\beta$. This is because the model-based prediction $Y_* = X_*^T\beta + \varepsilon_*$ has two sources of uncertainty:

- the uncertainty in the mean response, since we do not know the true β, and
- the uncertainty in the error term, since we do not know ε_*.

Combining the *individual* confidence *intervals* for all observations makes a confidence *band*. The joint coverage rate of the individual (marginal) confidence intervals with $(1-\alpha)$-confidence is not necessarilly $1-\alpha$ due to the dependence among them. It is optional in many software such as `R` to plot the confidence band with a confidence level. The confidence level specified in the software is for individual confidence intervals, and *not* for the whole dataset.

Example 8.6. Continue with the `Motor Trend Car Road Tests` example. In this example, we plot the *asymptotic* 95% confidence band of the mean response function $m(X) = X^T\beta$.



simple linear regression quadratic regression

Figure 8.5: Linear models of `wt` and `mpg` with error band

```
ggplot(mtcars,aes(x = wt, y = mpg)) + geom_point()+ #xlim(0,6)+ ylim(5,35)+
  geom_smooth(method = "lm", formula = y~x, se = TRUE, level=0.95)
ggplot(mtcars,aes(x = wt, y = mpg)) + geom_point()+ #xlim(0,6)+ ylim(5,35)+
  geom_smooth(method = "lm", formula = y~poly(x,2,raw=TRUE), se = TRUE, level=0.95)
```

```
5
6  proj = function(X){
7    return( X %*% solve(crossprod(X)) %*% t(X))
8  }
9  response = mtcars$mpg
10 design = cbind(intp = 1, x= mtcars$wt)
11 ## design = cbind(1,x = mtcars$wt, mtcars$wt^2) ## for quadratic regression
12 beta = solve(crossprod(design)) %*% t(design) %*% response
13 yhat = proj(design) %*% response
14 resid = response - yhat
15 diagPx = diag(proj(design))
16 s2 = sum(resid^2)/(length(resid) - ncol(design))
17 se = sqrt(s2 * diagPx)
18
19 plot(mtcars$wt, mtcars$mpg, pch = 20, xlim = c(0,6), ylim = c(5,35))
20 abline(a = beta[1],b=beta[2],lwd=1)
21 perm = order(design[,"x"])
22 lines(design[perm,"x"], yhat[perm] + 1.96 * se[perm] )
23 lines(design[perm,"x"], yhat[perm] - 1.96 * se[perm] )
```

♢

8.4 Gaussian linear regression

Reading(s) 8.3. Wasserman (2004) 13.2; Hogg et al. (2018)

Statistical inference of least squares is based on asymptotic theory. In contrast, finite sample sampling distribution can be obtained for Gaussian linear regression

$$Y_i = X_i^T \beta + \varepsilon_i, \text{ for } i = 1, 2, ..., n,$$

where $\varepsilon_i | X_i \sim N(0, \sigma^2)$, and $\{(Y_i, X_i)\}$ are independent. The distribution assumption $\varepsilon_i | X_i \sim N(0, \sigma^2)$ for $i = 1, 2, ..., n$, allows a likelihood function of the parameters in the (linear) model for the dataset. The inference theory developed for maximum likelihood estimation can be applied here.

8.4.1 Maximum likelihood estimation

With the linearity, iid, and Normality assumptions, the likelihood function is

$$\mathcal{L}_n(\beta, \sigma^2) \propto \prod_{i=1}^n (\sigma^2)^{-1/2} e^{-\frac{1}{2\sigma^2}(Y_i - X_i^T \beta)^2} \propto (\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2}(Y - X\beta)^T (Y - X\beta)}.$$

The log-likelihood function is

$$\ell_n(\beta, \sigma^2) = constant - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2}(Y - X\beta)^T (Y - X\beta).$$

By direction calculation,

$$\begin{aligned} \hat{\beta} = & \ (X^T X)^{-1} X^T Y \\ \hat{\sigma}^2 = & \ \frac{1}{n}(Y - X\hat{\beta})^T (Y - X\hat{\beta}). \end{aligned}$$

The maximum likelihood estimator $\hat{\beta}$ is identical to the least squares estimator $\hat{\beta}_{OLS}$. But the maximum likelihood estimator $\hat{\sigma}^2$ is not the unbiased estimator. The optimality properties of maximum likelihood estimation can be used to justify least squares estimation in finite sample, at the price of making more (restrictive) assumptions.

8.4.2 Confidence intervals

The Gaussian error assumption $\varepsilon_i | X_i \overset{iid}{\sim} N(0, \sigma^2)$ allows exact characterizations of the sampling distribution of $(\hat{\beta}, \hat{\sigma}^2)$.

Theorem 8.8 (Sampling distribution)**.** *Suppose data $\{(Y_i, X_i)\}$ are i.i.d. from the Gaussian linear regression*

$$Y_i = X_i^T \beta + \varepsilon_i, \text{ for } i = 1, 2, ..., n,$$

where $\varepsilon_i | X_i \sim N(0, \sigma^2)$ and $X_i \overset{iid}{\sim} \mathbb{P}_X$. Then the following holds:

(1) $\hat{\beta} | X \sim N(\beta, \sigma^2 (X^T X)^{-1})$;

(2) $\hat{\beta} \perp \hat{\varepsilon} | X$, hence $\hat{\beta} \perp \hat{\sigma}^2 | X$;

(3) $n\hat{\sigma}^2 / \sigma^2 | X \sim \chi^2_{n-p}$.

Proof. The assumptions imply $\varepsilon|X \sim N(0, \sigma^2 I_n)$. Note $\hat{\beta} = (X^T X)^{-1} X^T Y = \beta + (X^T X)^{-1} X^T \varepsilon$ is a linear transformation of a known multivariate Normal vector ε,

$$\hat{\beta}|X \sim N(\beta, \sigma^2 (X^T X)^{-1}).$$

The proof of (2) is similar to the Gaussian mean and variance estimation problem. To apply the multivariate Normal theory, note

$$\begin{bmatrix} \hat{\beta} - \beta \\ \hat{\varepsilon} \end{bmatrix}_{(p+n)\times 1} = \begin{bmatrix} \hat{\beta} - \beta \\ Y - X\hat{\beta} \end{bmatrix}_{(p+n)\times 1} = \begin{bmatrix} (X^T X)^{-1} X^T \\ I - P_X \end{bmatrix}_{(p+n)\times n} \varepsilon.$$

Then by a linear transformation of multivariate Normal random vector, the resulting random vector follows

$$\begin{bmatrix} \hat{\beta} - \beta \\ Y - X\hat{\beta} \end{bmatrix}_{(p+n)\times 1} |X \sim N\left(\mathbf{0}_{(p+n)\times 1}, \sigma^2 \begin{bmatrix} (X^T X)^{-1} & \mathbf{0}_{p\times n} \\ \mathbf{0}_{n\times p} & I_n - P_X \end{bmatrix} \right).$$

As the off-diagonal block is a 0 sub-matrix, $\hat{\beta} - \beta \perp Y - X\hat{\beta}$, and hence their functions, $\hat{\beta} = (\hat{\beta} - \beta) + \beta$ and $\hat{\sigma}^2 = (Y - X\hat{\beta})^T (Y - X\hat{\beta})/n$, are also independent.

To show (3), note the identity

$$\varepsilon^T \varepsilon = \varepsilon^T P_X \varepsilon + \varepsilon^T (I - P_X) \varepsilon,$$

where on the left-hand side $\varepsilon^T \varepsilon / \sigma^2 | X \sim \chi^2_n$, and on the right-hand side, $\varepsilon^T P_X \varepsilon \perp \varepsilon^T (I - P_X)\varepsilon | X$ because of perpendicular projection $P_X(I_n - P_X) = \mathbf{0}_{n\times n}$. With a similar trick used for the Wald test, (1) implies the "sandwich" form

$$(\hat{\beta} - \beta)^T (X^T X)(\hat{\beta} - \beta)/\sigma^2 | X \sim \chi^2_p$$

where $X(\hat{\beta} - \beta) = P_X \varepsilon$ and then $\varepsilon^T P_X \varepsilon / \sigma^2 | X \sim \chi^2_p$. By (conditional) independence of $\varepsilon^T P_X \varepsilon$ and $\varepsilon^T (I - P_X)\varepsilon$ and the additivity property of χ^2 distribution, the moment generating function of $\varepsilon^T (I - P_X)\varepsilon$ (conditional on X) implies

$$\varepsilon^T (I - P_X)\varepsilon / \sigma^2 | X \sim \chi^2_{n-p}$$

where $n\hat{\sigma}^2 = (Y - X\hat{\beta})^T (Y - X\hat{\beta}) = \varepsilon^T (I - P_X)\varepsilon$.

□

The sampling distribution (1) of Theorem 8.8 immediately implies

$$\mathbb{E}[\hat{\beta}|X] = \beta, \text{ and } V[\hat{\beta}|X] = \sigma^2 (X^T X)^{-1},$$

which are in line with Theorem 8.3. This is not surprising. Theorem 8.8 assumes the error distribution follows a Gaussian distribution whose mean and variance conditions are assumed in Theorem 8.3. With the Normality assumption, Theorem 8.8 brings us more fruitful sampling distributions.

In the proof of (2) of Theorem 8.8, we use the fact that zero correlation implies independence for Gaussian random variables and vectors. For general continuous random vectors, zero correlation does not imply independence.

Confidence intervals for σ^2 are immediate from (3) of Theorem 8.8. For instance, a two-sided $(1 - \alpha)$-confidence interval is

$$(n\hat{\sigma}^2 / \chi^2_{n-p}(1 - \alpha/2), n\hat{\sigma}^2 / \chi^2_{n-p}(\alpha/2)).$$

Other confidence intervals, e.g., one-sided, can be constructed similarly.

The sampling distributions derived in Theorem 8.8 implies the marginal sampling of individual regression coefficient estimator:

$$\frac{\hat{\beta}_k - \beta_k}{\sqrt{s^2[(X^TX)^{-1}]_{k,k}}}|X \sim t_{n-p}, \text{ for } k = 1, 2, ..., p,$$

where $s^2 = n\hat{\sigma}^2/(n-p)$ is the unbiased estimator for σ^2. The $(1-\alpha)$-confidence interval for regression coefficient β_k for $k = 1, 2, ..., p$, is

$$\hat{\beta}_k \pm \sqrt{s^2[(X^TX)^{-1}]_{k,k}} \times t_{n-p}(\alpha/2)$$

where $t_{n-p}(\alpha)$ denotes the α-upper quantile of t_{n-p}. It is important to note the difference between $[(X^TX)^{-1}]_{k,k}$ and $[X^TX]^{-1}_{k,k} = 1/[X^TX]_{k,k}$. They are equal only when X^TX is diagonal.

Example 8.7 (simple linear regression)**.** Suppose independent data $\{(X_i, Y_i)\}$ are generated from the simple linear regression

$$Y_i = \beta_0 + \beta_1(X_i - \overline{X}_n) + \varepsilon_i, \text{ for } i = 1, 2, ..., n.$$

Due to an earlier example, the maximum likelihood estimates are

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} = \begin{bmatrix} \overline{Y}_n \\ S_{XX}^{-1} S_{XY} \end{bmatrix}_{2\times 1}$$

where $S_{XX} = \sum_{i=1}^n (X_i - \bar{X}_n)^2$ and $S_{XY} = \sum_{i=1}^n (X_i - \bar{X}_n)Y_i$, and $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$.

To derive the confidence intervals, we also need the inverse of the cross product of design matrix:

$$(X^TX)^{-1} = \begin{bmatrix} 1/n & 0 \\ 0 & 1/S_{XX} \end{bmatrix}_{2\times 2}.$$

The $(1-\alpha)$-confidence interval for intercept β_0 is

$$\hat{\beta}_0 \pm \sqrt{s^2/n} \times t_{n-2}(\alpha/2),$$

where $s^2 = n\hat{\sigma}^2/(n-2)$, and the $(1-\alpha)$-confidence interval for intercept β_1 is

$$\hat{\beta}_1 \pm \sqrt{s^2/S_{XX}} \times t_{n-p}(\alpha/2).$$

♢

8.4.3 Hypothesis testing

In linear regression, the default null hypothesis is $\beta_k = 0$, that is, the kth covariate $\{X_{i,k}\}_{i=1}^n$, often a newly introduced effect, has *no* impact on the response. For instance, if we are interested in whether the mean response function is quadratic or linear, the null hypothesis is $\beta_2 = 0$, that is, the regression coefficient on the quadratic term is 0.

To test the null $H_0 : \beta_k = 0$ v.s. the alternative $H_1 : \beta_k \neq 0$, the default testing procedure is the $t-$test with test statistic

$$T_k = \left| \frac{\hat{\beta}_k}{\sqrt{s^2[(X^TX)^{-1}]_{i,i}}} \right|$$

and critical value is $t_{n-p}(\alpha/2)$ for a size α test. The p-value for a two-sided test is

$$2(1 - F_{t,n-p}(T_{obs}))$$

where $F_{t,n-p}(\cdot)$ denotes the CDF of t_{n-p}, and T_{obs} denotes the observed test statistic T_k. Most standard software, e.g. `R`, outputs the hypothesis testing results by default.

Example 8.8. Continue with the `mtcars` example. We first run the simple linear regression and the `R` command in blue. The estimated simple linear regression is

$$\texttt{mpg}_i = 37.2851 - 5.3445 \times \texttt{wt}_i + \hat{\varepsilon}_i.$$

The following standard output reports the t-test statistics, the $p-$values, $\sqrt{s^2}$ and its degrees of freedom (highlighted in red). For instance, we reject the null hypothesis that $\beta_0 = 0$ with a tiny p-value.

```
> lm(mpg~wt, data = mtcars) %>% summary()

Call:
lm(formula = mpg~wt, data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max
-4.5432 -2.3647 -0.1252  1.4096  6.8727

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  37.2851     1.8776  19.858  < 2e-16 ***
wt           -5.3445     0.5591  -9.559 1.29e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.046 on 30 degrees of freedom
```

Then, we run the quadratic regression, and the estimated quadratic regression is

$$\texttt{mpg}_i = 20.0906 - 29.1157 \times \texttt{wt}_i + 8.6358 \times \texttt{wt}_i^2 + \hat{\varepsilon}_i.$$

The following standard `R` output reports the t-test statistics, $p-$values, $\sqrt{s^2}$, and its degrees of freedom (highlighted in red). For instance, we reject the null hypothesis that $\beta_2 = 0$ with a p-value of about 0.003;

```
> lm(mpg~poly(wt,2), data = mtcars) %>% summary()

Call:
lm(formula = mpg~ poly(wt, 2), data = mtcars)

Residuals:
   Min     1Q Median     3Q    Max
-3.483 -1.998 -0.773  1.462  6.238

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   20.0906     0.4686  42.877  < 2e-16 ***
poly(wt, 2)1 -29.1157     2.6506 -10.985 7.52e-12 ***
poly(wt, 2)2   8.6358     2.6506   3.258  0.00286 **
---
```

```
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.651 on 29 degrees of freedom
```

♢

The t-test suffices if our main interest is in the individual coefficients. However, multiple testing (also known as multiplicity) issue arises when the goal is to *jointly* test multiple regression coefficients.

Example 8.9. Suppose in the linear regression with $p \geq 4$, the null hypothesis is

$$\beta_1 = \beta_2 = \beta_3 = 0$$

where the intercept β_0, and other regression coefficients $\beta_4, ..., \beta_p$ are nuisance parameters. ♢

If the H_0 is false, at least one of the regression coefficients is non-zero. When $\beta_k \neq 0$, the corresponding covariate is called *active*.

A few options are available in the literature. For instance, the most general remedy is the Bonferroni method, but it can be too conservative to be useful.

Hypothesis testing theory for Gaussian linear regression treats the linear constraints quite elegantly. Testing jointly multiple regression coefficients fits into the linear constraints framework. Suppose we have k many linear constraints,

$$R\beta = r,$$

where $R_{k\times p}$ and $r_{k\times 1}$ encode the linear restrictions.

Example 8.10. The null hypothesis $\beta_1 = \beta_2 = \beta_3 = 0$ is equivalent to $R\beta = r$ where

$$R = \begin{bmatrix} \mathbf{0}_{3\times 1} & I_3 & \mathbf{0}_{3\times(p-4)} \end{bmatrix}_{k\times p}, \text{ and } r = \mathbf{0}_{3\times 1}.$$

♢

Definition 8.4 (F distribution)**.** Suppose $X \sim \chi^2_n$ and $Y \sim \chi^2_m$ are independent, then their ratio follows the F distribution with degrees of freedom n and m,

$$F = \frac{X/n}{Y/m} \sim F_{n,m}.$$

The ratio $\frac{Y/m}{X/n} \sim F_{m,n}$.

Theorem 8.9. *Suppose data $\{(Y_i, X_i)\}$ are i.i.d. from the Gaussian linear regression*

$$Y_i = X_i^T\beta + \varepsilon_i, \text{ for } i = 1, 2, ..., n,$$

where $\varepsilon_i | X_i \sim N(0, \sigma^2)$ and $X_i \overset{iid}{\sim} \mathbb{P}_X$. The null hypothesis H_0 is

$$R\beta = r,$$

where $R_{k\times p}$ and $r_{k\times 1}$ are both known.

Let $W = (R\hat{\beta} - r)^T(R(X^TX)^{-1}R^T)^{-1}(R\hat{\beta} - r)$. Then, under the null hypothesis,

$$\frac{W/k}{s^2} \sim F_{k,n-p}.$$

Proof. By (1) of Theorem 8.8, the linear transformation $R\hat{\beta}$ has the sampling distribution,

$$R(\hat{\beta} - \beta) \sim N(0, \sigma^2 R(X^T X)^{-1} R^T),$$

where the null hypothesis further implies

$$W/\sigma^2 \sim \chi^2_k.$$

In the proof of Theorem 8.8 (2), we know $\hat{\beta} \perp \hat{\varepsilon}|X$, then W, as a function $\hat{\beta}$, and $\hat{\sigma}^2$, as a function $\hat{\varepsilon}$, are also independent conditional on X.

By (3) of Theorem 8.8 and the definition of F distribution,

$$\frac{W/k}{s^2} = \frac{W/k}{n\hat{\sigma}^2/(n-p)} \sim F_{k,n-p}.$$

□

When $k = 1$, the sampling distribution under the null is $F_{1,n-p}$. Though the F test statistic and t-test statistic are different numerically, the power, size, and p-value of the two tests are identical, as the underlying sampling distributions remain the same.

The F test for the null hypothesis that all covariates are inactive is done by default in most standard software, e.g. `R`. In the output, the F test statistic, degrees of freedom, and the p-value are reported.

Example 8.11. Continue with the `mtcars` example. We first run the simple linear regression. The F statistic is 91.38 and the null distribution is $F_{1,30}$ with only one regression coefficient being tested. Indeed, the p-value of the F test is identical to the t-test.

```
> lm(mpg~wt, data = mtcars) %>% summary()

Call:
lm(formula = mpg~wt, data = mtcars)
....
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  37.2851     1.8776  19.858  < 2e-16 ***
wt           -5.3445     0.5591  -9.559 1.29e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.046 on 30 degrees of freedom
Multiple R-squared:  0.7528,        Adjusted R-squared:  0.7446
F-statistic: 91.38 on 1 and 30 DF,  p-value: 1.294e-10
```

Then, we run the quadratic regression which has one more covariate. The F statistic is 65.64 and the null distribution is $F_{2,29}$ with two regression coefficients being jointly tested. With a tiny p-value, we should reject the null hypothesis. It is reasonable that at least one of the regressors is active.

```
> lm(mpg~poly(wt,2), data = mtcars) %>% summary()

Call:
lm(formula = mpg~ poly(wt, 2), data = mtcars)
```

```
Residuals:
   Min     1Q Median     3Q    Max
-3.483 -1.998 -0.773  1.462  6.238

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   20.0906     0.4686  42.877  < 2e-16 ***
poly(wt, 2)1 -29.1157     2.6506 -10.985 7.52e-12 ***
poly(wt, 2)2   8.6358     2.6506   3.258  0.00286 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.651 on 29 degrees of freedom
Multiple R-squared:  0.8191,        Adjusted R-squared:  0.8066
F-statistic: 65.64 on 2 and 29 DF,  p-value: 1.715e-11
```

♢

8.4.4 Predicted/confidence band

If σ^2 is known, (1) of Theorem 8.8 implies the (conditional) sampling distribution of $X_i^T\beta$ is

$$X_i^T\hat{\beta}|X \sim N(X_i^T\beta, \sigma^2[P_X]_{i,i}),$$

where $P_X \equiv X(X^TX)^{-1}X^T$ is the projection matrix onto $\mathcal{C}(X)$. If σ^2 needs to be estimated from data, Theorem 8.8 implies

$$\frac{X_i^T\hat{\beta} - X_i^T\beta}{\sqrt{s^2[P_X]_{i,i}}}|X \sim t_{n-p},$$

and the $(1-\alpha)$-confidence *interval* of the fitted value $X_i^T\beta$ conditional on X is

$$X_i^T\hat{\beta} \pm \sqrt{s^2[P_X]_{i,i}} \times t_{n-p}(\alpha/2).$$

Similar to the fitted value, the point prediction (a point estimator) for a new observation Y_* with the new covariate X_* is $X_*^T\hat{\beta}$, and the *exact* $(1-\alpha)$-confidence interval for the mean response $X_*^T\beta$ is

$$X_*^T\hat{\beta} \pm \sqrt{s^2X_*^T(X^TX)^{-1}X_*} \times t_{n-p}(\alpha/2).$$

However, the *exact* $(1-\alpha)$-confidence (prediction) interval for Y_* is

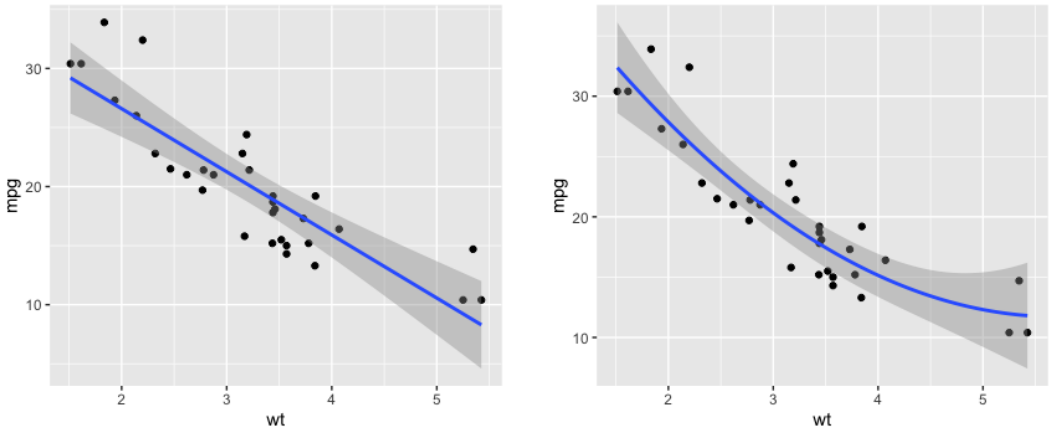
$$X_*^T\hat{\beta} \pm \sqrt{s^2\left(1 + X_*^T(X^TX)^{-1}X_*\right)} \times t_{n-p}(\alpha/2)$$

which is a wider interval than the prediction interval of the mean response $X_*^T\beta$. This is because the model-based prediction $Y_* = X_*^T\beta + \varepsilon_*$ has two sources of uncertainty:

- the uncertainty in the mean response, since we do not know the true β, and
- the uncertainty in the error term, since we do not know ε_*.

Combining the *individual* confidence *intervals* for all observations makes a confidence *band*. The joint coverage rate of the individual (marginal) confidence intervals with $(1-\alpha)$-confidence is not necessarilly $1-\alpha$ due to the dependence among them. It is optional in many software such as `R` to plot the confidence band with a confidence level. The confidence level specified in the software is for individual confidence intervals, and *not* for the whole dataset.

Example 8.12. Continue with the `Motor Trend Car Road Tests` example. In this example, we plot the *exact* 95% confidence band for the mean response $m(X_i) = X_i^T\beta$.



simple linear regression quadratic regression

Figure 8.6: Linear models of `wt` and `mpg` with error band

```r
ggplot(mtcars,aes(x = wt, y = mpg)) + geom_point()+ #xlim(0,6)+ ylim(5,35)+
  geom_smooth(method = "lm", formula = y~x, se = TRUE, level=0.95)
ggplot(mtcars,aes(x = wt, y = mpg)) + geom_point()+ #xlim(0,6)+ ylim(5,35)+
  geom_smooth(method = "lm", formula = y~poly(x,2,raw=TRUE), se = TRUE, level=0.95)

proj = function(X){
  return( X %*% solve(crossprod(X)) %*% t(X))
}
response = mtcars$mpg
design = cbind(intp = 1, x= mtcars$wt)
n_p = nrow(design)-ncol(design)
## design = cbind(1,x = mtcars$wt, mtcars$wt^2) ## for quadratic regression
beta = solve(crossprod(design)) %*% t(design) %*% response
yhat = proj(design) %*% response
resid = response - yhat
diagPx = diag(proj(design))
s2 = sum(resid^2)/(length(resid) - ncol(design))
se = sqrt(s2 * diagPx)

plot(mtcars$wt, mtcars$mpg, pch = 20, xlim = c(0,6), ylim = c(5,35))
abline(a = beta[1],b=beta[2],lwd=1)
perm = order(design[,"x"])
lines(design[perm,"x"], yhat[perm] + qt(0.975 , n_p) * se[perm] )
lines(design[perm,"x"], yhat[perm] - qt(0.975 , n_p) * se[perm] )
```

♢

9 Concluding Remarks

9.1 Statistical Wisdoms

Stigler (2016) describes seven statistical wisdoms which are paraphrased as follows,

1. data reduction or compression.
2. $\sqrt{n}-$rule, where n denotes the sample size, implies that the marginal value of obtaining additional sample diminishes.
3. likelihood principle that connects probability theory and inference.
4. Intercomparison which uses the same data to evaluate the estimation accuracy.
5. Regression to the mean and multivariate analysis.
6. Experiment design that leverages random assignments to provide causal answers to many scientific questions.
7. Residual diagnosis that help understand and assess the model fit.

Despite inference is only one component of the Box's loop, most statistical wisdoms are related to statistical inference. This course intends to provide a *modern* treatment of statistical inference and cover both classical (frequentist) and Bayesian inference. A side objective of this course is to better appreciate the statistical wisdoms.

9.2 Statistics study plan

Foundational courses

- Linear models
- Bayesian statistics
- Causal inference
- Statistical computing
- Stochastic processes
- Nonparametric statistics

Practical courses

- Categorical/discrete data
- Spatial data
- Time series data
- Biological data, genomics data

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